

Foreign Resources and Economic Development

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Sean Gallagher

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PREFACE

Economic development occurs when all segments of the society benefit from the fruits of economic growth through economic efficiency and equity. Economic efficiency will be present with minimum negative externalities to society, including agency, transaction, secondary, and opportunity costs. At the same time, disintegration of national sovereign states into more fragmented nations along the ethnic lines would not help these newly formed societies to accede to a formidable economic development regardless of their form of government.

Economic development is a term used to describe concerted actions taken for the purpose of economic progress within a specific area. It is the process of generating and sustaining wealth in a community. Every community has unique assets that can be used to attract employers and investment. Building on these assets is one of the best ways communities can stimulate their economies.

An underlying concept in planning tourism is that tourism should be viewed as an inter-related system of demand and supply factors. The demand factors are international and domestic tourist markets and local 'residents who use the tourist attractions, facilities and services. The supply factors comprise tourist attractions and activities, accommodation and other tourist facilities and services. Attractions include natural, cultural and special types of features - such as theme parks, zoos, botanic gardens and aquariums - and the activities related to these attractions. Accommodation includes hotels, motels, guest houses and other types of places where tourists stay overnight. The category of other tourist facilities and services includes tour and travel operations, restaurants, shopping, banking and money exchange, and medical and postal facilities and services. These supply factors are called the tourism product. Other elements also relate to supply factors. In order to make the facilities

and services usable, infrastructure is required. Tourism infrastructure particularly includes transportation (air, road, rail, water, etc.), water supply, electric power, sewage and solid waste disposal, and telecommunications.

The great revolution, from an economic standpoint, which followed the Palaeolithic Age with the beginnings of the Mesolithic and Neolithic periods, is clearly revealed by detailed economic analysis which has rarely been traced farther back than this new period by our economists. For it was then for the first time that the pattern of the so-called domestic economy of Professor Spiethoff began, and spread practically over the whole globe. Many archaeologists and ethnologists, on the other hand, have attempted to explain the various differences between the economic pattern of the Palaeolithic Age and that of the succeeding periods, and named as new and revolutionary moments in economic progress either such advances in the technique of labour as hoe agriculture, cattle economy, plough culture, the division of labour into agricultural, industrial, and trading professions, or new types of settlement which gradually developed to permanent villages. Furthermore progress could be traced in the improvements in technique which appeared in the hoe, plough, potter's wheel, cart, sledge, ship, and many other implements.

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An exploration of the differing impacts of and responses to tourism in a range of political, social, economic and environmental contexts within Europe, evaluating current trends and policy issues and options.

– *Sean Gallagher*

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State and Economic Development

THE ROLE OF PUBLIC ENTERPRISES IN THIS SYSTEM

Although other stages of this conference will cover this subject more fully, I shall add a few comments in an attempt to reconstruct the role of public enterprises within the now unlearned logic of pre-Ricardian economics. In this part, I shall largely follow the subject as treated by Werner Sombart. Unfortunately Sombart's profound understanding of the growth of the capitalist system is mainly accessible in German. This problem of inaccessibility is strengthened by the most unfortunate neglect of pre-war German economic theory, a neglect today apparently promoted by the Germans themselves.

This nation has all reason to be proud of their heritage in economic theory, but Germany seemingly attempts to unlearn this tradition collectively together with less fortunate aspects of the past. This sad state of affairs, coupled with the US misconceptions — bordering on denial — of their own nineteenth century economic theory and economic policy, in my view bereaves the world of the only existing fully-fledged and viable alternative to the neo-classical tradition.

Sombart makes the point that the reasons given for the establishment of State-owned industries in the historical record do not give justice to their importance:

- The importance of for the creation of modern industry cannot be overestimated. These served to set, not only a prototype example of industry, but also the pace and pattern for the new form of organisation. It was the State-owned

enterprises which, due to the demands they created, often served as catalysts for the development of capitalist industries. These enterprises are so essential that they cannot be left out of an account of the development of capitalism, which — although their conceptual roots lay elsewhere — they furthered in thousands of ways.

The early Adam Smith explains how government intervention is made to help neither the consumers nor the producers, but for the interest of the system, the State, the common weal. Similarly, state-owned enterprises were primarily established in the interest of the State.

The following were the main motivations:

- *Administrative Reasons:* On occasion the State administration found that, in certain industries, the only practical way of carrying out the necessary controls was by State ownership. This was most often done for reasons of financial policy, particularly in the case of mining of precious metals.
- *To Meet the Needs of the Ruler and the Court:* Many state-owned manufactures were set up seemingly with the sole objective of providing luxury goods to the King. Rulers and ruled took pride in showing off the skill level and production capacity of their nation — be it in producing china, silk or tapestries — seemingly much in the same way nations are proud of their athletes today. The role of the State in providing demand for skills — of the State as a Porterian demanding customer — the effect of the mass production of luxuries clearly had an effect of making such goods more accessible further down the social ladder, and, more importantly, of building up technical and managerial skills which spread to other activities.
- *For Reasons of Statesmanship/to Further the Common Weal:* The State intervention was often made under this heading. Reasons given were those of creating employment, increasing exports, reducing dependence on foreign nations, etc. Often State ownership appears as “entrepreneurship of last resort”. Also tariffs were used to promote industries out of the same logic, the State trying to establish industries with the minimal tariff which would attract investors. Certain industries were seen as being mandatory for the common weal, and this results in the obvious pattern that the most backward nations tended to have the highest tariffs.
- *For Reasons of National Defence:* The importance of national defence for technological development has been enormous — from the earliest history to the contribution of Reagan’s Star War programme to cellular telephone communication. A main challenge to statesmen of today is to devise mechanisms of financing technological developments other than as unintended by-products of warfare. To this point we refer to Sombart’s “Krieg und Kapitalismus”. Obvious State industries in this respect are gunpowder and armaments, but also industries closer to the core of industrial and commercial

development, like iron foundries and shipbuilding. However, the technologies and logistics needed to supply the huge armies also contributed to develop skills, scale, and scope in the State-owned production and distribution of goods like bread and clothing. The shortage of natural raw materials for the purposes of war is clearly a most important reason for research and inventions in the field of synthetic materials.

State-owned industries seems to have been part of the role of the “obligatory passage point” played by a strong and active State. These companies were — as Sombart emphasises — “places of learning”, not only for technical skills, but also for organizational and social capabilities. These are deep and elusive, but nevertheless indispensable skills for the process of economic development. Military demand seems to have had the same positive effect in the USA as in Europe. Ely Whitney financed the development of the revolutionary “cotton gin” from the profits from selling rifles to the Revolutionary Army, and the whole concept of production with interchangeable parts — paving the way for industrial mass production — originated in the production of rifles in the USA.

However, in the USA a much larger role was played by private entrepreneurs than in Europe. Although in their trade policy and in terms of “getting the nation into the right business” the USA was clearly in line with Europe, we here face a difference in historical development. We can only speculate if this was because of a considerably better supply of entrepreneurship in the USA — a crucial factor in economic development — if it was a result of a later industrialisation when a historical prototype of industrialisation was evident in Europe, of ideological preference, of having a more developed industrial structure before really threatening foreign wars appeared in US history or all of the above.

Clearly the starting point of many technological trajectories was statal or parastatal demand, not only from war but also from the construction of infrastructure, power generation, a national telephone system, etc. Examples abound: that the automotive industry in Japan has its origins in the demand for trucks for the Imperial Army is a typical one. Import substitution based on reasons of national security seems to have played an extraordinarily important role through the centuries. Over time this led to a convergence of industrial structure, establishing a common platform of technological knowledge in all advanced countries.

With this relatively similar industrial structure as a starting point — from this common platform — the process of specialisation and globalisation today mutually benefits the advanced countries. The need for national security supplies forced nations into increasing return activities, even though their natural comparative advantage may originally have been elsewhere. National security considerations helped create a new comparative advantage in the production of skill-intensive increasing return goods.

Our problem today is, however, that having unlearned this historical insight, we pretend that the nations which have not been through this process of cumulative knowledge-

building — the nations which have not reached this platform of knowledge — will equally benefit from the globalisation process. In terms of skills and scale, this common platform of knowledge creates the basis for a symmetrical pattern of international trade among the advanced countries, of a pattern of specialised trade between people with similar levels of skill, like that of lawyers trading with physicians. The asymmetrical trading relationships between the North and the South are more like a nation of skilled lawyers trading with a nation of unskilled lettuce pickers — there are no logical reasons why any factor-price equalisation should take place between the two, unless some technological development opens up for lettuce-picking profitably to absorb as much human capital as the study of law.

In the existence or not of a common “platform of technological knowledge and organisational skills” seems to lie the main historical difference separating the two convergence groups in the world economy — the rich “North” and the poor “South”.

EXOGENIZING THE ENGINES OF GROWTH: ADAM SMITH AND THE LOSS OF KNOWLEDGE, INSTITUTIONS AND SYSTEMIC EFFECTS IN ECONOMIC THEORY

At the time of the publication of Adam Smith’s *The Wealth of Nations* in 1776, no doubt the mercantilist system had developed from its Renaissance origins of a dynamic and wealth-producing quest for dynamic rents from new knowledge, to a system often characterised by static rent-seeking. Adam Smith’s prescription of a heavy dose of market was no doubt a correct one. However, Adam Smith did much more than prescribing more market and less intervention.

He produced a system which removed the very engines of growth from the previous world view:

- He efficiently argued that knowledge was of no value either to society or to the individual. This counterintuitive reasoning is retained in today’s economic theory when it predicts equalisation of wages between nations exchanging goods produced at extremely different skill levels.
- His theory removed human institutions. Exhibits a powerful aesthetic aversion to any type of collective action, a visceral distaste bordering on revulsion. For him “human institutions” so invariably produce “absurd” results that they have no presumptive legitimacy.
- His atomistic view removed all systemic effects and constructs a theory of individuals void of any uniqueness, all governed by their “human nature” which is the same in all human beings.

The paradox of Adam Smith is that he saw an important Renaissance-type systemic effect — the division of labour — as being the source of wealth, an idea which was

there both in Serra and very explicitly in Petty. Yet, Smith's other assumptions — atomism, no increasing returns, no institutions, no new knowledge — ousted this key systemic insight forever from classical and neo-classical theory. Says US economist George Stigler: "Almost no one used or now uses the theory of division of labour, for the excellent reason that there is scarcely such a theory... There is no standard, operable theory to describe what Smith argued to be the mainspring of economic progress".

The loss of human knowledge, of institutions, and of any notion of systemic common weal in economic theory is, in this writer's opinion, a fundamental reason for our failure to capture why the world polarises into two convergence groups of nations — one wealthy and one poor. We argue for the importance of recognizing that what got Mankind from the cold and draughty caves into economic welfare was more than the discovery of barter and of "getting the prices right" — that it required immense addition of knowledge and skills. A basic assumption running through pre-Smithian economics was that different economic activities contributed differently to the common weal — that the uneven advances of human knowledge were intimately tied to the process creating uneven growth.

The professions which absorbed more knowledge — embodied or disembodied — were seen as the carriers of the common welfare of a nation. On this background it is interesting to go through Adam Smith's works with one particular question in mind: What is the role of knowledge in Adam Smith's system? Adam Smith explains what causes differences in wages between professions — *i.e.* in a national economy. Wages constituting more than 70 per cent of GNP in a modern industrial nation; differences in national income between nations are principally a result of differences in wages, rather than differences in profits.

We would further argue that, with an international economy with increasing factor mobility, the distinction between a theory of international trade as opposed to a theory of national trade is losing its importance. The only aspect which divides a theory of international from a theory of national trade is the question of factor immobility. Therefore Adam Smith's answer to the question as to which circumstances, "either really, or at least in the imagination of men, make up for a small pecuniary gain in some and counter-balance a great one in others", is of great interest. Smith lists five reasons why some people are paid better than others. The list is remarkable in that in each and every point raised, Adam Smith goes out of his way to explain why human knowledge and human skills do not produce a higher standard of living than ignorance — neither to society nor to the individual.

If people with more knowledge and more skills have higher incomes — which was also observable at the time of Adam Smith — it is never due to the fact that skills and knowledge produce value, but due to one of the following five reasons:

- *Wages Vary with the Agreeableness of the Employment:* For this reason, "the most detestable of all employments, that of the public executioner, is, in proportion to the quantity of work done, better paid than any common trade

whatever". Under this point Adam Smith also discusses why human skills and talent are often very well rewarded — attempting to explain what he sees as the "exorbitant rewards" of artists, "opera-singers andc.". The rewards to these talents are to Smith a direct result of "the discredit which attends the employment of them as the means of subsistence."

To Smith the fact that society rewards extraordinary talent is a direct result of the fact that "we despise their persons". "While we do the one, we must of necessity do the other." "Should the public opinion or prejudice ever alter with regard to such occupations, their pecuniary recompense would quickly diminish". Smith argues that if we would just stop despising our actors, artists and sportsmen, their incomes would fall to the level of an agricultural labourer. His system does not allow for a pecuniary reward which is coupled with admiration — his natural system has to pair "high reward" with "despise".

- *Wages Vary with the Cost of Learning the Business*: Smith makes it very clear, however, that "the cost of apprenticeship accounts for the wages of manu-facturers being higher than those of country labour". There are therefore no advantages to manufacturing over agriculture; although the earnings in manu-facturing "may be somewhat greater, it seems, however, to be no greater than what is sufficient to compensate the superior expense of their education". In other words, the mercantilist tradition that nations who export the products from professions of higher skills will be wealthier than nations exporting products with low skills is here strongly denied. From the point of view of both society and the individual, adding knowledge to labour is, in Smith's system, clearly a zero-sum game.
- *Wages Vary with Constancy of Employment*: For this reason, professionals like masons and bricklayers who "can neither work in hard frost and foul weather", and who are not secured constant demand for their services, will have a higher wage than people who are permanently employed. "The high wages of these workmen, therefore, are not so much the recompense of their skill, as the compensation for the inconsistency of their employment". Again, any role of skill and knowledge is denied.
- *Wages Vary with the Trust to be Reposed*: Some professions — Smith mentions goldsmiths, physicians, lawyers, and attorneys — are higher paid because of the "great trust which must be reposed in the workmen". We have to have confidence in these people, says Smith, and the reason we pay them more is that we do not have confidence in people who are not well paid. "Such confidence could not safely be reposed in people of a very mean or low condition. Their reward must be such, therefore, as may give them that rank in society which so important trust requires". To Adam Smith, in other words, we do not pay lawyers and doctors better than people who wash dishes because

their skills are more valuable, but because we have to trust these people, and could not dream of having confidence in people from the lower classes of society.

- *Wages Cary with the Probability of Success*: “Put your son apprentice to a shoemaker, there is little doubt of his learning to make a pair of shoes: But send him to study law, it is at least twenty to one if ever he makes such proficiency as will enable him to live by his business.” For this reason, Adam Smith looks at the skilled professions like a lottery: “those who draw the prizes ought to gain all that is lost by those who draw the blanks”. Since only one in 20 lawyers makes something out of their profession, this one lawyer “ought to receive the retribution of his own so tedious and expensive education, but that of more than twenty others who are never likely to make anything by it”. Again, to society, knowledge is a zero-sum game.

As opposed to today’s economists, Adam Smith is consistent in carrying his anti-mercantilist theory down to the family level. Few economists today recommend their children to get a job washing dishes rather than to go to university, using the argument that “factor-price equalization is just around the corner anyway”. Privately — but not at the national level — today’s economists not only see the value of human capital, but they also accept US economist Daniel Raymond’s most important point from 1820: Different professions have different capacities profitably to absorb capital— different professions have different “windows of opportunity” for creating welfare.

One cannot profitably add as much human capital to the job of washing dishes as to the job of being a lawyer. For this reason economists would recommend to their children professions which require a university education — although by doing this they express what they — at the level of a nation — would describe as “a mercantilist preference for one profession to another.” Adam Smith, however, is very consistent on this point: all risks consider it is safer to let your son become a shoemaker’s apprentice than to become a lawyer.

Today’s economic theory addresses the problem of differing absorption capacities of knowledge by assuming that all activities have the same production function — that the necessary extension of skills is automatically brought to life by the addition of capital to labour. This describes a process of creating economic growth which is similar to the process of adding water to instant coffee: one assumes that the knowledge necessary in order to increase productivity is already there in equal measure in all human activities.

Adam Smith further argues that the choice of profession is of no importance, and he dispenses with skill and knowledge as economic factors based on two assumptions:

1. He assumes that labour comes from the same pool of people: “If, in the same neighbourhood, there was any employment either more or less advantageous than the rest, so many people would crowd into it in the one case, and so many would desert it in the other, that its advantages would soon return to the other employments.”

2. He assumes that skills can be learned extremely fast. Investing in machines and instruments may take a long time, says Smith, “but when both have been fairly invented and are well understood, to explain to any young man, in the completest manner, how to apply the instruments and how to construct the machines, cannot well require more than the lessons of a few weeks: perhaps those of a few days might be sufficient. In the common mechanical trades, those of a few days might certainly be sufficient.”

Adam Smith’s discussion on what causes a difference in retribution between professions leads up to a severe criticism of the English statutes of apprenticeship. These dated from Elizabeth I — from the time of the cultivation of knowledge *per se*, which provided apprenticeships of up to seven years. Adam Smith saw apprenticeships as an extreme waste, since he is of the opinion that everything could be learned “in a few days”. Unlike some later economists, like Alfred Marshall, Adam Smith was not a practical man — he “seemed the unlikeliest of guides to the practical world” in the words of Harvard Business School professor Thomas McCraw. In both Adam Smith and in neo-classical theory there is in some fundamental way a contradiction between the notion of perfect markets and the way the economy adds knowledge. This problem spills over to how economic theory today explains profits.

There is no incentive to produce new knowledge in perfect markets — the possibility of appropriating the fruits of new knowledge is absent. By letting new knowledge enter the system like “manna from heaven”, the very engine of growth — the search for new knowledge which creates imperfect competition — is excluded from mainstream theory. For this reason, an understanding of how different degrees of imperfect competition is caused by conditions of production is at the very core of any understanding of economic growth. Economic theory, however, does not have a relevant theory of production, and of the role of human knowledge in this process.

Thorstein Veblen in his “The preconceptions of economic science” says it this way: “To sum up: classical economics, having primarily to do with the pecuniary side of life, is a theory of the process of valuation”. Production is left out. In the words of Werner Sombart: “There is like a tacit agreement in that one has reached the conviction that the science of economic life, in so far as this is studied by the economics profession, is a science of the circulation and distribution of goods”. Did Adam Smith believe that the invisible hand would produce factor-price equalization? Hardly.

In *The Theory of Moral Sentiments* Smith explains how the invisible hand promotes income distribution. However, to a modern ear, his arguments are hardly convincing. Smith bases his argument on the principle that the rich have a limited capacity to consume, and therefore are forced to share with the poor: “It is to no purpose, that the proud and unfeeling landlord views his extensive fields, and without a thought for the wants of his brethren, in imagination consumes himself the whole harvest that grows upon them. The homely and vulgar proverb, that the eye is larger than the belly, was never more fully verified than with regard to him. The capacity of his stomach bears no

proportion to the immensity of his desires... The rest he is obliged to distribute... The rich... consume only little more than the poor. They are led by an invisible hand to make nearly the same distribution of the necessities of life, which would have been made, had the earth been divided into equal portions among its inhabitants...”.

THE LOSS OF THE STATE AND THE REVENGE OF THE CENTAUR

One of the most moving paintings of the Renaissance is Sandro Botticelli's Pallas and the Centaur, painted in the late 1480s. This work shows Pallas — the ancient goddess of wisdom — mildly and somewhat condescendingly caressing the head of a centaur, symbolising Man's past of ignorance and violence. The expression of embarrassment on the centaur's face is precious — he is so obviously ashamed of his wild and ignorant past.

The symbolism — the victory of reason, wisdom and peace — comes across in an atmosphere densely filled with harmony, a feature which makes Botticelli's works so unique. The economists of that time — and pre-Smithian economists in general — seem to have been aware of this ongoing battle between the wisdom and peace of Pallas and the violent ignorance of the centaur. Channeling the passions and energies of the centaur into productive — rather than destructive — activities was a main challenge of the Renaissance. This was a matter which could not be left to any self-organising invisible hand.

The prominent and influential German economist J.F. von Pfeiffer — an ardent antiphysiocrat — puts it this way: “You can make of human beings what you want. The way he is governed commends Man to good, or to evil. The advantage of the nation must be the advantage of its members. He who separates these advantages is an ignorant who damages the public body. This paragraph in Pfeiffer summarises two important aspects of the pre-Smithian development state. The quality of the “system” makes Man good or evil, and as in Chandler's works on managerial capitalism, there is a conscious and visible hand — Man's wit and will — which holds this system together. The second point, which Pfeiffer saw as being connected, belies the present Anglo-Saxon notion that pre-Smithian economics — and especially German cameralism — collectivistically subdued the interest of the individual to the interest of the State. Just as in the quote from the early Adam Smith on the cover of this chapter, the overall growth of the system prevented the need for any static concerns for Pareto optimality: “The incoming tide lifts all boats”.

The invisible hand starts working when conditions have been created bringing the vested interests looking for private rate of return in line with the interests of the public rate of return — when, like Pfeiffer says, what is in the interest of the individual also is in the interest of the collectivity. This happens when a critical mass of labour in a nation is employed in activities subject to increasing returns, and when the fruits of these

increasing returns are allowed to filter out, to a large extent, as higher monetary wages inside the producing nation, rather than as lowered prices for foreign customers. In this way the core of society escapes from the static economics of zero-sum games: where the gain of one individual is the loss of another. Such activities — which “raise all boats” — are the starting point of what the historians of technology call the process of “historical snowballing” — of the virtuous circles of development.

Under such conditions — in increasing return activities and when appropriate mechanisms for the distribution of dynamic rents have been established — the saying attributed to Alfred P. Sloan is true to a surprising extent: “What is good for General Motors is good for the United States.” The metaphor of “the invisible hand” was frequently used by Adam Smith in his works on astronomy and in *The Theory of Moral Sentiments*.

It is only once mentioned in *The Wealth of Nations* — and in a context which can be taken to indicate an awareness of just such a process of historical snowballing. The chapter in question starts with a description of how protective duties have created a number of industries in Great Britain. Smith thereafter describes a centripetal system whereby merchants — because they prefer to trade closer to home — bring their capital there. This argument is really the description of the start of a virtuous circle. It is this tendency, for reasons of risk/distance, of capitalists to support domestic rather than foreign industry — implicitly assuming that such an industry exists — which is the starting point of the beneficial work of the invisible hand in Smith’s system.

It can be argued that Smith uses the “invisible hand” in the context of historical snowballing. But it is most important to keep in mind that the starting point of this system is — in Smith’s narrative — the fact that all industries of Great Britain were created by mercantilist protectionism. The negative counterpart of historical snowballing produces poverty, violence and ignorance. In the 1490s, Botticelli — who was also a novelist — experienced the Renaissance backlash of Savonarola’s violent revolution against the Medicis in Florence, when the books authored by him were burned.

In the late twentieth century we are also experiencing the comeback of the centaur — of Man’s ignorant and violent past: in Somalia and Liberia the State as a civilizing agent, for all practical purposes, has ceased to exist. In Rwanda and Burundi tribal wars commit mutual genocide. A similar “system of unreason” bothered Keynes during the First World War: “When reason dies, monsters are born.” No doubt these processes are partly produced by diminishing returns in agriculture in a situation of population pressure, when there is no employment outside the diminishing return sector.

In this case, the failure of neo-classical economics to distinguish between increasing and diminishing return activities seems to be closely linked to the failure of the international community to improve the situation. This retrogression of civilization suggests the fragility of the veneer of what we would call a civilized society which holds the primitive centaur at bay. Today seemingly more and more countries are threatened by the process of “Somalisation” — the return to power of what the

international press descriptively labels as “war lords.” The same development threatens part of the former communist nations. In Russia we can only be baffled by the fact that a notoriously inefficient planned economy managed to produce a GNP per capita which was clearly considerable higher than what the “free” market is able to produce today.

We probably cannot continue to explain this through the new science of “transitology” for much longer. This proves the point of nineteenth century mainstream US and German economics: having a relatively inefficient industrial sector — in the process of being educated to graduate into participation in a bigger market — is infinitely better than not having an industrial sector at all. There is no doubt that Renaissance economists would have put together a very different strategy for the transition of the former communist nations from that which has been done in the 1990s: all they would have established an appropriate legal system and adopted long-term mechanisms to save and upgrade the manufacturing industries.

Deindustrialising nations is equal to removing most of the increasing return activities which increase the common weal system-wide — the activities which not only create isolated pockets of welfare. Reducing nations to resource-based activities and to a traditional service sector only is tantamount to “bombing them back to the Stone Age.” Today economic policy too often lets the ideological criterion of a nation’s “openness” to foreign imports overrule the concern for keeping increasing return activities in business.

This not only reduces local GNP and world welfare; such a policy also creates the kind of nature-based poverty which is the main threat to global sustainability. People and nations who are not in the position to make a living from their knowledge may easily end up with no other alternatives for survival than making a living raping the environment. And these retrogressing countries are “market economies.”

It is difficult to blame the state of affairs on excessive government intervention or on their “having got the prices wrong”. Cohn’s nineteenth century explanation, essentially that we have unlearned that societies have to go through a process of Renaissance-type State building, is still fundamentally correct. We argue that a strong State — leading a nation into increasing return activities — historically has been an obligatory passage point in the development of any nation. If this is the case, to what extent does a world economic order based on an ahistorical and atomistic neo-classical economic theory — with its thesis of futility, perversity and jeopardy — contribute to the “barbarisation” which we are witnessing? In our view, quite a lot.

We see the present zeitgeist as being a product of a few very fundamental influences — of a “handful of metaphors”, to use McCloskey’s term. We are still victims of the excesses of Adam Smith’s criticism of a mercantilism which had lost much of its original content and had degenerated into protecting monopolies rather than creating and protecting knowledge. Our ideas are still locked into a pattern shaped by a 100-year fight with communism, when the Utopia of communism fought the Utopia of factor price equalisation — a fight which led to the demise of all alternative economic theories

after the Second World War. The glorification of Adam Smith's primitive faith in the invisible hand of Providence — rather than in the rational will of Man — is in our view the product of a most unfortunate combination of events: the assumptions needed in order to turn Adam Smith into an unassailable theoretical bastion against communism coincided fully with the assumptions which were needed in order to convert economics into mathematics. In a world where the increasing disparities between rich and poor are clearly attributable to imperfect knowledge, imperfect competition and increasing returns, the assumption of perfect competition is powerfully cemented.

The welfare of the rich nations is clearly a product of a dynamic "industry rent" which, in the short term, is not hurt by neo-classical adjustment policies. However, the same adjustment policies in a relatively backward economy will kill off what little there is of advanced economic activities in these nations — through what we have elsewhere called the Vanek effect — and destroy welfare. The World Bank and the IMF tend to use the criterion of "openness of the economy" as the main criterion for success. Clearly an advanced nation with high skills is better off in an open economy than in a closed one. A favourite example is the success of Chile as an open economy — after a century of natural and State-sponsored protection.

To this Friedrich List and the nineteenth century US economists would have unanimously responded: of course, Chile's advanced economy — especially when seen in relation to those of her neighbours — was mature to graduate into the world economy. But "Openness" as a criterion for success is meaningless without a reference to the degree of skills contained in a nation's exports. "If the rod be bent too much one way, says the proverb, in order to make it straight you must bend it as much the other", says Adam Smith. This proverbial maxim seems to apply to the general attitude towards the State after the fall of the planning paradigm: since the market economy was clearly "wrong" and is dead, we can continue to assume that markets must be perfect and atomistic. In such a framework it is difficult to find a place for the State in the promotion of economic development.

It is not clear today how a reasonable balance can be restored. It is a very long way from the appearance of theoretical "toy models" showing the importance of increasing returns — as we see today — to a World Bank interference with the market through a systematic nourishing and protection of such activities in the Third World. The ideological barriers seem difficult to overcome. When science is wrong, the burden is unequally distributed. During the centuries, when bleeding the patient was a cure-all in the medical science, the weakest were the ones whose situation deteriorated the most.

Similarly, in a world economic order based on an atomistic economic theory void of human institutions, on perfect competition and the absence of increasing returns, where "deregulation" and "openness" to imports are seen as a cure-all, the poorest nations are again the ones who suffer. We are now again facing a battle which will prove to be at least as long and unrelenting as Keynes' battle against the monetary orthodoxy. Many nations of the world are trapped in an "underdevelopment equilibrium" which is not

unlike the “unemployment equilibrium” of the interwar period. To get the world out of “the unemployment equilibrium”, Keynes had to slay the dragon defending the gold standard. Belatedly, economists came to recognize, with Keynes, that national welfare can be positively influenced by monetary stimuli: the thesis of futility, perversity and jeopardy had been proved wrong.

What is the formula to get a large part of the world population out of this “poverty equilibrium”? E.A.G. Robinson recalls the “burning sense of the world’s stupidities” which animated Keynes as a lecturer on the economic situation of his day. Keynes, slayer of the foregoing dragon, the recipe for gaining long-term influence in the face of massive theoretical disagreement is “violent and relentless truth-telling”, which will work “even if slowly”.

This time the battle is carried inside the “black box” of economics, in order to show that the contents of the black box — the “quality” and diversity of the activities in which a nation is engaged — are perhaps the main determinant of the standard of living of nations. In addition to this, a relatively equitable income distribution is an indispensable factor, but income distribution without the nation being in the right activities is a futile exercise. Economic development should be seen as the product of a set of “reactants”, all of which have to be present in order to set off the desired reaction leading into “autocatalytic” — self-reinforcing — systems.

The dragon which has to be slain is the dragon defending the assumptions of perfect information, perfect competition and constant returns to scale — a set of assumptions which must be got rid of en bloc — not cautiously investigated one by one and then put back into the theory as is done in new growth theory.

The result will, in my view, confirm the common sense of Daniel Raymond’s 1820 thesis on the US economy updated to the global economy: the factors which determine the differences in standard of living between nations are essentially the same factors which determine the differences in standard of living within nations: levels of knowledge and skill which create successions of temporary oligopoly power based on dynamic imperfect competition.

However, the spread effects of the leading sectors are such that the standards of living of the vast majority of wage earners of a nation, in the traditional service sector—whose efficiency is virtually the same in rich and poor nations — are determined by the skill level of the export sector.

A Bolivian barber and bus driver are just as efficient as their Norwegian or German counterparts, but the systemic effects of the world economy is such that their real wages are just a fraction of that of their First World counterparts. This is a result of what we have called the collusive spread of the benefits from technological change — the fact that these benefits in large part are taken out as higher real wages in the producing countries rather than being distributed in the classical way, as lowered prices to the world markets. For this reason, a more equitable distribution of world income cannot be achieved by distributing income as such — which inevitably will result in putting

whole nations on the dole. A better world income distribution can only be achieved by distributing production of skill-intensive “high quality” activities — those which produce systemic feedback systems in the local labour market — to poor countries.

This is also the activity-specific lesson from The Asian Tigers. This is a dynamic consideration beyond any concern of static “Pareto optimality” — by increasing the skills of the “laggard” countries by distributing production, they will buy more, not less, from the First World.

This was the point made by the defenders of the nineteenth century “American System of Manufacturing” in their discussions with English economists: the US demand for imports would increase as a result of more industrialisation, this demand would be upgraded to more skill-intensive thesiss.

They proved to be right, and proved their point that mutually optimal solutions — what Henry Carey called the harmony of interests — in the case of asymmetrical economic development, can only be found with the help of a visible hand of the State, which makes a conscious economic policy based on theories which have internalised the causes of uneven economic growth. Successful State interventions have created dynamic imperfect competition — a process of dynamic and collective national rent-seeking — which increases enormously the size of the pie.

The State has given temporary help, and with clear strings attached. Unsuccessful State interventions may at a first glance look similar to the successful ones, *e.g.* because also they protect national manufacturing industries. However, the less successful State interventions — as in India and many Latin American nations — have created static rent-seeking through relatively permanent protection with no strings attached, leading to a “shallow” industrialisation, to inefficient monopolies with a limited potential to increase the pie.

In spite of their inefficiencies, however, it is important to keep in mind that a relatively inefficient industrial sector produces a much higher GNP than having no industrial sector at all — this is one of the lessons of the “transitology” of former communist states. Deindustrialising inefficient nations is not a viable option if one has a minimum concern for human welfare.

This dilemma is very similar to that facing Germany in the 1830s; a situation with more than 30 small states, each with an extremely protective tariff.

The solution in Germany was a first stage which removed the tariff between the 30 German states, and created an enormous competition among nations of fairly similar skill levels.

Within this symmetrical trade pattern the nation continued to build skills, finally to “graduate” to compete openly in the world markets. In this way the State completes its most important historical function — to establish the nation solidly in economic activities subject to increasing returns, and to set up a dynamic national system where innovations are an essential byproduct of the way its national industry competes.

MECHANISMS CAUSING AND DIFFUSING ECONOMIC GROWTH AND WELFARE

A chapter purporting to trace the influence of the State on economic growth must — implicitly or explicitly — be based on a model of the mechanisms which cause economic growth. We shall attempt to clarify how our assumptions differ from those of Adam Smith and of the neo-classical system. Partly using the terminology of Werner Sombart, we base our analysis on a productioncentered, activist-idealistic set of assumptions, in contrast with today's mainstream, which in this context can be labeled barter-centered and passivist-materialistic.

Our set of assumptions differs from that of today's mainstream economic theory, and therefore also concludes with a different view of the role of the state in economic growth. To clarify: if the conditions of "real world" did correspond to the assumptions of neo-classical theory, we would have shared the view that any attempt to influence income growth and distribution would be futile. However, as we shall argue, the factors causing uneven economic development — where State intervention may play a role — are the very factors which neoclassical theory assumes away. When neo-classical theory is "right", it is, on key points, "right for the wrong reason".

We would for example argue that the benefits from international trade are mainly caused by what Schumpeter called historical increasing returns — a blend of increasing returns and technical change — rather than on the static gains in the Ricardian theory. The need for "governing the market" arises from such factors which are left out of mainstream theory.

There seem to be three important aspects to this question:

1. How economic growth is created;
 2. The alternative mechanisms through which growth and welfare are diffused between and within the nation-states, and to the individual; and
 3. How this alternative understanding is based on a different philosophical basis.
- These three aspects are briefly discussed in the next three sub-sections.

ASSUMPTIONS ABOUT THE CAUSES OF ECONOMIC GROWTH

One important starting point is that, in our view, today's mainstream economics — as a cluster of variants around the neo-classical paradigm — fundamentally lacks a theory of economic development beyond seeing it as a process of adding capital to labour. In 1956 Stanford economist Moses Abramowitz showed that capital accumulation only accounted for 10-20 per cent of US economic growth — which he then referred to as "a measure of our ignorance about the causes of economic growth".

What are the forces creating welfare? It seems that search process for "the approximate causes of growth", as Abramowitz calls them, follows the path of Ibsen's Peer Gynt, to whom the onion — and his inner self — only revealed layer after layer, but no core. We would argue that the factors should be seen as "reactants", factors which, when they are

all present — and only then — produce economic welfare. These “reactants” have been uncovered layer after layer in modern economics, starting at the top of the following list.

The causes of increased economic welfare are:

- Markets.
- Capital.
- Technology.
- Technology.
- The attitude to new knowledge.
- Systemic aspects giving rise to positive feedback mechanisms.
- Man’s rational will.

Today’s mainstream economic theory basically limits itself to the first two factors only. Markets and capital are clearly necessary, but far from sufficient building blocks for a theory of economic growth. New research under the heading of “evolutionary economics”, as sponsored by the OECD under the title TEP also includes factor 3, and increasingly also factor 4. Another striking feature of the economics of Adam Smith is his long and deliberate proofs that the production of knowledge, from the point of view of the individual as well as for society, is a zero-sum game.

This is probably where Adam Smith most clearly breaks with the Renaissance tradition. Today — partly as an effect of the TEP programme — the opposite message of Adam Smith’s is slowly sinking in: the driving forces of the economic system are innovations created by new knowledge. Nations which stop innovating do not keep their standard of living, they lose their standard of living even though they keep the same efficiency.

ASSUMPTIONS ABOUT THE MECHANISMS WHICH DIFFUSE GROWTH AND WELFARE

Another, and most important, dimension to the phenomenon of economic growth is that it seems to be a product of a number of systemic effects which are not captured by theories atomistically studying only the single actors in the system. These aspects — synergies, economies of scope, or systemic economies of scale — are the starting points of positive feedback mechanisms — virtuous circles — in the economy, and they only appear in view when the system as such is the object of analysis.

The presence or absence of such factors determines how economic growth spreads in the economy — essentially in one of two forms: either as higher wages to man-the-producer or as lowered prices to man-the-consumer. In a closed system, the two modes of collecting the benefits from technical change make no practical difference. In international trade between two different labour markets with an asymmetrical trade pattern which of the two models operates makes all the difference. The collusive mode of income distribution took place system-wide and in parallel in all industrialised countries behind the barriers to entry created by Fordist mass production. Demands for higher

monetary wages to labour were met, while at the same time money supply was increased at least at the pace of the increase in national physical productivity.

In the perfectly competitive classical model of income distribution, people should have got richer by having their purchasing power increased through falling prices — just as Adam Smith and Ricardo predicted — not through the raising of monetary wages. We have also argued that the term “competitiveness” and its historical equivalents have been used to describe the existence of such systemic results of dynamic imperfect competition and of a collusive spread of the benefits from technological change. The practical consequences in world income distribution, created by the two different modes of distributing the benefits of technical change, can best be seen in the traditional service sector.

In this huge sector in any economy there is normally very little difference in productivity between First World and Third World workers. A bus driver, a barber, or a chambermaid are about as productive in Bolivia or Haiti as they are in Norway or Italy. Why then do the barbers or bus drivers in Bolivia or Haiti have real wages which are only a fraction — 10-20 per cent — of the real wages of a barber or a bus driver in Norway or Italy? Why does the invisible hand reward people with the same productivity so widely differently in different economies? We would argue that this is the result of *Strukturzusammenhänge* — of systemic effects — created by the fact that a large part of the benefits resulting from technological change in increasing return activities are spread in the collusive mode, *i.e.* as higher monetary wages in the local labour market rather than as lowered prices internationally.

In the perspective from a poor sub-Saharan nation, what we see as “economic development” basically appears as a system based on “industry rent” from increasing return activities which spreads throughout the national labour markets. The existence of increasing return activities — and their accompanying barriers to entry — creates a national common weal which gives the Norwegian bus driver real wages which are five times higher than his equally productive Bolivian counterpart.

We would argue that the origin of such systemic effects is found in the fixed costs which accompany Man’s tools: the increasing sophistication of tools and of specialised human skills to work the tools, leads to increasing minimum efficient sizes of operations in an ever-increasing number of professional specialisations and in an ever-increasing diversity of products and services. This causes cumulative causations and path dependency.

As an example: the village which had enough volume to support its own blacksmith had a clear advantage over a smaller village with no blacksmith in the next cumulative step up the ladder of technological sophistication. This is the reason why “the division of labour is determined by the size of the market”, Adam Smith’s deep insight which later disappeared from neo-classical economics. This is also the explanation for the great importance of a large population which is held unanimously by pre-Smithian economists.

THE DIFFERENT PHILOSOPHICAL UNDERPINNINGS OF THE ACTIVISTIC-IDEALISTIC TRADITION

Promoting new knowledge was a fundamental driving force for the economic policies of the Renaissance — and later of all realoÈkonomisch mercantilism, cameralism and Colbertism. If one puts together a list of economic interventions in Renaissance economics, it becomes clear that the element uniting these seemingly most diverse economic policies is the promotion and protection of new knowledge.

These policies further rested on the assumption, implicit or explicit, that some knowledge was more valuable than other. Consequently, it was in the interest of the State to create and protect such knowledge. On a philosophical level, this valuable knowledge emanates from the mind of Man.

Therefore, any theory not taking Man's mind and Man's soul into consideration will not understand the increase in Man's material condition, *i.e.* economic growth. In the activist-idealistic tradition, economics and the social sciences require a different kind of understanding from the natural sciences. The social sciences, concerned with the ends and values instead of laws, should aim to understand.

The natural sciences, on the other hand, aim mainly to describe and conceptualise. Qualitative relationships are at the core of the Geisteswissenschaften — the humanities, or literally, the sciences of the mind. A crucial aspect of these sciences is therefore their irreducibility to natural science. Although economics is also about numbers, there is also a basic incommensurability between numbers and some fundamental aspects of economic science, like Man's wit and will. The German economic tradition continued the Renaissance tradition.

Here the inner unity of the Geisteswissenschaften — *e.g.* the refusal to isolate "economic man" from the rest of the human being — shines through from Leibniz to Werner Sombart. The common weal in Renaissance economics is also the result of such a synergetic Strukturzusammenhang. The fact that our two equally efficient bus drivers — one in Bolivia and the other in Norway — have such widely different real wages, is a product of economic Strukturzusammenhänge, of which today's atomistic economic theory and economic policy are totally ignorant.

The economic policies resulting from theories of this kind were carried out with varying degrees of understanding of the underlying principles. These policies were, in our meaning of the word, not based on what we would normally call scientific analysis. These theories were based on "clues", on a mode of inference called abduction — or phronesis, Aristotle's third form of knowledge. This tradition is continued by the Italian philosopher Giambattista Vico by the US philosophical tradition of C.S. Pierce, and in economics in Kaldor's "stylised facts".

Pierce, "*can never originate any idea whatever. No more can deduction. All the ideas of science come to it by the way of Abduction. Abduction consists of studying facts and devising a theory to explain them. Its only justification is that if we are ever to understand things at all, it must be in that way*".

Pierce here describes the role played by the formulation of hypotheses as the fundamental element in the creation of new knowledge, in the German eighteenth century philosophers Leibniz and Wolff. We shall explain how policy is generated through abductive reasoning with a parallel from the history of medicine: starting in the twelfth century sailors in the Mediterranean used lemons to prevent scurvy. This was a very effective policy. However, the explanation as to why this policy worked only appeared in 1929, with the discovery of Vitamin C. Likewise, we would claim that it is entirely possible to establish good economic policies for a time, without fully understanding the factors involved.

For example, identifying “progress”, or ability to pay more taxes, with the use of machinery in an increasing number of industries would result in a beneficial public policy, even if the causal relationship between the use of machinery and wealth were not clearly established, or had been “unlearned”. The intuitive abduction often precedes what we would think of as a more “scientific” type of knowledge. This view that abduction anticipates “science” was expressed by the English economist Edward Misselden in 1623: “Wee felt it before in sense, but now wee know it by science”.

We would argue that business decisions are made — mostly under high degrees of uncertainty — based on such intuitive abduction about what is good for the firm, on what at business schools is called “management by gut feeling”. Early statesmanship seems to have worked in a similar way. Going through the voluminous letters and instructions of Colbert, one is struck by his role as being that of a businessman in charge of a huge empire: as an entrepreneurial input-coordinator for France Inc., in a venture to get into knowledge-based activities, he was faced with what historians of technology call “reverse salients”— of dynamic bottlenecks — retarding the system and demanding managerial attention.

By differentiating intuitive abduction from “science”, the position of writers like Friedrich List comes in a new and different light. List is full of interesting ideas and has an intuitive understanding of the mechanisms at work, but he fails to develop concepts which explain to us clearly what’s going on. As Werner Sombart says about List: “His concepts levitate like undelivered souls on the banks of Hades”. However, just as lemons helped against scurvy 800 years before the exact mechanisms through which they work were established, economic growth was successfully promoted using “new knowledge” and “use of machinery” as proxies for the underlying factors causing systemic economic growth.

THE THREE ROLES OF THE STATE

For the purpose of the chapter we find it useful to divide the roles of the State into three broad categories:

1. The State as a provider of institutions — in the widest sense.
2. The State as a provider of income distribution and as an “insurance company”.
3. The State as a promoter of economic growth.

There are, of course, other alternative classifications of the roles of the State. Wilhelm von Humboldt says: “A State then, has one of two ends in view; it designs either to promote happiness, or simply to prevent evil”. Another German economist, Adolph Wagner, divides the role of the State into two different categories of objectives: Law and Power and Culture and Welfare. However, for our purpose it is important to divide the welfare aspect expressed in the other classification into two very different categories: the creation vs the distribution of income.

The State as a provider of institutions”. These institutions may, on the one hand, be seen as preconditions required by the invisible hand in order to get on with its job of creating economic growth. On the other hand, they may be seen as part of a wider and more active strategy as that under point 3, the State as actively “increasing the size of the pie”.

However, we choose to put these fundamental institutions — which hardly come under the attack of the thesis of futility, jeopardy and perversity — in this separate category. These basic institutions include private property and the well-worn German concept of a Rechtsstaat. As standards of living grow, secondary demands are created and the State is extended into a Kulturstaat, providing institutions for education, science, charity, sanitation, etc. Summing up, one German author says: “Food, drink, clothing, shelter, amusement, social intercourse — these are the primary wants with the covering of which private economy is mainly occupied; peace, order, security, culture, relief — these are the higher needs which are mainly served by the public economy”.

To point 2 — “Sharing of the pie and distributing risk”: The thought of distributing the inevitable risks of life is a very old one — the participants of the camel caravans of the Near East 2200 BC had a risk sharing system. Modern insurance traces its root to the middle of the fourteenth century. The same ideas of risk sharing are present during the beginnings of the welfare state, already in the Byzantine Empire, where the idea of sharing the risks of life gradually develops into a role of income distribution. Thoughts around welfare are found very clearly in the writings of German philosophers Gottfried Wilhelm Leibniz and Christian von Wolff. Leibniz suggested that a national health system should be established.

The health system should be the basis for a welfare guaranteed by the State. Wolff specifies clearly that he wants a State which secures welfare for the individual, but not one which automatically provides for him — a most difficult balance to this very day. Clearly “sharing the pie” is not a wholly independent exercise from that of “increasing the size of the pie”. Economic arguments for income redistribution include the argument that a certain income distribution is required to create and maintain a mass market.

This argument seems to be one used to defend the enormous transfer payments within the European Community; *i.e.* “Poverty is bad for business”. One can further argue that an unequal income distribution threatens the social fabric of society, and therefore the nation itself and its economy. A most important argument — similar to that used by nineteenth century US economists — is that a continuous increase in the price of labour

relative to the price of capital is a key factor in the virtuous circle created by State intervention: the increasing relative price of labour is a strong incentive for further mechanisation, which again allows for even higher wages and higher profits, and so on. By making labour more skilled, it is made more valuable.

This was also an important argument of nineteenth century US economists against the dismal pauperisation theories in English classical economics, and formed the basis for the dual policy of “the high wage strategy” and “the American system of manufactures” giving protection to increasing return activities. We would see this argument for income distribution as part of the virtuous circles which are created by State intervention under the next point.

To point 3 — “The role of the State in increasing national wealth”. Clearly the institutions in points 1 and 2 are necessary, but in our view not sufficient, conditions for economic growth. The objective of this chapter is to look at the more active role of the state in economic development — “the developmental State”. We must look behind the State as a provider for institutions for which there is a “natural demand” into another role of creating demand for things which are necessary, but for which there is no articulated demand.

If people do not want to educate their children — because they are themselves not educated, and because they need their daily labour — the State plays a new role: the State uses “its power of coercion in order to create a general demand for the institutions which it has established”, says Cohn. Here enters a more visible hand — the role of “the State as a factor of production” to use Luigi Einaudi’s term, originally from Adam Müller. It is the role of this visible hand which forms the core of this chapter. The role played by the State is contained in the felicitous term coined by Robert Wade: “governing the market”. It is clearly particularly to this role of the state that the message of futility, perversity, and jeopardy is directed.

Historically the roles of the State as a promoter of economic growth can be listed under the following headings:

- Getting the nation into “the right business”, *i.e.* recognising, as a historical starting point, the activity-specific nature of economic growth. We intuitively understand that Japan could not have reached her present position by making inexpensive shirts, rather than inexpensive cars or electronics, and that Pavarotti could not have made the same fortune washing dishes as singing, but this intuitive knowledge is not compatible with today’s barter-centered economic theory. Historically all presently industrialised nations— for the right or for the wrong reasons — have passed through an initial stage with a policy based on the understanding that not all economic activities are equally feasible as starting points for the self-enforcing positive feedback system which we call development.
- Creating a comparative advantage in “the right business”. A common element of all successful strategies for catching up with richer nations is the conviction

that free trade is not wanted until the nation has created a comparative advantage in the “right” economic activities. There is an important underlying perception in this that in a world economic system, if there is world demand both for skilled and for unskilled labour, a nation may end up locked into a permanent comparative advantage in being poor and unskilled.

- A very strong emphasis on the role of the State as a supplier of infrastructure. This is a unifying element of most “enemies” of classical and neo-classical economists, from Colbert to Friedrich List and to Al Gore/Robert Reich. Infrastructure is, as all other systemic elements of the economy, conspicuously absent from today’s economic theory. Infrastructures are key factors in extending markets and are “highways” which the positive feed-back mechanisms need in order to displace themselves geographically.
- Setting standards has been a very important task of the state, from the neo-classical point of view to lower transaction costs, from the evolutionary point of view to form a basis for standardised mass production. A visitor to the Renaissance towns of Italy can to this very day observe the iron bars fastened to the church wall on the main square, establishing the standard units of measurement valid in the city. Each city had different measurements, which made information and conversion tables on these issues an important task for early economic books. Today the setting of standards is important to hi-tech products like mobile telephony. Clearly the state also has an important role in setting legal standards and providing a social virtue which is crucial to prosperity. This was a point much emphasized by Renaissance philosophers, and is rediscovered today in Fukuyama’s book *Trust*.
- A responsibility of the State to provide skilled labour and entrepreneurship if in short supply. Early policy measures — often helped by religious wars — were aimed at bringing in skilled labour and entrepreneurs from abroad by granting exclusive rights for a limited time, or through bounties, tax relief, etc. The State is also an entrepreneur of last resort.
- Unless one firmly believes in Say’s Law — that supply creates its own demand — the State clearly has an important role in creating demand in general. Out of the Victorian slums with only subsistence demand, a better income distribution of a growing economic pie paved the way for the mass markets. This development made industrial production and “Fordism” possible, *e.g.* through the establishment of minimum wages. Nineteenth century US economists saw particularly well the role of raising labour skills to increase their market value — the policy which was called “The High Wage Strategy”. There is reason to believe that the sharp fall in economic welfare experienced by weakly industrialized nations in the wake of “structural adjustment” in part was due to the elimination of demand created by the State. In this context it is important to keep in mind that what we

call “economic development” is little more than a collective “industry rent”, the national distribution of which — in spite of our laissez-faire mythology — is the result of decades of “artificial” redistribution, partly due to the activities of labour unions. Our understanding today of the connection between the key variables: growth of demand, income redistribution and higher wages pushing mechanisation, is very limited. In several nations today — as, for example, in Peru — there is evidently a break in the positive circuit created by these factors.

- The State has played a very strong role pushing the technological frontier by being a supplier of high-quality demand for national production, of demand for goods at the borderline of what, at any time, was technically feasible. Key mechanisms here have been infrastructure projects and warfare. Werner Sombart, in 1913, describes the destruction of war as being the starting point both for the creative spirit of Man — for searches for synthetic substitutes for raw materials in short supply — and also for the organizational capabilities of the State. This is a most important argument in the spirit of Chandler and Lazonick. Not only was State demand crucial for purposes of war and infrastructure, but also the personification of the State — royalty and nobility — served as Porterian demanding customers for luxuries, and provided the basis for future technological expansion into lower-cost mass production. Historically the demand for luxury has played a role similar to that played by people who, ten years ago, paid \$4,000 for a cellular phone, enabling later mass production to deliver better phones for \$80 today. Again, the function of State demand for luxury is brilliantly described by Werner Sombart.
- An emphasis on the value of knowledge and education *per se*.
- A long-term goal similar to what in the nineteenth century USA was called The High Wage Strategy — *i.e.* seeing high wages *per se* as a goal. At the core of this economic thinking was Man and his Needs, rather than a “dead equilibrium” which reduces Man to one of several factors of production, the high or low reward of which has no place in the system. The importance of the wage level in explaining GNP is illustrated by the fact that today wages are typically 70 per cent of GNP: *i.e.* maximising wealth essentially means maximising national wages.
- An understanding of the importance of a legal system built to strengthen the above structures. To Christian Wolff — the eighteenth century German economist and philosopher of law — a system of property rights assisting the dynamic symbiosis between “persons who collected knowledge and people who collected money” was seen as the core of a system creating — like Francis Bacon’s system before him — a never ending frontier of development.
- The State as an entrepreneur and capitalist of last resort. There is no reason to assume a “sufficient” supply of entrepreneurship at any point in time and in

any culture — the poorer the nation, the less so. Moving the State into a role of capitalist or entrepreneur resulted from “reverse salients” in the system, rather than from any ideological preference.

We would argue that — as economic agents — States exist for fundamentally the same reasons that firms exist, both of them for reasons not well captured by today’s economic theory, which focuses on barter and atomism. We suggest that the absence of a theory of the firm in modern economic theory is caused by the same reasons which cause the absence of the theory of the State in the same body of theory.

Both firms and states are institutions which are brought to life by the kind of systemic synergies which are excluded in neo-classical theory through the assumptions of full divisibility of resources, of perfect information, perfect competition, and of the absence of increasing returns. To continue this analogy between the State and the Firm, we would argue that the size and strength of a firm to a large extent are determined by the sum of the countervailing forces of economies of scale and diseconomies of scale. Large chemical firms are the product of huge economies of scale in production, advertising and finance; coupled with the opposite effect — the diseconomies of bureaucratisation — in administration and coordination.

Similarly the strength and size of the State are also torn between two opposite effects, well captured by Humboldt: “The highest good” which arises from associations, like the State, “is the very variety arising from the union of numbers of individuals”. This variety is a core part of the synergies creating the common weal, and in my view also the underlying principle why, as in Adam Smith’s argument, “the division of labour” is so important. However, “this variety is undoubtedly merged into uniformity in proportion to the measure of State interference”.

Thus, the existence of a State tends to produce “uniformity and inertness”, which is the very reverse of what caused State intervention in the economy in the first place, which was the need for “variety and activity” — *i.e.* the establishment of a large number of professions through entrepreneurship. The extent and size of a firm are a result of competitive powers — which are industry- and product-specific — on the underlying trade-off of increasing and decreasing returns to scale. Similarly, the extent and activity of the State should, at any point in time, reflect the paradox that, having united in order to exploit the synergies and economies of scope which result from the variety and activity of Mankind, the potentially perverse effect of producing uniformity and inertness is likely to occur.

This uniformity and inertness threaten the variety and activity, the synergies of which the State was established to foment and exploit in the first place. In the case of both the firm and the State, one important answer to the diseconomies of scale produced by size and growing complexities lies in the concept of organizational capabilities. The Managerial Revolution was, in the world of business, the reply to these challenges. The role of the state in industrialised countries is often seen as one of protecting “civil liberties”, or the form of freedom which we would call “freedoms to”. However, the

role of the State in the early stages of economic development is one of jump-starting the systemic effects which were to secure the “freedoms from” — freedom from hunger, freedom from injustices, freedom from ignorance.

With time, however, another aspect of the dynamic balancing creating an “optimal state” appears when the actions of the State to provide “freedoms from” are gradually seen as encroaching upon Man’s “freedoms to”. The Smithian revolt against Renaissance economic policy can be seen as such a conflict, as seen in Turgot’s arguing against the excesses of Colbertism. In England at the time, the policies of the “development state” had entered into an area of diminishing returns of static and individual rent-seeking rather than the synergetic collective rent-seeking. Clearly a big dose of “markets” was needed.

However, as German, US, and Japanese economists in the nineteenth century were so eager to point out, this did not mean that nations which had not reached the level of England could use the same policies there and then. The role of the State — like the particular balance of emphasis on freedoms to and freedoms from — is highly context-specific.

NEW KNOWLEDGE, SYSTEMIC EFFECTS AND POSITIVE FEEDBACK-LOOPS IN RENAISSANCE ECONOMICS: THE CONSEQUENT ROLE OF THE STATE

Renaissance was brought into reality by philosopher-statesmen like Francis Bacon who consciously used his Utopian tract *The New Atlantis* in forecasting the future in order to inspire and influence it, a process he himself called “feigned history”. Bacon’s *New Atlantis* and his *Essay on Innovations*, describe the development of society as being propelled by new inventions and innovations, and should, in my view, be considered as the earliest tracts of what we today would call “Schumpeterian economics”. In this chapter, we shall attempt to show that the economic policies carried out by the European nation-states starting with the Renaissance, although making little sense from the perspective of neoclassical economics, make eminent sense when seen in the perspective not only of Schumpeterian economics, but also of other recent developments in economic theory.

Not only in Bacon’s “feigned history”, but also in real history, these early inventions and innovations came about as the result of what, using a Chandlerian vocabulary, ought to be called the organizational capabilities of the State. During this century’s conflict between planned economies and market economies, the relationship between the State and private entrepreneurs is still frequently seen as one of natural animosity. Historically, however, the roles of the State and of private entrepreneurs have been more in the spirit of complementarity and partnership, where the State facilitated and coerced private entrepreneurs into business, and — this failing — also moved in as an

entrepreneur of last resort. The motivating force behind the European Commission's 1995 "Green Paper on Innovation" — 30,000 copies of which have been distributed — was precisely again to promote the organizational capabilities of the modern European states in maintaining the "competitiveness" of Europe by strengthening its National Innovation Systems. The foreword to the most recent advanced textbook in economic growth states that "economic growth comes largely from the accumulation of knowledge". We shall attempt here to show that a principal historical role of the State from the Renaissance onwards has been precisely one of promoting and protecting new knowledge and innovations.

We have assembled what we consider the most important government interventions in economic policy starting in the Renaissance. We would argue that the common theme of these economic policies — the thread which unites them — is just this concern for the creation and protection of new knowledge. Looking at history from a simple perspective of barter, not production, and under diminishing returns/single equilibrium/perfect information, the importance of these policies is lost. In the diminishing return/equilibrium perspective, any and all factors causing unequal economic growth are lost, creating the world of artificial harmony and world-wide factor-price equalisation.

As we shall attempt to show later, a most important historical role of Adam Smith's was precisely that of laying the ground for "perfect markets" and "natural harmony" by making the quest for knowledge into a zero-sum game — using the metaphor of a lottery — from the point of view of both the individual and the State. In this way Adam Smith effectively removed the quest for imperfect competition through new knowledge which was so important to Renaissance thinking. This is the root of why new knowledge and new technology hits today's mainstream economics as "manna from heaven". Pre-Smithian economic thinking takes a holistic starting point — the People, the State.

A fundamental underlying idea exists that the situation of each individual can be improved by measures which take into consideration the collectivity of individuals. In other words, there are systemic effects which cannot be found if one limits the horizon atomistically to study individuals alone. We shall argue that the existence of such systemic effects is the fundamental reason why the State has a role to play in economic growth. We still find the strong belief in this systemic approach in the early Adam Smith — before his conversion by the physiocrats. In the later Adam Smith the atomism of John Locke takes over, but Smith's fundamental insight that "the division of labour is limited by the size of the market" is the description of one such systemic effect.

We argue that the division of labour is only important if systemic economies of scale are at work. "The division of labour" is, of course, lost in neo-classical theory. In this part, we shall discuss the theories of the time as they relate to economic growth, and also try to see these theories in the perspective of modern evolutionary economics. As we have said, the goal of the economic policy of the State was to increase the common weal, *i.e.* the prosperity of the community. This is the starting point of virtually all economic writing of the period. To the Renaissance economists systemic effects seem

first to have arisen from the observation that widespread wealth seemed to accumulate in the cities — not in the countryside.

This was the fundamental observation of one of the earliest best-selling books in economics, *Delle Cause della Grandezza delle Città* written by Giovanni Botero. The English translation, published in London in 1606, is entitled *The Cause of the Greatnesse of Cities*. But there were also huge differences between cities, between the opulence of Venice and the poverty of Naples, and this issue would be thoroughly discussed by Antonio Serra in 1613.

In the best theoretical works of the time, this difference between the wealth and poverty of cities and countryside, and between cities, is explained in terms of the following main factors:

- The size and density of population.
- The different “qualities” of economic activities.
- The presence or absence of diversity of economic occupations, and the different capacities of economic activities to initiate “virtuous circles” or positive feedback mechanisms.

THE SIZE AND DENSITY OF THE POPULATION

One key aspect of this set of theories is the importance of having a large population. “The power of a State is not the result of its territorial extension, but of its number of people, and its good government.” This phrase is found with little variation in the writings of virtually all economists and philosophers of the time, by Mun, Child, Petty, and Davenant in England, by MontchreÂtien, Forbonnais and Necker in France, and by Leibniz, Wolff, Becher, and Justi in Germany.

One of the reasons for the great seventeenth century interest in China was its large population density which was seen as proportionate to the degree of knowledge and wisdom. This way of thinking is very far from today’s zeitgeist, which has completely internalised the dismal science of Ricardo and Malthus in this respect. Because we have excluded knowledge and technology from today’s economics, we fear overpopulation in Bolivia and in Peru but we are not worried about Holland with 400 persons per square km. The population argument makes sense if, as Adam Smith claims, “the division of labour is limited by the extent of the market”. If there are fixed costs and minimum efficient scale involved in establishing new professions, which is certainly the case, then the argument makes sense that a larger population contributes to a larger division of labour.

We shall see that Antonio Serra’s 1613 work clearly stresses the importance of the division of labour, thus anticipating Petty by about 70 years and Adam Smith by about 160 years. One observation which can be made here is that Adam Smith’s key contribution of the division of labour is not really compatible with the assumptions of neo-classical trade theory: If there are no economies of scale — *i.e.* no fixed cost — and perfect information, there is little reason for any division of labour at all. Samuelson’s

theory which proves factor price equalisation seems to be based on assumptions which take away the very incentives which cause the division of labour.

THE DIFFERENT “QUALITIES” OF ECONOMIC ACTIVITIES

“Renaissance economics” presented several levels of arguments for policy intervention, which may roughly be classified in three categories as follows: from observations of isolated and activity-specific differences in welfare, through the description of systemic synergies emanating from these activities, to the description — particularly perfected in one case — of fully-fledged positive feedback systems:

1. Observations of higher welfare in some economic activities rather than in others, a static and non-systemic observation of welfare being activity specific.
2. Some economic activities as core of systemic synergies which produce and spread welfare locally or nationwide.
3. There are degrees of understanding how these systemic synergies develop into positive feedback systems, but the top performance is that of Antonio Serra in 1613, who has a description of Venice as a true autocatalytic system where increasing returns and diversity — the latter expressed as the number of different professions in a nation— are identified as being at the core of virtuous circles which generate wealth. Naples is the example of the opposite effect in Serra’s system, because the production of raw materials is not subject to increasing returns.

Just as we today would see a career of washing dishes in a restaurant as having a limited potential for creating income compared with a career as a lawyer, the Renaissance economists extended this argument to apply to the common weal as well.

In other words, they believed that the factors which created differences in welfare within an economy were the same factors which created differences in income between nations. As a result of the process of pre-Ricardian common sense, no factor-price equalisation would be achieved by putting all the people washing dishes in one nation and all the lawyers in another and open up for free trade between the two nations.

In these theories economic growth is “activity-specific”; it is only available in some economic activities subject to dynamic imperfect competition, and not in others. Adam Smith, in his late work, goes out of his way to prove that differences in income between professions do not exist. We are here talking about differences in wages — which constitutes more than 70 per cent of GNP in an industrialised economy — and not in profits. What, Giovanni Botero asked in 1590, is more important for making a nation wealthy, the fertility of its soil or industry? No doubt, industry, says Botero.

First of all “because the things produced by the able hands of man are many more, and have a much higher price, than the things produced by nature. Nature gives the material, but the object... is the work of Man. Wool is a simple, coarse material of nature. How many beautiful objects, varied in form and shape, the Arts can produce from this...” We find the same line of reasoning among the German cameralist

economists. The German writers of economics were — as opposed to their English counterparts — mostly employed by the rulers of the small German states. One of their main tasks was to increase the income from taxes to the regent.

Their observations coincided with those of the Italian economists: the subjects who worked with machinery were able to pay much higher taxes than those working without machines. In this way the use of machinery became a proxy for a type of profession desired in a nation. The beginning of the industrial revolution in England — Henry VII's establishment of textile industries — was based on this same line of reasoning. Understanding economic development as being activity-specific leads to two different policy measures by the State. One is a set of positive measures, taken in order to bring domestic economic activities into the “right” industries, those where the particular time period allowed for the build-up of new knowledge, the use of machinery, etc. The same understanding of the activity-specific nature of economic growth opens up for a set of negative measures, taken in order to prevent other nations from getting into that “right” type of activities. Prohibiting the use of machinery in the Colonies was one such measure. Other measures aiming at the destruction of already established high-quality activities include the British policies in Ireland starting 1699, when the prosperous exports of high-skilled woollen manufactures were prohibited, and the more labour intensive and less mechanizable linen manufacture was brought in.

In the beginning of the nineteenth century similar policies were in force against Indian textile manufacture. At the same time contemporary US sources describe English commercial policy at the time as being aimed at preventing the USA's industrialisation. As late as in 1904, the official history of the US Republican Party, which carries a laudatory foreword by President Theodore Roosevelt, comments: “We had suffered quite enough before the Revolution from the policy of Great Britain in checking our industrial and mechanical aspirations, and keeping us a buying instead of a making people...”. The English debates made no secret of the fact that their strategy was one of keeping other nations from industrialising.

In 1816 Brougham explained the English strategy in the House of Commons:

- It is well worth while to incur a loss upon the first exportation, in order, by the glut, to stifle in the cradle those infant manufactures in the United States which the war has forced into existence.

We would argue that this is a fairly typical nineteenth century comment. Also Richard Cobden — the champion of free trade — saw the elimination of the Corn Laws as means to weaken other nations' manufactures. Cobden saw the high price of corn as the basic reason why England had not been able to hold on to her near-monopoly of world manufacturing.

To Cobden free trade in corn was above all a means to prevent other nations from threatening the British nearmonopoly of manufacturing exports:

- The factory system would, in all probability, not have taken place in America and Germany; it most certainly could not have flourished, as it has done, both

in these states, and in France, Belgium, and Switzerland, through the fostering bounties which the high-priced food of the British artisan has offered to the cheaper fed manufacturer of those countries.

Historically the economic policies recommended in production-based economics follow two alternative lines of argument:

1. The risks and perils of a resource-based economy — being specialised in economic activities not requiring high labour skills and subject to diminishing returns. These two effects may combine into a lock-in effect trapping a nation in poverty, which technical change will not solve. Exporting natural resources constituted “bad trade” in the orthodoxy of English economic policy until long after Adam Smith and David Ricardo. The most clear statement of this orthodoxy is contained in three volumes published by Charles King in 1721. In King’s Taxonomy exporting raw materials was “bad trade”. We later find this same line of argument in nineteenth century USA, Germany, and Japan, and in twentieth century Australia and Canada.
2. The benefits of concentrating a nation’s economic activities in mechanisable, skill-absorbing activities, subject to increasing returns — a cluster of characteristics which for a long time could be efficiently shorthanded as manufacturing. Exporting goods from these activities qualified as “good trade” in King’s Taxonomy — the English orthodoxy which built her national strength. The Canadian and Australian arguments later added to this by showing that the presence of a national manufacturing base upgraded skills in the resource-based activity and prevented these activities from going into Diminishing Returns — making raw material exports from a manufacturing country a viable strategy.

What, then, are the characteristics of growth inducing — “good” — economic activities?

We have, in several publications, provided a “quality index” of economic activities, listing the characteristics which, in a system of dynamic imperfect competition, ranks economic activities according to their ability to provide increasing economic welfare to a nation. This “quality index” is reproduced in Figure.

Differences in wage levels, both nationally and between nations, seem to result from varying degrees of imperfect competition — caused by both static and dynamic factors.

The factors at work have long been identified both by businessmen and in industrial economics, and they are correlated.

Figure attempts to create an area from light to dark grey where “the quality” of economic activities at any time can be roughly plotted on a scale from white: “perfect competition” — to black: “monopoly”.

The latter is only a temporary state, as new technologies fall towards a lower score as they mature.

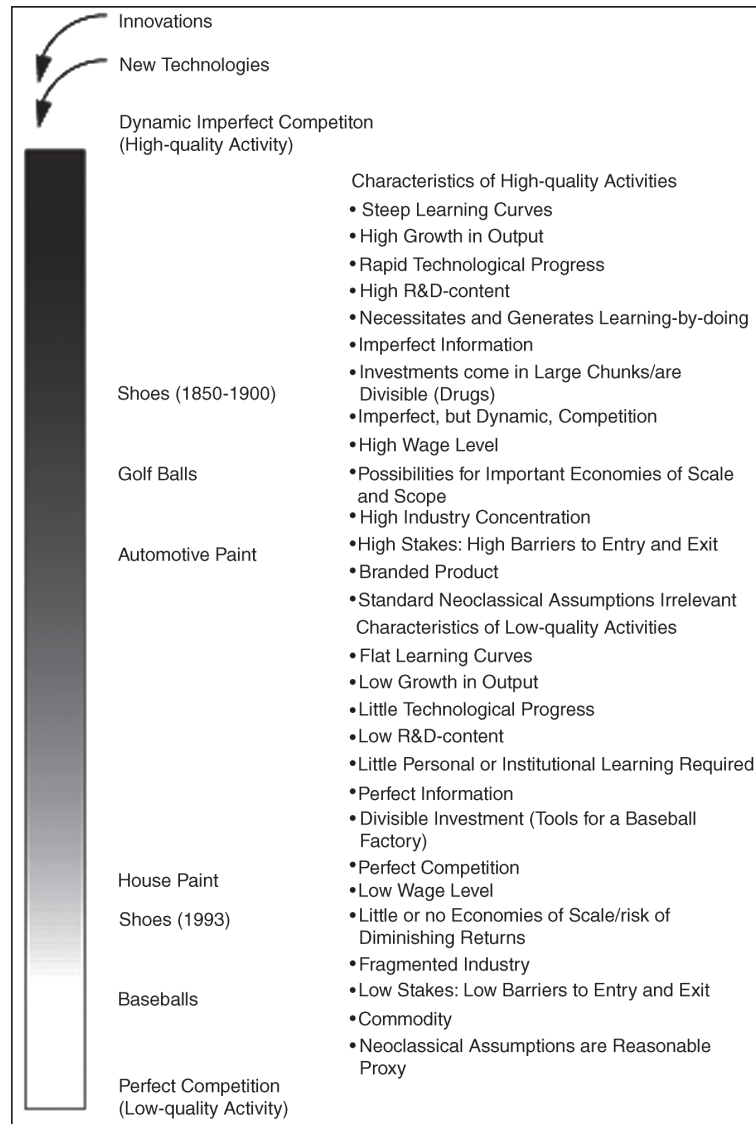


Fig. The Quality Index of Economic Activities

DIVERSITY, SYNERGIES AND POSITIVE FEED-BACK MECHANISMS IN RENAISSANCE ECONOMICS

The static observation of higher welfare in some economic activities than in others is very frequent. Daniel Defoe gives us a systemic extension of this argument as the line of reasoning which persuaded Henry VII to start an English textile industry when he came to power in 1485: while living with his aunt in France, the future King of England had observed not only that the French textile producers were much richer than their English providers of raw materials, but also that wealth spread to the whole community: where there was manufacturing, also the shopkeepers were richer. There

were synergetic effects between manufacturing industry and the common weal of people outside the manufacturing sector.

There are several arguments founded on this kind of systemic synergy caused by manufacturing. Adam Smith's *The Theory of Moral Sentiments* — from before his meeting with the French physiocrats — shows him as a relatively traditional mercantilist in this aspect. The reasons given by German philosophers and statesmen Leibniz and Wolff for why a State is needed include an emphasis on learning which triggers positive systemic effects.

The reason why there is so little conflict between the interest of the individual and the common weal in their system is precisely that increased knowledge produces more of both individual and collective profits, something like: “The incoming tide raises all boats”. Wolff observes that “Some people collect knowledge like other people collect money”, and indicates the benefits to society of putting these two types of people together.

The most remarkable of all economic treatises before Adam Smith is, in this author's opinion, no doubt the 1613 book by Antonio Serra, *A Brief Treatise on the Causes which can make Gold and Silver Plentiful in Kingdoms where there are no Mines*. The title corresponds to our stereotypes of mercantilist tracts, that they are only about gold and silver. In fact Serra produces a most sophisticated model, producing, on the one hand, systemic economic development and, on the other, underdevelopment. Serra's starting point is knowledge. He denounces “ignorance as the cause and starting point of all evil”. He further comments on “everybody's innate desire for knowledge”.

He outlines the plan of his work as:

- Understanding why some nations, even though they have no mines, are very rich, and
- Based on this understanding, to explain the apparent paradox that his own nation, the Kingdom of Naples, although abounding in natural resources has reached such an abysmal level of poverty that “it does not leave us to breathe nor to enjoy what nature has given us”.

Serra is the first economist to describe increasing returns and with the increasing returns as his starting point, he describes positive feedback mechanisms which lead to virtuous circles of development in a national system.

Serra has two types of factors which cause the wealth of nations:

1. Particular factors, which can occur only in one nation and not in others; and
2. Common factors which may occur in any nation.

Particular Factors

The first particular factor in Serra's system is a surplus of products for export. His phrase “The surplus of goods which are produced in a kingdom in excess of its own needs and conveniences” reminds us of an Adam Smith type of “vent for surplus” theory of international trade, but this is only the beginning of Serra's long and sophisticated reasoning. Serra explains that he lists this as a particular factor — rather

than as a general or common one, by pointing out that a surplus — or a positive balance of trade — cannot apply to all nations. His second particular factor is the geographical position of the nation “relative to other kingdoms and parts of the world... being a potent occasion, and almost a cause, of extensive trading of a kingdom”. Rating nations according to their geographical position, “Venice holds the first place”.

General Factors

Serra lists four common or general factors which bring wealth, and, most importantly, how these factors interact with each other and with the particular factor of the geographical position of a nation listed:

1. *The number and variety of industrial professions:* We see the “number of professions” as fundamentally the same concept as “the division of labour”, in which case Serra anticipates this concept in William Petty’s watch factory by 64 years and in Adam Smith’s pin factory by 173 years. Clearly the number of industrial professions in a nation is a symptom of — and a proxy for — a variety of economic factors: technological sophistication, a sophisticated pattern of demand, a large diversity of skills, and — due to a minimum efficient scale of production in each profession — a large market. Serra rates this factor higher than the “vent for surplus” factor which he has listed under particular factors. This is because to Serra industrial professions, most importantly, behave differently from agricultural.
2. *The quality of the population:* The quality of a population is good “when the inhabitants thereof are by nature industrious, or diligent and ingenious in building up trade not only in their own industry, but outside, and on the watch for opportunities to apply their industry.” On this factor Genoa gets the highest score, followed by Florence and, only third, Venice, which “though it has more commerce than all the cities of Italy together, will nevertheless hold third place with respect to this factor.” Serra clearly relates the barrenness of the Genoese republic to their industriousness and their wealth.
3. *The presence of a great commerce:* Here we find Serra’s description of how the various factors creating wealth interact with and mutually reinforce each other in creating virtuous circles of development. In the case of Venice, “she is aided by her extensive manufactures; a factor which brings a great many people there, not only because of the trades themselves, in which case the effect would be attributed to them, but also as a result of the concurrence of these two factors together, because one gives strength to the other, the great concourse due to commerce and due to the geographical situation being increased by the manufactures, and the manufactures being increased by the great concourse due to commerce, while commerce is made greater by the same concourse of people”. It cannot be emphasized enough that the starting point for the virtuous circles described by Serra are to be found in the increasing returns of manufacturing.

4. *The regulations of the State*: Here Serra emphasizes the role of government policy in order to create national wealth. This is a most difficult task, he says, because one policy measure can have very different effects in different industries: “like the sun which makes clay hard, but makes wax soft, like a low whistle which irritates the dog, but quietens the horse.” In spite of these difficulties, Serra makes it clear that economic policy is the most important factor causing the wealth of nations.

THE RENAISSANCE STATE VS. NATURAL HARMONY

In 1338 Ambrogio Lorenzetti finished his frescoes *Allegory of Good and Bad Government* in the Town Hall of Siena. The fresco symbolising good government shows thriving shops, fine buildings and dancing citizens enjoying their leisure. Bad government is shown as ruin, rape, robbery and murder. The *Allegory of Good and Bad Government* represents the optimistic Renaissance view of Man’s untapped potential to improve his own situation. Theirs was a view of history being a continuous optimisation process where “Man’s wit and will”, applied to harnessing the forces of Nature, held enormous potential for improving his lot: “the never ending frontier of knowledge.”

The starting point for Renaissance economics, and the birth of the modern State, was an acute awareness of the suboptimality of the present situation of Mankind — steeped in the ignorance and poverty of the Middle Ages. This situation could clearly be improved, and this optimisation was chased as an ever-moving target in the distance. The propellant of this process was learning — the acquisition of new knowledge. This process resembles today’s evolutionary economics at its best.

The usefulness of a State in this process arises out of the Renaissance concept of the common weal — or the “common good” — a systemic dimension which is lost in the atomistic and static structure of today’s mainstream economics. We use the term “Renaissance State” for a type of activist and idealistic State which, we shall attempt to show, has been an “obligatory passage point” for all presently industrialised nations, bringing the nation into economic activities creating a common weal through increasing returns and self-enforcing feedback mechanisms. At a very simple level, a common weal arises out of the synergies stemming from the sharing of fixed costs — resulting either from specialised tools or from specialised knowledge, like the old story of the blind man and the deaf man whose weal was improved by acting together. We shall argue that the growth of complex economies has important similarities to the growth of complex technological systems — and that, in both cases, increasing returns are at the core of positive feedback mechanisms which increase welfare.

Such systemic synergies are further based upon diversity — just as both the very existence and the common weal of a household fundamentally rest on synergies arising from Man and Woman being different. A common weal, then, is systemic and synergy-

based — it is a dynamic concept in a process which increases the size of the economic pie — much as the process described by Adam Smith, who prior to his meetings with the French physiocrats in *The Theory of Moral Sentiments* recommends government intervention to promote a certain industry to help neither the consumer nor the producer, but because it benefited the system as a whole — the “common weal”.

At this point it is important to point out that the actions emanating from an understanding of a “systemic common weal” are very different from the idea of distributive collective action — in a setting of static rent-seeking and zero-sum games — in modern Anglo-Saxon economics. The economics of State involvement in the Renaissance was both immensely activist and idealistic. Albert Hirschman discusses the arguments which, since the late eighteenth century, have been used against this type of activist and idealistic interference with the “natural harmony” created by the market mechanism. Hirschman gives us the history of ideas listing the arguments why Ambrogio Lorenzetti’s optimistic frescoes expressing Man’s ability to improve his own destiny were, at best, naïve and futile. Hirschman has collected the arguments in favour of passivity as a strategy — a natural corollary to Ricardo’s “dismal science”. We shall discuss the role of the State in economic growth and in the history of economic thought as being torn between two fundamentally different economic outlooks: a production-centered and activist-idealistic Renaissance tradition and a barter-centered and passive-materialistic tradition of Adam Smith, David Ricardo and neo-classical economics.

Hirschman divides the arguments against any active strategy on the part of the State in three categories, and finds to his surprise that both the traditional “right” and the traditional “left” gradually started to make the same kind of arguments:

- *Perversity*: Any attempt at improving the economic or social order will have the opposite effect of that intended. This argument is clearly present already in Adam Smith’s late works.
- *Futility*: Any attempt at changing the social or economic order is doomed to fail.
- *Jeopardy*: Any attempt at changing the social or economic order will carry with it costs that are so high as to jeopardise what has previously been achieved.

The zeitgeist of the late 1990s is clearly closer to that described by Hirschman than to the optimism of Lorenzetti and his times. But the fall of the Berlin wall now gives us an opportunity to re-examine the role of the State in economic development under less ideological pressure than, not only since the start of the Cold War, but since the Ghost of Communism entered the stage 150 years ago.

However, we are seriously hampered by the fact that at the core of present-day mainstream economics — as a result of the standard assumptions of neoclassical theory — there still lies a “natural harmony”, in a world void of any systemic effects, of Samuelson’s factor price equalisation: the natural harmony which will make all wage earners of the planet equally rich — if we can only “get the prices right” and “provide

a level playing field”. Out of these philosophies of “natural harmony” rises the rejection of the State as such, as in the “Civil disobedience” of Henry David Thoreau and its present-day manifestations, as in the 1995 bombing of the US Federal Building in Oklahoma City. We live in a society caught between, on the one hand, the wish for the simple individualistic life of Thoreau, with its roots in the late eighteenth century, whose ideal is living outside any society. On the other hand, we are addicted to a standard of living which can only be kept up in the network of a fine grained specialization, the synergies and scale of which are essential to the production of systemic effects, to the common good of Renaissance economists.

These economists observed the wealth of populous and economically diversified cities — like Venice — in stark contrast to the poverty of the undiversified economic base in the countryside and in agricultural/administrative cities like Naples. Even the pessimistic Machiavelli, who “wants to present us with Mankind in its most negative and depressing aspects”, says it this way: “Il bene commune e’ quello che fa grandi le città ” — “The common good is what makes the cities great”. However, today considerable tension is created by the fact that any systemic effects in the economy — and consequently any role of the State — are external to the core of the ruling economic theory.

This fact is all the more harmful because the experiences of the presently industrialised countries indicate that the need for State interventions is stronger, the poorer the country. Those who produce economic theory all live in nations where a strong State is taken for granted — where the “obligatory passage point” of a Renaissance-type state is long history. The assumptions of neo-classical theory correspond to a world which fits Henry Thoreau’s ideal: no human institutions and no systemic effects.

Reading Thoreau is the key to understanding the dissatisfaction of the average American with Governments of any kind. Thoreau shares with Adam Smith a strong aversion to any type of human institutions and collective action, a view — clearly inspired by Rousseau — that the institutions of civilised society have corrupted Mankind. To Adam Smith all human institutions — private and public — “so invariably produce ‘absurd’ results that they have no presumptive legitimacy”. Neo-classical economics has kept Thoreau’s and Adam Smith’s myth alive by failing to internalise the systemic synergies of societies, among multitudes of professions, each with a minimum efficient size of operation, which, in turn, also cause societies themselves to have a minimum efficient size. This minimum efficient size of societies grows as more knowledge is added and more professions are formed — increasing the standards of living — and forming the fundamental connection between geography and economics.

These same factors led to the creation — in succession — first of the medieval city economies, then of national economies, and finally of “globalisation”. The needs for a State essentially arise from the same synergies and interdependencies, and from the differing abilities of economic activities to provide the increasing returns which are at the core of this system. One of the problems of today’s mainstream is, then, that —

through its assumptions of a complete absence of increasing returns to scale and of perfect information — it has produced a theory which is as individualistic as Henry David Thoreau's visions: there are no systemic external effects present at the core of the theory.

Economic theory today fails to tell us why we cannot have our cake and eat it too. This is essentially the reason why theories of the State, of the Firm, or of any other human institutions are external to the core of economic theory. Renaissance economists tell us that the State exists because of the systemic effects in an economy, effects which also the early Adam Smith glorifies. Today's practice of labeling — in a rather ad hoc manner — all unexpected economic effects either as "externalities" or "market failures" contributes little to the understanding of economic systems.

Ad hoc exceptions are more easily seen, and acted upon, close to home, in the industrialized countries, rather than far away in the Third World. For this reason, the fundamental argument for the single market in the European Union is the existence of increasing returns observed in practice but without any ties to trade theory. On the other hand, the theoretical foundation for the EU policy against the Third World is that such increasing returns do not exist. This frequent and inconsistent "assumption-juggling" in economic theory was denounced already by Joan Robinson.

From the point of view of the Third World this may be seen as an alternative version of "The Golden Rule" — the one who has the gold makes the rules. We see it as an important task ahead for economic theory to internalise the externalities which produce welfare: the systemic synergies of scale and scope which have their origin in the creation and implementation of new knowledge in those production processes which are subject to increasing returns. The World Bank 1997 World Development Report — to be entitled "The State in a Changing World" — will focus on the role of state in economic development. Predictably it will continue the discussion on the role of the State which was started with the publication of *The East Asian Miracle*.

This chapter discusses the historical role of the State in a different framework of assumptions from that of the World Bank. In our view, the successful East Asian nations essentially follow very similar strategies to those followed first by England and later by all other presently industrial nations in the early stages of development. World Bank studies do not go back more than 50 years, so these similarities are unlikely to be uncovered in their normal process. Likewise, the history of economic thought having today largely been reduced to a genealogy of neo-classical economics, the record of the past economic policies of the presently industrialised countries has to a large extent been "unlearned".

There is no such thing as an academic sub-discipline called "History of Economic Policy", and the industrial powers of Europe and North America seem to be unified in a common misconception about their own past, about the role of the State in bringing them out of poverty. We attempt to lift the field of vision beyond the post-World War II era, in order to show that in this 500-year perspective, the active role of the State in the

East Asian “miracle” — as described by authors like Amsden and Wade— is essentially business as usual: “an obligatory passage point” in the transition from underdevelopment to development. We provide a different set of assumptions from that of mainstream economics as to how economic welfare is created and distributed. We attempt to produce an inventory of the roles historically played by the State to promote the common weal.

THE ROLE OF THE RENAISSANCE STATE IN THE LIGHT OF RECENT ECONOMIC THEORY

We argue that the pre-twentieth century involvement of the State in economic growth, starting in the Renaissance — while making virtually no sense in the “fatally simple structure” of neo-classical economics — is eminently compatible with new approaches in the evolutionary economics of complex systems of, for example, W. Brian Arthur and Paul David. It is also compatible with the key elements of the new trade theory, with the theories of Paul Krugman from the 1980s and of Grossman and Helpmann and Krugman, and to the debate on strategic trade policy. This is precisely because Renaissance economists did not see the economy as being deterministic, predictable, and mechanistic. As we have attempted to show, synergetic and systemic economic effects, which modern economists see as being typical of activities subject to increasing returns, were typically recognised — with varying degrees of sophistication — by a vast number of pre-Smithian economists, and used to argue for specific State intervention.

The mechanisms creating these cumulative causations were much more clearly described by Antonio Serra in 1613 than by Friedrich List in the 1840s or by Gunnar Myrdal in the 1950s. Today the most surprising aspect of the new theories emphasising increasing returns is that they are applied either to technological systems — where income distribution is not an issue — or to what we would call symmetrical trade, to trade between groups of nations all involved in activities subject to cumulative causations. In such cases the logic of State intervention is infinitely less than in the cases of asymmetrical trade, in the cases of Third World countries which are historically locked into exporting products produced under conditions of diminishing returns, and importing goods produced under conditions of increasing returns.

The existence of symmetrical trade between nations at the same level of development, mutually in increasing return activities, is an argument for free trade and no intervention. Charles King had argued already in 1721 that trading manufactured goods for other manufactured goods was “good trade”, beneficial to both trading nations, whereas exporting raw materials and importing manufactured goods was “bad trade”. These new theories, based on increasing rather than diminishing returns, open up for a process-dependent, organic world, where historical cumulative causations, positive feedbacks, and lock-in effects give multiple solutions — also clearly suboptimal ones.

The external economies of Alfred Marshall, the growth pools of Perroux, the linkages identified by Albert Hirschman and the observation of Michael Porter that innovations

thrive in clusters are also recognitions of important systemic effects of economic development — in all these theories past history reinforces into positive feedback systems and creates path-dependence. Other recent works on technological change — of authors like Christopher Freeman and Giovanni Dosi — also describe the path dependent nature of technological change. Surprisingly many early economic writers also recognise such systemic effects, and describe them. An important insight from these modern theories is that small differences early in the history of an industry may have huge effects further on — like the “butterfly effect” in chaos theory — whether the nation is going to be specialised in increasing or diminishing return industries.

This clearly opens up a huge area for State intervention, and we argue that this is exactly the kind of intervention that what we have labeled Schumpeterian mercantilism has provided from England in the 1480s to Korea in the 1980s.

Joshua Gee, in his 1738 treatise, presents a similar argument to the modern one:

- The Trade of a Nation is a mighty Consequence, and a Thing that ought to be seriously weighed, because the Happiness or Misfortune of so many Millions depend upon it. A little Mistake in the Beginning of an Undertaking may swell to a very great one. A Nation may gain vast Riches by Trade and Commerce, or for want of due Regard and Attention, may be drained of them.

We argue that this dynamic view of the world finally opens up for a much larger role of “human wit and will” — private and collective — in influencing the wealth of nations, rather than this wealth being a product of the invisible hand of Providence. Since increasing returns are most typical in manufacturing, these new theories inadvertently create theoretical support for the century-long mercantilist preference for manufacturing over the production of raw materials.

In these early theories the use of machinery was often a proxy for “good trade”. Again, this theory is being vindicated by recent theoretical development on the important role of machinery in economic growth. The fundamental problem of neo-classical theory, and of the thinking behind the policies of the World Bank today, is that they are based on the implicit assumption that all economic activities are “alike”. These theories fail to open up for the presence and absence of increasing returns and consequently of asymmetrical trade — trade between nations exporting products from increasing return activities and nations exporting diminishing return activities. “New Growth Theory” explicitly combines increasing returns with perfect competition, which, in my view, is based on a fundamental misrepresentation of how new knowledge is produced and spread.

In this way, although opening up for the benefits of increasing returns, most of these economists perpetuate factor-price equalisation. Some few thess in new growth theory open up for the world view on which this chapter is based — *i.e.* for inherent differences between economic activities as being the starting point for explaining the present wealth and poverty of nations. The most notable exception is Robert Lucas Jr — last year’s winner of the Nobel Prize in economics. Borrowing our terminology from the study of

complex technological systems, we argue that these differences between economic activities made a strong state into an “obligatory passage point” for economic development.

Seeing a nation as a complex system — like a firm or a technological system — the need for input coordination arises. In the development of a technological system — just as in the development of nations — obligatory passage points “represent instances of power, because they discipline the interactions of actors”. As nations evolve — just as when large technological systems evolve — there are “components” in the system which may fall behind others, limiting its potential efficiency.

In the study of large technological systems, these are called “reverse salients”— a dynamic conception of what in a static system would be called a bottle-neck, and where, at certain stages, innovative energy has to be focused. We would argue that, in all presently industrialised countries, an active State has historically been an “obligatory passage point” providing the necessary function of removing “reverse salients”, and of coordinating and coercing the various “reactants” which together produce economic wealth. We would argue that the role played by the State of Henry VII of England starting in 1485, of coercing entrepreneurship in order to establish a textile industry in England, is essentially an expression of an understanding of the same obligatory passage point to development as could be observed when the Korean government coerced Samsung from trading activities into the manufacturing of semiconductors in the early 1980s. Also the other policy measures — some types of bounties, cheap credit, temporary protection of local market, etc. — are strikingly similar in both instances.

The same reasoning underlies the two State actions 500 years apart: a fundamentally Renaissance notion that new learning is the basis for economic development, and that this new learning is able to create development — through what we would call virtuous feed-back systems — in some economic activities rather than in others. In both cases, resource-based activities — in the absence of a manufacturing base — were not seen as being able to do the job of lifting the nation out of poverty.

This is what we refer to as the activity-specific element of economic growth; the fact that only nations which include a large enough percentage of activities subject to dynamic increasing returns have been able to develop, and that, in this light, the phenomenon that we call “development” is essentially a “dynamic rent” which labour, capital and governments collect from such a dynamic feed-back system. Based on the abundance of evidence of similar production- and learningbased, activity-specific strategies for early economic development since the late 1400s — they were clearly at the core of the now forgotten nineteenth century development strategies of the USA, Germany and Meiji Japan — we argue for the existence of an alternative 500-year canon of economic theory.

In the perspective of this alternative knowledge-based canon, the period of dominance of classical and neo-classical barter-based canon stands out as a 200-year old parenthesis. In this system where growth is activity-specific, the State has played the necessary role

of active midwife for all developed nation-states, much in the same way as entrepreneurs have played the initial role of strong input coordination and midwifery in all big corporations. However, both companies and States grow up, and the role of the autocratic leader and champion is reduced in both cases. Once positive feed-back systems have put a nation on the path of dynamic increasing returns — having started what in the study of complex technological systems sometimes is called “historical snowballing” — the role of the State is reduced to a more indirect one of maintaining the national innovation system, and of generating demand for new skills.

In other words, the closer a nation finds itself to the virtuous circles of knowledge generation operating near the world “frontier of knowledge”, the more the nation-state has to depend on indirect measures of influencing the economy. The biggest “latecomer advantages” of being able to catch up to a known technological frontier have been lost. For this reason the mythical MITI plays a very different role in the Japanese economy today from that after World War II. Once the vested interests of the ruling classes have been moved from resource-based activities and into manufacturing, the positive feedback system also embraces the political system: private rentseeking has been brought into the activities which also brings with it collective rent-seeking, and “what’s good for General Motors is good for the United States.”

With this, we explicitly suggest that in some economic activities, much more than in others, there is a community of interest between the vested interests of the entrepreneur and the vested interests of the nation as a whole. This point was very clearly made by nineteenth century US economists, particularly by Henry Carey, who also made it clear that such effects could not be produced in the absence of manufacturing industry. Because of this, perhaps the most important historical role of the State has been to bring the economic activities of the nation into such positive feedback activities. This factor is the fundamental common element between England in 1485 and Korea in the 1980s. This was also the core argument of the “American System of Manufactures” and of the “High Wage Strategy” which completely dominated the economic policy of nineteenth century USA. The US Civil War was a war between the free traders of the South and the politicians of the North which argued that no national wealth could be created without the protection of domestic manufactures. Reading of the contemporary US sources leaves little doubt that “Free Trade, Slavery, and Secession were... sworn allies”.

In this context, Latin America is best seen as a group of nations where the “South” — the groups with vested interests in resource-based activities — won the civil wars. This kind of argument — and the strategies of Henry VII and Korea 500 years apart — are essentially only dynamic versions of an industrial policy which Alfred Marshall recommends in the first edition of his *Principles*: “A tax... on the production of goods which obey the Law of Diminishing Return, and devoting the tax to a bounty on the production of those goods with regard to which the Law of Increasing Returns acts sharply”. We would argue that Marshall here describes the most typical economic policy of what we today despise as “mercantilism”. We find the same kind of reasoning again

as the fundamental mechanism causing uneven economic development in Frank Graham's 1923 theory of international trade. Most unfortunately, with the mathematisation of general equilibrium theory all such elements were gradually thrown out of economic theory as being "not compatible with equilibrium" or "as inconsistent with free competition, and therefore as outside the scope of the present discussion". In this way economics moved increasingly further into the realm of pure ideology, the ideology being hidden in the assumptions, as Joan Robinson and others have claimed.

The nations producing behind the barriers to entry created by the largest effects of increasing returns and with the most exclusive new information were the very nations forcing upon the world a model assuming no increasing returns and perfect information. In our view, the seriousness of the long-term practical consequences of this on the poor world will only dawn upon us over the next decade. The further removed an economy operates from the assumptions of neo-classical economics, the greater its vested interests in making other nations operate as if this theory were a true reflection of reality. We would argue that, since the end of the Second World War, "Cold War Economics" has contributed to strengthen this development even further: The political need for an economic theory providing an ideological defence line against communism, and the further mathematisation of general equilibrium economics mutually reinforced each other.

The historical coincidence is that both these forces demanded that the elements which created anything but a harmonious and even economic growth be eliminated. Economic theory developed, as Paul Krugman put it, along "the perceived line of least mathematical resistance". At the height of the Berlin Blockade Paul Samuelson proved the universal harmony of "factor-price equalization" based on the standard assumptions of neo-classical theory. At the height of McCarthyism, Milton Friedman defended the use of these seemingly counterfactual assumptions as the basis of economic theory.

The crossfire of mathematisation of a static neo-classical theory and of political demands — both needing perfect markets — cleaned out of economic theory all the factors which cause uneven economic growth, and consequently also most of the factors which bring about a need for a State intervention. The following is an example of the poverty-equilibrium which the historical absence of a development state is likely to create: the world's most efficient producers of baseballs for the American sport are located in Haiti: their wages are 30 US cents per hour.

Their technology is the same as the US producers' — all baseballs are sewn by hand. All the skills and capital of the USA have not managed to mechanise this production — it is a relic left over from a technoeconomic paradigm long past. Just as in Keynes' unemployment equilibrium, such poverty-equilibria may lock nations into a comparative advantage of being poor and ignorant, in industries requiring low skills. This in spite of their being the most efficient producers in the world with the best available technology. Historically the "development state" has taken a nation into economic activities which require higher skills, and thus established a platform from which successive growth

becomes more self-sustaining with much less intervention. We consequently argue that the presence of a “development state” — taking the nation-state into increasing return activities — as an obligatory passage point is determinant for economic welfare. The most interesting aspect of such poverty traps is how the systemic aspects of development create widely differing wages in professions where productivity of labour and capital are essentially the same the world over. The cumulative effects, which today generally have their roots in the export sector, spread through the labour market also to activities which, by themselves, are not subject to increasing returns: the vast majority of workers in any nation are employed in the traditional service sector. Productivity in these activities vary very little from nation to nation, as in our example with the bus drivers in Norway and in Bolivia.

The systemic effects accompanying economic growth — which are created by increasing returns — produce what we in the next part shall call a collusive distribution of the gains from technological change: in the Fordist mass production paradigm — centred on the nation-state — the gains from technological change spread to a large extent as higher domestic monetary wages — rather than as lowered prices internationally.

THE TWO CANONS OF ECONOMIC THEORY

Next year’s European Conference on the History of Economics — in Athens — is based on the “role of the canon in the history of economics.” The call for papers notes that in most sciences the authority of the “classics” has been challenged — not so in economics, “where the canonical sequence of Smith, Ricardo, Marx, etc. still constitutes the skeleton for most teaching and scholarly discussion”. The question of whether alternative “canons” exist is also raised. In this chapter, we argue that a parallel tradition — an alternative canon — exists. The theoretical conflict between the forefathers of today’s mainstream economics and the forefathers of the alternative canon has existed since the 1622-23 debate between Gerard De Malynes and Edward Misselden, where Malynes represented a static theory rooted in barter and Misselden represented a theory centred on learning and production.

In the history of economic thought, their debate is interpreted as being about exchange controls and the balance of trade. However, by going back to the sources, one finds that the main line of attack by Misselden against Malynes is his “mechanical” view of man — Malynes has left out Man’s “art” and “soul”. Misselden quotes at length a paragraph from Malynes, where Malynes reduces trade to three elements, “namely, Commodities, Money, and Exchange”. Objecting to this definition, Misselden says: “It is against Art to dispute with a man that denyeth the Principles of Art”. Misselden scorns Malynes for not seeing the difference between a heap of stones and logs and a house — because Man’s productive powers produce the house but his soul has been left out.

A similar criticism can be made of neoclassical economics. Misselden represents the acute Renaissance awareness of the enormous territory to be covered between

Mankind's present poverty and ignorance, and the enormous potentials. This released enthusiasm and energy. The situation recalls Keynes' frustration with the suboptimal situation of the world under the Great Depression.

We shall attempt to show that both to the Renaissance philosophers/economists as well as to Keynes, the formula needed to "free" society from its suboptimal position was what Keynes called "salvation through knowledge". In the late eighteenth century a new type of economic theory came into being, focusing on the "natural harmony" of Nature. This was truly a paradigm shift in Kuhn's sense. The incentives of Renaissance economics to produce knowledge through a process which we have labeled dynamic and knowledge-producing rent-seeking — or Schumpeterian Mercantilism — had in many cases degenerated into static rent-seeking. Whereas the optimistic theory of the Renaissance focused on the limitless potential of "Man the producer", the new economic theory came to focus on "Man the trader and consumer".

The two theories were steeped in very different realities — the old one in Man's ability to create and produce, and the new one in a world of barter, based on the mechanics of an *ordre naturel* — the "Natural order". The old theory was dynamic and organic and centred on "thought" and "werden", the new theory was mechanical and static, centred on "matter", and "sein". In the old theory the market was present in the role of a servant of active human beings who knew where they were going, in the new theory the market acquired many of the characteristics of "Providence", as the manifestation of the *ordre naturel*. Werner Sombart fittingly calls the Renaissance economics activist-idealistic, and the economics from Adam Smith onwards passivist-materialistic.

Just as Renaissance economics sees no limits to progress — they truly see "a never ending frontier of human knowledge" — in Adam Smith's system nations reach a stationary state where they can "advance no further", when that "full complement of riches which the nature of its soil and climate... allowed it to require" had been reached. It is only here that we see the practical consequences of Adam Smith and the "deep ecology" movement sharing the same assumptions — no new knowledge entering the system. The only logical consequences of a theory which does not allow for the production of new knowledge is either a stationary state or an "ecological disaster."

This disaster can be predicted by simple extrapolations. However, each level of knowledge carries with it its own level of "sustainability." As we shall discuss later, "knowledge" and "institutions" are the conspicuously and "actively absent" factors in Adam Smith's system, *i.e.* these factors are not only ignored, but it is actively argued that they have no relevance. Just as the focus of Renaissance economics was on production, the focus of neo-classical economics is on barter and exchange. Leibniz, in 1671, sees the origin of barter as being in production, and quotes Aristotle: "Nam Mercator transfert tantum, Manufactura gignit" — Trade can carry only as much as the factories produce. To Leibniz, the poverty of the producing artisans was an important argument for the establishment of an active State. "After all, is not the entire purpose of Society to release the artisan from his misery? The farmer is not in need, since he is

sure of his bread, and the merchant has more than enough". The fundamental characteristic of the new "classical" economic theory was its conscious de-emphasis on the role of human knowledge which comes across very clearly in Adam Smith's late work, *i.e.* The Wealth of Nations. The new economics of the late eighteenth century was to produce the "dismal science" of Malthus and Ricardo, and fundamentally to change Man's view of the role of the State. The old theory emphasized systematic effects emanating from *das Ganze* — from the whole as a system. The new theory, starting with Adam Smith and reaching its apex with Paul Samuelson's factor-price equalisation in 1950, had the individual as its sole unit of analysis.

However, not even the whole individual became the unit of analysis, only a part of the individual labeled *Homo Oeconomicus* who engaged in barter. Excluded was not only the state, but also *Homo Faber* — man the innovator and producer. In economics, the mechanical world view is centred on barter, accumulation, physical metaphors, equilibrium, and optimality. In this mechanical view, a fundamental characteristic of Man is his propensity to barter. The organic view in economics is centred on inventions, production, evolution, biological metaphors, and disequilibrium.

In an evolutionary system, almost by definition, no optimality exists, except as an ever-moving target in the distance. In this organic view a fundamental characteristic of Man is his propensity to explore, to invent, and cumulatively and continuously to create new knowledge. In this perspective, today's evolutionary economics still may be seen as being too "mechanistic" — substituting biology-envy for "physics envy" in a process where the role of conscious human effort — individual and collective — as the main engine propelling evolutionary change is still missing. One fundamental difference between the two world views appears when authors from the two schools describe how Man differs from animals.

The most famous representative of the barter-centred theory of development says it this way:

- The division of labour arises from a propensity in human nature to... truck, barter and exchange one thing for another... It is common to all men, and to be found in no other race of animals, which seem to know neither this nor any other species of contracts... Nobody ever saw a dog make a fair and deliberate exchange of one bone for another with another dog.

The reply from the production-centred theory of development:

- ... Beavers build houses; but they build them nowise differently, or better, now than they did five thousand years ago... Man is not the only animal who labours; but he is the only one who improves his workmanship. These improvements he effects by Discoveries and Inventions....

The roots of the problems of today's mainstream economics can be traced back to these conflicting views on Man. We would argue that it is time for economic theory to find again the knowledge-based and production-based alternative to today's economic tradition. This tradition is traced over the last 500 years in Figure.

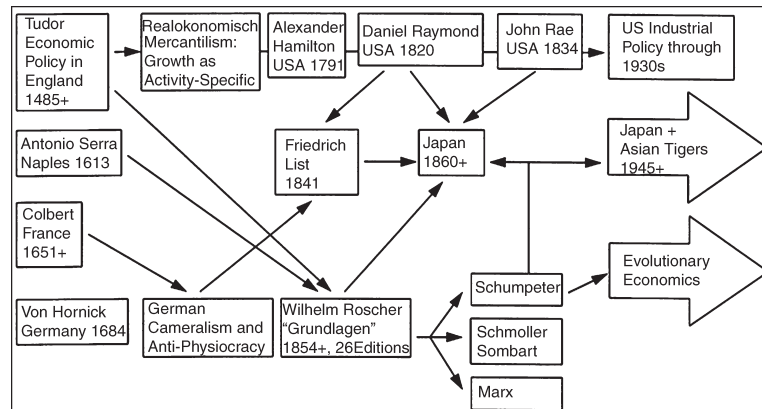


Fig. The Knowledge- and Production-based Canon of Economic Theory

Adam Smith's nineteenth century enemies in the USA would ask how Adam Smith's theory explains economic growth. How do you get more bones into the economy by teaching dogs to barter and to make contracts? And, how do you explain with Adam Smith's theory why dogs today eat canned dog food and not bones? Although Adam Smith has a "general tendency of things to improve" in the background, these improvements descend on mankind as "manna from heaven" — not as a result of organised conscious effort — and hit everybody at the same time.

The term *laissez-faire* is today used as a slogan for no government intervention. However, from the context of how and when the remark was made, a different interpretation is possible. The term is known through the French physiocrats, but the term was not invented by them. Quesnay perpetuated the term as a slogan for the non-intervention of government. However, the physiocrats picked up an expression which long before had been used by a merchant called Legendre, who was consulted, around 1680, by Colbert, as to what could be done to promote trade and welfare.

This merchant then dropped the words which became famous: "*laissez-nous faire*", to which later Quesnay added: "*Ne pas trop gouverner*" — "Do not govern too much". However, it seems important to keep in mind that Legendre at this point had the whole arsenal of Schumpeterian mercantilism at his disposal — he may very well have expressed the feeling that this was all the government could be expected to do — that the maximum point of government intervention had been reached, that the rest was up to the entrepreneurs.

“UNITED BY A COMMON MISCONCEPTION ABOUT OUR PAST” — THE DECLINE AND FALL OF RENAISSANCE ECONOMICS

"A nation", said Harvard political scientist Karl Deutsch, "is a group of persons united by a common error about their ancestry and a common dislike of their neighbours".

This saying is presumably valid for all tribes — including academic tribes. The same maxim seems most appropriate when applied to our understanding of the role of the state in bringing Mankind out of poverty and ignorance. The history of Renaissance economics in today's textbooks is not unlike a history of Rome written by the Huns. Just as the Black Legend — *la leyenda negra* — has disfigured Spanish history, our view of mercantilism has been deformed by the “Midas legend” — the legend that the mercantilists were only interested in gold. Modern civilized society was formed during the Renaissance; all artists and scientists of the Renaissance — from Galileo to Michelangelo — are our heroes. Renaissance Man is still a term denoting profound knowledge in many fields.

Yet, we are told by virtually all modern histories of economics that the economic theory behind all this was only about collecting gold. In the field of Cold War Economics, just as in the cold war politics of a Latin American banana republic, those not siding clearly with either of the two superpowers were the ones most at risk; they would — literally — be shot at from both sides. Cold War Economics was in many ways a battle of Utopias, where, as Colin Clark once put it, “theory” was mistaken for “science”. This was a relatively unsophisticated scientific war between “good guys” and “bad guys”, as in the Western movies of the time, where the market was seen as either “good” or “bad”.

This led to a polarisation of economic theory — the combined effect of the Cold War and the process of mathematisation of a static system crowded out other and most relevant aspects. Focusing on the quantifiable factors in their barter-based models, the economics profession came to be prisoners of their own limited toolbox. As the history of economic thought also slowly came to be filtered through neoclassical lenses, the debates of the past were reinterpreted to fit the Cold War mold.

Historical authors were given labels as free traders or protectionists while both the historical context and the other often very sophisticated aspects of these older theories were lost. This is the history of economic thought with which we still live. Another feature of this is that some nations come to be curiously at odds with their own past. Nowhere is this more clear than in the USA. US banknotes provide a portrait gallery of people whose economic policy today would be ridiculed both by the World Bank and in American university textbooks in economics — from the views of Benjamin Franklin that “protection and competition” was the only way to make manufactures cheap in the USA to George Washington who, for the same reasons, insisted on a necessarily primitive American-made suit at the first presidential inauguration, via the protectionists Hamilton and Jackson, to Thomas Jefferson's attempt to prohibit the works of Ricardo in the US, to the view of Abraham Lincoln that at the origin of wealth there are inventions and innovations.

The uncomfortable fact that the US Civil War — to a large extent a war between the free traders of the Southern Confederacy against the protectionist North — was won by the “bad guys”, who represented despicable mercantilist theories of economic

development, is today seriously played down. With this development the real treasure of US economic theory, which even today is very relevant — a century of thought from Daniel Raymond to John Rae, E. Peshine Smith and Simon Patten — is all but deleted from the national collective memory, and from the history of economic thought. Most nineteenth century US economists made it very clear that their economic policy was specific to the situation of the US economy then; they would not have recommended protectionism as a solution for the problems of the USA today.

One of several studies puncturing the myth that US economic policy during its first 100 years was one of a weak State and of laissez-faire was written by a Harvard historian: *Economic Policy and Democratic Thought*. We seem to be still suffering from the illusion in nineteenth century economic theory of an automatic development — a view also largely shared by Marx. However, this view had its strong opponents also in the nineteenth century, particularly in Germany and the USA. German economist Eugen DuÈhring, the great defender of US economist Henry Carey in Europe, scorns die Karrikierer des Merkantilismus — the caricature makers of mercantilism — who “only too often spoke as if the businesspeople and the statesmen of the day almost believed that precious metal could be used as food for the human body”. The important systemic aspects of this theory — the promotion of the common weal — are mostly left out of today’s accounts.

Welfare is clearly a product of conscious human will, not of any invisible hand of Providence. Johann Gottfried Hoffman in his *Theory of Taxation* puts it this way: “The delusion that security of life and property, the productivity of labour, and the consequent possibility of acquisition and enjoyment, and even the elevation of the spiritual and the ennobling of the moral nature — that these goods came to Man in the gift of gratuities, is itself a proof of the advanced stage of culture which the greater part of Europe at present occupies. As the grown man has long since forgotten the pains it cost him to learn to speak, so have the peoples, in the days of their mature growth of the State, forgotten what was required in order to free them from their primitive brutal savagery”.

Another German economist, Gustav Cohn, who was born in the year Hoffman’s *Theory* appeared, picks up the argument and continues: “In point of fact, how significant was the involuntary testimony which the eighteenth century, with its repudiation of the historic State and its yearning after the primordial state of nature, bore to the blessings of the inherited culture which it ungratefully enjoyed”. This description — written more than 100 years ago — also fits the zeitgeist of today. Again today “repudiation of the State” and “the end of the nation-state” — based on English classical and neo-classical economics — are mixed with the “yearning after the primordial state of nature” from the “deep ecology” movement.

In fact such anti-modernist movements seem to occur towards the end of techno-economic paradigms. Rousseau and Adam Smith represented the eighteenth century version of this movement. The nineteenth century fin de siècle movement comprised nihilism, and towards the end of the twentieth century we see a postmodernism bordering

on nihilism. Out of the late nineteenth century fin de siècle mood grew both the Blut und Boden movement and the conservative revolution of the 1920s and 1930s, paving the way for irrational Nazism. The new type of economic theory brought with it a very different view of the role of government. There were good objective reasons for this.

The old mercantile system had degenerated, and Adam Smith's basic idea that the English and French economies now needed a heavy dose of market was no doubt correct. However, Adam Smith was so successful in ridiculing all economists before the Physiocrats that the whole production-based tradition in economics was slowly lost. Reconstructing a new economic theory which exogenised both knowledge and State was clearly much less risky to advanced nations than to the more backward ones. Adam Smith's only mention of the invisible hand in *The Wealth of Nations* can easily be read as the description of a virtuous circle which takes on the character of a perpetuum mobile — of "historical snowballing" — once it has properly been put into motion.

The closer a nation is to the frontier of human knowledge, the less the State has to contribute in direct interventions. Or, as Keynes said in 1923: "The more troublous the times, the worse does a laissez-faire system work." The issue was therefore not, said Keynes, one between collectivism and laissez-faire, but between targeted state action and a socialism which was out of date and contrary to human nature. This requires a strong State, but not a big State. The Midas Legend is more deeply entrenched than ever.

2

Global Economy

The development of the global economy can be traced back many hundreds of years when traders from the east and west came together to exchange goods. However, the growth of the modern global economy is marked by a number of features as follows:

LEGACY OF MERCANTILISM 1500-1750

The prevalent wisdom was one of nationalism, that is, that one nation prospered at the expense of another. Nations like the UK, Netherlands and later France and Germany, with powerful navies which ruled the waves in the West, and the traders of the East, dominated that area. Over time, nationalism gave way to bullionism, where gold and silver, rather than other raw materials, became the basis of wealth. Still later, domination took another form, where countries were believed to be powerful if they had a favourable balance of trade - an excess of exports over imports. Mercantilism died with the development of the United Nations (UN) and the General Agreement on Tariffs and Trade (GATT), along with Adam Smith's tome on the "Wealth of Nations" which advocated market forces as the principal driving force to development and wealth.

WORLD TRADE

Economic progress is linked to world trade and those who preach trade restrictions are denying this fact. Countries like the old communist bloc have not developed as fast as those with more outward orientation. The same can be said of African nations, where the inability to industrialise and export in volume has locked them into, generally, primary product producers. Economic Structural Adjustment Programmes (ESAP) are

supposed to remedy this situation by giving “command economies” a market oriented focus. Another argument concerns whether marketing has relevance to the process of economic development. Less developed countries (LDCs) have traditionally focused on production and domestic income generation. Also, marketing addresses itself to needs and wants and it could be argued that where LDCs’ productive capabilities are far less than unsatisfied needs and wants, then marketing is superfluous. However, adopting “marketing” could lead to the more efficient and effective use of productive and marketing resources and it may be able to focus on current needs and find better solutions.

For example, techniques developed in the West for optimising transport resources could well be transferred to effect. Similarly, adopting new methods of marketing may give better results.

A good example is the Cold Storage Company of Zimbabwe (CSC). By changing from the current system of marketing cattle (the CSC takes in cattle, at fixed prices and slaughters) to an auction system by description, all actors in the system could benefit. Decisions in product, price, communications and merchandising can stimulate economic development. Changing from fixed price systems to market based pricing could lead to the faster achievement of development objectives (for example “higher incomes”). In current drought conditions in Africa, governments could well benefit from advertising other forms of nutritious food, for example, fish, rather than let the populace be left uninformed and disgruntled about the lack of maize.

Composition of World Trade

Agriculture, minerals, fuels and manufactured goods figure most in world trade. However shifts are occurring.

Table Shift in Commodity Trade - % of World Trade

Product	1980	1985%	1988%
Agriculture	22.5	1	14
Minerals	14	-5	14
Fuel	41	-3.5	1
Manufactures	17	4.5	1

Interestingly enough, those economies which have divested themselves of agriculture (or made it more efficient) and invested in manufacturing are those which have shown spectacular growth. Table compares Zimbabwe with Thailand⁶.

Patterns of Trade

Most industrialised nations trade with each other. This had led to their continued domination, particularly the USA, Western Europe and Japan which between them have 66% of world GNP and trade. In 1985 industrialised trade to other industrialised countries accounted for 47% of trade, next came developing countries to industrialised and finally industrialised to developing countries (13%). Political influences can also be seen between trading partners, for example Zimbabwe’s trade with China. Marketers

need to identify trading patterns between nations and product trading patterns. East-West trade and West to the former communist bloc is likely to grow at the expense of North-South trade.

Table. Structure of Production

Country	GDP \$ m		Distribution of GDP %							
			Agriculture		Industry		Manufacturing		Services	
	1970	1992	1970	1992	1970	1992	1970	1992	1970	1992
Zimbabwe	1415	5350	15	22	36	35	21	30	49	43
Thailand	7087	110337	26	12	25	39	16	28	49	49

This pattern is repeated throughout Africa and Asia in general.

Comparative Costs - comparative Advantage

Price has been called the immediate basis for international trade - cheaper prices based on different cost structures, especially labour. Countries trade because they produce and export goods in which they enjoy a greater comparative advantage and import goods in which they have a least comparative advantage.

Balance of Payments

This is the measure of all economic transactions between one nation and another. The balance of payments is made up of the current account, showing trade in goods and services; and the capital account, which shows financial transactions. In 1989, after official transfers, the USA had a US\$ 109,242 million deficit on its current account, Japan had a \$ 131,400 million surplus, Tanzania a \$ 778,5 million deficit and Zimbabwe a \$ 2,783 million deficit.

The balance of payments account helps marketers select the location of supply for foreign markets and the selection of markets.

The capital account may show the nations which have control restrictions and hence be difficult to deal with. In this regard, African nations are generally disadvantaged.

Government Policy

This refers to the government measures and regulations which have a bearing on trade - tariffs, quotas, exchange controls and invisible tariffs. These can cause formidable barriers to marketers and will be dealt with at length later.

World Institutions

Institutions like GATT and the United Nations Conference on Trade and Development (UNCTAD) have been of help to countries in their development. GATT had over 120 members and associated and accounted for 80% of world trade. Its intention was to create a general system of preferences and negotiate tariffs for members' products on a nondiscriminant basis and provide a forum for consultation.

The Kennedy Round of the 1960s was superseded by the Tokyo round of the 1970s and that by the current Uruguay round signed in 1994. UNCTAD furthers the

development of emerging nations. It seeks to improve the prices of primary goods exports through commodity agreements. It also established a tariff preference system favouring developing nations.

Regionalism

Regionalism is a major and important trade development. Some regional groupings have either market (EU) or command (China) or mixed economies (former communist countries and The Preferential Trade Area (PTA) and The Southern African Development Community (SADC). With these developments, free trade zones have occurred (all internal barriers abolished) economic unions (the EU), export pricing zones (Mauritius) and other plans.

The major regional economic organisations are: Acuerdo de Cartagena (Andean Group), Association of South East Nations (ASEAN), Asian Pacific Rim countries (APC), Caribbean Community and Common Market (CARICOM), Central American Common Market (Mercado Comùn Centro Americano), Council of Arab Economic Unity, Economic Community of West African States (ECOWAS), the European Union (EU), Latin American Integration Association, Organisation Commune Africaine et Mauricienne, Preferential Trade Area (PTA) and the Southern African Development Conference (SADC).

A principal collapse has been the Council for Economic Assistance (COMECON) with the disappearance of the communist bloc in Eastern Europe. Of these blocs, the EU (reporting 33% of world trade) and EFTA are very important. To counteract the growing power of the EU, the USA and Canada have entered into an agreement with Mexico as a willing partner and created the North American Free Trade Agreement (NAFTA). These blocs are of various form, power, influence and success. ASEAN is a collaboration of industry and agriculture, PTA in tariffs. SADC and PTA have had historically little impact but are now beginning to grow in importance in view of the normalisation of South Africa. The EU, North American Union and the Pacific Rim Union will pose the greatest power blocs in future years.

Many developing countries have entered into trading blocks as a reaction against loss of developed country markets or as a base to build economic integration and markets. The development of trading blocs can bring headaches and advantages to trade. It is worth comparing the European Union, a relatively well developed bloc, with SADC and the PTA which are well developed.

International Monetary System

Historically, any commodity that was generally accepted as a medium of exchange could be used as money. Various objects have been used as money over time, including human skulls, cows, wine, shells, and mainly metals. Even today in the Yap island of the Pacific, large stones (fei) are used as money. Cows were used as money in ancient Greece or Hellas, Egypt, China, Italy, and Britain. In Southern Sudan, even today, people

use cattle and goats as a standard of value. Thus, “a reasonably sound wife costs about 40 head of cattle, with perhaps a few goats and chickens thrown in.”

From recent discoveries in Dispilio, near Kastoria, Greece, the most ancient signs of written words, fishing and other instruments, and vases were dated back to 5260 B.C. Also, shapes of coins, similar to talents used in ancient Greece and Rome and existing now in the Museum of Heracleion, Crete, were found to belong to the same ancient times. Such discoveries put an end to the myths about the Phoenician alphabet and the Indo-European race. Hellenes were living in that area and the rest of Greece at that time. The research continues and is financed by the European Union. Moreover, human fossil bones and tools were discovered by French and Greek archeologists in Triglia, Halkidiki, as ancient as 11–12 million years old, compared to 2.5-million-year-old tools found in Africa.

Lending operations for commodities predates money itself, as for example during the Minoan and Mycenaean civilizations, when values were expressed in terms of cows and oxen. Also, the Code of Hammurabi deals with mortgages and debts. However, the use of coins and other monetary units, away from the inefficient barter system, later necessitated the establishment of banks.

In addition to cattle, during the Homeric period, certain pieces of gold called talents were used as money. Recent research in Mycenae discovered a copper ingot 2.5 feet long in a shape of ox-hide, dated around the fourteenth century B.C. Moreover, certain pieces of gold were in circulation but were expressed mainly in terms of cows.

An alternative name to euro, suggested in the European Union, was the talent, used in ancient Greece and classical Rome, which are considered as the common roots of the EU member states. It was considered as a hard currency with a value of 26 kg of silver.

In ancient Sparta, large iron disks were introduced by Lycurgus as money units. The Spartan monetary units had a face value higher than the intrinsic or metal value. This system of debased money is used even today by advanced money societies. Later, gold and silver were deposited by the Lacedaemonians with the Arcadians for safekeeping.

Silver coins replaced previous tortoise-shell currency in Corinth, the island of Aegina, and mainly Athens. They were stamped to indicate their value on the one side, and with an owl on the other side. Aristotle, commodities have a use value and an exchange value measured in drachmas, which can be used for exchanges and storage of wealth. Plato proposed, in the *Laws*, that token money be used for domestic transactions and gold and silver for transactions with foreigners. In that way, money was internationalized by the Hellenes at that time. However, Aristophanes noted in “*The Frogs*” that “in our Republic bad citizens are preferred to good, just as bad money circulates while good money disappears.” This concept is very near to Gresham’s law, after Sir Thomas Gresham, Queen Elizabeth’s Chancellor of the Exchequer.

As domestic and foreign trade increased, mainly between Athens-Piraeus and Phoenicia, Egypt, Sicily, and other neighbouring countries, banking transactions developed. A number of arcades, the most famous called *Degma*, were used for

exhibitions and exchanges of commodities from all the markets of the eastern Mediterranean and beyond. Moreover, regulations similar to our antitrust laws were enacted by the city government.

Other important monetary units used in ancient times were the Pu money, in the shape of a spade, as well as knives and other metallic instruments, in China around 700 B.C. Also, the first paper or parchment money was issued in China in 140 B.C. In Egypt, the first coinage was established under the Ptolemies at the end of the fourth century B.C.

Money and credit opened new freedoms and opportunities during the period of the Roman Empire. Financiers and profiteers flourished while trade enterprises were growing, and money interest began to play an important role in trade and finance. The Romans used a common currency to help military cooperation and to stimulate free economic transactions with the regions under their domain. Such common currencies were used later by the United States and the European Union.

Copper bars or ingots were the first monetary units in ancient Rome. Silver coins, called denarii, similar to those previously used in Greece and mostly debased, were introduced after 269 B.C. Furthermore, with the use of gold later, three metals were used together and serious problems of trimetalism or bimetalism appeared as the relative values of the various metals changed over time. During the Roman Empire, the creation of a new commercial class, the patricians, increased transactions in the cities and necessitated a body of laws that later had a profound influence on legal and economic institutions, including banks and other financial firms. The recognition of juristic or artificial persons by the Roman laws was important for the creation and expansion of the corporate form of modern enterprises. The Romans, with their efficient administration of justice and the security of life and property, created many trading, financial, and shipping centres in their dominion and promoted international transactions.

Nevertheless, Rome suffered at times from too little or too much gold and silver and the frequent debasement of its metal money. The rudimentary banking system was inadequate to deal with price changes in the absence of a paper or check system. Moral corruption, slavery, and decline of the silver and gold mines of Spain and Greece were the main reasons for the fall of the Roman Empire.

During the Byzantine Empire and the Middle Ages, gold was the main currency used, but there were problems of debasement of currencies and price fluctuations, mainly from changes in the supplies of gold and silver. Tax reductions and other administrative reforms stimulated trade and made Constantinople, the capital of Byzantium, a commercial and financial centre for southeastern Europe and the Middle East. However, commercially minded Venetians, who were better organized in foreign trade and finance, managed to settle in a number of ports around the Balkan and other areas and to control customs revenue and banking activities.

In competition with the Genoese, another commercial people, they exploited the lucrative trade and financial markets of the Golden Horn, the Black Sea, and other

neighbouring regions. In the Middle Ages, early financial operations were carried on by monasteries, but any interest was considered immoral by Church law and was considered unnatural by Aristotle. After the Renaissance, modern banking activities were conducted at the old Fairs and monasteries and by the Lombard bankers in England, as well as by financial houses such as the Bardi, the Medici, the Peruzzi, and the Strozzi of Italy, based mainly in Florence, and the Fuggers and the Rothschilds in Germany. They were exempted from the ban on usury and made loans to kings and even to the Pope. Goldsmiths or jewelers in Lombard Street and other places accepted deposits for safekeeping, giving the proper receipts, which were payable to the bearer. The depositors could hand the receipts over to their creditors, performing the same functions as in modern banking. Thereafter, important banks appeared, such as the Bank of Sweden, the Bank of France, and the German Reichs Bank.

The British silver coin initially was a pound of standard silver. Also, the French livre was one pound of silver, and both represented one of the early international monetary units. Other countries, mainly Mexico, China, and India, developed a silver standard.

During the period of Mercantilism, emphasis was given:

- On foreign trade as a means of accumulating precious metals, primarily gold and silver, as the most desirable form of national wealth,
- On the supremacy of the state over the individual. Gold and silver could be used to finance trade or pay large armies for foreign wars or be retained as luxuries.

As Columbus said, with gold one can get even souls into paradise. The state should support foreign trade and establish colonies to increase national wealth. Commercial expansion occurred by monopolies, such as the Merchant Adventurers, the Eastland Company, the Muscovy Company, and the East India Company.

Towards the end of the Mercantilistic period, the physiocratic school appeared in France. The school was based on the natural order, which prevailed over all the economic activities in a harmonious way, similar to that of the planetary system. This *laissez-faire* system was further developed by Adam Smith and other classical economists and is considered as the basis of capitalism. Later, Eugen Von Böhm Bawerk and other Austrian economists stressed the justification of interest payment, based mainly on the productivity of capital, which helps the worker and the entrepreneur to increase production. The gold standard, first introduced in Britain in 1821, was also adopted by Portugal (1854), Germany (1871), and other European countries later. Britain abandoned gold from 1914 to 1925 and permanently in 1931.

During the classical gold era, roughly from 1870 to 1914, there was a high capital mobility, mainly from Britain, France, and Germany (about 3–9 per cent of the GDP) to Australia, Canada, New Zealand, and Sweden. Such capital flows were used primarily for investment but also as official reserves (gold and foreign exchange), mainly by the United States, Russia, India, Belgium, and Sweden. A similar phenomenon can be observed in our day with the capital flow in emerging nations, which is used primarily

for reserve holdings. In all these cases, the role of investment banks in 1870–1914 was important in stimulating global capital flows, as it is now.

Monetary Systems in the Americas

In the Western Hemisphere, the Incas, covering most of the coast and Highlands of western South America, had no monetary system of any kind, whereas in the American colonies, coins brought over by European immigrants and Indian wampums (ornamental cloth) were initially used as money. Corn was mainly used as a medium of exchange (1631) in Massachusetts, whereas tobacco and rice were used in the south. In order to facilitate trade with the Spanish colonies, the Spanish silver dollars, or fractions of them, were also used in the Americas. After George Washington's inauguration (1789) as the first president of the United States, gold and silver full-bodied coins were used.

In the United States, for practical purposes, silver was used from 1792 to 1834 and primarily gold from 1834 on, although the Coinage Act of 1792 provided for a mint and stipulated that the principal coin would be the \$10 Eagle (weighing 270 grains and containing 247.5 grains of pure gold out of 480 grains in a troy ounce). In 1837, the price of an ounce of gold was fixed by the U.S. Congress at \$20.67. In 1832 and later, Robert Owen of England introduced a paper money in "the Owenite communities in America," mainly in Cincinnati, Tennessee, and Ohio, with nominal values in labour hours. These labour notes entitled "National Equitable Labour Exchanges," based on the utopian teachings of Charles Fourier of France, were short-lived, primarily because of the difficulties in exchanges.

Nevertheless, bank notes and U.S. Treasury notes were mostly used because they were less costly and more convenient than gold. But "In issuing paper money...it is only necessary that its quantity should be regulated." Moreover, international payments were made in bills of exchange, comparable to postdated checks. Thus, a U.S. exporter would receive gold or more usually bills of exchange from the importers of other countries and sell them for U.S. dollars in New York or other cities. Because of the Civil War and the increase in prices, payments in gold in the United States were suspended in December 1861. The convertibility of gold for the dollar was restored at \$20.67 in January 1879 and remained so until 1933. The pure gold standard system was introduced in 1900 in order to avoid relative price fluctuations in the prevailing bimetallic system of gold and silver. The United States managed to hang onto gold during World War I, but prices increased by more than double from 1914 to 1920, and then by 1932 they fell more than they had risen.

However, new discoveries of gold led to changes in prices and economic instability. Also, when a country had a deficit in the balance of payments, a net outflow of gold would take place, and under a gold standard system, money supply and eventually prices would be reduced, with the possibility of a recession or depression and unemployment. *Ceteris paribus*, lower prices would lead to more exports and less imports, thereby restoring the balance-of-payments equilibrium. The opposite is expected

in countries with surplus in the balance of payments, which would lead to a net inflow of gold, an increase in money supply and prices, less exports and more imports until the balance-of-payments equilibrium is restored. Metals, primarily gold and silver, were used as money mainly in international transactions by different societies. Metals are generally accepted as money because they are homogeneous commodities, easily portable, divisible, storable, and relatively scarce and have a cost to produce and a value in industrial use. In 1933, the United States abandoned the pure gold standard system, in which anybody could get gold. In its place, the gold reserve system was introduced, in which a certain percentage of the money supply was kept in gold. The price of gold was determined at \$35 per ounce in January 1934, but in practice, it was used only for the settlement of international transactions. In March 1973 the price of gold was left to the free market of supply and demand.

The Bank of North America, established by the Continental Congress in 1781, and the Bank of the United States, established by Alexander Hamilton in 1791, were the first U.S. banks. In order to manage money and credit more effectively, the Federal Reserve Bank ("Fed"), as an institution independent of the federal government.

The Nature of Economic Activity

Economic activity is often correlated to the type of economic activity. Various methods have been derived to classify economies.

Stages of Market Development

Global markets are at different stages of development which can be divided into five categories based on the criterion of gross national product per capita.

- *Preindustrial countries*: Incomes less than US\$ 400 GNP per capita. Limited industrialisation, low literacy rates, high birth rates, heavy reliance on foreign aid, political instability. Parts of Sub-Saharan Africa. Little market potential.
- *Less developed countries*: Per capita between US\$ 401 and US\$ 1,635. Early stages of industrialisation, growing domestic market, mature product markets, increasing competitive threat.
- *Developing countries*: Per capita income between US\$ 1,636 and US \$ 5,500. Decrease in percentage of agricultural workers, industrialisation, rising wages, high literacy rates, lower wage rates than developed countries, formidable competitors.
- *Industrialised countries* - per capita income between US\$ 5,501 and US\$ 10,000. Moving towards post industrialisation, high standard of living.
- *Advanced countries*: Per capita income in excess of US\$ 10,000. Post industrialisation, information processors, knowledge based, less machine based. Product opportunities are in new products, innovations and raw materials plus fresh foods.

World Bank Classification

The World Bank has drawn up a classification of economies based on GNP per capita.

- Low income economies, China and India, other low-income-GNP per capita income of between US\$ 675 or less, 41 nations including Tanzania, Kenya, Zambia and Malawi.
- Middle income economies, lower middle income, GNP per capita of between US\$ 676 and US\$ 2,695, 40 nations including Zimbabwe, Mexico and Thailand.
- Upper middle income, GNP per capita of between US\$ 2,676 and US\$ 8,355, 17 nations including Brazil, Portugal and Greece.
- High income economies, OECD members and others, GNP per capita of between US\$ 8,356 or more, 24 nations including UK and the USA.
- Other economies - communist bloc.

Mozambique and Switzerland are the two extreme ends of the spectrum with US\$ 80 per capita and US\$ 29,880 per capita respectively.

Rostow: Whilst economic in nature, Rostow produced a five stage model of economic takeoff:

- Stage 1 traditional society, little increase in productivity, no modern science application systematically, low level of literacy.
- Stage 2 the preconditions of takeoff, modern techniques in agriculture and production, developments in infrastructure and social institutions.
- Stage 3 the takeoff, normal growth patterns, rapid agricultural and industrial modernisation, good social environment.
- Stage 4 the drive to maturity, modern technology applied to all fronts, international involvement, can produce anything.
- Stage 5 the age of high mass consumption, production of durable goods and services, large amounts of.

These classifications enable marketers to assess where and how to operate in countries which may display the stage characteristics. For example African exporters would look to stage 4 and 5 economies to obtain the greatest revenue opportunities for other produce. Another way to assess the market alternatives to a potential global marketer is to look at the origin of its national product - is it farm or factory generated? Farm workers tend to have low incomes. Input-output tables provide other insights into a country's potential, that is, what inputs go into a particular industry's output? What combination of labour, materials and equipment?

Infrastructure

Infrastructure is a very important element in considering whether to market in a country or not. Transportation, for example, is vital. Zambia and Zimbabwe are landlocked and have relatively poor transport facilities. Tanzania, whilst having direct

access to the coast, has also a relatively poor internal rural infrastructure. Chaos can therefore ensue, especially during the rainy season. Without being able to get produce to the point of exportation, countries will suffer poor export performance accordingly. Energy consumption shows the overall industrialisation of a society as does its infrastructure. The less energy is consumed, the less likely the development of the market resulting in a not too attractive market proposition.

Communications are essential. India has only some 10 million telephones to a population of 1 billion people. Media availability is important. Zambia has 680 radios per 1000 population, France 2,059 per 1000.

Malawi has no domestic television service but access to satellite television. Commercial infrastructure is also vital - banks, accountants, advertising agencies and other services. Without these “ transaction “ facilities, exporting cannot take place.

Urbanisation

Differences exist between “urban” and “country” dwellers. City dwellers may have more income, more developed communications and access to new products. Developing countries tend to suffer from rural drift, but without the accompanying incomes characteristic of developed countries. So when assessing market opportunities widespread urbanisation is no guarantee of a good market potential.

Other

Inflation causes havoc with economies and foreign exchange. For example Zambia has an unofficial inflation rate of over 100%, which makes it difficult and expensive to access capital for investment and obtain pre-export finance. The role of Government is essential. Some encourage joint ventures and investment, others do not. The number of international companies operating in an economy can be both good and bad. Japan’s investment in the USA and UK is high, creating jobs, but gives rise to negative feelings because access to Japan is not so easy. This has led to calls for protectionism. Similarly, flows from developed countries to less developed ones are generally one way. This leads to instability in the underdeveloped country because it has no “hostage” leverage. Repatriation or transfer of dividends can be an issue which can detract from investment if negative facilities exist. This can seriously undermine economic development and trade.

Many African countries are undergoing structural adjustment and trade liberalisation programmes. In some cases, these have met with limited success. They can create market opportunities, but they also can cause internal economic upheavals for long periods of time, detracting from investment by outsiders and limiting the export opportunities, especially if interest rates rise, as is often the case. The economic environment is one of the major determinants of market potential and opportunity. Careful analysis of this, particularly income and the stage of economic development is essential. Failure to do so will lead, at best, to sub optimal opportunity and, at worst, to disaster. Less developed

countries like Africa, are at a disadvantage, due to their primary material export dependence. It behoves these nations in the continent to derive policies and strategies for rapid industrialisation, or forever to be at the mercy of world demand and prices.

Entrepreneurship and Economic Development

Entrepreneurship helps in the process of economic development in the following ways:

- *Employment Generation:* Growing unemployment particularly educated unemployment is the problem of the nation. The available employment opportunities can cater only 5 to 10% of the unemployed. Entrepreneurs generate employment both directly and indirectly. Directly, self employment as an entrepreneur and indirectly by starting many industrial units they offer jobs to millions. Thus entrepreneurship is the best way to fight the evil of unemployment.
- *National Income:* National Income consists of the goods and services produced in the country and imported. The goods and services produced are for consumption within the country as well as to meet the demand of exports. The domestic demand increases with increase in population and increase in standard of living. The export demand also increases to meet the needs of growing imports due to various reasons. An increasing number of entrepreneurs are required to meet this increasing demand for goods and services. Thus entrepreneurship increases the national income.
- *Balanced Regional Development:* The growth of Industry and business leads to a lot of Public benefits like transport facilities, health, education, entertainment etc. When the industries are concentrated in selected cities, development gets limited to these cities. A rapid development. When the new entrepreneur grow at a faster rate, in view of increasing competition in and around cities, they are forced to set up their enterprises in the smaller towns away from big cities. This helps in the development of backward regions.
- *Dispersal of economic power:* Industrial development normally may lead to concentration of economic powers in a few hands. This concentration of power in a few hands has its own evils in the form of monopolies. Developing a large number of entrepreneur helps in dispersing the economic power amongst the population. Thus it helps in weakening the harmful effects of monopoly.
- *Better standards of living:* Entrepreneur plays a vital role in achieving a higher rate of economic growth. Entrepreneurs are able to produce goods at lower cost and supply quality goods at lower price to the community just as to their requirements. When the price of the commodities decreases the consumers get the power to buy more goods for their satisfaction. In this way they can increase the standard of living of the people.

- *Creating innovation:* An entrepreneur is a person who always looks for changes. Apart from combining the factors of production, he also introduces new ideas and new combination of factors. He always try to introduce newer and newer technique of production of goods and services. An entrepreneur brings economic development through innovation.

Entrepreneurship also helps in increasing productivity and capital formation of a nation. In short, the development of the entrepreneurship is inevitable in the economic development of the country.

Entrepreneurship and Economic Growth

The idea that entrepreneurship and economic growth are very closely and positively linked together has undoubtedly made its way since the early works of Schumpeter. An increase in the number of entrepreneurs leads to an increase in economic growth. This effect is a result of the concrete expression of their skills, and more precisely, their propensity to innovate.

Schumpeter has already described this innovative activity, "the carrying out of new combinations", by distinguishing five cases:"

1. The introduction of a new good - that is one with which consumers are not yet familiar - or of a new quality of a good.
2. The introduction of a new method of production, that is one not yet tested by experience in the branch of manufacture concerned, which need by no means be founded upon a discovery scientifically new, and can also exist in a new way of handling a commodity commercially.
3. The opening of a new market, that is a market into which the particular branch of manufacture of the country in question has not previously entered, whether or not this market has existed before.
4. The conquest of a new source of supply of raw materials or halfmanufactured goods, again irrespective of whether this source already exists or whether it has first to be created.
5. The carrying out of the new organisation of any industry, like the creation of a monopoly position or the breaking up of a monopoly position".

Through his innovative activity, the Schumpeterian entrepreneur seeks to create new profit opportunities. These opportunities can result from productivity increases, in which case, their relationship to economic growth appears quite clearly.

Moreover, the disequilibrium created by the entrepreneur can be propitious for additional innovations and profit opportunities. Therefore, more entrepreneurs means more growth, which in turn leads to more entrepreneurs... The phenomena seem to be self-feeding. Although in some way relatively limited in regards to the interaction between entrepreneurship and economic growth, economic theory is not without enlightening arguments on the subject. As suggested, use some of these arguments in our examination of two fundamental questions, which have already received some

treatment in the economic literature. The first of these questions is the assessment, by means of a non-formalised discussion, of the interdependent nature of entrepreneurship and growth. This assessment will permit us to set certain limits on this relationship. As suggested, grant special attention to particular concepts, as well as to the description of entrepreneurial supply, by referring to individual decisions to undertake economic activities.

These decisions can appear as the result of an arbitrage between different professional occupations based on relative expected rewards just as to the specific abilities of the agent. As suggested, then suggest the endogenous relationship between entrepreneurship and growth and how labour market reactions operate to create some boundaries on their symbiotic interactions. The second question that is examined is in fact threefold. It deals with the types of activities to which the entrepreneur directs his talents.

These activities will be distinguished just as to the social productivity of their character. As suggested, introduce, at this time, rent-seeking behaviour, followed by an examination of factors explaining the allocation of entrepreneurial skill. Finally, respective to this allocation, as suggested, see that an assessment of its impact in terms of economic growth must be appreciated not only in a static framework - the importance of positive effects for the society as a whole depends upon the proportion of productive projects -, but must also take into account the interactions between entrepreneurship and rent-seeking behaviour.

Occupational Arbitrage and Economic Growth

This part is articulated as follows: first, as suggested, reappraise, from the introduction, the relationship between entrepreneurship and economic growth by going a step further into conceptualisation. As suggested, point out skills that are generally associated with the entrepreneur. The individual arbitrage between different professional occupations will then be sketched. An entrepreneurial supply will be defined. Finally, taking into account the aggregated impact of individual choices on expected pay-offs, will allow us to underscore the endogeneity of growth and entrepreneurial decisions.

Innovative Entrepreneurship and Economic Growth

Among the possible sources of economic growth and development, innovation has received a special treatment in the economic literature because it has been more or less linked with the figure of the individual entrepreneur. Relatively to this question, the early Schumpeter appears to be particularly original and pervasive.

Since the beginning of the twentieth century however, the way in which economic theory brings together economic and innovative processes through entrepreneurship, must certainly be revised. In fact, this has been in that economic analysis has undoubtedly reduced the "noise" between the articulation of simple concepts and recent observations. Schumpeter himself predicted that, due to irresistible trend of concentration in the capitalist system, innovation could no longer be the realm of the entrepreneur, but

would in fact primarily be the job of innovation professionals and laboratories controlled by big companies, themselves being directed by managers.

If indeed there still remains a place for innovative entrepreneurship, embodied by the action of individuals, it can no longer be seen as the sole vector of innovation. In a recent and comprehensive survey of the literature, Wennekers and Thurik emphasise the decomposition of the concept of entrepreneurship in order to gain a better understanding of the links between entrepreneurship and economic growth. Although their "pragmatic distinctions" are formulated for operational and not theoretical purposes, they remain helpful in grasping the question at hand.

Associating, on the one hand, entrepreneurial with "the manifest ability and willingness of individuals, on their own, in teams, within and outside existing organisations, to perceive and create new economic opportunities and to introduce their ideas in the market, in the face of uncertainty and other obstacles, by making decisions on location, form and the use of resources and institutions", and, on the other hand, managerial with "organising and coordinating", Wennekers and Thurik cross these definitions with the distinction between self-employed and employee. Thus, the authors are able to define four different types of situations. Three of these are types of entrepreneurs: Schumpeterian entrepreneurs (entrepreneurial and self-employed), entrepreneurs (entrepreneurial and employee) and managerial business owners (managerial and self-employed).

Both Schumpeterian entrepreneurs and intrapreneurs contribute actively to feed a creative destruction process. Although managerial business owners play an important role in the economy for the achievement of production and trade, and can even reveal themselves as entrepreneurial and develop innovative projects, their activities are more closely tied to routine work. So, "(t)hey include many franchisees, shopkeepers and people in professional occupations" and generally represent the great majority of the self-employed.

By introducing new ideas, new processes, new products and services, Schumpeterian entrepreneurs and intrapreneurs affect and ultimately renew the economic activities: the activities not only of the firms and industries, but also those of the region in which they are situated. When these entrepreneurial initiatives are aggregated and evaluated in terms of economic impacts at a territorial level, it could mean economic growth. This process tends to be a diverse one. "At the aggregate level of industries, regions and national economies, the many individual entrepreneurial actions compose a mosaic of new experiments.

In evolutionary terms this can be called variety. A process of competition between these various ideas and initiatives takes place continuously, leading to the selection of the most viable firms and industries. Variety, competition, selection and also imitation (...) expand and transform the productive potential of a regional or national economy (by replacement or displacement of obsolete firms, by higher productivity and by expansion of new niches and industries)".

Greater productivity, as well as firm exits, allow for the possibility of resource reallocation into new combinations. When particular to a given territory, this greater productivity could mean an increase in competitive advantages and localised economic growth. However, this process should be globally followed by a positive net effect. In other words, the competition case that we describe is not a zero-sum game. From entrepreneurial initiatives to economic growth, a number of conditions and multiple effects, at different levels, ultimately determine the impact of one phenomenon in regards to another. However, a working assumption will be: the larger the share of innovative entrepreneurs in the workforce, the more elevated the rhythm of economic growth.

Towards the Definition of an Entrepreneurial Supply

A simple approach in determining the share of innovative entrepreneurs in the active population is to begin by considering each individual as a potential entrepreneur, characterised by the specific capacity to become one. That is, we recognise the heterogeneity of the population. These individuals are then confronted with heterogeneous career opportunities (different projects) with different expected pay-offs.

The individual choice is finally an arbitrage between pairs of project and pay-offs, for given skill sets. These projects include the choice of a career as an employee. What then, briefly, are the individual characteristics that determine the entrepreneurial potentialities? Knight introduces a distinction between risk and uncertainty. Risk enters into computations with known probabilities.

On the contrary, uncertainty although it takes into account the notion of risk, refers also to non-foreseeable events. In other words, events whose probabilities are at this time unknown. Estimating the degree, to which a particular event is weighted with uncertainty, depends upon individual judgement expressed in terms of subjective probabilities. This understood, an entrepreneurial candidate is the one who, taking into account his judgement, would accept to bear the uncertainty of production and trade.

Expected profits would be payment for this specific activity. We note, by the way, that, within a Schumpeterian framework, what characterises an entrepreneur is not a particular ability to bear the consequences of unforeseeable events. The capitalist plays this role. Undertaking economic activities and innovating require skills that make the entrepreneur a person of exception. One of his motives continues to be profit-seeking. Lucas, Murphy, Shleifer and Vishny, Jovanovic come closer to a Schumpeterian concept of entrepreneurship in the sense that, in their models, individuals are not different in their attitude towards risk, but in their competence, intelligence, creative capacities...

It is in fact possible to see in the enumeration, a wide range of skills that are corresponding to “talents”, valuable traits in an entrepreneurial project. A synthesis of the different approaches runs through the tight definition of an entrepreneur offer curve. This offer sums up, amongst other arguments, the previously evoked traits of the working population. It meets an entrepreneurial demand that represents a variety of profit

opportunities. What can be considered as the advantageous aspects of the offer-demand synthesis, at the same time comprises some of its weaknesses. The general framework on which it is based, can disguise heterogeneous situations that each previous model translates, although only in part. We must recognise that what is following an individual arbitrage is in fact rather coming out of a black box.

More Entrepreneurs, more Growth, more Entrepreneurs

We can consider an individual's career as a chain of individual arbitrages between pairs of project and pay-offs given individual abilities. Aggregating individual situations at a given time t , reveals the share of entrepreneurs in the working population. However, for the purposes of assessing more carefully the links between entrepreneurship and economic growth, this is not a sufficient result.

Time must be reintroduced as a period, allowing in that case interactions between each term. By fostering economic growth, entrepreneurship affects the whole economy. Particularly hard to ignore are its effects on labour markets.

More growth might, on one hand, signify an increase in profit opportunities for potential entrepreneurs. On the other hand, it could lead to inflationary pressures on wages. It must be noted that raises in wage cannot necessarily be derived as the result of disequilibrium on the labour markets. In fact, raises in wage may indeed be the consequence of the application of profit-sharing schemes within the firms. It follows that growth in turn can seriously affect the individual arbitrage that we have discussed. So, appears the problem of endogeneity by way of markets and pay-offs adjustments.

Entrepreneurship Versus Rent-seeking

Different rhythms of growth between nations, that have been theoretically associated with entrepreneurship, are not fully explained by different entrepreneurial supply, measured in quantitative terms. The allocation of this supply between more and less socially productive projects must also be taken into account. This question introduces the idea of rent-seeking behaviour. After having briefly defined the general framework and the employed concepts, as suggested, discuss the allocation of entrepreneurs between different types of economic projects, as well as the explicative factors of this allocation. A dynamic set-up will follow, where as suggested, examine interactions between project categories and their relations to growth.

Definitions

To this point, the individual arbitrage between different remunerative occupations ended up, in all cases, in the development of socially productive activities. Therefore, the described economy has been a strict income economy. Individuals were exploiting, as best they could, their skills entrepreneurial or otherwise and given the exercised occupations, private interests and social benefits were successfully coinciding.

As suggested, introduce another type of activity, which is remunerated by transfers. We may note that, by definition, transfers do not imply a productive counterpart. This assumption will result in the possibility that a distortion between private and collective interests may occur. This distortion becomes undoubtedly effective when rent-seeking is involved. The rent-seeking refers to “the socially costly pursuit of wealth transfers”. In other words, rent-seeking is manifested when the bottom-line of its social consequences is negative.

The fact that this definition allows for an expression of the social opportunity cost that represents a diversion of resources towards this type of activity should be emphasised. Examples of rentseeking are: corruption, stealing, bribery, as well as seeking abusive judicial compensation or protection-seeking with the express purpose of limiting economic competition and promoting particular interest... Rent-seeking can originate both from the public or private sector.

In regards to the connection between entrepreneurship and growth, the occurrence of unproductive but yet remunerated activities means not only that projects with socially positive or negative impacts are in competition, but also that there is a direct potential diversion of entrepreneurial talents. For this diversion to take place, we need indeed to assume that the skills and abilities required by entrepreneurship and by rent-seeking correspond. Following these considerations, at least two remarks can be formulated. The first one refers to public services.

These are generally financed by transfers, but are not included in rent-seeking, given their productive contribution. We note furthermore that discussing the redistribution role of the state is outside the framework. This is different, however, as soon as we look at legal institutions that organise productive activities, as they might, by limiting competition for example, create rent-seeking situations. The second remark concerns the entrepreneurial initiatives. Through innovation, the entrepreneur seeks to create a monopoly position, from which he will assume overprofit. In the Schumpeterian model, this position is necessary fore it motivates the innovation activity. It is however temporary, as competition will quickly reduce this position to zero in favour of a new monopoly position created by a new innovation... The institutional framework, that of competition, is then of primary importance. In this case, a dynamic assessment of events will justify the existence of an abnormal profit, as it will distinguish its positive net contribution to the social benefit. Rent-seeking behaviour thus being defined, as suggested, address in the remaining discussion the allocation of talents and its links to growth.

Explaining the Allocation of Talents

The allocation of talents between socially productive and unproductive projects and its impact on economic growth have already been subject to theoretical studies. The static results of these studies scarcely elicit further commentary, when we take into account the preceding arguments. Indeed it appears quite easily that the diversion

of entrepreneurial talents towards unproductive activities will have a negative impact on growth rate. However, the explicative factors of the allocation deserve greater consideration. Attempting to explain the allocation of talents implies both reconsidering the arguments that have been exposed with regard to the individual arbitrage between remunerative activities and considering the non-pecuniary factors that may also have some impact. Murphy, Shleifer and Vishny, the point is the following: “talent goes into activities with the highest private returns, which need not have the highest social returns”. These authors assume increasing returns on talent.

In other words, the greater the ability of an individual, the greater his private benefits will be. Because the exercise of talent is physically limited, talented individuals tend to invest themselves and their abilities into reward maximising occupations. It follows that their occupational choice will be determined by the size of the market, the compensation contract and the technology. Furthermore, the allocation of talent can be linked both with institutional context, as well as non-pecuniary explicative factors.

The legal framework and its effective use define a propitious environment for entrepreneurship and, contrarily, for rent-seeking behaviour. Property rights, the conditions of their application, and the respect of these rights, appear to be crucial factors. They contribute particularly and decisively to precise the compensation circumstances that have already been quoted, as well as the fiscal organisation.

Information is also important as it determines the efficiency of allocation and the possibility of linking the application of talent and its results. Social esteem might play a role. The question becomes therefore how much entrepreneurship is socially valuable and valorised, over another less socially productive occupation. Finally, entrepreneurs or rent-seekers might influence, by voting or lobbying, political organisation and political decisions. The idea is that the political equilibrium, replying to one or another group's interests, will make decisions that favour its maintenance.

Entrepreneurship and Rent-seeking in Motion

The interaction between entrepreneurship and rent-seeking is without doubt an interesting question to examine. Formalised models show that multiple equilibria – an equilibrium being defined by an entrepreneur share in the population and a rate of growth – may exist. As suggested, not insist on model developments. The interested reader may consult the references we provide. However, briefly, we note that the results are grounded on the specification of two functions, both with negative slopes, and consequently a potential for multiple intersections.

Because it places burdens on entrepreneurial rewards, rent-seeking negatively affects entrepreneurship. Moreover, given competition in the rent-seeking sector itself, rent-seeking rewards will depend negatively on the number of rent-seekers. Baland and Francois formalise as well the effect of entrepreneurial activities on rent-seeking. Their model can apply to an economy with import license. The production of direct substitutes by local entrepreneurs tends to turn off the rents obtained by importers. Here also,

results suggest the existence of multiple equilibria. Murphy, Shleifer and Vishny deal with rent-seeking affecting the innovation sector. The rent-seeking, whether from private or public origins, can undoubtedly burden the profits of established productive sectors. The innovation sector, however, might be described the reserved hunting ground, for the most part, of public rent-seeking. Their arguments rely particularly on the nature of innovation. In his project developments, the innovative entrepreneur is confronted with legal and environmental constraints. Innovation can imply indeed production permits, licenses, dispensations, as well as amending the local zoning regulations.

This results in demand for government intervention and it provides opportunities for corruption. Moreover, the socially unproductive transfers that corruption implies, might have some inhibitive effects on innovation activity, given that innovators may not have equal lobbying power compared to that of established firms, or the same financial resources to pay bribes. Important funds may then be consumed instead of being invested, to avoid expropriation.

The ex post existence of rent-seeking should raise the project risk and effective cost. The authors mention, following these arguments, that the negative effects of rent-seeking could be limited if the rent-seeker himself becomes a stakeholder in the innovation project. In the long run, he would find his interest in such an involvement.

We note moreover that this idea can be generalised as: rent-seeking, by burdening current entrepreneurial profits, limits its future transfer opportunities. Thus, the interaction between entrepreneurship and rent-seeking can generate multiple equilibria that correspond to an allocation of talents and an economic growth rate. Starting from a dynamic extension of his basic model, Acemoglu discusses the history dependence of an economy.

Past and current allocations of talents influence the future structure of rewards. Given historical circumstances, the economy can be locked in low or high steady state equilibrium. Under these circumstances, only a shock will have any positive and sustainable effect.

3

Development Economics Theories

A number of theories are concerned with how economies develop over time. Some of these theories include:

- Comparative advantage: Predicts all countries gain if they specialise and trade the goods in which they have a comparative advantage. This is true even if one nation has an absolute advantage over another country.
- Rostovian take-off model: A linear theory of development that argues that economic modernization occurs in five basic stages of varying length-traditional society, preconditions for take-off, take-off, drive to maturity, and high mass consumption.
- Harrod–Domar model: Explains an economy's growth rate in terms of the level of saving and productivity of capital.
- Dual Sector model: Explains the growth of a developing economy in terms of a labour transition between two sectors, a traditional agricultural sector and a modern industrial sector.

THEORIES AND MODELS LINKED TO DEVELOPMENT

This tutor 2u revision note introduces some of the main theories and models linked to the subject of economic development.

Links are provided to further revision notes that explain each theory or model in greater detail.

WHAT DO THEORIES AND MODELS TRY TO DO?

Economic development theories and models seek to explain and predict how:

- Economies develop (or not) over time.
- Barriers to growth can be identified and overcome.
- Government can induce (start), sustain and accelerate growth with appropriate development policies.

Theories are generalisations. While LDC's share similarities, every country's unique economic, social, cultural, and historical experience means the implications of a given theory vary widely from country to country.

There is no one agreed 'model of development'. Each theory, like Rostow, gives an insight into one or two dimensions of the complex process of development. Eg Rostow helps us to think about the stages of development LDC's might take and Harrod Domar model explains the importance of adequate savings in that process.

Table. The Main Models.

Comparative advantage	Economic theory predicts all countries gain if they specialise and trade the goods in which they have a comparative advantage. This is true even if one nation has an absolute advantage over another country.
Rostow	This is a linear theory of development. Economies can be divided into primary secondary and tertiary sectors. The history of developed countries suggests a common pattern of structural change:
Harrod-Domar	The Harrod-Domar model developed in the 1930s suggests savings provide the funds which are borrowed for investment purposes.
Lewis	The Lewis model is structural change model that explains how labour transfers in a dual economy. For Lewis growth of the industrial sector drives economic growth.
Dependency Theory	Dependency refers to over reliance on another nation. Dependency theory uses political and economic theory to explain how the process of international trade and domestic development makes some LDC's ever more economically dependent on developed countries.
Balanced Growth Theory	Balanced growth (or the big push) theory. argues that as a large number of industries develop simultaneously, each generates a market for one another.
Unbalanced Growth Theory	Unbalanced growth theorists argue that. sufficient resources cannot be mobilised by government to promote widespread, coordinated investments in all industries.

Open Directory Project

ODP uses a hierarchical ontology scheme for organizing site listings. Listings on a similar topic are grouped into categories, which can then include smaller categories.

Project Information

ODP was founded as Gnuhoo by Rich Skrenta and Bob Truel in 1998 while they were both working as engineers for Sun Microsystems. Chris Tolles, who worked at Sun Microsystems as the head of marketing for network security products, also signed on in 1998 as a co-founder of Gnuhoo along with co-founders Bryn Dole and Jeremy

Wenokur. Skrenta was already well known for his role in developing TASS, an ancestor of tin, the popular threaded Usenet newsreader for Unix systems. Coincidentally, the original category structure of the Gnuhoo directory was based loosely on the structure of Usenet news groups then in existence. The Gnuhoo directory went live on June 5, 1998. After a Slashdot article suggested that Gnuhoo had nothing in common with the spirit of free software, for which the GNU project was known, Richard Stallman and the Free Software Foundation objected to the use of Gnu. So Gnuhoo was changed to New Hoo. Yahoo! then objected to the use of “Hoo” in the name, prompting them to switch the name again. ZURL was the likely choice. However, before the switch to ZURL, New Hoo was acquired by Netscape Communications Corporation in October 1998 and became the Open Directory Project. Netscape released the ODP data under the Open Directory License. Netscape was acquired by AOL shortly thereafter, and ODP was one of the assets included in the acquisition. AOL later merged with Time-Warner.

By the time Netscape assumed stewardship, the Open Directory Project had about 100,000 URLs indexed with contributions from about 4500 editors. On October 5, 1999, the number of URLs indexed by ODP reached one million. According to an unofficial estimate, the URLs in the Open Directory numbered 1.6 million in April 2000, surpassing those in the Yahoo! Directory. ODP achieved the milestones of indexing two million URLs on August 14, 2000, three million listings on November 18, 2001 and four million on December 3, 2003.

From January 2006 the Open Directory published online reports to inform the public about the development of the project. The first report covered the year 2005. Monthly reports were issued subsequently until September 2006. These reports gave greater insight into the functioning of the directory than the simplified statistics given on the front page of the directory. The number of listings and categories cited on the front page include “Test” and “Bookmarks” categories, but these are not included in the RDF dump offered to users. The total number of editors who have contributed to the directory as of March 31, 2007 was 75,151. There were about 7330 active editors during August 2006.

MEASURING ECONOMIC DEVELOPMENT

GDP

North America, even though one of the slowest growing continents, has stable growth. Most of the faster growing economies are in the Caribbean. South America has a Boom and Bust growth with high followed by recession growth, most notable in Brazil, however growth has been stabilizing and the whole continent is growing. Africa has seen the fastest growing but also the slowest growing/declining. From the oil fields which made Angola the 3rd fastest growing country in the world, to Zimbabwe the slowest growing and declining country in the world. Oil in Africa has created ‘wealth spots’ were a few

countries have exceeded their neighbours in wealth. Out of the 10 fastest growing countries in the world, 3 were African. Some countries have in the past been the fastest growing in the world. Equatorial Guinea reached 75% growth in 2004 because of oil reserves.

Europe has one of the most stable growth rates. After the fall of the Soviet Union, there was a period of economic decline in Eastern Europe over the 1990s, followed by recovery in the 2000s. The region is now experiencing growth, particularly in those countries that have recently joined the European Union. If the Caucasus were included, Europe would be one of the fastest growing continents in the world. Most countries are growing at a medium speed; however, many smaller countries exceed 7% and grow exceptionally faster than their neighbours. Out of the 10 fastest growing countries in the world, only 1 is in Europe.

Overall in the 20th century Asia was seen as the area with most growth; however, in the 21st century, most of this has been dominated by China. Some spots of growth are starting to appear in East and even South Asia. Most nations with high populations have seen high growth especially. Out of the 10 fastest growing countries 3 were directly in Asia, and 3 indirectly or partially. Meanwhile Oceania has seen moderate growth. The only exceptional growth in Oceania has been on Vanuatu. Some countries have negative growth, most often due to ongoing wars or hyperinflation. These countries include Palestinian territories, Zimbabwe, Fiji and Chad.

Regional Policy

In its broadest sense, policies of economic development encompass three major areas:

- Governments undertaking to meet broad economic objectives such as price stability, high employment, and sustainable growth. Such efforts include monetary and fiscal policies, regulation of financial institutions, trade, and tax policies.
- Programs that provide infrastructure and services such as highways, parks, affordable housing, crime prevention, and K–12 education.
- Job creation and retention through specific efforts in business finance, marketing, neighbourhood development, small business development, business retention and expansion, technology transfer, and real estate development. This third category is a primary focus of economic development professionals.

Economic Developers

Economic development, which is thus essentially economics on a social level, has evolved into a professional industry of highly specialized practitioners. The practitioners have two key roles: one is to provide leadership in policy-making, and the other is to administer policy, programs, and projects. Economic development practitioners generally work in public offices on the state, regional, or municipal level, or in public-private partnerships organizations that may be partially funded by local, regional, state, or

federal tax money. These economic development organizations (EDOs) function as individual entities and in some cases as departments of local governments. Their role is to seek out new economic opportunities and retain their existing business wealth.

There are numerous other organizations whose primary function is not economic development work in partnership with economic developers. They include the news media, foundations, utilities, schools, health care providers, faith-based organizations, and colleges, universities, and other education or research institutions.

With more than 20,000 professional economic developers employed world wide in this highly specialized industry, the International Economic Development Council [IEDC] headquartered in Washington, D.C. is a non-profit organization dedicated to helping economic developers do their job more effectively and raising the profile of the profession. With over 4,500 members across the US and internationally, serving exclusively the economic development community. Membership represents the entire range of the profession ranging from regional, state, local, rural, urban, and international economic development organizations, as well as chambers of commerce, technology development agencies, utility companies, educational institutions, consultants and redevelopment authorities. Many individual states also have associations comprising economic development professionals and they work closely with IEDC.

There is intense competition between communities, states, and nations for new economic development projects in today's globalized world, and the struggle to attract and retain business is further intensified by the use of many variations of economic incentives to the potential business such as; tax incentives, help with investment capital, donated land and many others.

IEDC places significant attention on the various activities undertaken by economic development organizations to help them compete and sustain vibrant communities. Additionally, the use of community profiling tools and database templates to measure community assets versus other communities is also an important aspect of economic development.

Job creation, economic output, and increase in taxable basis are the most common measurement tools. When considering measurement, too much emphasis has been placed on economic developers for "not creating jobs."

However, the reality is that economic developers do not typically create jobs, but facilitate the process for existing businesses and start-ups to do so. Therefore, the economic developer must make sure that there are sufficient economic development programs in place to assist the businesses achieve their goals. Those types of programs are usually policy-created and can be local, regional, statewide and national in nature.

Development Theory

Development theory is a conglomeration of theories about how desirable change in society is best to be achieved. Such theories draw on a variety of social scientific disciplines and approaches.

System Failure and Editing Outage, October to December 2006

On October 20, 2006, the ODP's main server suffered a catastrophic system failure that prevented editors from working on the directory until December 18, 2006. During that period, an older build of the directory was visible to the public. On January 13, 2007, the Site Suggestion and Update Listings forms were again made available. On January 26, 2007, weekly publication of RDF dumps resumed. To avoid future outages, the system now resides on a redundant configuration of two Intel-based servers.

Competing and Spinoff Projects

As the ODP became more widely known, two other major web directories edited by volunteers and sponsored by Go.com and Zeal emerged, both now defunct. These directories did not license their content for open content distribution. The concept of using a large-scale community of editors to compile online content has been successfully applied to other types of projects. ODP's editing model directly inspired three other open content volunteer projects: an open content restaurant directory known as ChefMoz, an open content music directory known as MusicMoz, and an encyclopedia known as Open Site.

Gnuhoo borrowed the basic outline for its initial ontology from Usenet. In 1998, Rich Skrenta said, "I took a long list of groups and hand-edited them into a hierarchy." For example, the topic covered by the comp.ai.life news group was represented by the category Computers/AI/Artificial_Life.

The original divisions were for *Adult, Arts, Business, Computers, Games, Health, Home, News, Recreation, Reference, Regional, Science, Shopping, Society, and Sports*. While these fifteen *top-level* categories have remained intact, the ontology of second- and lower-level categories has undergone a gradual evolution; significant changes are initiated by discussion among editors, and then implemented when consensus has been reached. In July 1998, the directory became multilingual with the addition of the *World* top-level category. The remainder of the directory lists only English language sites. By May 2005, seventy-five languages were represented.

The growth rate of the non-English components of the directory has been greater than the English component since 2002. While the English component of the directory held almost 75% of the sites in 2003, the *World* level grew to over 1.5 million sites as of May 2005, forming roughly one-third of the directory.

The ontology in non-English categories generally mirrors that of the English directory, although exceptions which reflect language differences are quite common. Several of the top-level categories have unique characteristics.

The *Adult* category is not present on the directory homepage, but it is fully available in the RDF dump that ODP provides. While the bulk of the directory is categorized primarily by topic, the *Regional* category is categorized primarily by region. This has led many to view ODP as two parallel directories: *Regional* and *Topical*. On November 14, 2000, a

special directory within the Open Directory was created for people under 18 years of age. Key factors distinguishing this “Kids and Teens” area from the main directory are:

- Stricter guidelines which limit the listing of sites to those which are targeted or “appropriate” for people under 18 years of age;
- Category names as well as site descriptions use vocabulary which is “age appropriate”;
- *Age tags* on each listing distinguish content appropriate for kids (age 12 and under), teens (13 to 15 years old) and mature teens (16 to 18 years old);
- Kids and Teens content is available as a separate RDF dump;
- Editing permissions are such that the community is parallel to that of the Open Directory.

By May 2005, this portion of the Open Directory included over 32,000 site listings.

Since early 2004, the whole site has been in UTF-8 encoding. Prior to this, the encoding used to be ISO 8859-1 for English language categories, and a language-dependent character set for other languages. The RDF dumps have been encoded in UTF-8 since early 2000.

Maintenance

Directory listings are maintained by editors. While some editors focus on the addition of new listings, others focus on maintaining the existing listings. This includes tasks such as the editing of individual listings to correct spelling and/or grammatical errors, as well as monitoring the status of linked sites. Still others go through site submissions to remove spam and duplicate submissions.

Robozilla is a Web crawler written to check the status of all sites listed in ODP. Periodically, Robozilla will flag sites which appear to have moved or disappeared, and editors follow up to check the sites and take action. This process is critical for the directory in striving to achieve one of its founding goals: to reduce the link rot in web directories. Shortly after each run, the sites marked with errors are automatically moved to the unreviewed queue where editors may investigate them when time permits.

Due to the popularity of the Open Directory and its resulting impact on search engine rankings, domains with lapsed registration that are listed on ODP have attracted domain hijacking, an issue that has been addressed by regularly removing expired domains from the directory. While corporate funding and staff for the ODP have diminished in recent years, volunteers have created editing tools, such as linkcheckers to supplement Robozilla, category crawlers, spellcheckers, search tools that directly sift a recent RDF dump, bookmarklets to help automate some editing functions, and tools to help work through unreviewed queues.

License and Requirements

ODP data is made available for open content distribution under the terms of the Open Directory License, which requires a specific ODP attribution table on every Web

page that uses the data. The Open Directory License also includes a requirement that users of the data continually check the ODP site for updates and discontinue use and distribution of the data or works derived from the data once an update occurs. This restriction prompted the Free Software Foundation to refer to the Open Directory License as a non-free documentation license, citing the right to redistribute a given version not being permanent, and the requirement to check for changes to the license.

RDF Dumps

ODP data is made available through an RDF-like dump that is published on a dedicated download server, where an archive of previous versions is also available. New versions are usually generated weekly. An ODP editor has catalogued a number of bugs that are/were encountered when implementing the ODP RDF dump, including UTF-8 encoding errors (fixed since August 2004) and an RDF format that does not comply with the final RDF specification because ODP RDF generation was implemented before the RDF specification was finalized. So while today the so-called RDF dump is valid XML, it is not strictly RDF, but an ODP-specific format. Software to process the ODP RDF dump needs to take account of this.

Content Users

ODP data powers the core directory services for many of the Web's largest search engines and portals, including Netscape Search, AOL Search, Google, and Alexa. Other uses are also made of ODP data. For example, in the spring of 2004 Overture announced a search service for third parties combining Yahoo! Directory search results with ODP titles, descriptions and category metadata. The search engine Gigablast announced on May 12, 2005 its searchable copy of the Open Directory. The technology permits search of websites listed in specific categories, "in effect, instantly creating over 500,000 vertical search engines".

As of September 8, 2006, the ODP listed 313 English-language Web sites that use ODP data as well as 238 sites in other languages. However, these figures do not reflect the full picture of use, as those sites that use ODP data without following the terms of the ODP license are not listed.

Information-led Development

Information-led development (ILD) most commonly refers to a development strategy whereby a developing country makes as a primary economic policy focus the creation and development of a national information technology (IT) sector with the express aim of relying on this sector as an engine of growth. Notable examples of such countries are India and the Philippines.

More recently, a new formulation of ILD has emerged. With origins in community economic development in the United States, the new ILD model describes the use of data to generate actionable information or information solutions to development

challenges. Examples of this include the inclusion of non-financial payment obligations in consumer credit files, also known as alternative data, and the use of this information in underwriting, as a means to reduce financial exclusion in the United States, where an estimated 54 million Americans are shut out of mainstream credit access as there is insufficient information about them in their credit files to be scored by a credit scoring model. This variant of ILD was pioneered by PERC, a non-profit policy research organization and development intermediary headquartered in Chapel Hill, North Carolina. Other US-based organizations, including Social Compact and the Local Initiatives Support Corporation, employ variants of ILD, but none has applied this internationally except for PERC. This development model is gaining traction in emerging markets such as Colombia and South Africa, where the data is being used to reduce financial exclusion and facilitate credit access as a means to build wealth and form assets. It is also attracting increasing attention from development agencies, including USAID, the International Finance Corporation, the World Bank Group, and the Consultative Group to Assist the Poor.

Policies and Procedures

There are restrictions imposed on who can become an ODP editor. The primary gatekeeping mechanism is an editor application process wherein editor candidates demonstrate their editing abilities, disclose affiliations that might pose a conflict of interest, and otherwise give a sense of how the applicant would likely mesh with the ODP culture and mission. A majority of applications are rejected, but reapplying is allowed and sometimes encouraged. The same standards apply to editors of all categories and subcategories. ODP's editing model is a hierarchical one. Upon becoming editors, individuals will generally have editing permissions in only a small category. Once they have demonstrated basic editing skills in compliance with the Editing Guidelines, they are welcome to apply for additional editing privileges, in either a broader category, or in a category elsewhere in the directory. Mentorship relationships between editors are encouraged, and internal forums provide a vehicle for new editors to ask questions.

ODP has its own internal forums, the contents of which are intended only for editors to communicate with each other primarily about editing topics. Access to the forums requires an editor account, and editors are expected to keep the contents of these forums private.

Over time, senior editors may be granted additional privileges which reflect their editing experience and leadership within the editing community. The most straightforward are *editall* privileges, which allow an editor to access all categories in the directory. *Meta* privileges additionally allow editors to perform tasks such as reviewing editor applications, setting category features, and handling external and internal abuse reports. *Cateditall* privileges are similar to *editall*, but only for a single directory category. Similarly, *catmod* privileges are similar to *meta*, but only for a single directory category. *Catmv* privileges allow editors to make changes to directory ontology

by moving or renaming categories. All of these privileges are granted by admins and staff, usually after discussion with *meta* editors. In August 2004, a new level of privileges called *admin* was introduced. Administrator status was granted to a number of long serving metas by staff. Administrators have the ability to grant editall+ privileges to other editors and to approve new directory-wide policies, authorities that had previously only been available to root (staff) editors. A full list of senior editors is available to the public, as is a listing of all current editors.

All ODP editors are expected to abide by ODP's Editing Guidelines. These guidelines describe editing basics: what types of sites may be listed and which may not; how site listings should be titled and described in a loosely consistent manner; conventions for the naming and building of categories; conflict of interest limitations on the editing of sites which the editor may own or otherwise be affiliated with; and a code of conduct within the community. Editors who are found to have violated these guidelines may be contacted by staff or senior editors, have their editing permissions cut back, or lose their editing privileges entirely. ODP Guidelines are periodically revised after discussion in editor forums.

Site Submissions

One of the original motivations for forming Gnuhoo/Newhoo/ODP was the frustration that many people experienced in getting their sites listed on Yahoo! Directory. However Yahoo! has since implemented a paid service for timely consideration of site submissions. That lead has been followed by many other directories. Some accept no free submission at all. By contrast the ODP has maintained its policy of free site submissions for all types of site — the only one of the major general directories to do so. One result has been a gradual divergence between the ODP and other directories in the balance of content. The pay-for-inclusion model favours those able and willing to pay, so commercial sites tend to predominate in directories using it. Conversely, a directory manned by volunteers will reflect the aims and interests of those volunteers. The ODP lists a high proportion of informational and non-profit sites.

Another consequence of the free submission policy is that the ODP has enormous numbers of submissions still waiting for review. In large parts those consist of spam and incorrectly submitted sites. So the average processing time for a site submission has grown longer with each passing year. However the time taken cannot be predicted, since the variation is so great: a submission might be processed within hours or take several years. However, site suggestions are just one of many sources of new listings. Editors are under no obligation to check them for new listings, and are actually encouraged to use other sources.

Controversy and Criticism

There have long been allegations that volunteer ODP editors give favourable treatment to their own websites while concomitantly thwarting the good faith efforts of their

competition. Such allegations are fielded by ODP's staff and meta editors, who have the authority to take disciplinary action against volunteer editors who are suspected of engaging in abusive editing practices.

In 2003, ODP introduced a new *Public Abuse Report System* that allows members of the general public to report and track allegations of abusive editor conduct using an online form. Uninhibited discussion of ODP's purported shortcomings has become more common on mainstream Webmaster discussion forums. Although site policies suggest that an individual site should be submitted to only one category, as of October 2007, Topix.com, a news aggregation site operated by ODP founder Rich Skrenta, has more than 10,000 listings. Early in the history of the ODP, its staff gave representatives of selected companies, such as Rolling Stone magazine or CNN, editing access in order to list individual pages from their websites. Links to individual CNN articles have been added until 2004 and have been entirely removed from the directory in January 2008 due to being outdated and not considered worth the effort to maintain. Such experiments have not been repeated later.

Ownership and Management

Underlying some controversy surrounding ODP is its ownership and management. Some of the original GnuHoo volunteers felt that they had been deceived into joining a commercial enterprise. To varying degrees, those complaints have continued up until the present.

At ODP's inception, there was little thought given to the idea of how ODP should be managed, and there were no official forums, guidelines, or FAQs. In essence, ODP began as a free for all. As time went on, the ODP Editor Forums became the *de facto* ODP parliament, and when one of ODP's staff members would post an opinion in the forums, it would be considered an official ruling.

Even so, ODP staff began to give trusted senior editors additional editing privileges, including the ability to approve new editor applications, which eventually led to a stratified hierarchy of duties and privileges among ODP editors, with ODP's paid staff having the final say regarding ODP's policies and procedures.

Editor Removal Procedures

ODP's editor removal procedures are overseen by ODP's staff and meta editors. According to ODP's official editorial guidelines, editors are removed for abusive editing practices or uncivil behaviour.

Discussions that may result in disciplinary action against volunteer editors take place in a private forum which can only be accessed by ODP's staff and meta editors, and volunteer editors who are being discussed are not given notice that such proceedings are taking place.

Some people find this arrangement distasteful, wanting instead a discussion modelled more like a trial held in the U.S. judicial system. In the article *Editor Removal Explained*,

ODP meta editor Arlarson states that “a great deal of confusion about the removal of editors from ODP results from false or misleading statements by former editors”.

The ODP’s confidentiality guidelines prohibit any current ODP editors in a position to know anything from discussing the reasons for specific editor removals. However, a generic list of reasons is for example given in the guidelines. In the past, this has led to removed ODP editors wondering why they cannot login at ODP to perform their editing work.

4

World Economic Resource Management

AN OVERVIEW

In the past fifty years the global economy has changed rapidly. Particularly marked has been the development of world economic integration and standardised products. Coca Cola, Nissan and Marlboro cigarettes are examples of products which serve nearly every market.

Generally there have been four major changes:

- Capital movements rather than trade have become the driving force of the global economy
- Production has become “uncoupled” from employment
- Primary products have become “uncoupled” from the industrial economy and,
- The world economy is in control - individual nations are not, despite the large world economic share of the USA and Japan.

Taking each of these changes in turn, world trade is about some US\$ 3 trillion, however, capital movements are much higher. The London Eurodollar market is worth about US\$ 75 trillion per annum and foreign exchange transactions are US\$ 35 trillion per annum.

Another change is the decoupling of employment from production. Employment is in decline whilst manufacturing output is growing or remaining static at 20-25% of GNP. Sectors such as agriculture, are achieving higher productivity through

mechanisation but this is at the expense of employment. Still another change is the decoupling of the primary product market from the industrial economy. Many commodity prices have collapsed, *for example tea*, yet industrial economies have been relatively affected. Unfortunately the prime producers have been dramatically affected.

Finally, the most significant change is the change of focus from domestic to the world economy as the chief economic unit. This has been grasped by Japan and Germany, but not really by the USA, or Africa. These factors have repercussions on exporting by developing countries. Firstly with developing countries' emphasis on the export of primary products, they are at the mercy of world supply and demand movements, with the resultant fluctuations in prices.

Depressed world market prices can have a deleterious effect on developing economies. Secondly the rapid globalisation and focus away from domestic economies has created global competition and in turn, this has pushed up quality.

Generally speaking, unless developing countries can break into non-comittally based products they are being further left behind in the global economic stakes. However positively, whilst developed worlds concentrate on industrial and service products it leaves opportunities for developing countries to export more food based products.

THE INTERNATIONAL FINANCIAL SYSTEM

Global financing operations based on the gold standard gave rise to instability, so Bretton Woods, post World War II, saw the nascence of the International Monetary Fund (IMF) and World Bank. The IMF deals with the International Monetary System. Involved countries joined IMF to establish a par value for other countries in terms of the US dollar and maintain it with +/- one per cent of that value.

The system fell down because large corporations were holding more funds than banks and so a "float" set in. IMF began to fade somewhat. However it still lends, on a short term basis, to countries with payment problems to help them continue trading. The World Bank, or International Bank for Reconstruction and Development (IBRD) deals with international capital.

It provides long term capital to aid economic development. Currently it has about US\$ 22 billion annually for this operation. The role of the World Bank has often been criticised especially on its conditionalities for loans to Africa in funding structural adjustment and trade liberalisation programmes. However many developing countries require institutional funding to help them with trade and balance payment problems.

Other major lenders include the EU and bilateral donors and agencies who have provided money for developmental projects. A principal donor is the United States Agency for International Development (USAID).

THE UNITED STATES OF AMERICA

Since the Gulf War of 1991, the USA has played an increasingly important role in the economic affairs of the world. Since that time, itself, and its agency USAID, have

increasingly flexed their muscles. However, the balance of economic power in recent years, has shifted towards the Pacific rim, especially Japan and the Asian Tigers.

Individual Economies

Whilst the global factors have aided the development of a world economy, marketers must consider carefully individual economies. A study of these helps answer the questions - how big is the market and what is it like? Currently there are over 200 individual countries in the world.

Size of Market

General indications of market size include population (growth rates and distribution) and income (distribution, per capita, GNP).

- *Population:* In general, the larger the population, the bigger the market. However there is no correlation between income level and population. China has 2 billion plus people, India 1 billion, Zimbabwe 8 million. However, they do not have the same income per capita as the USA or UK. In 1993 the USA population of 252.2 million, the UK 57.4 million and Africa 400 million, were respectively 6%, 1.5% and 9% of the world's population. However the USA and UK had an infinitely higher GNP per capita income than Africa, US\$ 22,520, UK \$17,300 and Africa \$ 270 respectively.
- Different countries experience different population growth rates. In the early 90s, the UK had an annual growth rate of 0.1%, the Ivory Coast 6%, and Africa in general, 3% per annum. Low income countries and oil rich countries have the largest growth rates. Growth rates have a dual edge - they are good for sales but bad for world resources. The world population, currently standing at 5 billion is experiencing a rapid growth rate. It is expected to reach 7 billion by the end of the century. The strain on world resources is likely to be very large. The distribution of the population is also important. Different age groups have different needs and population density should mean good market potential, the higher the better. The Netherlands have 1000 persons per square mile, Bangladesh 1,791 but the USA only 65 persons per square mile. However, the USA spends more per capita than Bangladesh
- *Income:* No one has yet been able to assess accurately the impact of the AIDS pandemic on world population and economic activity. South Africa estimates AIDS will cost South African industry R16.7 billion by the year 2000. Suffice to say, unless a cure or prevention is found, it could be serious, especially in Africa and South East Asia, the world's "hot spots"

Income is the most important variable affecting market potential. Markets are not markets without money to spend. Interestingly, there is an inverse correlation between GNP per capita and income elasticity of demand for food. Asia has a 0.9 income elasticity of demand and the USA 0.16.

The distribution of income is very uneven. In Kenya the lowest 20% of the population receive less than 3% of national resource. This bimodal distribution of income means marketers must analyse two economies in a country. Per capita measures have therefore, many limitations. Per capita judges a country's level of economic development and its degree of modernisation and progress in health, education and welfare. Half of the world's population lives with an average per capita income of only US\$ 270. Per capita is usually reflected in US dollars and is only valid for comparison if exchange rates are equal. Exchange rates reflect international goods and services in a country but not domestic consumption.

Another limitation of per capita measures is the lack of comparability with the figures themselves. The US budget contains food, clothing and shelter.

In many of the less developed nations these items may be largely self provided and therefore not reflected in national income tables. Also in the UK, snow equipment is included, and this is not, obviously, in Africa and parts of Asia.

Other limitations are that sales of goods are not well correlated with per capita income and if there is great unevenness in income distribution, per capita figures are less meaningful. Product saturation can be equally troublesome in affecting market potential.

A vacuum cleaner in the Netherlands has a 95% household penetration rate, but only 7% in Italy.

Gross National Product is a better indicator of potential than Gross Domestic Product as GDP includes more than "product". World GNP figures reveal the concentration of wealth in the three nations, the USA, Japan and Western Europe. Africa trails far behind.

However, when evaluating markets it is wise to consider individual product areas. For example, Belgium's GNP is better than India's but India's, consumption of steel is 3 times that of Belgium's.

The United Nations International Comparison Project (ICP) developed a sophisticated method for measuring total expenditure, which has been used to derive more reliable and directly comparable estimates of per capita income. The World Bank has published a comparison of ICP findings with its own Atlas figures based on the exchange rate conversion. The use of exchange rates tends to distort real income or standard of living measures.

The Nature of Economy

More than money makes up an economy's economic environment. Natural resources -raw materials now and in the future are important. If synthetic gold or tobacco were developed or, in the case of the latter, became unfashionable, Zimbabwe's economy would be ruined. Topography may produce two, three or more submarkets in a country.

Zambia, for example, has "rural" and "urban" areas with different needs and wants. Extremes of climate - like the Southern African drought in 1992 can devastate economies and derail any economic development plans and exports. Simply, products are not available to export, because they are being consumed by the domestic economy.

INTERNATIONAL MONETARY FUND

The most effective international organizations, formed by agreements among a large number of nations, are the International Monetary Fund (IMF), the International Bank for Reconstruction and Development or World Bank, and the General Agreement on Tariffs and Trade (GATT), recently named the World Trade Organization (WTO). The IMF and the World Bank were established by forty-four countries at Bretton Woods, New Hampshire (United States), in July 1944, whereas the GATT was negotiated in Havana, Cuba, and established in Geneva in 1948.

The International Monetary Fund (IMF), which is a pool of central-bank reserves and national currencies, was established at Bretton Woods in 1944 under the auspices of the United Nations, but it actually started operations on March 1, 1947. In order to facilitate international finance and help correct fundamental balance-of-payments disequilibriums, it tied all currencies to gold and hence to each other. The IMF, with headquarters in Washington, D.C., was created to help finance balance-of-payments deficits and to reform the international monetary system towards mutual cooperation and freely convertible currencies among the member nations. The waves of devaluations in the currencies of a number of countries before World War II, mainly in 1931–1936, as a way to increase exports and reduce imports at the expense of their trading partners (beggar-thy-neighbour policies) was the rationale of the creation of the IMF. These policies resulted in frequent changes in exchange parities, economic instability, and high unemployment.

The IMF makes its own stock of currencies available (gives loans) to the members. It functions as a pool of currencies or an international clearing institution, using the assigned quotas of the member countries to arrange international payments. Also, it provides short-term loans and advice to member countries, primarily emerging nations, with balance-of-payments difficulties. Member nations were committed to keep their currencies within plus or minus 1 per cent of parity, although further changes in exchange rates were permitted if needed. From that standpoint, the system can be characterized as an “adjustable peg” exchange rate system. Moreover, because the dollar, being the key currency, was convertible into gold (1/35 ounce of gold), every currency had a gold value related to the dollar, which may be considered as a yardstick of value. The Bretton Woods system then may also be characterized as a gold exchange standard system.

Each member nation was responsible to subscribe to the capital of the IMF, paying one fourth in gold and three fourths in its currency (in demand notes), depending mainly on its share of world trade. The member countries were responsible to sell their currencies to the IMF, which could also borrow currencies from its members and their subscribed quotas. Initially the IMF had a total quota of \$8.8 billion, but that was increased gradually to about \$260 billion (198 billion Special Drawing Rights or SDRs) and 182 members presently. Immediately after World War II, the United States had about 33 per cent of the voting power in the IMF and 35 per cent in the World Bank, because of its dominant economic position. Thereafter, the U.S. voting power steadily declined to the current

levels of less than 20 per cent in the IMF and around 17 per cent in the World Bank. As we enter a new millennium, international institutions, mainly the IMF, are expected to play a more important role in regulating financial cooperation and reducing tensions among member nations, particularly between rich and poor nations, which have heavy debts.

Nevertheless, gold is an expensive reserve asset, as human resources are used to dig it in distant areas and bury it in heavily guarded places in central banks or in the IMF in Washington. That is why Robert Triffin of Yale University suggested that central bank deposits with the IMF be denominated in a new international unit of account or a form of paper gold. A similar system was proposed by John M. Keynes, as an alternative to the U.S. plan for the IMF at Bretton Woods in 1944. Moreover, the use of gold for the determination of money supply may lead to excessive liquidity and global inflationary pressures, when excessive amounts of gold are available, as well as to inadequate liquidity, when not enough gold is available, resulting in recessions or even depressions, as happened with the severe recession of 1907 (rich man's panic) and the Great Depression of the 1930s.

Although the Bretton Woods system performed well in the 1950s, the fact that the United States had large deficits in its balance of payments in the 1960s increased the dollar reserves of other countries and consequently the demand for gold, forcing the United States to apply first voluntary controls and then mandatory controls to capital outflows. At the same time, Japan, West Germany, and other European countries with surpluses refused to revalue their currencies, suggesting that the United States eliminate its balance-of-payments deficits. In an attempt to further discourage European companies from raising capital in the United States, the interest-equalization tax was imposed in the mid-1960s to equalize the cost of raising funds in the United States, where interest rates were lower. Moreover, the U.S. Treasury issued medium-term bonds with an exchange-rate guarantee, known as Roosa bonds, to discourage foreign central banks from cashing their excess dollars for gold. Also, U.S. short-term interest rates were raised to attract foreign funds, but long-term interest rates remained low to encourage domestic investments, a policy known as "operation twist." Nevertheless, the "dollar glut" continued in the 1960s, mainly because of the U.S. trade deficits, in contrast to the "dollar shortage" in the 1950s.

In 1969, a special quota (of \$10 billion) was created by the IMF to be used for automatic drawings. They are Special Drawing Rights (SDRs), which are gradually replacing gold as a reserve asset. The acceptance of SDRs is an obligation for members, subject to some limitations. Each member can buy currencies directly from other countries up to the amount allocated by the IMF, without going through the regular Fund operations. In the case of SDRs the IMF acts as an intermediary or guarantor. Although large amounts of SDRs are allocated to developed countries, a growing number of emerging nations receive SDR allocations lately. The IMF carries on its books \$5 billion gold, but at market prices the gold the fund owns is worth more than \$30 billion.

Therefore, the fund can write off some \$8 billion that forty poor countries owe it, but under conditions that would not give them incentives for further borrowing with the expectation of not paying back future loans.

Each member paid one fourth of its quota in gold or in dollars and the rest in its own currency. This pool may be considered as an extension of the central-bank reserves and aimed at stable exchange rates. It was accepted that drawings of member countries could be made on permission, as Harry D. White, the U.S. representative, insisted, and not automatically, as John M. Keynes, the British representative, suggested.

The quotas, which are revised every five years to reflect economic changes of members, determine the number of votes and the borrowing limits of the member countries.

The United States has almost 20 per cent of the total voting power, followed by Germany and Japan with 5.7 per cent each, and then Britain and France with 5.1 per cent each. These five developed countries with more than 40 per cent voting power dominate the IMF, which operates as a central bank for central banks. Member governments select governors, who constitute the highest governing body of the Fund and meet every September in Washington, whereas power is vested in the Board of Executive Directors (some twenty persons).

A member of the Fund can borrow 25 per cent of its quota without restrictions (gold tranche) and higher amounts with restrictions (credit tranches), but repayments are normally to be made within three to five years. Approximately, the quotas (in billions of Special Drawing Rights or SDRs) (one SDR is equal to about \$1.3) are: 27 for the United States; 8 each for Germany and Japan; 7 each for Britain and France; 5 for Saudi Arabia; 4 each for Canada, Italy, and Russia; 3 each for Belgium, China, India, and Netherlands; 2 each for Australia and Brazil; and close to 2 each for Mexico, Spain, and Venezuela, and so on.

The U.S. official reserve assets in March 1998 were \$69.35 billion, \$30.22 billion of which were foreign currencies, \$11.05 billion gold, \$10.11 billion holdings of IMF SDRs, and \$17.98 billion was the reserve position, that is, the U.S. ability to draw foreign currencies from the IMF. As the cure for sick economies, the IMF usually prescribes tough austerity measures, which lead to deep cuts in public spending, tight credit, high unemployment, sharp currency depreciation, severe stock market decline, and even social unrest. Some of these symptoms occurred in Mexico and the Southeast Asian countries.

Some countries blame Washington and other Western capitals for their strong influence on the IMF. Such austerity measures may result in high unemployment, and many people blame IMF for worsening their misery and for losing their jobs (arguing that IMF stands for I'M Fired?). Moreover, the loans from the IMF are relatively small compared to about \$2 trillion traded daily in the international currency market. Although it provided more than \$150 billion for the Asian and Mexican bailouts lately, as a lender of last resort, it is criticized that in its effort to save weak economies, it discourages the

private sector and the related governments to take the necessary painful fiscal and monetary measures to correct the problems of crony capitalism. That is why the U.S. Congress is skeptical on providing additional financing for the IMF.

WORLD TRADE ORGANIZATION

The World Trade Organization (WTO) is an organization that intends to supervise and liberalize international trade. The organization officially commenced on January 1, 1995 under the Marrakech Agreement, replacing the General Agreement on Tariffs and Trade (GATT), which commenced in 1948. The organization deals with regulation of trade between participating countries; it provides a framework for negotiating and formalizing trade agreements, and a dispute resolution process aimed at enforcing participants' adherence to WTO agreements which are signed by representatives of member governments and ratified by their parliaments. Most of the issues that the WTO focuses on derive from previous trade negotiations, especially from the Uruguay Round.

The organization is currently endeavoring to persist with a trade negotiation called the Doha Development Agenda (or Doha Round), which was launched in 2001 to enhance equitable participation of poorer countries which represent a majority of the world's population. However, the negotiation has been dogged by "disagreement between exporters of agricultural bulk commodities and countries with large numbers of subsistence farmers on the precise terms of a 'special safeguard measure' to protect farmers from surges in imports. At this time, the future of the Doha Round is uncertain."

The WTO has 153 members, representing more than 97% of total world trade and 30 observers, most seeking membership. The WTO is governed by a ministerial conference, meeting every two years; a general council, which implements the conference's policy decisions and is responsible for day-to-day administration; and a director-general, who is appointed by the ministerial conference. The WTO's headquarters is at the Centre William Rappard, Geneva, Switzerland. The WTO's predecessor, the General Agreement on Tariffs and Trade (GATT), was established after World War II in the wake of other new multilateral institutions dedicated to international economic cooperation - notably the Bretton Woods institutions known as the World Bank and the International Monetary Fund.

A comparable international institution for trade, named the International Trade Organization was successfully negotiated. The ITO was to be a United Nations specialized agency and would address not only trade barriers but other issues indirectly related to trade, including employment, investment, restrictive business practices, and commodity agreements. But the ITO treaty was not approved by the United States and a few other signatories and never went into effect. In the absence of an international organization for trade, the GATT would over the years "transform itself" into a de facto international organization.

The GATT was the only multilateral instrument governing international trade from 1948 until the WTO was established in 1995. Despite attempts in the mid 1950s and 1960s to create some form of institutional mechanism for international trade, the GATT continued to operate for almost half a century as a semi-institutionalized multilateral treaty regime on a provisional basis. Seven rounds of negotiations occurred under the GATT. The first real GATT trade rounds concentrated on further reducing tariffs.

Then, the Kennedy Round in the mid-sixties brought about a GATT anti-dumping Agreement and a part on development. The Tokyo Round during the seventies was the first major attempt to tackle trade barriers that do not take the form of tariffs, and to improve the system, adopting a series of agreements on non-tariff barriers, which in some cases interpreted existing GATT rules, and in others broke entirely new ground. Because these plurilateral agreements were not accepted by the full GATT membership, they were often informally called "codes". Several of these codes were amended in the Uruguay Round, and turned into multilateral commitments accepted by all WTO members. Only four remained plurilateral (those on government procurement, bovine meat, civil aircraft and dairy products), but in 1997 WTO members agreed to terminate the bovine meat and dairy agreements, leaving only two.

Well before GATT's 40th anniversary, its members concluded that the GATT system was straining to adapt to a new globalizing world economy. In response to the problems identified in the 1982 Ministerial Declaration (structural deficiencies, spill-over impacts of certain countries' policies on world trade GATT could not manage etc.), the eighth GATT round - known as the Uruguay Round - was launched in September 1986, in Punta del Este, Uruguay.

It was the biggest negotiating mandate on trade ever agreed: the talks were going to extend the trading system into several new areas, notably trade in services and intellectual property, and to reform trade in the sensitive sectors of agriculture and textiles; all the original GATT substances were up for review. The Final Act concluding the Uruguay Round and officially establishing the WTO regime was signed during the April 1994 ministerial meeting at Marrakesh, Morocco, and hence is known as the Marrakesh Agreement. The GATT still exists as the WTO's umbrella treaty for trade in goods, updated as a result of the Uruguay Round negotiations. GATT 1994 is not however the only legally binding agreement included via the Final Act at Marrakesh; a long list of about 60 agreements, annexes, decisions and understandings was adopted.

The agreements fall into a structure with six main parts:

- The Agreement Establishing the WTO
- Goods and investment - the Multilateral Agreements on Trade in Goods including the GATT 1994 and the Trade Related Investment Measures
- *Services*: The General Agreement on Trade in Services
- Intellectual property — the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS)
- Dispute settlement (DSU)
- Reviews of governments' trade policies (TPRM)

The topmost decision-making body of the WTO is the Ministerial Conference, which usually meets every two years. It brings together all members of the WTO, all of which are countries or customs unions. The Ministerial Conference can take decisions on all matters under any of the multilateral trade agreements. The inaugural ministerial conference was held in Singapore in 1996.

Disagreements between largely developed and developing economies emerged during this conference over four issues initiated by this conference, which led to them being collectively referred to as the “Singapore issues”. The second ministerial conference was held in Geneva in Switzerland. The third conference in Seattle, Washington ended in failure, with massive demonstrations and police and National Guard crowd control efforts drawing worldwide attention. The fourth ministerial conference was held in Doha in Persian Gulf nation of Qatar. The Doha Development Round was launched at the conference. The conference also approved the joining of China, which became the 143rd member to join.

The fifth ministerial conference was held in Cancún, Mexico, aiming at forging agreement on the Doha round. An alliance of 22 southern states, the G20 developing nations (led by India, China and Brazil), resisted demands from the North for agreements on the so-called “Singapore issues” and called for an end to agricultural subsidies within the EU and the US. The talks broke down without progress.

The sixth WTO ministerial conference was held in Hong Kong from 13 December – 18 December 2005. It was considered vital if the four-year-old Doha Development Agenda negotiations were to move forward sufficiently to conclude the round in 2006. In this meeting, countries agreed to phase out all their agricultural export subsidies by the end of 2013, and terminate any cotton export subsidies by the end of 2006. Further concessions to developing countries included an agreement to introduce duty free, tariff free access for goods from the Least Developed Countries, following the Everything But Arms initiative of the European Union — but with up to 3% of tariff lines exempted. Other major issues were left for further negotiation to be completed by the end of 2010. The WTO General Council, on 26 May 2009, agreed to hold a seventh WTO ministerial conference session in Geneva from 30 November–3 December 2009.

A statement by chairman Amb. Mario Matus acknowledged that the prime purpose was to remedy a breach of protocol requiring two-yearly “regular” meetings, which had lapsed with the Doha Round failure in 2005, and that the “scaled-down” meeting would not be a negotiating session, but “emphasis will be on transparency and open discussion rather than on small group processes and informal negotiating structures”. The general theme for discussion is “The WTO, the Multilateral Trading System and the Current Global Economic Environment”

The WTO launched the current round of negotiations, the Doha Development Agenda (DDA) or Doha Round, at the fourth ministerial conference in Doha, Qatar in November 2001. The Doha round was to be an ambitious effort to make globalization more inclusive and help the world’s poor, particularly by slashing barriers and subsidies in farming.

The initial agenda comprised both further trade liberalization and new rule-making, underpinned by commitments to strengthen substantial assistance to developing countries.

The negotiations have been highly contentious and agreement has not been reached, despite the intense negotiations at several ministerial conferences and at other sessions. Disagreements still continue over several key areas including agriculture subsidies.

FUNCTIONS

Among the various functions of the WTO, these are regarded by analysts as the most important:

- It oversees the implementation, administration and operation of the covered agreements.
- It provides a forum for negotiations and for settling disputes.

Additionally, it is the WTO's duty to review and propagate the national trade policies, and to ensure the coherence and transparency of trade policies through surveillance in global economic policy-making. Another priority of the WTO is the assistance of developing, least-developed and low-income countries in transition to adjust to WTO rules and disciplines through technical cooperation and training. *The WTO is also a centre of economic research and analysis*: regular assessments of the global trade picture in its annual publications and research reports on specific topics are produced by the organization. Finally, the WTO cooperates closely with the two other components of the Bretton Woods system, the IMF and the World Bank.

PRINCIPLES OF THE TRADING SYSTEM

The WTO establishes a framework for trade policies; it does not define or specify outcomes. That is, it is concerned with setting the rules of the trade policy games. Five principles are of particular importance in understanding both the pre-1994 GATT and the WTO:

- *Non-Discrimination*: It has two major components: the most favoured nation (MFN) rule, and the national treatment policy. Both are embedded in the main WTO rules on goods, services, and intellectual property, but their precise scope and nature differ across these areas. The MFN rule requires that a WTO member must apply the same conditions on all trade with other WTO members, *i.e.* a WTO member has to grant the most favourable conditions under which it allows trade in a certain product type to all other WTO members. "Grant someone a special favour and you have to do the same for all other WTO members." National treatment means that imported goods should be treated no less favourably than domestically produced goods (at least after the foreign goods have entered the market) and was introduced to tackle non-tariff barriers to trade (e.g. technical standards, security standards *et al.* discriminating against imported goods).

- *Reciprocity*: It reflects both a desire to limit the scope of free-riding that may arise because of the MFN rule, and a desire to obtain better access to foreign markets. A related point is that for a nation to negotiate, it is necessary that the gain from doing so be greater than the gain available from unilateral liberalization; reciprocal concessions intend to ensure that such gains will materialise.
- *Binding and enforceable commitments*: The tariff commitments made by WTO members in a multilateral trade negotiation and on accession are enumerated in a schedule (list) of concessions. These schedules establish "ceiling bindings": a country can change its bindings, but only after negotiating with its trading partners, which could mean compensating them for loss of trade. If satisfaction is not obtained, the complaining country may invoke the WTO dispute settlement procedures.
- *Transparency*: The WTO members are required to publish their trade regulations, to maintain institutions allowing for the review of administrative decisions affecting trade, to respond to requests for information by other members, and to notify changes in trade policies to the WTO. These internal transparency requirements are supplemented and facilitated by periodic country-specific reports (trade policy reviews) through the Trade Policy Review Mechanism (TPRM). The WTO system tries also to improve predictability and stability, discouraging the use of quotas and other measures used to set limits on quantities of imports.
- *Safety valves*: In specific circumstances, governments are able to restrict trade. There are three types of provisions in this direction: substances allowing for the use of trade measures to attain noneconomic objectives; substances aimed at ensuring "fair competition"; and provisions permitting intervention in trade for economic reasons. Exceptions to the MFN principle also allow for preferential treatment of developing countries, regional free trade areas and customs unions.

ORGANIZATIONAL STRUCTURE

The General Council has multiple bodies which oversee committees in different areas, and they are the following:

- *Council for Trade in Goods*: There are 11 committees under the jurisdiction of the Goods Council each with a specific task. All members of the WTO participate in the committees. The Textiles Monitoring Body is separate from the other committees but still under the jurisdiction of Goods Council. The body has its own chairman and only ten members. The body also has several groups relating to textiles.
- *Council for Trade-Related Aspects of Intellectual Property Rights*: Information on intellectual property in the WTO, news and official records of the activities

of the TRIPS Council, and details of the WTO's work with other international organizations in the field.

- *Council for Trade in Services*: The Council for Trade in Services operates under the guidance of the General Council and is responsible for overseeing the functioning of the General Agreement on Trade in Services (GATS). It is open to all WTO members, and can create subsidiary bodies as required.
- *Trade Negotiations Committee*: The Trade Negotiations Committee (TNC) is the committee that deals with the current trade talks round. The chair is WTO's director-general. The committee is currently tasked with the Doha Development Round.

The Service Council has three subsidiary bodies: financial services, domestic regulations, GATS rules and specific commitments. The General council has several different committees, working groups, and working parties. There are committees on the following: Trade and Environment; Trade and Development (Subcommittee on Least-Developed Countries); Regional Trade Agreements; Balance of Payments Restrictions; and Budget, Finance and Administration. There are working parties on the following: Accession. There are working groups on the following: Trade, debt and finance; and Trade and technology transfer.

Voting System

The WTO operates on a one country, one vote system, but actual votes have never been taken. Decision making is generally by consensus, and relative market size is the primary source of bargaining power. The advantage of consensus decision-making is that it encourages efforts to find the most widely acceptable decision. Main disadvantages include large time requirements and many rounds of negotiation to develop a consensus decision, and the tendency for final agreements to use ambiguous language on contentious points that makes future interpretation of treaties difficult.

In reality, WTO negotiations proceed not by consensus of all members, but by a process of informal negotiations between small groups of countries. Such negotiations are often called "Green Room" negotiations (after the colour of the WTO Director-General's Office in Geneva), or "Mini-Ministerials", when they occur in other countries.

These processes have been regularly criticised by many of the WTO's developing country members which are often totally excluded from the negotiations.. Richard Harold Steinberg (2002) argues that although the WTO's consensus governance model provides law-based initial bargaining, trading rounds close through power-based bargaining favouring Europe and the United States, and may not lead to Pareto improvement.

DISPUTE SETTLEMENT

In 1994, the WTO members agreed on the Understanding on Rules and Procedures Governing the Settlement of Disputes (DSU) annexed to the "Final Act" signed in Marrakesh in 1994. Dispute settlement is regarded by the WTO as the central pillar of

the multilateral trading system, and as a “unique contribution to the stability of the global economy”. WTO members have agreed that, if they believe fellow-members are violating trade rules, they will use the multilateral system of settling disputes instead of taking action unilaterally.

The operation of the WTO dispute settlement process involves the DSB panels, the Appellate Body, the WTO Secretariat, arbitrators, independent experts and several specialized institutions.

ACCESSION AND MEMBERSHIP

The process of becoming a WTO member is unique to each applicant country, and the terms of accession are dependent upon the country’s stage of economic development and current trade regime. The process takes about five years, on average, but it can last more if the country is less than fully committed to the process or if political issues interfere. As is typical of WTO procedures, an offer of accession is only given once consensus is reached among interested parties.

Accession Process

A country wishing to accede to the WTO submits an application to the General Council, and has to describe all aspects of its trade and economic policies that have a bearing on WTO agreements. The application is submitted to the WTO in a memorandum which is examined by a working party open to all interested WTO Members. After all necessary background information has been acquired, the working party focuses on issues of discrepancy between the WTO rules and the applicant’s international and domestic trade policies and laws. The working party determines the terms and conditions of entry into the WTO for the applicant nation, and may consider transitional periods to allow countries some leeway in complying with the WTO rules. The final phase of accession involves bilateral negotiations between the applicant nation and other working party members regarding the concessions and commitments on tariff levels and market access for goods and services. The new member’s commitments are to apply equally to all WTO members under normal non-discrimination rules, even though they are negotiated bilaterally. When the bilateral talks conclude, the working party sends to the general council or ministerial conference an accession package, which includes a summary of all the working party meetings, the Protocol of Accession (a draft membership treaty), and lists (“schedules”) of the member-to-be’s commitments. Once the general council or ministerial conference approves of the terms of accession, the applicant’s parliament must ratify the Protocol of Accession before it can become a member.

Members and Observers

The WTO has 153 members (almost all of the 123 nations participating in the Uruguay Round signed on at its foundation, and the rest had to get membership). The 27 states of

the European Union are represented also as the European Communities. WTO members do not have to be full sovereign nation-members.

Instead, they must be a customs territory with full autonomy in the conduct of their external commercial relations. Thus Hong Kong became a GATT contracting party, and the Republic of China (ROC) (commonly known as Taiwan, whose sovereignty has been disputed by the People's Republic of China or PRC) acceded to the WTO in 2002 under the name of "Separate Customs Territory of Taiwan, Penghu, Kinmen and Matsu" (Chinese Taipei).

A number of non-members (30) are observers at WTO proceedings and are currently negotiating their membership.

As observers, Iran, Iraq and Russia are not yet members. Russia is the biggest economy outside WTO and after the completion of Russia's accession, Iran would be the biggest economy outside the WTO. Observers must start accession negotiations within five years of becoming observers. Some international intergovernmental organizations are also granted observer status to WTO bodies. 14 states and 2 territories so far have no official interaction with the WTO.

AGREEMENTS

The WTO oversees about 60 different agreements which have the status of international legal texts. Member countries must sign and ratify all WTO agreements on accession. A discussion of some of the most important agreements follows. The Agreement on Agriculture came into effect with the establishment of the WTO at the beginning of 1995.

The AoA has three central concepts, or "pillars": domestic support, market access and export subsidies. The General Agreement on Trade in Services was created to extend the multilateral trading system to service sector, in the same way the General Agreement on Tariffs and Trade (GATT) provides such a system for merchandise trade. The Agreement entered into force in January 1995. The Agreement on Trade-Related Aspects of Intellectual Property Rights sets down minimum standards for many forms of intellectual property (IP) regulation.

It was negotiated at the end of the Uruguay Round of the General Agreement on Tariffs and Trade (GATT) in 1994. The Agreement on the Application of Sanitary and Phytosanitary Measures - also known as the SPS Agreement was negotiated during the Uruguay Round of the General Agreement on Tariffs and Trade, and entered into force with the establishment of the WTO at the beginning of 1995. Under the SPS agreement, the WTO sets constraints on members' policies relating to food safety (bacterial contaminants, pesticides, inspection and labelling) as well as animal and plant health (imported pests and diseases).

The Agreement on Technical Barriers to Trade is an international treaty of the World Trade Organization. It was negotiated during the Uruguay Round of the General Agreement on Tariffs and Trade, and entered into force with the establishment of the

WTO at the end of 1994. The object ensures that technical negotiations and standards, as well as testing and certification procedures, do not create unnecessary obstacles to trade”. The Agreement on Customs Valuation, formally known as the Agreement on Implementation of Article VII of GATT, prescribes methods of customs valuation that Members are to follow. Chiefly, it adopts the “transaction value” approach.

5

Models of Economic Development

Together with a saving function, growth rate equals s/a (s is the saving rate, and a is the capital-output ratio). Assuming that the capital-output ratio is fixed by technology and does not change in the short run, growth rate is solely determined by the saving rate on the basis of whatever is saved will be invested.

HARROD–DOMAR MODEL

The Harrod–Domar model delineates a functional economic relationship in which the growth rate of gross domestic product (g) depends positively on the national saving ratio (s) and inversely on the national capital/output ratio (k) so that it is written as $g = s / k$. The equation takes its name from a synthesis of analysis of growth by the British economist Sir Roy F. Harrod and the Polish-American economist Evsey Domar. The Harrod–Domar model in the early postwar times was commonly used by developing countries in economic planning. With a target growth rate, and information on the capital output ratio, the required saving rate can be calculated. It was the best system.

EXOGENOUS GROWTH MODEL

The exogenous growth model (or neoclassical growth model) of Robert Solow and others places emphasis on the role of technological change. Unlike the Harrod-Domar model, the saving rate will only determine the level of income but not the rate of growth. The sources-of-growth measurement obtained from this model highlights the relative importance of capital accumulation (as in the Harrod–Domar model) and technological change (as in the Neoclassical model) in economic growth. The original Solow (1957)

study showed that technological change accounted for almost 90 percent of U.S. economic growth in the late 19th and early 20th centuries. Empirical studies on developing countries have shown different results.

Even so, in our post-industrial economy, economic development, including in emerging countries is now more and more based on innovation and knowledge. Creating business clusters is one of the strategies used. One well known example is Bangalore in India, where the software industry has been encouraged by government support including Software Technology Parks.

However, when looking at the growth rate put forward from the neoclassical growth model, it seems to suggest that countries with same characteristics and technology will eventually converge to the same rate of growth. However, one should know that the knowledge presented in countries that promotes technological advancement is not stationary. Meaning that knowledge are linked to individual and not to the country.

According to Lucas Jr (1988) to compensate the movement of knowledge, we should implement factors such as labour factor to predict immigration flow. With labour movement coming into factor, we can then predict the flow of knowledge which can then successfully lead to increase in technology.

HISTORICAL DEVELOPMENT THEORIES

MODERNIZATION THEORY

Modernization Theory is a theory of development which states that the development can be achieved through following the processes of development that were used by the currently developed countries. Scholars such as Walt Rostow and A.F.K. Organski developed stages of development through which every country develops. Samuel Huntington determined development to be a linear process which every country must go through. Modernization Theory, in contrast to Classical Liberalism, viewed the state as a central actor in modernizing “backward” or “underdeveloped” societies. Talcott Parsons’ functional sociology defined the qualities that distinguished “modern” and “traditional” societies.

Education was viewed as key to creating modern individuals. Technology played a key role in this development theory because it was believed that as technology was developed and introduced to lesser developed countries it would spur economic growth. One key factor in Modernization Theory is the belief that development requires the assistance of developed countries to aid developing countries to learn from their development. In addition, it was believed that the lesser developed countries would develop and grow faster than developed countries.

Thus, this theory is built upon the theory that it is possible for equal development to be reached between the developed and lesser developed countries.

Dependency Theory

While Modernization Theory understood development and underdevelopment as a result from internal conditions that differ between economies, dependency theory understood development and underdevelopment as relational. It saw the world's nations as divided into a *core* of wealthy nations which dominate a *periphery* of poor nations whose main function in the system is to provide cheap labour and raw materials to the core. It held that the benefits of this system accrue almost entirely to the rich nations, which become progressively richer and more developed, while the poor nations, which continually have their surpluses drained away to the core, do not advance. Developed in the 1950s, dependency theory shared many points with Rosa Luxembour's and V.I. Lenin's earlier, Marxist, theories of imperialism; and dependency theory was embraced by many Marxists and neo-Marxists. Dependency theorists held that for underdeveloped nations to develop, they must break their ties with developed nations and pursue internal growth. One type of policy crafted from this insight was Import substitution industrialization.

World Systems Theory

In response to some of the criticisms of Dependency Theory came World Systems Theory, which the division of periphery and centre was further divided into a trimodal system consisting of the core, semi-periphery and periphery. In this system, the semi-periphery lies between the core and periphery and is exploited by the core and exploits the periphery. This division aims to explain the industrialization within lesser developed countries.

World Systems Theory was initiated by Immanuel Wallerstein in, among other writings, *World Systems Analysis*, and focuses on inequality as a separate entity from growth in development and examines change in the global capitalist system. One distinguishing feature of this theory is a distrust for the state and a view in which the state is seen as a group of elites and that industrialization cannot be equated with development. Out of this theory stem anti-systemic movements which attempt to reverse the terms of the system's inequality through social democratic and labour movements.

State Theory

In response to the distrust of the state in World Systems Theory, is state theory. State Theory is based upon the view that the economy is intertwined with politics and therefore the take-off period in development is unique to each country. State Theory emphasized the effects of class relations and the strength and autonomy of the state on historical outcomes. Thus, development involves interactions between the state and social relations because class relations and the nature of the state impact the ability of the state to function. Development is dependent upon state stability and influence externally as well as internally. State Theorists believe that a developmentalist state is required for development by taking control of the development process within one state.

Allegations that Editors are Removed for Criticizing Policies

David F. Prenatt, Jr. (former ODP editor *netesq*) and the former editor known by the alias *The Cunctator* claim to have been removed for disagreeing with staff about changes to the policies, with special regard to the ODP's copyright policies. According to their claims, staff used the excuse that their behaviour was uncivil to remove bothersome editors.

Blacklisting Allegations

Senior ODP editors have the ability to attach “warning” or “do not list” notes to individual domains, but no editor has the unilateral ability to block certain sites from being listed. Sites with these notes might still be listed, and at times notes are removed after some discussion.

Hierarchical Structure

Recently criticism of ODP's hierarchical structure emerged. Many believe hierarchical directories are too complicated. As the recent emergence of Web 2.0, folksonomies began to appear. These people thought folksonomies, networks, and directed graph are more “natural” and easier to manage than hierarchies.

Software

Search: The ODPSearch software is a derivative version of Isearch which is open source, licensed under the Mozilla Public License.

Editor Forums

The ODP Editor Forums were originally run on software that was based on the proprietary Ultimate Bulletin Board system. In June 2003, they switched to the open source phpBB system. As of 2007, these forums are powered by a modified version of phpBB.

Bug Tracking

The bug tracking software used by the ODP is Bugzilla, and the web server Apache. Squid web proxy server was also used but it was removed in August 2007 when the storage servers were reorganized. All these applications are open source.

Interface

The ODP database/editing software is closed source, although Richard Skrenta of ODP did say in June 1998 that he was considering licensing it under the GNU General Public License. This has led to criticism from the aforementioned GNU project, many of whom also criticise the ODP content license. As such, there have been some efforts to provide alternatives to ODP. These alternatives would allow communities of like-minded editors to set up and maintain their own open source/open content Web

directories. However, no significant open source/open content alternative to ODP has emerged. Economic development is the development of economic wealth of countries or regions for the well-being of their inhabitants. This is the short definition of Economic Development.

Economic Growth & development are two different terms used in economics. Generally speaking economic development refers to the problems of underdeveloped countries and economic growth to those of developed countries.

By Economic Growth we simply mean increase in per capita income or increase in GNP. In recent literature, the term economic growth refers to sustained increase in a country's output of goods and services, or more precisely product per capita. Output is generally measured in terms of GNP. The term economic development is far more comprehensive. It implies progressive changes in the socio-economic structure of a country. Viewed in this way economic development involves a steady decline in agricultural shares in GNP and continuous increase in shares of industries, trade banking construction and services. Further whereas economic growth merely refers to rise in output; development implies change in technological and institutional organization of production as well as in distributive pattern of income.

Hence, compared to the objective of development, economic growth is easy realize. By a larger mobilization of resources and raising their productivity, output level can be raised. The process of development is far more extensive. Apart from a rise in output, it involves changes in composition of output, shift in the allocation of productive resources, and elimination or reduction of poverty, inequalities and unemployment.

In the words of Amartya Sen "Development requires the removal of major sources of unfreedom poverty as well as tyranny, poor economic opportunities as well as systematic social deprivation neglect of public facilities as well as intolerance or over activity of repressive states...." Economic development is not possible without growth but growth is possible without development because growth is just increase in GNP It does not have any other parameters to it. Development can be conceived as Multi-Dimensional process or phenomena. If there is increase in GNP more than the increase in per capita Income then we can say that Development is possible. When given conditions of population improves then we can say that this is also an indicator of economic Development.

6

Economics of Planter Civilizations

The practice of investing human labour in cultivating plants with the help of seeds or cuttings which yielded a permanent revenue began about 10.000 B.C. Very little further output of labour was required to tend and harvest them in subtropical and tropical climates. This new practice of hoe agriculture probably developed in the “old planter civilizations” and began earlier than the age of tending cattle that may be presumed to have begun near 7500 B.C. or so as radiocarbon dates from the Belt Cave on the southern shores of the Caspian Sea have made probable recently. As the European Palaeolithic Age ended several millennia later than that in the Middle East, it is easy to explain, with influences from the Southeast, an otherwise surprising find of dwarf wheat in the Ice Age cave of Merkenstein in Austria.

The oldest archaeological planter remains, which we can place with any certainty, are found in an area of Eurasia and Africa, that is bounded by the Atlantic Ocean, the Sahara and South East Africa, the mountains of Central Asia, by tundra and desert. The ancestors not only of the grains, but of practically every kind of plant which was important to the Ancient World before the discovery of America, are found in this part of our globe. The final change-over to plant cultivation is best accepted according to Professor von Eickstedt to have been somewhere around Turkestan. From there its influence may have extended to Eastern Asia and further to America, to India, Polynesia, and particularly over Northern Europe to Middle and Southern Europe, as far as Eurafrica. At the same time the planter practices of Central Asia may have reached Africa and Europe *via* The Middle East in a wide circle also. Part of this extension of planter civilization may have been the result of tribal migrations. In the same sense biological discoveries and

archaeological finds seem to indicate that plants which influenced Northern Europe, the Middle East, and Eastern Asia were first cultivated in regions comparatively near to Turkestan. This would support Professor von Eickstedt's hypothesis. If the present political situation would make it possible to excavate systematically in and near Turkestan, certainty in the question of the origin of the planter and especially the grain planter civilizations could possibly be reached. The only independent centres of plant cultivation, as far as we can discover, seem to be Egypt where there are very early finds of grains and other plants, Southeast Asia, and Peru where we find a remarkable number of native trees and plants grown under hoe cultivation.

At present it is difficult to decide whether an independent development to the earliest plant cultivation must not have taken place in the case of at least these three regions. Southeast Asia had easy opportunities to cultivating tropical plants, and this area is consequently considered by many to have been the scene of the beginnings of the oldest plant farming on this earth and as the region from which came this revolutionary advance. It is quite conceivable that from a fisher-planter centre in Southeast Asia those of Central Asia, East Africa, and even Peru were stimulated. The advance may have been first inspired by tribal migrations and the consequent spread of ideas. The result has been a system of farming that leads directly to our own more complicated husbandry and horticulture.

Archaeological and ethnological sources indicate at present that in the cases of Peru, Southeast Asia, East Africa, and Central Asia, a threefold pedigree may be drawn up at will. It is left to the tact of the historian to decide whether the foundation for the various locally earliest cultivations of plants were laid in Southeast Asia, or whether a fourfold independent development is more probable. Perhaps instead the credit has still to be given to a creative development in Central Asia, so that all other centres consequently were influenced from here to take up the practice of cultivation, only the choice of plants being varied by reason of the changing climate. Outside of these four regions archaeology and ethnology seem to indicate that it is improbable, from finds and later folklore, that an independent change-over to hoe culture took place, or only perhaps in a quite primitive form that has left no trace today.

Feminine intuition and intelligence are more responsible for the beginnings of this tending of plants than the masculine. We probably owe to woman one of the greatest advances of the history of man. She was already accustomed, through her traditional plantgathering activities, to digging, and it cannot have been a big step for the plantgatherer of the late Palaeolithic Age to uproot cuttings with the help of her hands, a digging staff, pointed stick, or scraper, and to plant them near her own dwelling place, so that she would not have to go so far afield to search for them. And again, this was only a short way from sowing seeds and rootcrops with such tools. It may be that such seeds had been left in graves or in storage places or she may have noticed the chance appearance of eatable plants and fruits which had grown from the seeds and plants left behind by earlier tribes, who had made that region their home. So was born a new

society, not in a rational way, as it appears, but full of myths and symbols and by a searching enquiry into the relations of life. It seems that the religious concepts of the mysteries of death and rebirth that concern us to this day are connected in some way with primitive ideas on the life of the plant world arising in the early planter civilizations.

At any rate, as soon as the plantgatherer took the step deliberately to plant and sow seeds of plants so that this work became one of her permanent occupations, there arose a new many-sided economic force of primary production, the garden, from digging or hoe culture. Both these last terms are of a technical nature. Professor Eduard Hahn has differentiated for the first time between the methods of cultivation of the soil with the help of the plough, and those with other tools. The plough works with the help of animal strength, and agriculture had in consequence to be more rationally planned and developed until it was perfected. With the digging staff, pick, hoe, or spade, the cultivation of individual plants, especially the seeds, was undertaken in one garden. There was not a large area necessary for the setting of plants, as is the case in plough culture, but a mixed group of beds, with or without furrows to separate them, served for all types of plants which society needed. This is the identical system that is still practised today in all the gardens of the world although it varies in the concepts and plants with the climate. Hoe culture and gardening is at the same time the most ideally organized and developed feminine occupation.

There is no single harvest in this type of cultivation. The woman cared constantly for her garden and took from it day by day all the food which she needed. Vegetables, grasses, berries, roots, and flowers grew together and were planted among wild vegetables that were not entirely without use either. For the first time, grafting and deliberate attempts to improve the crop were made, originally inspired by magical considerations as Professor Netolitzky has suggested some time ago in agreement with Professor Hahn. He interpreted grafting as some kind of attempt to consummate a mystic marriage between plants supposedly male and female. Gradually through careful tending wild plants became cultivated, repayed their care by an increased production and, economically speaking, higher rent. Ethnologically it is certain that in exclusive planter civilizations, or those where hunting is combined with hoe culture, the garden is the real sphere of woman's work even today. Men usually help with work that requires physical strength, especially in the cultivation of fruit trees. In native hoe cultures, if male slaves were employed in the garden, they usually had to wear women's clothing, as modern anthropologists inform us.

Hoe cultivation produced for the first time more or less settled living conditions. Because of it, the house like the garden became, as it is today, in many cases the property and in most cases at least the province of woman. New economic relationships also altered at one stroke the social position of woman. The foodgatherer of the Palaeolithic Age, compared with man's activities as a hunter, had had, on the whole, an equal position economically. For hers was the more secure chance, whereas man's occupation had greater possibilities of gain. Now woman was far superior economically to man owing

to her garden, as long as man remained a hunter. Her permanent plant resources fed the whole family much more competently than had the united efforts of man and woman formerly during the Palaeolithic Age. And so grew a social and political ascendancy of woman over man, and is witnessed, up to today, by the so-called matriarchy existing in many tribes to quite extreme forms of civil and social rule of womankind over men and children, polyandry, the bearing of arms by women alone, and feminine chieftains and priestesses. Female governments and legal jurisdiction within the family and over her kin are especially noticeable here, and remains have survived in many societies which finally adopted a patriarchal pattern.

This control of society by women was certainly not necessarily the only constitutional pattern, the so-called hunter-planter period tended to by reason of its economic structure. For woman had become now economically an object of considerable value for man, to some extent his most valuable economic possession. She was in a position to feed everybody without any great expenditure of toil by the men. Very often more or less secret unions of men not only kept down alien prisoners of both sexes at the point of the spear, but even terrorized and kept down the women of their own tribes, as being the weaker in body, and reduced them, more or less, to the status of working slaves. So that the economic success of women led just as often to their social degradation as to their social advance. Warlike expansion, in which one tribe killed or ate the adult men of a hostile tribe, and brought its women into a combination of marriage and slavery has also often enough been witnessed in the legends and early history of the Ancient World. Columbus arrived in the West Indies at the exact moment when the seafaring Caribees, who gave their name to cannibalism, had just achieved a barbarous subjection of this kind, by attacking a peaceful gardening tribe.

At the beginnings of the Planter Age the balance of power and equality between the sexes was seriously disturbed for the first time. The pendulum has swung in this field to and fro between extremes to this day, influenced as it has been by constant social changes and new situations brought about by the different periods of civilization. The men's superior ability to earn, in cattle breeding, plough culture, mining, and most of the town civilizations has benefitted them most. On the other hand the latest Capitalist and Bolshevik development of our Age of Machines, in which the capacity for work of both the sexes has played an equally prominent part in production, contributing to a much improved satisfaction of human needs, seems to make again for the Young Palaeolithic equality between man and woman.

The transition to hoe cultivation was far-reaching in its consequences, and its effect can be seen most easily in the increase of the size of the tribes. The number of children who reached adult age was greater. Large tribes and small groups increased in size. Definite camping sites were founded and inhabited. Cultivation of the plants in each of these new tribal areas followed a planned traditional cycle of events, so that a change in the place of settlement was only made after the harvest once a year or so, and in particularly favourable climates no change at all was necessary. Building technique,

home crafts, and trades had to meet higher demands. One group settled near another. Hunting, fishing, and plant gathering tribes of the earlier days were often pressed back to the seacoast, forests, and inhospitable regions, or were taken prisoner to work as slaves. Finally they themselves adopted the new ideas and the methods inherent in them, trading plant products for game and fish; or because of their small numbers they were easily weakened by fighting and perished altogether.

Hoe culture still forms, in the world today, the main basis of many important economic forms of primary production, e.g. of our vineyards, the cultivation of potatoes, cotton, tobacco, dates, and figs. It is up to now in India, East Asia, and Polynesia the foundation of agriculture, not always to the advantage of the populations of these regions. The transition to this economic stage took on a different form in almost every small stretch of the earth's surface because of the different resources prevailing. For man had to adapt himself quickly to the conditions of the region in which he intended to settle, and to take into consideration all possible methods of cultivation of its plants. And so arose many differences in tribal life and tradition, as a result of this development into a widespread diversity from the few simple and general social patterns of the Palaeolithic Age.

The taming of animals is in the main a masculine achievement, and is considered to have arisen from the Palaeolithic hunting traditions. The first domesticated animals appear early in very different regions of the earth. But similar to the beginnings of hoe cultivation, their wild ancestors came, as a rule, from Central Asia or East Africa. There are very many early remains of domesticated animals in Northern Europe. It appears that as early as the Neolithic Age, practically all those animals were domesticated here, which are used today in this climate. These European regions were comparatively near to the centres of domestication in Central Asia and Northern East Africa and this fact explains, perhaps, their progress so far and so early. These regions have even been attributed with some independent first attempts at domestication, although this remains doubtful.

The dog became, perhaps, a favourite pet and domestic animal as a hanger-on at the camp fire, until gradually it was completely tamed. There were probably two centres involved. The earlier one in Asia and Europe was in regions inhabited by wolves, and the whole area stretched from Central Europe eastward into Asia and embraced the whole of Siberia and Northern China. Near 8160 1400 B.C. (the earliest radiocarbon date) in North and Central Europe, in Northern France, possibly even as far as Spain in the West and far into Asia in the East, the earliest tamed dogs are found soon after the end of the Palaeolithic Age. The dog can certainly be considered to be the earliest domestic animal tamed by man. Sledges for dogs are preserved from a somewhat later period, perhaps proving the prehistoric use of the dog as a draught animal. Its meat and its skin were probably used before this animal was found useful in hunting or guarding.

Whether there was a Late Palaeolithic domestic use of dogs descended from wolves in Europe is not yet sufficiently clear. But the hypothesis of an European origin of the

domesticated dog is not too convincing. Dogs were used in early Eastern Siberia and Tibet, although this practice may have developed separately. The natives of America and Australia, after their very early immigration into their continents which were still easily accessible from Asia, brought with them dogs tamed from Northeast Asia wolves. This fact could not be easily explained by suggesting a European centre for the first domestication of a breed of wolves to dogs, and it seems more likely that this centre was in Central or East Asia. In the South a second complex of breeds of dog developed in a somewhat later period from the shakals of Egypt, Mesopotamia, and of the Sahara which was still habitable in the Mesolithic and Early Neolithic Ages. These southern breeds include especially our greyhounds and bulldogs.

The date of the domestication of the reindeer and possibly of the elk too, recently tamed again in Russia with success, has been in great doubt. Some place it very early, others much later. And unfortunately in the excavations we have made, there are still no definite early traces of these animals as used in domestication, although today they are the most important meat and sledge animals of the regions around the North Pole. Consequently any attempts to fill in the gaps in our discoveries by constructions on paper must be pure hypothesis.

But we are much more certain about the domestication of the horse. Two centres are now fairly well established. In Eurasia as far as Scandinavia in the West and Turkestan in the East lay the dwelling sites of the tarpan, the ancestor of our ponies and other small breeds of horse, in Mongolia those of the "Przewalski" horse, the ancestor of our so-called Arab bloodstock. Many other breeds of horses originate from a combination of these two primeval groups. Already the regions where they were found untamed were intermixed. But the heavy Belgian horse is probably descended from a third species. At any rate, an untamed breed of this kind appears on Palaeolithic drawings and lived exclusively in Europe. The first tamed wild horse was the tarpan. Its domestication appears to have begun in Turkestan and Iran between 6000 and 5000 B.C. Between 4000 and 3000 B.C. the tarpan horse reached slowly the regions of the Indus, Mesopotamia, and the rest of the Middle East; but it did not appear in Egypt, Africa, and South Western Europe until after 2000 B.C. The same tarpan appears as a domesticated animal of Europe in the early Neolithic Age and there was probably some connection with the centre in Turkestan, even if we cannot completely exclude its domestication from wild forms in Eastern and in Northern Europe. The "Belgian" breed appears some time afterwards. There were tamed Przewalski horses in the Black Sea and Caspian steppes about 2000 years B.C., that is, during the Bronze Age, possibly even going back to the Copper Stone Age of Iran, sires of the cavalry horses of the Hittites, Mitanni, and Meder.

The tamed donkey appeared in Nubia and Egypt during the 5th millennium B.C. It was originally used as a baggage animal and was later ridden or used as a draught animal. The breeding of mules followed, as soon as contacts with regions of horse breeding were established. The Asiatic "onager" has been left to run wild until today,

although there were parts of Iran and Mesopotamia where horses could not be imported from the original centre of domestication, and so the onager had to be used. The domestication of this breed was early, but was not continued. However, it may have served to touch off that of the more useful Nubian and Egyptian donkey in Africa. We have discovered breeds of domesticated onagers from the Middle East in Ur and Elam between 4000 and 3000 B.C. or so. Remains of donkeys, mules, and, it seems, horses have been found there also, but these finds are much disputed. It seems significant that the donkey was known in India from the earliest time of town civilization of which we have any record. This may or may not point to connections between the Hittites in Asia Minor and perhaps Minoan Crete and the Indus civilization, as Professor Hrozný recently has tried to establish. The domestication of the zebra and similar related animals has occasionally been successfully attempted in Central Africa.

It is well known that the elephant was tamed and used in India as far back as the Indus civilization and, not so early, in Africa also. The first centre of domestication of the camel, on the other hand, cannot be traced with any accuracy because there is one untamed species, the Arabian dromedary, in the Middle East, and a Bactrian two humped camel found in Central Asia, both being used in early prehistoric times. There are remains of the domesticated dromedary, dating from 4000/3000 B.C., found from Egypt in the West to India in the East; but this breed seems to have vanished in Egypt again, and perhaps even in Arabia, to be reintroduced near 1100 B.C. or so. The taming of the llama, the alpaca, and guinea-pig in Peru is not too well explored and may be rather late. The rabbit was tamed even later, perhaps in the Spain of the Iron Age.

The ox originates from the urus (wild ox), yak, Indian bull, buffalo, zebu, and other related wild taurids in Europe and East Africa, West and Central Asia which were tamed in different periods. The idea seems to have come from Central Asia. This can be explained from ethnological sources. If they can be trusted, the yak of Tibet is the earliest domesticated cattle. On the other hand, the earliest finds of tamed cattle from excavations come from the Mesolithic Age of the Northern sphere of Europe, of contemporary Egypt and, slightly later, from the whole Mediterranean Region, the Near East, and Northern India. It should be noted that the sites of these earliest remains lie in a half circle around Central Asia, but that the origin of cattle breeding in this region is nevertheless not yet established with any certainty.

The sheep is descended from a Trans-Caspian wild race which appeared domesticated very early in Europe, but apparently only in the North and then sparsely. This original species was interbred with the Mediterranean wild mufﬂon and an African species at least after 5000 B.C.; but it is not excluded that the mufﬂon and the African ancestor of the Egyptian sheep were domesticated independently in the Mediterranean or in the Caucasus, and in the Nile Valley or the Sahara. The breed was improved and differentiated in the Bronze Age of Europe and the Near East, as in that of India and China.

The taming of the goat succeeded near or in the Altai with that of the so-called Markhur, and spread from there under the influence of various wild races, especially

the Bezoar breed of the Near East, and perhaps a European prehistoric species, until it had taken in the whole of the Ancient World. The goat and the sheep occur as early as before 6135 ± 1400 B.C., and are common in Neolithic Europe. Goats appear, and to some extent in greater numbers than the sheep, in the Neolithic Age of Malta in 4000/3000 B.C., in Central Germany, in Switzerland, and more often in the Neolithic Age proper of Europe, North Africa, and Egypt. The earliest remains of domesticated sheep and goats come from the Belt Cave on the southern shores of the Caspian Sea, as radio carbon research has proved, of goats from Jericho.

The pig is considered by modern research to have been domesticated in Europe at the beginnings of the Neolithic Age in Sweden and Italy independently from two different kinds of wild pig. Many other localized attempts at domestication are witnessed from North Africa to East Asia or very probably so. Two main varieties, the wild boar of Europe and Asia, and a species in Southern East Asia found preference in this process. From whence and whether the idea of taming this animal originated has not yet been clearly established, although in this case, too, Central Asia is at least worthy of careful consideration.

In Egypt cats, ducks, and swarms of geese were tamed from wild varieties of the Nile country relatively late. The domestic fowl is not met very early with in archaeological finds, but first in the Indus civilization between 3000 and 2000 B.C., and then advanced slowly in historical time further to the East and West. Since the Iron Age, if not earlier, there was no part of Europe where this domestic bird was completely unknown. The probable first centre of domestication may have been contemporary and identical with that of the peacock, and may be found in Indo-China or farther to the East in the Malayan and Indonesian Archipelago. The dove, on the other hand, appears much earlier. We find this bird in Mesopotamia after 6000 B.C.. This breed originated probably from one wild species; and early spread through Asia, Europe, and Africa.

The turkey was domesticated in Mexico, before the Europeans arrived, at which approximate date, is not yet established. The breeding of hawks and falcons, of pheasants and other rare birds, and of different species of carps and other fish does not lead back, in our finds, to Neolithic times, except perhaps in East Asia. The town civilizations of the Bronze Age from China to Egypt, and probably the European Bronze Age also, are mainly responsible for these luxury experiments. Domestic use of bacterias and other diminutive living beings was unconsciously made, from very early times, for the preparing of cheese, alcoholic beverages, and other food and medicines. Only the Machine Age has succeeded to domesticate such beings exactly like cattle, e.g. in the case of the penicillin mould.

Among the insects, the bee is the only one to be mentioned whose connection with early agrarian industry concerns both the Old World and the Americas in many kinds of local species. There was no general attempt at domestication in our findings. Yet the conscious care of bees may have reached back as far as the domestication of the dog, if not farther, i.e. to the Late Palaeolithic Age. The culture of silk-worms began with the

collecting of the product of wild species for clothing purposes and succeeded near 3000 B.C. in China, but spread only slowly into the rest of the Ancient World. Europe was reached in the Byzantine Period.

Concerning the psychological aspect of the domestication of animals, we have an interesting, sharply contested hypothesis by Professor Eduard Hahn which more recently has been synthesized and substantiated by Professor Frobenius in a form that is worthy of note. That the hunter from the beginning would have known how useful the taming of animals might be is completely excluded, except where he already had the analogy of tamed animals of a similar species to guide him. Therefore domestication must have originated from other considerations. It is conceivable that young animals were tamed to play with them. In the case of modern primitives, all kinds of animals are kept in the camp for decoration and ornament—so-called pets—, occasionally with religious significance.

On the other hand, such customs do not seem, as a general rule, to have led to domestication. Under these circumstances, Professor Eduard Hahn has framed the following much-discussed hypothesis, that the taming of animals arose originally as a consequence of religious rites, and from that technical magic which played, as we have already seen, so great a role in the Palaeolithic Age. Wild cattle and, with them, goats, sheep, pigs, horses, donkeys, reindeer, camels, bears, cats, and birds, man had to sacrifice, as his tribal customs required, to divinities at fixed dates according to this hypothesis. Considering the uncertainty of a catch immediately before the date of a sacrifice, the hunters preferred to capture such animals alive some time in advance and to let them graze in enclosures of the vicinity, so that they could not escape again, until the day of sacrifice had arrived. From the setting out on such religious hunts to the final sacrifices, religious play-acting according to the different tribal customs must be supposed and these games affected the animals more than the mere hunting and holding in captivity. Man became, at the same time, accustomed to the neighbourhood and special characteristics of all these animals, and the animals to the influence and nearness of man, so that domestication gradually developed and took the place of ritual hunting.

In the Palaeolithic Age, the earliest sacrificial animal for the rite leading to domestication was probably the bear, who is certainly easy to tame, but does not reproduce itself sufficiently in captivity. In a still later day, in many hunter tribes of Northern Eurasia, we find sacrificial and ritual treatment of the bear in the form of imprisoning young animals, which at first are tended and taught games before they are used in the end as sacrificial animals. According to the hypothesis formulated by Professor Hahn, other animals finally replaced the bear, and cattle played a predominant role in this domestication. Man tried originally to use oxen and cows for drawing the images of the gods in procession on carts through the country before their actual sacrifice, and so the cart was invented and improved. Gradually, in the period following this religious use of the cart, it was turned to secular purposes, and other animals were used also. By this, the life of animals was gradually co-ordinated with the life of man and his

economic sphere. However that might be, it cannot be doubted that an irrational factor appears both in the tending of plants and the domestication of animals. Even the opponents of Professor Hahn do not dispute its existence. Here we have before us a historical phenomenon which appears similarly again and again in the course of human history. Desire to conquer the dangerous economic forces surrounding man manifests itself in the spiritual sphere a long time before it actually succeeds. The magical practices of the Palaeolithic Age show us our ancestors' wishes for economic certainty and stability. At the beginnings of the Mesolithic and Neolithic Ages this desire achieved its first and greatest break-through to success. The whole of the social life of mankind was revolutionized with the help of planting and animal breeding. According to modern religious research, a most intense and superstitious veneration for gods, in the planter, animal breeder, and plough civilizations, took the place of the much clearer and simpler religious forms of the hunter and foodgatherer ages. In fact that the Palaeolithic religious desire had led to tremendous economic success, explains convincingly the mass psychology of the subsequent societies.

If we consider the cattle farming purely from an economic standpoint, it appears to be a still more powerful social force than the hoe culture. It is no mere chance that so many words which are used in later money economies, as for example our own word capital, or the terms for interest in very many languages, have developed directly by analogy with the tending of cattle. The animal, even after it had been tamed, lived with a herd; and although its original way of life had been affected by man's subjection, it needed only a limited expenditure of labour to tend it. Its fodder it found for itself without any help and only in exceptional cases did it need any other assistance. But the animal, in exchange, provided its master with most of his material needs. The new economic advance created, without much human labour, for man a true capital rent which did not end for a long time. It took the form of food (meat, milk, eggs, honey), clothing (wool, silk, fur, leather), tool materials (horn, bones, feathers), or additional labour force to help with his work, from beasts for transport or riding, to war or hunting. So that cattle capital is the first true working capital among the possessions belonging to tribes or individuals. It did not only provide provisions and tools, as most plantations do, for one or a few harvests, but was lasting and could increase, unless climate and health conditions were unfavourable. The herd can increase the wealth of their owner continuously, and under favourable conditions, in the steppeland, can grow to an almost unbelievable size. In consequence, the difference in the possessions of the animal breeding families of the same tribe caused social differences which were unusual in the planter-hunter and the specialized planter civilizations.

Wealth, therefore, was judged among the animal breeders, according to convention, as of utmost importance for society. The number of his heads of cattle and the species of his animals were usually the foremost factors in deciding the social position of an individual, or a clan. It is remarkable that nomads, up to our own day, kill the animals which have grown to maturity under their care with great unwillingness. so as not to

destroy the status and the income gained by the possession of the herd. They have already developed an almost pure “Bedarfsdeckungswirtschaft” from an instinct to preserve their property, and this has only changed its object in modern times into the urge to save. Usually only the young cattle was used for the daily meals and then only to a moderate extent. If the herd were larger, the number of herdsmen had to be increased, and so all the tribes endeavoured to increase their birthrate. For children, and particularly sons, were of the greatest value. Man was by far superior, as a herdsman, to woman as a foodgatherer at this stage or during the mixed herdsmen and planter cultures that followed somewhat later. In a nomadic life, woman very often became and becomes a subject being or at least takes second place in the patriarchal¹⁴⁾ social system. Polygamy was considered inevitable because of the increasing demand for men to look after the increasing herds. To have many wives and enormous families was therefore of the highest economic value.

The former irregular life of the hunter-foodgatherer civilizations was preserved by the cattle breeders in its external forms only. Man cannot exist without eating vegetables of some kind and the herdsmen were able to get them by a system of barter, by robbery, or as the usual tribute from foodgatherers, planters, or agricultural communities which used the plough. Often herdsmen went to bring women from the planter tribes into their harems and enslaved them to care for their own gardens. Man was compelled by the herds under his protection to take up a completely new and different form of social and economic life. Many prisoners were made and used as slaves. They had to look after some part of the herd and became a subject class apart in the social order.

Epidemics which man was quite powerless to prevent, and other natural or political blows of fate could weaken the herds with startling suddenness, and may even today make tribes who were formerly rich, so poor that they almost starve to death. Under these circumstances differences in wealth and possessions and consequently a rigid class distinction grew up, even within the free upper class of these nomad civilizations, and the more warlike they became, the more intensively it developed. The animal breeding society became in fact, to use a phrase of the late Professor Franz Oppenheimer, almost a “wandering state” with a threefold social caste system of the nobles and rich owners, the poor and free mass of the people, and the despised slave class. This was the structure to which the shepherd races had developed, and now travelled through their enormous pasture lands proudly and arrogantly.

ECONOMICS OF PLANTER, BREEDER AND PEASANT CULTURE

If we attempt to understand the pattern of the last townless period from our surveys of economic facts, we shall see quickly that we have here the ideal type of Professor Spiethoff’s domestic economy before us, no other period of actual human history being closer to this pattern of economic theory. The ethical aims of Neolithic peasant economy

were universal and directed to promote the welfare of larger or smaller family groups. Compulsion and fear of punishment in the case of subject races and social groups; needs for food, delight in work, and desire for honour, in the case of the ruling classes, were the main spiritual incentives of man to act economically. After the revolutionary period of economic advance to a higher level of civilization, at the end of the Palaeolithic hunter and foodgatherer stage, techniques and customs were based on traditional experiences and changed very slowly. The density of population was very much greater than in the Palaeolithic Age. But the number of inhabitants of the new agricultural territories remained almost stationary, after the new levels had been reached, or moved only very slowly upwards.

There was already a form of division of labour according to technical methods, but mental and physical efforts were not yet divided up between different individuals. The technique was adapted to the organic rhythm of human life and Nature. The collaborating social units were small, united by relationship of blood or by compulsion. In spite of trends which usually led to the formation of upper classes, the Neolithic peasant cultures were essentially democratic. There was a professional division of labour; but it was still weak, unstable, and remained connected with agriculture as the main profession. Peasant economy is planned. The soil and other means of production were usually the property of a tribe or, by special grant, of a noble family father. Articles of consumption, occasionally the lesser means of production, could be individual property. The production of goods was practically limited to satisfy the needs of the small producing communities, and was organized from one centre. The exchange of goods, by trade and warfare, moved only a very small part of the general production. The customs which pertained to the distribution of the social product tended to a general and traditional reward. Labour was usually considered to be a communal and collective occupation, and the course of economy was steady, with the exception of natural catastrophes. Never in the later city civilizations was there such a well balanced and nevertheless effective economic pattern in every sphere of social life, as we find in prehistoric peasant cultures. It reached over the whole world; and we must recognize these archaic and early economic forms as satisfying, even compared with the more efficient recent forms of society, and as responsible for the formation of the city cultures of today. We have progressed so far, that we are now able to realize how creative have been the economic effects of the prehistoric planter, animal breeder, and plough cultures, in their production of new forms of society. But at the same time, we begin to understand the actual achievements of all the town cultures which surpassed the level of the peasant stage. Again we shall have to ask us here, which were the economic causes and new forms of life which enabled mankind to overcome the stage of domestic economy.

ECONOMICS OF MIXED FARM CULTURE

There followed an economic development of combined animal breeding and planter-hoe cultures according to the record of excavations and ethnological information. The

new cultures spread through the world like streams following the opportunities which Nature offered. Many different planter civilizations settled in regions suitable for the plants, tools, and working methods that they used. Similarly a great number of different nomadic cultures sprung up in steppe regions. Pig farming, the rearing of horned cattle, and of riding animals began in the Eurasian steppe lands. Planter settlements grew plants by various methods in the Near East, in the Balkans, down the length of the Nile from Nubia to its mouth, on the coasts of the Aegean Sea, in Syria, and North Africa, where fishing and pirate communities settled too, in the regions of the Danube and in Western Europe. In the Northern Circle as in the Mediterranean, fishers and groups of pirates formed enclaves in the planters' civilizations of the North Sea and the Baltic. Similarly we can prove the existence of hoe culture in India and East Asia during the Mesolithic and Neolithic Ages.

Between animal breeders and hoe cultures there was an almost necessary interconnection that sometimes took a peaceful, but more often a more warlike turn. In the Northern parts of Europe only, where steppe, heather, wooded and grass grown areas are still close to each other, hunter and fisher tribes found a peaceful symbiosis with cattle breeders and hoe culture planters, and finally all arrived at a form of community life which gradually moulded widely different groups of people into secondary tribes. At least we have here very few traces of any considerable early belligerent subjection. Those elements of the population who had dwelled in Young Palaeolithic North Europe already were apparently assimilated peacefully by the "Euroid" tribes that came from the East so that conformity and unity prevailed.

But the greater part of the rest of the world was less fortunate. The gigantic Eurasian steppes became particularly important with the beginning use of the horse in these regions. Warlike horsebreeders have left traces of their conquests over this whole territory and to the South, East, and West. The warrior nations of these vast stretches of land have been a menace to their neighbours from the end of the Ice Ages to the present day. There were nomadic movements from similar centres in Arabia and even from the Sahara desert, which were at this time gradually drying up. We can see from archaeological evidence that such cultural waves spread over the borders of North and East Africa, the Near and the Far East, the Indian Peninsula, Polynesia, Australia, and America at different times. In consequence of this development many different mixed civilizations were created. Most of the new arrivals, who formed an upper class, came from the steppelands of the nomads where animals were reared or from the homes of seafaring nations. It was not by chance that the Proto-Indogermanic and Indo-Germanic migrations to the South of Europe and into Asia, and the similar migrations of the Semito-Hamitic nations, which between them conquered the larger part of the Old and finally the New Worlds, consisted of such secondary mixed groups which had from necessity to return to nomadic life occasionally again. In fact, we do not need to probe any further to discover that primary, secondary nomads, and most fisher tribes, seem to have been inclined to almost continuous political expansion in particular. While the

planter and the agriculturalist had land at their disposal for additional cultivation during our period, and so were not compelled to make attacks on territory owned by other tribes, the shepherd tribes were compelled to conquer mighty stretches of land for pasture, whenever their herds increased in size. Similarly if the seafaring communities wished to keep to their own way of life, they had to encroach on further stretches of coast as their population increased. Accordingly the symbiosis between planter and animal breeder developed very early over the whole stretch of the Eurasian steppes, whether there was war or peace between the two groups. This brought in its train a very important economic change, an increase of agricultural efficiency made possible only by new economic forms, a stabilization in pattern of agricultural civilization which has lasted up to the present day. Animal and plant capital of every kind were combined in their economic effects by the new mixed animal breeding and planting civilizations which were finally forming. In future the men of a tribe worked at their former tasks, tending the cattle. But simultaneously they intensified and revolutionized this work by tending plants also,—no longer considering this a task for women alone. The so-called “food-producer” revolution began.

Soon an appreciable diminution of the natural risks in agriculture was noticeable, when, in the same region, as a result of the mixture of civilizations and tribes, plantations, pastures, hunting, and fishing spread and mingled in such a way that exchange of products was easy and regular. This production was, as a rule, sufficient to supply the needs of all members of the societies involved, and to make good occasional failures of the one or other economic sector. Already very full use of the possibilities which Nature offered could be made, by means of specialization and division of labour according to the economic practice of the day, so that gradually, by peaceful and warlike means, several of the different branches of primary production were drawn more and more into the hands of one owner or one family.

These tendencies, further developed, led to a mixed economy in many districts, even although planting and animal breeding were already many-sided enough. In Europe, the pollen analysis shows that such a system of mixed animal breeding and hoe culture began at least as early as 5000 B.C., in the North and South. In Egypt, the Near East, and Turkestan, the chronology of archaeological levels points perhaps to slightly earlier times, while the development of the Far East seems to have been much later than that of the European regions. Among the new improvements of the millennia between 6500 and 3600 B.C. were better fertilization by means of animal manure, a planning organization of traditional field-grass economy by the tribe, extensive canals and drainage in suitable countries. Animals were used, too, to increase the yield and to refine the products of the plantations. Tools were made from better materials, and improved by technical inventions. Farming methods were advanced far beyond the range of the exclusive hoe culture period. In the earlier stages of the Ancient Orient they occasionally revealed themselves even as the suitable foundations of fully developed city civilizations. In the later stage of the combined animal breeder-planter civilizations the different

grain varieties achieved the importance for human food, which they have always held since. Permanent villages sprang up in consequence in suitable climates. The earliest among them in the Middle East are Jericho before 6200 B.C., Badari, and Iarmo in Iraq going back to 4756 B.C. according to radiocarbon dating. The Egyptian Neolithicum A followed Badarian between 4249 and 3839 B.C. Tools which man used then were the digging spade with a rest for the foot, the so-called cashrom, and, in addition, the hoe and the drawn spade. The plough must have been a refinement of one of these forms, although the actual details of its origin have been much under dispute. At any rate, the plough was used over much of the inhabited world from the early Neolithic Age onwards.

This new symbiosis of planting and animal breeding led finally to the building up of a further branch of man's primary production, mining. This was a completely new and extraordinarily vital development. Here the food surpluses of the planter, and on the other hand the animal labour of the cattle breeder were continually needed. The use of animal strength that began in these fields of economic activity in the Neolithic Age was only replaced by that of machines in the 19th century. Mining, improving on the experience gained from the Palaeolithic pits, was prospecting for flint and soon for many metals that were already known to the so-called Cuprolithic Age. It was unthinkable that mining could be carried on without draught and baggage animals, any more than without stores of plant products for men and animals. This new branch of primary production was not too important at first and only gradually advanced in its economic pattern when the town civilizations of the Bronze and Iron Ages needed its products urgently and developed it to its present significance. For mining could only become fully efficient through the quick and wide movements of its products, in caravan and naval trade, so characteristic for the Ancient Orient and for China. But it should be noted that its beginnings can be traced, noticeably earlier, back to the mixed agriculture civilizations of the sixth millennium B.C.

Still more important for the possibilities of agricultural production in the oikumene was, that between 5000 and 4000 B.C. as far as we can tell, man began to invent new tools to put animal strength at the service of plant cultivation; and from this first arose a completely new method of agriculture and with it the most important step towards present day agrarian methods of production. Plough cultivation or agriculture in its narrower sense was now fully discovered. It differed from planting and hoe culture in that large areas were planted with one crop, usually grain. Many of the new soils could not be easily used for hoe culture because the earth would be too difficult to work. The new plough land could be sown and harvested, as a unit, with the aid of improved tools and greater power at man's disposal. So that by means of extensive and rationalized economy this type of agriculture produced more paying results than mere planting, and more intensive crops have not been obtained even with the most modern machinery in quite a number of regions. The plough is the most important tool in agriculture, as the hoe is in horticulture. But the cart and the sleigh play almost as important a part. Many scholars think that the progress made here was again connected with magic, as is probable

for the beginnings of hoe culture or the domestication of animals. According to this attractive working hypothesis, originally, a magician blessed the seed when it was sown with the aid of a phallic instrument to encourage the fertility of Mother Earth, and used holy cattle or horses to draw his chariot to the field where the ceremony was to take place. So that possibly this custom suggested, to an inventive mind, a source of increased power for these tools, with the aid of animal strength. At any rate, the use of the plough and the other new tools had an early economic effect, and they were quickly employed over the whole of the Ancient World, but entered rarely the Americas, perhaps from Polynesia. Ergologically, there was an earliest form of the plough, and two very old types developed from it. The latter were the Krumel plough and the four-sided plough. As far as we can discover from several archaeological finds it is practically certain that the communities of Northern Europe and of the Near East had a wooden plough drawn by animals, as early as the Neolithic Age. A plough of this type was found at Georgsfeld, a Krumel plough, and therefore certainly not the oldest kind of plough ever known. This monument of early agriculture is judged from pollen analysis to have been in use near between 3500 and 2500 B.C. Ploughs appeared in Mesopotamia as early as between 4000 and 3000 B.C. Yet in the Indus civilization of between 3000 and 2000 B.C. they seem to have been completely unknown. The predynastic Pharaoh "Scorpion" of the period near 3000 B.C. is represented in a relief as cultivating the fields of Egypt with the sole use of a hoe and a seed basket. This seems to indicate that either the plough was still not known in Egypt at the end of the Predynastic period, or that it had been introduced shortly before, and was considered as far too "modern" to be used for sacred rites of the state.

Mesopotamia, however, already had a plough which was much further developed, and its plough culture seems to have been an early one. Its territory, too, lay nearer to the original centre of plough culture within the Ancient World than Egypt. The cart, sleigh, and potter's wheel may be closely connected with the invention of the plough according to our ethnological and ergological sources. Their use spread gradually from Central Siberia to Europe and to the Near East. The origin of the cart form, the sleigh, and the potter's wheel can be traced back to Anau in Turkestan, an important historical pointer. From there these utensils spread, occasionally at different stages of their development, to the East, South, and West of the oikumene up to the early Iron Age, very much as many cultivated plants and domesticated animals had done many generations earlier. The cart and the potter's wheel reached Mesopotamia near 3000 or even 4000 B.C., but were only used in Egypt and India at a later date. I think there is at least a probability strong enough to ascribe the discovery of the oldest three forms of plough used and of the three related tools to an old Turanian and Southwest Siberian centre of development. It was from here that numerous agrarian plants, domesticated animals, tools, and working methods actually originated, among them the principles of the yoke and of using wheels for human needs which are so essential for the development of the plough and cart forms. Possibly future excavations will bring to light definite

remains of the most primeval ploughs and wheeled vehicles from this completely neglected stretch of earth, where strategical reasons prevent any real progress in archaeological research. A Southwest Siberian centre of invention would explain the fact that people in Northern and Central Europe in the Neolithic Age used ploughs, carts, and sleighs at the same time as the nations of Mesopotamia, or perhaps even earlier, as excavations have revealed. These European ploughs were of the best types known at that time and even simple models of the wooden potter's wheel were known in Northern Europe at the same period. This progress seems to have taken place without the North influencing the South or the South influencing the North. It would seem the simplest explanation of the facts at our disposal, if a third region in Northwest Asia had influenced both territories independently.

With the invention of the plough village civilization in its three branches, plough cultivation, tree and horticulture, and animal breeding was finally ready in its principles, and man had only to experiment with the essentials of these forms of agriculture to adapt them to the potentialities of the different regions. In the following c. 6000 years agriculture was intensified as far as possible under the changing influence of city civilizations; but there was no structural change possible in the number of basic elements. Mankind has only been able to add to the six basic sectors of primary production which existed from the Neolithic Age onwards, that is plough culture, hoe culture, animal breeding, hunting, fishing, and mining, three more, the exploitation of water, of air power, and of chemistry. We were able to make use of water and air power, at first, during the time of the Ancient Oriental, Graeco-Roman, and Islamic town civilizations by means of ingenious sailing ships, water and windmills, and in modern times through all kinds of power-driven machines, chemical processes, and materials for all the three more recent spheres of primary production. Otherwise the Neolithic Age had completed the primary foundations of present day economy.

For the first time in human history there was an extensive surplus of production above the more primitive and essential daily needs to be obtained from the intensive investment of human labour, in tending gardens, fields, and cattle simultaneously in almost every region of the earth. This new economic potential had to be used and distributed, and its distribution gave a powerful impetus to man's activities which had been at too primitive a level during the Palaeolithic Age. Mankind's outlook on life broadened and took definite shape, a shape which has lasted on to our modern times, according to the same structural pattern, with no very considerable changes. Various professions were differentiated for the first time, and more or less removed from primary production, crafts, art, trade, religious and political hierarchy, war and privateering by sea and on land. The men and women in such secondary jobs and production remained connected, in the mixed agricultural stage, with some branch of primary production, outside their normal occupations, until town civilizations arose. Nevertheless, a sharply defined differentiation of the various professions and stations followed; and by it they were already valued differently, so that the way was open for the formation of positive

castes politically and socially. In the course of this development a characteristic social pattern was formed. It is found in all townless civilizations and has preserved a social and economic importance in world history up to the early capitalistic era of modern times.

This differentiation may be ideally considered to have developed from the social order at the height of the period of plough cultivation and its immediately preceding stages. The highest stratum in peasant states and societies has manifold charismatic offices and dignities like those of kings and headmen. Kingship and related positions are often restricted to individual clans and families.

Then follow, in the social order, clans and families of priests and similar servants of the gods, who hold religious sway and are closely connected with families, small tribes, and professional groups who are considered to be imbued with magical properties. The third highest group in the social scale consists of the nobles. When plough cultivation was thoroughly developed, the nobles were the animal farmers with large herds, among them horses, elephants, or camels.

The nobles represented the inner circle of society and shared their position with the most successful warriors, pirates, hunters, fishers, and slave owners. In planting, and nomadic culture too, there was still only a nobility, which rose and declined quickly in consequence of victories in time of war, through changes in the domestic organization of the tribe, or through the difference in the wealth of their possessions.

But the main group of all cattle breeder, hoe, and plough culture societies consisted of the numerous clans, who owned herds of a moderate size, tilled plots of a moderate extent, and nevertheless were usually the strongest political force in the community.

The remaining free people of the region came next. They did neither own land nor larger animals, and as a rule were, or were considered to be, foreigners. They found their living in crafts, trading, and free farm labour, and were practically at the mercy of the four upper classes, who controlled primary production.

The only exception was if they were believed to be protected by religious influences or needed in a political situation.

These social groups of landless free people fluctuated often over wide stretches of land and among many nations. It was this social class, who introduced to the small particularized peasant regions of townless society many cultural and economic innovations from neighbouring areas, using their experience by passing on goods produced by other nations to those who required them.

The lowest stratum in the rural social scale consisted of the slave class. Its members were either personal servants or serfs. Both were compelled to place their labour potential completely, or to an overwhelming extent, at the disposal of other people without reward or hope of escape. The serfs belonged usually to alien groups of people who had been conquered, or they were originally free people of their own tribe who had fallen into a dependent position. All were now bound to the soil, compelled to serve the upper classes and the clans above them as hereditary communities. The slaves were usually individual

booty from war, piracy, and robbery. Women particularly, and experts in trades and crafts who could not really be exploited as serfs were pressed into these despised classes.

With the differentiation of professions began a stable class distinction, based on the increased production potential of the new societies. It was already capable of great achievements, and it has existed, practically unchanged, up to the beginnings of the capitalist and coinage age of Greece and Rome. Where a war destroyed it, or disturbed the balance, this social order necessarily reformed immediately on the former basis. Only Greece and Rome, Islam, Russia, and Western Civilization in the Middle Ages and in modern times have produced new social scale systems west of India and China that might better sustain heightened economic possibilities of the pattern of town communities. Even here, there followed a wide spread decline in the economic structure and intensity at times like the early Iron Age migrations, the Dark Ages of Europe, certain centuries of the Byzantine period, in the Islamic and Russian civilizations after the different Mongolian and Turkish attacks, in Abyssinia, Armenia, Georgia, or similarly as a consequence of revolutions and world wars in certain countries of our own 20th century. The ancient form of the class order of the peasant civilizations reappeared more or less widespread and lasting in such periods of decline of town civilizations. The economic crisis of the 20th century has already resulted in attempts to create a new nobility consisting of political, military, and scientific experts, a new priesthood of the state and its servants, a new dictatorial monarchy, and a new class of serfs and slaves, the permanence of which in world history can not yet be fully known. Everything considered, the social scale of the peasant cultures, much as it has been romanticized as a symbol of eternity, had certainly not such an enormous expectation of life, nor was it as indestructible as the simpler social systems of the Palaeolithic hunter and foodgatherer Ages had been. A better system followed and replaced it in the town civilizations until modern times. This system was able to guarantee an advance towards the higher development of mankind with less compulsion and brutality. But for the time being it has succeeded unfortunately only during periods, relatively short in world history, and over comparatively small areas of our globe.

The relative position of the sexes took on a somewhat more stabilized form from the Neolithic Age onwards. After the initial vacillations in its balance, the final position remained unchanged up to modern times in peasant regions. Animal breeding, hunting, and fishing had always been man's domain, as we have seen. And with the climax of townless economy in the Neolithic Age, with plough cultivation, mining, and the widespread use of water power for irrigation, the general use of cattle being something essential everywhere, the masculine sphere of influence increased considerably. Man had control of animal breeding, of the most promising part of plant cultivation, the spheres of mining, and of water economy because great physical strength was required in such activities. Woman's sphere, on the other hand, remained the garden alone from primary production, by tradition. For the rest she was pressed into secondary production.

A part of domestic industry and craftsanship, particularly pottery, textile, embroidery, and cooking, remained her province for a long time to come. She also took some part in religious practices, medicine, and art. In this way, even where the social balance of both sexes wavered, as it sometimes did, woman was economically dependent on man for about 7500 years up to the 19th century. Consequently she was excluded from the sphere of government, as a rule, and even sometimes from that of masculine society, and was compelled to assume at least an outward passivity outside of the home. This position is, in fact, only beginning to disappear now under the impetus of the Age of Machines.

The form of class distinction which we have outlined as already set in the period of plough cultivation, and even on general lines in the hoe culture and animal breeding ages, was so well organized that on that basis we must suppose that regular states may have been formed from the Mesolithic period onwards in certain regions. For if we consider the definition of the constitutional schools of Professors Jellineck and Giercke for a state as justified, groups of men were now to be found everywhere in areas, which were well defined.

These groups were organized according to certain ideologies and were sovereign in their rule. These states guaranteed the order of law with the help of the power and authority of their magistrates. Their order may have been erected either by conquest and the subsequent formation of a social upper stratum, whose members exploited those they had conquered by force; or it was formed by making use of magical charismata. The different participants in the ruling classes of the state divided among themselves the goods which they acquired from warfare, piracy, tributes, communal ownership, or sacrifices to certain state gods.

A legal order, after the changeover of mankind to more permanent settlements, regulated the main economic processes of both village and town civilizations. We can judge as much at least from what we know about the early Sumerians, and of Proto-Indogermanic, Indogermanic, and Hamito-Semitic peoples before the division of these tribes into nations as they were later on. We are able to supplement and interpret the legal remains of all these nations which date back beyond the beginnings of the Bronze and Iron Age town civilizations and to the townless stages of the Neolithic, Bronze, and Iron Ages, by ethnological comparisons with the primitive tribes of today, where detailed facts are at our disposal. Dominion over the soil was the focal point of all such archaic legal government. The details of administration were decided according to locality, popular traditions, and historical development. But yet there were basic trends in the laws of all townless civilizations which conformed with the pattern of domestic economy. There were practically always feudal elements, for the advantage of the three highest orders of society, the princes, priests, and nobles, implied in the legal arrangements of such civilizations. The clans of the three highest orders usually had the right to administrate large areas which were ceded by the whole tribe and were tended by serfs. In exchange for these concessions, revocable in theory, they had to give presents and

special services to the community. With the exception of these large estates, the agricultural soil was divided up carefully and as equally as possible, among the families of free small holders, originally for one or several years, thereafter permanently. This division formed the normal basis for the land laws of most townless civilizations.

The most important means of peasant production, the agricultural land, could be, in some town civilizations of the Ancient World, in completely free individual possession. But such legal possibilities did practically never exist in early prehistoric communities, as far as we are able to judge. Either the tribe or the clan had the supervision or some final property rights over all small farmsteads. These were, nevertheless, worked by individual small farmers who, in addition, leased much of the larger estates of the nobles and temples, as far as serfs could not be used on them. Similarly some cattle and most pastures usually remained the possession of the tribal community. Otherwise, these belonged to a family, but practically never to individuals. Mines, salt and mineral workings, fenlands, wells, fishing streams, and hunting grounds also remained, in practically every case, communal property which could be divided only for temporary uses. The balance, which had prevailed during the Palaeolithic hunter and foodgatherer Ages between individual and collective ownership, now swayed in a remarkable manner towards the collectivist side and to planning for groups of people. But this effect was only apparent in the case of very small political and social units, so that it is, in my opinion, preferable to speak of a “communard” and not of a communist primitive peasant economy in such cases. Later the individualistic Mediterranean Cultures of the first two millennia of the Iron Age from Persia and Israel to Greece, Rome, and Byzantium, and our own Western Civilization too, were able to bring new power to the individual in the course of their spiritual, social, and economic refinement. But the balance between individualism and collectivism has rarely remained undisturbed in the world historic development up to the present day.

So the appearance of Neolithic plough culture some centuries before 4000 B.C. means the end of a total revolution in the primary production of the human race. Such a revolution has never occurred before nor since, and its consequences direct our fate today still. It may be by chance, but it is somewhat surprising, that the change-over from light hoe cultivation and foodgathering to mixed agriculture in Palestine actually dates, according to archaeology, from the period for which the chronology of the Bible assumes the expulsion of the first men from Paradise.

The economic foundations of human society were modified and reconstructed, by using means of primary production which were only now attained in the hierarchy of inventing working economic possessions. In consequence there succeeded a highly intensive organization of human labour and of the forces of Nature exploited by it. The foundations for the primary production of six or so millennia until modern times and for the outlines of professional differentiation even in secondary production were now laid. Here originated a new balance between the sexes, and with it the importance of clan and family as economic factors was stabilized, which is so characteristic for all

agricultural and most city cultures. Just as the social and political class distinctions of this stage of civilization in its broad outlines were to form the foundation of social life for the succeeding 6500 years also. The factors leading to the formation of nations and states, as they arise, change, and decay in turn, created first the Semito-Hamites, thereafter the Indogermans.

We have now to describe the social pattern of the village and plough cultures in more detail. In general, their basic principles have been put into practice by most nations throughout vastly different rural areas in a very satisfactory and usually the best possible way. However where this did not happen, we may attribute such failure to unfavourable climate, racial inferiority, and degeneration, or to restrictions of an ideological nature, which were caused by magical and religious considerations. There are non-economic factors implied in this, which have to be explored, for all countries and nations, by theology and geography, the science of heredity, and the history of religious and spiritual development. But their influence does not really come within the sphere of this stage, except, as far as exceptional developments in remote regions helped to preserve the traces of earlier prehistoric conditions. The empirical cases of such primitive societies are too rare in history to claim our detailed consideration. They are too restricted in numbers and dwelling grounds, and form continuously decreasing and shrinking islands in the flood of human progress. For beyond the Neolithic and Chalcolithic zenith of the plough culture civilizations which developed between 4500 and 4000 B.C. there followed, remarkably soon afterwards, the first city civilizations of world history in the Ancient Orient.

APPEARANCE OF CURRENCY

A sphere of early capital development in domestic economy had already grown up after 10,000 B.C. or so, and it is important as a focal point from which all forms of capitalist economy in every town civilization have developed up to modern times. During the Neolithic Age man not only already knew how to use his labour power to have means of production at his disposal at a later time. For this purpose he domesticated, tended, and used plants and animals, investing most of his labour potential into them. His surplus products he often also preserved, when he did not use them completely for his daily needs. But he also found further and more powerful means, to implement the labour potential at his disposal. He increased human labour for his personal needs by hiring servants against payment with surplus products or goods which he could procure with the help of barter. Furthermore, he took over slaves and dependent serfs by conquest in battle. We may even say that violence was the most useful means at this stage of acquiring slaves, animals, pasture, and arable land. Yet robbery, tribute to buy off violence, and barter, were not the only means society provided to increase the possessions of an individual, or those belonging to the clan as a whole. Pure barter entailed a long search until the trader found goods of which his partner could make use. Consequently, society very soon began, perhaps as early as in the Palaeolithic stage, to give preference

to certain wares, which were desirable to all from religious or economical reasons. These are the beginnings of money, that is, according to the definition of "Classical" Economics, of means of valuation and exchange which were generally recognized as standards for certain fields of economic activity.

We can use the cases of modern primitive tribes as examples here. The exchange of certain goods inside the same community follows a ritual, according to a traditional recognition of the value of most goods. Similarly wealthy owners hire the services of free men and women in a conventionally fixed manner by means of a meal of well defined valuation, or a conventional gift for special services. This custom has survived in the payment of agricultural labourers and wandering craftsmen in kind among the peasants and land owners of the Romance and Germanic nations, and has survived among Slave nations outside Russia, until recent times, as a system which was not originally adapted to a money economy. Similar concepts of payment and of reward were already appearing, if we follow the evidence of language archaeology, among early Indogermanic, Semito-Hamitic, and Ural-Altaic tribes, and are, in this way, made certain for the millennia between 5000 and 2000 B.C. In simpler forms, we are able to find this barter with preferential means of exchange even among hunter and foodgatherer communities of recent times in the ethnological material we have. We can, therefore, even conjecture that there was a similar, but more primitive system in existence, at the latest, in the Young Palaeolithic Age.

The next highest form of money economy, on the lines of our scientific system, which we use for the purposes of this book, developed when one standard of value and means of exchange found more or less exclusive preference generally. This occurs, again, already in the stages of civilization of the hunter and foodgatherer tribes of our own day. The material used as such money varies according to tribe and locality, and takes the form of plates of ostrich shells, shellfish, lumps of salt, bracelets or chains of conventional designs and material, pearls of all kinds, stones of every form, boar's tusks, elephant's tusks, or dog's teeth. It is remarkable that we have found such objects and materials among Young Palaeolithic remains, unfortunately, without being completely certain about their purposes. In any case, after the end of the Palaeolithic hunter and foodgatherer period, materials which were sought after as jewellery were used, at the same time, as money.

This custom has not completely died out in our own day, the kawri shell money of the Indian Ocean, the South Pacific, and many other regions, being unchanged a means of exchange which was already known in earliest China.

Other forms of money were used in many different regions, the so-called "cloths money", for example, pieces of textile, fur, leather, or similar materials of a definite quality and quantity. We can surmise that they were used among the original Indogermanic tribes, from etymology. In addition, such money was always known in the town civilizations of China, Egypt, and Mesopotamia, and dates back, in my opinion, to one of the millennia of prehistoric townless civilizations. In its case we find that

certain tools used by every individual, including jewel tools and jewellery, serve as money, in a fixed area. With them wages are paid, and goods are bought, sold, and valued. The character of such tool money was not constituted by the quantity or quality of the material used, but rather by its traditional form, an important economic innovation, from which later Greek coin money, the old Chinese coinage, and even modern European coin forms have developed. Tool money was characteristic for Neolithic and Bronze Age Europe in the village stage. We know of Japanese stone arrow heads, often made from semi-precious stones, which are thought to have originated in the Neolithic Age. There are fish hooks of mother-of-pearl from the South Sea and stone axes from New Guinea which were used as a money as late as the 19th century. Some scholars have even surmised with great probability that there were certain decorative stone axes from the Neolithic Age, metal rings, wheels, and axes from the Bronze Age in Europe which must be interpreted as money, because they are too flimsy to have ever been used as tools.

All the forms of money which have been mentioned so far differ from the forms of money we use today, because they could not be lent out for interest. Money lending only occurs as an exception (fish hooks) in townless civilizations as far as these types of money are concerned. The most ancient forms of money which were lent out on interest were “food money”.

If we make use of etymology and of ethnological parallels, we find that this type of money was known to the original Indogermanic, Semito-Hamitic, Ural-Altaic, and Sumerian nations, and we are led back, in this way, to the period near 5000 B.C., if not earlier. Dates, olives, figs, nuts, or seeds of grain were probably lent out, even as early as these periods, to serfs, poorer farmers, and dependents, to be sown and planted, and naturally an increased portion of the harvest had to be returned in kind. Economically, this has to be considered as a new characteristic of this money, supplementing its purpose as a means of exchange and standard for valuation.

“Animal money” should be included here also. It was used not only for definite trade transactions, or for penalties, sacrifices, and tribute, but animals could be borrowed too for a fixed time limit, the loan being repaid according to a fixed percentage from the young animals born subsequently. Fruits, seeds, and animals, were the most important forms of money used by the original Sumerian, Indogermanic, and Semito-Hamitic races, and were even known in Egypt, Mesopotamia, America, India, and China before town civilisations developed.

So that here we have the first forms of money, that man could use as a capital for investment, in the narrower sense. On the other hand, emergency credits and friendly loans were yet more common than seed and animal loans at this time and did not require interest. But even as relatively early as this, rich owners must have given out their surplus stocks regularly to poorer farmers and herdsmen, and gained interest in kind. They had to demand a higher return in view of possible losses from bad harvests or animal diseases. Loans with a rate of interest in kind were certainly well known among

the early Sumerian, Indogermanic, and Semito-Hamitic races and occur in written documents from the earliest known periods of the town civilizations of Mesopotamia, Syria, Asia Minor, and Egypt onwards. And so, in my opinion, this evidence, without us being unmethodical, can be extended to include all the more highly developed hoe, animal breeder, and plough culture civilizations in the townless Neolithic and Cuprolithic Ages of 5000 B.C. or so, if not earlier.

There was still a certain weakness inherent in this primitive form of usury, it is true. But this was soon overcome in the economic pattern of the later city civilizations. The surplus production from plants and animals could be used for the purpose of this type of interest bearing investment each year, but only, as yet, in very small quantities. Agrarian investment was, in addition, practically the only form of investment possible at this time. It was still risky enough; for the vagaries of Nature could have unpredictably disastrous results. A more general use could therefore not be contemplated at this time, that is to say, one in which there was a clearer, or more continuous economic advance, by means of increased and increasing long term investment of capital on the part of many rich owners or the whole community. So that the possibilities of this form of capital investment, even if they were exploited to the full and not restricted by social and religious taboos as they often were in fact, reached their limit fairly quickly.

Nevertheless, although seed and animal money were, as we may safely conclude, usually unimportant for the economic expansion of the townless civilizations of the Neolithic and Bronze Ages, a step was taken here towards the more efficient formation of capital in the narrower sense, and this development was to have an extraordinarily important effect on the later town civilizations. From the earlier periods of the Ancient Oriental civilizations onwards, we find that not only animals and seeds, which were measured out and used as money from prehistoric times, but even metals and other inanimate materials, which could be valued according to weight and measure, could now be loaned out, legally, against interest, after the model of the two earliest forms of usury money.

The Ancient Oriental states accepted this convention in the legal systems which they upheld. Thus for the first time a small completely capitalist economic sector, in the modern sense, was formed which provided for the transactions connected with the luxury trade and for a considerable part of agrarian and industrial investment.

Only a small upper class had been able, in townless civilizations, to assure itself of enough rents to keep up its dignity and position, from enforced labour, taxes, and serfdom. But in some regions, before and after the plough had been introduced, there grew up, with the wider use of the new money and capital forms, an additional class of private rent capitalists and state functionaries who received emoluments from communal property. The latter collected the taxes and passed capital on to merchants and craftsmen, whose jobs were now, at last, completely separated from primary production. A new movement to settle in cities began in consequence, with all its possibilities for the development of a higher form of civilization, in the Middle East after 3600 B.C. or so.

MAN AND THE FORCES OF NATURE

As we have seen, towards the end of the Palaeolithic Hunter and Foodgatherer Ages, man was able to revolutionize the use he had made of the soil so far. In the place of forest, desert, meadow, and steppe, as Nature had created them, he began to produce his own humanized landscape, according to the manner in which the agricultural and later the urban spheres of his economic systems increased in strength and efficiency. Finally these artificial landscapes included almost all the usable soil on this earth. Some meadows and steppes were left, to be used as pasture land and improved by planned burning. The flora of even such regions changed completely, as the steppe was fertilized by manure, and domesticated animals ate and destroyed certain plants. The wild animals in these territories changed even more. Their continued existence was by now completely dependent on man's indulgence. A completely new world appeared. A large proportion of the former meadow land, tropical and subtropical swamp land, a considerable area of steppe and forest changed to garden plots and ploughed fields, which were sown, manured, and tended by man according to a definite plan. This new landscape has been occasionally called "the steppe of civilization". The extent of the cultivated soil has been gradually increased in many regions up to the modern Machine Age, but not everywhere. The surprisingly numerous peasant homesteads which air photographs have revealed both in Neolithic Bulgaria and Apulia were perhaps not permanent settlements, but prepared for the continuously shifting field-grass economy of prehistoric tribes. But in Central and Southern Germany, in certain parts of Scotland, and other areas, where a very favourable climate must have been predominant in the Neolithic Age, the farmer had already pushed the forests back into the smallest possible compass, to an extent which can not always be profitably maintained in modern times. Similarly the deserts of the Sahara, Syria, Central Asia, and Arabia were small between 10000 and 5000 B.C., and still were used by cattle breeders and planters. But, as a rule, peasant and town civilizations have proved to be stronger than the natural conditions of most territories of our globe, and have victoriously overcome considerable climatic obstacles.

Forests, high mountains, deserts, lakes, swamps, jungles, rivers, and the ocean, were the only surface features of our globe which were able to keep much of their original appearance. The chaos of uncontrolled natural conditions, which man had banned from his own sphere of existence, was restricted to these areas. But even here improved hunting, tree culture, and fishing methods, irrigation, and drainage gradually had effect on the flora and fauna of mountain, desert, forest, water courses, lakes, and the sea. So that many plants and animals died out and made way, by accident or by man's design, for those which man favoured. The landscape, as geology had created it, influenced and changed in future not only everywhere the character and the planned intentions of its inhabitants. Each nation modified the landscape in which it lived according to fixed ideological, social, and economic points of view, which became a part of that nation's very being. In a certain sense such secondary humanized landscapes influenced and still influence the descendants as their ancestors intended. In addition, canals, mines,

and settlements, with their huts, temples, ditches, roads, and graves, changed the landscape more considerably than Palaeolithic man could ever have imagined in his wildest dreams of harnessing the strange and unlimited powers of the mysterious forces of Nature surrounding him.

THE HUMAN LABOUR

The Pattern of Labour

The factor capital was able to change the surface of the earth completely, even in its most archaic form. The pattern and methods of man's working practices were no less affected. Certainly an economic history does not need to trace every detail in the technical prowess of that time; but it has to give an adequate impression of the highest level of advance that the civilization of these days was able to reach by reason of its economic resources. In our period these were principally gained by investing capital in agriculture, animal breeding, and occasionally mining, and by the actual lending out, though still without interest as a rule, of inanimate and animate forms of money. Our conclusions from modern social and economic research theories, if we begin to survey the remains of the townless civilizations after the end of the Palaeolithic hunter and foodgatherer stage, may seem surprising to some readers. They are an antithesis to the traditional nineteenth century economic and historical views, which attempted to begin human history with that of the city civilizations, and so portrayed whatever existed before as unimportant, inferior, and undeveloped. Even in the townless agricultural Ages, however, the standard of culture appears to be unexpectedly high. Some consider that, because they are city dwellers, the city as they know it today, was responsible in the first place for the main characteristics in technique and spiritual development of peasant life also. Others have been taught in school that we have to thank the city civilizations of the Ancient Orient, Israel, Greece and Rome, Europe in the Middle Ages and modern times, for our many blessings exclusively. Instead we shall see from this stage that most, what is valuable in the cities, developed from the prehistoric village communities, and that in peasant culture the strong and promising basis of all town civilizations was established back to which they later often returned.

In addition the actual cultural achievements of the city civilizations can only become completely clear to us when we compare them in every point with the older townless stages. Two economic moments are important from this point of view for the development of all townless civilizations since the Mesolithic Age. The strongest assimilation to regional conditions of all possible economic elements of primary production, gardening, pasturing, agriculture, mining, hunting, and fishing, is characteristic for peasant civilization without Nature becoming a tyrant of man. There was an assimilation to Nature in secondary production also, and in the whole political and spiritual superstructure, to the conditions of climate and territory. Constant variations and differences in all the basic elements, which are otherwise identical over the whole face

of the world, were the consequence. In one region, for instance, horned cattle, so useful for many purposes, became predominant, in others, horses, camels, reindeer, pigs, sheep, or goats. Cultivation and the use of plants changed correspondingly, methods of crafts, and trade customs also. There were changes in the social and dress conventions, which were widely varied in all planter, animal breeder, and plough cultures according to our sources. Cross currents of influence spread constantly backward and forward, and there grew up a creative interchange between the social forms of life of different kinds, which, until modern times, is typical for the balanced pattern of village civilizations in their regional intercourse.

There were important consequences, on the basis of the Law of Natural Selection, for man's survival. Agricultural and city civilizations disappeared, prevented from satisfying this law by their climate political misfortune, ideological reasons, or by decadence. Their physical decay, in the final instance, was almost always caused by their ideology, that is, the ideals of their most refined periods were too widely separated from life and from their natural foundations. This is a phenomenon which gives us something to think about now, too. The adaptability of the planters, cattle breeders, and peasants first appeared in the Mesolithic and Neolithic Ages, and it has never been surpassed in any city civilization, although it is already inferior to that of the hunter and foodgatherer stages. Its basic elements have continued, in pattern, without any decisive change, but with increasingly efficient working methods and undiminished vital force, right through the townless Bronze Age, the Ancient Oriental civilizations, the townless Iron Age, the Classical Ages of Israel, Greece, Rome and Byzantium, Islam, and even through the Age of Capitalism in the West. In modern times the village has not lost its importance. For the essential elements of peasant culture still appear to be even stronger than Americanization, planning, economic rationalism, and Bolshevism, especially in times of crisis.

And right from the beginning there were in peasant civilization potentialities to supply all human ambitions simply and sufficiently, without that separation from Nature which is so apparent in city life. It is still capable of meeting those needs, and even has adapted itself well to our Age of Machines, which it may survive. The variety in human nature leads, in village life, to a widespread division of labour, which does not exclude very high honours; even though, under peasant conditions, these honours are gained only by very strong and particularly gifted personalities. Man's many-sided nature is satisfied by the combining of the numerous activities, which all began to appear as different professions in the townless civilizations. Weakness or lack of self-control, apparent in quite a number of individual members of all national communities, is deprived of its dangers for society by a certain rigidity in behaviour forms, convention, and custom of village life. The basis of these conventions has been extraordinarily well fortified by leaders of weak and cowardly or, otherwise, undisciplined and barbarous races. Such patterns may be hieratic. But spiritual freedom is not destroyed by them, except where city ideology penetrates the peasant world. There are more natural and

primitive urges and instincts in man. The need for social distinction and striving for position is satisfied by family, village, and tribal community life. Our urge for adventure, for gain, for a change, is satisfied in turn by trade, travel, robbery, and war. Our inventive spirits find an outlet in many-sided activities, crafts, for instance, and domestic pursuits. By all these means, man is able to satisfy his primitive and practical desires in life. Yet enough leisure still remains to a member of such a civilization in average climates, after he has satisfied all his material instincts. He may find his enjoyment in love as an art, dancing, ecstasies, peasant art, fairy tales, myths, drama, painting, plastics, architecture, and music. Or if he so desires, he may indulge in religious enthusiasms, the use of mathematics to measure space and time, observation of the skies and of Nature. Rural cultures already have their own intellectual activities and are in contact with practically all our own spiritual spheres, if our sources inform us correctly.

The forms of work too resembled those known to city civilizations, in spite of some differences in their pattern. We have been able to conclude that this is so from a study of etymology, excavations, language archaeology, and ethnological evidence. On the other hand, these activities were not continuous nor directed at one aim finally. Ambition and competition, but not the iron rule of society or an economic incentive to sell, drove man on to be more industrious than the climate required as a minimum for gaining a living. We can infer from written and ethnological sources that planned servant labour was practised for small social units, and that here slaves, serfs, and woman labour were used. It is a characteristic of rural civilization that work is completely adapted to the rhythm of the life of the community. Even individual work was practised preferably in society, as indeed is a custom with many farmers today and, not by chance, a Russian national characteristic. The solidarity of neighbours, tribal or blood relationships are age old too, and lead to planned communal labour in all jobs which are too heavy for individuals or small family groups. The construction of individual graves, houses, large ships, of roads, ditches, fortifications, holy stone circles, and communal buildings, the felling of trees, clearing of woods, and the digging of canals are usually communal tasks in village civilizations. Even the white immigrants in Australia, Africa, America, or recently the Jewish settlers in Palestine, in spite of their individualistic Western education, had to revive these primitive forms of labour for their new societies, because they were unable to replace collective labour by capital reserves, in the beginnings of their new civilizations. It is still true today that working in society and communal labour are favoured, even within individualistic city populations, subconsciously as a direct consequence of mankind's earlier peasant background. These forms of labour were and are often surrounded by a certain magical charisma, that occasionally has had fatal political consequences up to the present.

Remuneration in village civilizations had a particular pattern also. It disregarded originally demand and supply, but followed almost inflexible conventional rules, exactly similar to the traditionally fixed rewards which are found, still today, in rural communities all over the world. Each kind of work had a fixed conventional value in each community,

and this value was adhered to, practically without any change, through decades, and often even through centuries. When a canal was constructed in native New Guinea, for example, each labourer received food of a traditional quality and quantity each day. When the work was done, the natives had a great feast, again with conventional gifts for the workmen who attended it. We do not, however, need to go so far, to find this primitive economic pattern. If we live in the countryside or in a small town, where a seamstress, charwoman, or washerwoman comes to the house to work, then conventions like these appear also. The woman worker would be very dissatisfied, and her professional honour would be affronted, if the family in question abandoned the traditions, which mother and grandmother had adhered to, and would replace the usual gifts in kind, beverages, and dishes by money payments.

So the pattern of peasant life from the Mesolithic Age onwards shows a marked difference from what we usually read in popular books about it. Many details about the earlier periods of the peasant stages are still under dispute; but I have not discussed any fact here which has not been well substantiated and proved by archaeological research, etymology, or by ethnological parallels. But yet, any one who has excavated a Neolithic or Bronze Age settlement in Europe, Asia, Africa, or America, and compares the remains and finds with the tools and houses of a neighbouring Chinese, Hindoo, Negro, Red Indian, or European village, which has not very much yet been affected by modern capitalism and rationalization, will be astonished. What man has established on the same grounds from 6000 B.C. or so onwards, shows, as a rule, how close the level of the life of the ancestors was to that of their descendants. And this means, in almost all cases, a positive evaluation of both. The question arises at this point:—"Where is the intellectual boundary between peasants and higher forms of culture, if we consider this problem from the point of view of world history?" For peasant civilizations already offered, and, indeed, used to full advantage, surprisingly many potentialities. Economically speaking:—"Which of the greater demands of human dreams and necessities could not be met with by the methods of village civilization according to which man invested practically his total labour potential in animal and plant capital?"

And here I must state that, as far as I can see, it is only man's urge towards total organization of large territories which cannot be satisfied in village civilizations. It is impossible, with the economic means of this cultural stage, to be able to direct the whole of man's life over a wide area, to enclose it in a single overpowering hierarchic and ideal order. For such an order directs, in city civilization, art and philosophy, the state and society, intellectual and religious life, law and custom, economic and social structure, the individual, his family and tribe,—in short all spheres of human life. And this unifying force not only directs all these different spheres, but moulds them together and develops them, in a spiritual sense, to classical heights. It is in this instance that Egypt, Mesopotamia, the other Ancient Oriental town civilizations, India, China, Peru and Mexico, Israel, Persia, Greece and Rome, Islam, Russia, and the "Western" lightfires of civilization, differ from the peasant cultures, even though they developed out of

them. The economic basis of village life was too weak for such a mighty task. A highly developed city civilization is, therefore, only possible on broader economic foundations. The man who threatens these foundations, from ignorance or with criminal intent, threatens at once not only the physical existence of masses of men who have adapted their lives to a directed order. But he threatens, at the same time, the spiritual symbols and achievements which have been made since the rise of the city from village civilization. Even the cultures of the world in our own day are not as strong and eternal, as it may appear to the optimist. On the contrary, our Age of Machines is probably still more sensitive, in its interplay of civilizing influences, than were the great city civilizations of past ages.

For the economic pattern of the peasant cultures after the end of the Palaeolithic hunter and foodgatherer stages, I can only offer, in this book, a general picture without any strongly marked geographical subdivisions. It is true, that scholars have occasionally been able to discover that in certain regions different ways of living were experimented with, until the one that best suited the region was found, and that some of the creative discoveries made by small communities were found to suit larger ones, and were finally adopted over the whole world in the form of working techniques, new seeds and plants, tools, or methods of animal breeding. The already discussed discovery and general adoption of the cart, potter's wheel, and the plough, or the creation of new animal and plant forms belong here. There was a tendency to adopt and adapt, which exists up to this day in an undefined form, and appears to be immanent, almost like a natural law, in all agricultural civilizations. We can trace this trend through the last five millenia, where not only achievements of other village cultures have been accepted and assimilated by the rural populations of our globe, but even experiences and techniques taken from the pattern of life of great city civilizations, that have disappeared long ago. It is possible to distinguish regional differences even in the economic structure of the Mesolithic and Neolithic Ages. But they are still shadowy and, in detail, hypothetical, although the general picture of the material culture of the earlier townless periods can be established already in essence. And, in this connection, it is important to notice that all the different factors in the surveys of this stage can hardly be said to have existed in every region. The climatic, biological, historical, and racial potentialities of the world and its human societies differed too widely even then.

PRIMARY PRODUCTION

Hunting, fishing, foodgathering

The economic forms of the Palaeolithic Age had not lost all importance after the end of the exclusive foodgatherer and hunter periods. Hunting was still for man one of his means of finding food. In many regions it was important, in others less so. And in many cases, from the archaeological remains of meals, it has been fairly well established that 50% of the meat eaten by the communities in some regions was provided by hunting.

Northern Europe and Northern Siberia were bound to be districts of this kind. In other regions, however, bones of domesticated animals have been practically the only remains of meals discovered. The Indogermanic tribes of the Neolithic Age hunted much, as language archaeology makes obvious. But yet, hunting does not seem to have played a great economic role with them any longer, and was mainly pursued in man's defence, hunting being the privilege of all full members of the tribe. The hunting territories, the "preserves", remained the property of the tribe, and were not divided up. The main function of hunting in the Neolithic Age—and one which has continued until today to some extent in agricultural areas—was to provide protection for agricultural crops, and to a lesser extent to obtain clothing. Hunting for food was less widely practised than in the Palaeolithic Age. Hunting for sport occurs usually only in town communities, where there is a numerous upper, leisured class.

Individual hunters and archaic hunter tribes continued to exist in areas which were yet little used for farming and cattle rearing during the millennia after 10.000 B.C., especially in mountain regions, forests, deserts, and jungles. But even these people changed from being members of small separate groups to useful functionaries of the new peasant societies—except where racial differences were too great—, and they were strongly influenced by other communities in consequence. Ethnological evidence proves that these hunters were not, as a rule, treated as enemies, except where they were thought to be animals or sub-humans. Usually, they were considered to be useful friends of the planters, animal breeders, and ploughmen, whose crops and animals they defended from destruction by wild animals, and whose surplus products they willingly took over in exchange for their services. A symbiosis of various professional groups to form a simple united tribe began quite early in this way.

When hunting lost in importance, fishing in the ocean, rivers, and lakes became a much more important source of food for the new age than it had been before ⁵. For fishing could now be prepared and undertaken at greater leisure and with better tools at man's disposal. The ships of the time were even used here and there for deep sea fishing. Better angling tools than before were used, hooks and nets, which were thrown out into the richly populated fish world, on which man had as yet hardly touched. Even as early as this, fish hooks were looked upon, like animals and seeds, as a primitive kind of capital, that could be loaned out on interest. It is not surprising that hooks made of bone, metal, or some kind of stone, were used as currency, in some regions. In Denmark plaice, cod, herring, and eels were generally, as the "*Kjökkenmöddinger*" prove, used as food, in the Swiss pile-dwellings similarly eels, perch, pike, carp, and salmon. The original Indogermanic races used sturgeon and shadfish as food too. The herring, as early as the Neolithic Age, created a paying profession in the Nordic Circle, likewise the fishing for eel and salmon. Indogermanic etymology serves as an indication here, especially the fact that the herring has been given a great many names by Germans and Celts, varying widely with its external appearance. With other nations, only domesticated quadrupeds were usually given names in this way, horned cattle, for instance, in many

languages, or the more than thousand names given to the camel by the Arabs. The name of Scandinavia is derived directly from one of the Germanic names for herring, just as that of Schonen comes from another. The carp, which live in ponds, were perhaps domesticated or at least tended by Swiss pile-dwellers. This fish's European name seems to have come from that region. It is difficult to establish today whether dried fish was used as money in the Nordic Circle of Europe, as it was in Iceland from the 10th century A.D. onwards; but other remains of primitive civilization in Iceland which go back to primitive customs in the North European sphere make this plausible.

In North Africa, too, in the Aegean, and on the coasts of the Red Seas, fish, especially tunny, has been a characteristic food of these lands until now. There are many rubbish heaps still left from very early times, which show this clearly. And we can be sure, too, that cultural relations were established between the seafaring nations of the Atlantic Ocean and the Mediterranean Sea. For identical and similar archaeological remains of early times are spread along the shores and islands of both oceans. The most characteristic witnesses of this intercourse are tombs, the so-called Megalith Graves, and Menhirs which are found from Britain to Palestine, if not China and Polynesia. States founded by conquest from the sea can be surmised here, or waves of technical know how and spiritual propaganda. Clearly, fishing was of greater importance than it had been in the Young Palaeolithic Age, on coasts and in inland water courses. Fishing and shipping were more valued than hunting in the Neolithic period.

The profession of the woman foodgatherer lost even more in importance than the hunter's in this new time. The foodgatherer was reduced to the position that she has had ever since in village communities. The plants which were difficult to find were gathered only for the purposes of medicine and magic.

She confined her attention otherwise to wild plants which could be found easily, like herbs, flowers, fungi, berries, wild fruit, wild hay for the cattle, firewood, wood to build houses or to fence gardens. Not only women, but also children, old people, and "outsiders" were now occupied in this work.

Only in the times of scarcity, bad harvests, and cattle epidemics the Palaeolithic foodgathering methods were revived, to support life as they had formerly, and to supplement the plants used for food. So that the tradition of foodgathering, as we have noticed already for the Palaeolithic Age, has been preserved in some medieval chronicles and occasionally has had effect in more modern times. We know that a number of fruits were gathered in the Neolithic Age because archaeology and language archaeology inform us about it.

There were crab-apple, horse chestnut, wild pears, several kinds of cherry, mushrooms, rowanberries, hazelnuts, beechnuts, acorns, loganberries, raspberries, strawberries, elderberries, bilberries, whinberries, and others. It is not surprising that folklore is bound up with these fruits. But their part in providing man's food has not been so important in normal times and under normal climatic conditions as that of agricultural food sources, since the Neolithic Age.

Garden and tree culture, the ploughed field, animal breeding

The garden was the domain of woman in the Neolithic Age as it is today, a direct consequence of its whole evolution, as we have already seen. Only tasks which required heavy physical toil, e.g. planting fruit by using cattle, fell to the lot of man, as we know from ethnological evidence. Garden plants varied widely in different climates, as a matter of course. Many plants which we still grow were long established in Central Europe and are found with prehistoric remains here and in other parts of the world too. Broadbeans come within this category, peas, lentils, fennel, chicory, asparagus, and the wild poppy whose seeds were used, and still are, in the preparation of dishes for medical and magical purposes. There were oil yielding plants too, like chickweed, flax, lettuce, hemp, linseed. There were vegetables like cabbage, carrots, lettuce, turnips, onion, sage, and thyme. Among the trees and bushes there were apple, and perhaps even pear trees, cherry and plum trees. More and more garden plants were transferred to neighbouring areas with new waves of culture and tribal migrations. In the South some plantations developed in such a way that finally only one type of fruit was cultivated, or perhaps other plants were used as secondary crops only. Datepalms, figs, olives, and vines were grown in this way.

We still have plantations of this specialized type in all modern vineyards. They originally came from Western Asia and Southeastern Europe, and are tended by hoe culture, as they were thousands of years ago. It is significant that in the vineyards women are on terms of equality with men or, at least, that they play a more important part here than in the cultivation of the ploughed fields. The garden was enclosed. It became occasionally the communal property of the clan or tribe, but more often of large or small families and, in matriarchal societies, of the mother of the house. The enclosure gave the garden its Germanic name. It was fertilized, in many regions, by means of scorched earth practices, in which parts of the neighbouring woods were burnt down for the plantations and ploughed fields of each year, and the ground for planting had to be changed annually, or after a few years. The Neolithic farmsteads had already their manure heaps, so that we can imagine that animal and human excrements were put to a similar use then as now.

As the garden was woman's province, the ploughed field cultivation was man's. Certainly the strong woman of the European and Asiatic North took her place here too. Quite often she worked with the men, as Russian, Canadian, and Scottish women do today. The earlier ages, between 5500 and 4500 B.C. or so, had only known a mixed economy, which combined garden, specialized plantation, and animal breeding. The cultivation of grain developed here from furrowed beds in special plantations of hoe cultivation to agriculture on larger fields with tools which man had to draw along himself, and which preceded the plough. Near 4000 B.C. or so this most important agricultural instrument was invented, which was drawn by animals under the yoke. Originally the plough was made almost entirely of wood. But marked differences in the type used were apparent at a very early date as we have pointed out already. Scholars have attempted

to differentiate between a first stage of development and two different plough forms which branched off from it, as early as in the Neolithic Age. These began to vary widely in the regions where the new type of agriculture was accepted, as did the carts and sleighs, drawn by animals, which became essential working tools in the ploughed field. Hammers to break up the clods of earth and similarly useful tools developed too in this period, among them the so-called “*Schuhleistenkeile* “. These enormous stone instruments have wrongly been interpreted as primitive ploughs in earlier literature. They were of particular importance in the “Bandceramic” culture in Central Europe, and were often exported from the Danubian region to neighbouring tribes. Instructive analogies are here, in our survey, hammers to break clods of a similar type, which, it appears, had a wide circulation, reaching, according to our finds, as far as Northwestern India, the Gobi desert, and the Mesolithic and Neolithic farmlands which are now dried up in the Sahara.

Man chose steppe and meadow lands to settle in, after his transition to hoe and plough culture, that is, those lands which Nature had kept free from trees and bushes, because he could destroy trees and jungle with great difficulty only at this stage. He was however, able to penetrate the wooded area, bushland, and jungle slowly, relinquishing then the “grassland economy” of the original plough culture, in which agriculture was restricted to unwooded land. The Indian climate is more favourable to “grassland economy” than other subtropical and tropical countries, as the Indian subcontinent has extensive and fertile natural clearings. Therefore grassland economy has remained a prominent form of cultivation in many Hindoo provinces and states, in the face of the difficulties encountered in penetrating the jungle. Spiritual reasons, connected with Hindoo religion, are at work here too, and this and other conservative trends (neglect of the potato and related vegetables, cattle taboo of the Hindoo, Islamic distrust of productive, interest bearing credits) are responsible for the permanent food deficiencies and the so-called overpopulation of India which may become a menace to economic world stability. The produce of plough culture varied with soil and climate. It consisted, in Central and Eastern Europe, and in Transuralian Asia, of wheat, emmer, spelt, einkorn, barley, oats, and rye. In Mediterranean regions millet was added, and rice in Southern Asia and China. Everywhere special varieties developed in different climates and regions, to replace other varieties which were less suitable for the preparation of certain foods, or not adaptable to the climate of the area in question.

In this early period of Neolithic agriculture the most important economic technique, which was combined with “grassland economy”, was the so-called “field-grass economy”, perhaps of an even earlier date than the plough. The main stress in agriculture lies, under the influence of this technical method, on pasture land for animals, the first stage of transition from pure nomadic to more settled forms of life. Land for cultivation, which remained the communal property of the clans and tribes, was always put under the plough for only about one year, and taken out of the communal pasture for this purpose. Then there came a change. The domestic animals of the community were

allowed to graze on this land for several years. Until finally, one needed the fallow land again for a year for growing crops, and changed back to pasture afterwards. Woodland which would make good pastureland for pigs, goats, and sheep, or could be cleared easily, was taken over and included in the field-grass economy (“Hauberg” economy). It is easy to see that an agriculture of this kind will need very large areas, and that the land used in this way will not be able to provide food for so many people as it would under more progressive systems of farming. Settlements, too, at this time, were by no means permanent. They had to be abandoned in turn, so that the people could be nearer to the land under present cultivation, and eventually would be moved back again, together with the arable plots. There are remote regions, even in Germany, which have remained partly under “wild”, and partly under “regulated” fieldgrass economy until very recent times, and more territories of our globe have preserved remains of such old-fashioned agricultural methods than is good for the food reserves of the modern world.

The “regulated” field-grass economy gave magistrates, in a significant way, political power, which depended on the yearly redivision of communal land. Temporary plots had to pass into the hands of various agricultural clans and families in a traditional and fairway, often through many generations. The leader of the state, a king, headman, or priest, acquired here a supervisory function over the agricultural land. “Wild” field-grass economy, on the other hand, was still without any such rule. But it was this form of agriculture which finally changed over to individual full ownership, by single families, of ploughed fields, gardens, and meadowland. The fertile land in Egypt had, and has, to be measured and distributed each year, after the Nile has overflowed its banks, by the state magistrates of the Egyptian provinces, proceedings emerging as early as the Predynastic Age. The agricultural regions in early Mesopotamia were controlled, by the Sumerian centralized states, as planned economic units. In China, the Indus civilization, in Peru, and Central America, a similar agricultural organization prevailed in the Neolithic, Cuprolithic, and Bronze Ages. One should, in my opinion, explain this early rural organization as having developed organically from prehistoric regulated field-grass economy. The planter and peasant nations who occupied all these countries, obviously had land laws of this pattern, before their final settlement. They probably adapted afterwards their original agricultural technique to regional conditions in which a prolonged fallowing period would not have been reasonable, but river inundations had to be provided for. A “wild” field-grass economy has perhaps changed, under the influence of the Palestinian climate, to the well known Sabbatical year of the Bible. Here each piece of agricultural land seems to have been left fallow for one year after six years’ cultivation; but this custom was probably not yet put in practice simultaneously throughout the whole country, as it was later in the time of Israelite town civilization, something against all economic reason in a townless stage. “Regulated” field-grass economy was characteristic for early Greek, Italian, Slav, Celtic, Hittite, Hindoo, Iranian and Germanic agricultural techniques, royal and noble permanent states being exempt, forerunners of individualistic Iron Age economy.

Triennial farming, in which a field belonging to the tribe was divided into three strips, created a precarious family ownership in the course of time. The strips changed annually from winter- to summerfruit or to pasture and fallowness. Unfortunately the details of this early development cannot yet be clearly established, although this practice seems to have begun very soon in Central and Northern Europe, at least in its main principles. There was an increasing number of growing nations in these territories, who had used up those land reserves which could easily be taken over, and now had to intensify their rural economy or had to emigrate. The Greeks seem to have taken the methods of primitive triennial farming with them already in their migration from Central Europe to the South. In Classical Hellas triennial farming appears in our written sources with certainty. In the Near East, the Mediterranean region, India, and China the same intensifying process took place in agriculture by means of irrigation and canalization. Egypt and Mesopotamia had, in addition, a two harvests' system in their period of Bronze Age town civilization. This highly developed agricultural technique, which was, in a sense, a Southern analogy of triennial farming, may lead back to the time of townless village civilization of the Sumerian and Semito-Hamitic nations; but this cannot be proved with certainty. The Neolithic peasant used animal manure, too, to fertilize his land, as we have seen in the case of garden and plantation in hoe culture. The scorched earth practices were applied to plough culture also, such fertilization being very favourable for field-grass economy. If we survey all these forms of plough economy in the Neolithic Age, primitive as they may be, we are bound to give judgment in their favour. Compared with later periods, there was only lacking in this the rationalization and the intensifying of agricultural economics over wide regions which was brought to the fore in the later city civilizations with the application of more progressive methods. Such an intensification was hardly possible without a ready market for surplus crops, which only cities were able to supply. Nor was this likely to occur in a period where each clan was expected to provide for their own needs.

Animal breeding reached a good standard of economic advance in the Neolithic period also. One was still content with simple meadow and grass land economy, and pasture in woods. In addition, careful tending and the introducing of new special strains were in practice. A system of autarchy, which expected each clan and family to provide for themselves, meant that such economical methods were greatly needed. The ideology of the nobility in all townless civilizations, where cattle was tended, was influenced to an extraordinary degree by the basic principles of the selection for animal breeding. Customs derived from this sphere influenced strongly man's choice of his bride, and led up to certain well known racial concepts in some city civilizations, like those of India, Israel, Persia, and Greece. We can be sure, from etymology, that in the case of the early Indogermanic and Hamito-Semitic peoples, great importance was placed on animal breeding; but that the cultivation of gardens and ploughed fields were known to them too and just as carefully practised. We have here two of the many Neolithic tribal complexes with a characteristic field-grass economy. At the outset both national units,

if we follow their early customs, seem to have tended rather to “wild” than to “regulated” field-grass economy. Later their tribal subdivisions gave changing preference to hoe culture, plough culture, or animal breeding, “wild” or “regulated” field-grass economy, in accordance with the various climatic and political conditions of the regions where they finally settled. The cattle of the village civilizations in the Neolithic Age had not yet, as far as we can discover, the advantage of planned feeding in stables. Here it was again rather a city market that was lacking, than the technical means to intensify production, which delayed further development.

A particular form of animal breeding developed in the Cuprolithic period in East Asia, the care of silk worms. This technique has a place in the traditional and hieratic order of life of the early Chinese people. It was probably very old, and leads back, in my opinion, to the townless periods of the Chinese peasant culture. Here there was a remarkable combination between the tending of animals and plant cultivation with the hoe. For the care of the silk worm and of the mulberry were combined more closely in China than hoe culture and animal breeding have ever been in townless civilizations. The cultivation of the silk worm reached Europe only at a late date from China. But its care and attention have contributed, as far as we can see, decisively to the formation of the East Asiatic character, the Chinese patient philosophical outlook, and the particular forms of Chinese economy, in no less a revolutionary way than cattle culture did in the rest of the world.

Mining

Man thrust his way even into the earth, in the Mesolithic and Neolithic Ages, in his successful attempts to improve his condition, surpassing the very restricted efforts he had made, in the same direction, in the Young Palaeolithic Age. The village civilizations of the Neolithic Age had the economic strength to give an impetus to his efforts by their requirements. Flint, salt, and bitumen were close at hand and were easy to obtain. Archaeological sources show that, in numerous places in Europe, North Africa, East Africa, the Near and the Far East, flint was regularly dug for. We know of flint mines in Belgium, France, the British Isles, Sweden, Italy, Spain, Portugal, the U.S.S.R., Syria, Egypt, the Sahara, tropical Africa, Iran, Asia Minor, Palestine, India, and China. And man did not confine his attentions merely to the surface of the earth in his search. He was able to build shafts as deep as 20 metres and more. And even as early as this, he knew how to use piles to support the shafts which he had sunk, and even to build horizontal galleries. Naturally, mining was more dangerous in the Neolithic Age and the following prehistoric periods than it is today. We have dug up, thousands of years later, the bodies of men who were buried by falls of earth. The main equipment of these prehistoric miners were picks made from antlers, trees which they could climb in the place of ladders, ropes, and leather bags. And man did not stop at flint. In our finds we can trace cultures which, although based on using stone articles, had some luxury products, like jewellery and ornaments, occasionally made from copper, or of valuable

metals, or even of meteoric iron. This change occurs quite early, and particularly in those countries in which minerals can be found easily, like Hungary, certain districts of India, or certain parts of Egypt. This seems a clear proof that man learned, how to obtain and work with these minerals, as early as in the course of the Neolithic Age. Some of the gold, silver, tin, and copper shafts in the Europe of the Bronze Age, in Asia Minor and the Near East, on the Sinai peninsula and in other areas in Egypt, may actually have been worked in the Neolithic Age, earlier than one might expect; although it is true that these attempts were less intensive than later.

The earliest Sumerians and Pre-Sumerians, Proto-Indogermans, Indogermans, Hamito-Semites, Ural-Altaians, and Predynastic Egyptians knew and made use of gold, silver, copper, and, meteoric iron for luxury purposes, according to etymological sources and excavations.

Yet their ordinary life remained, as far as we can discover, almost entirely within the bounds of Stone Age civilization. After 4000 B.C. or so tin was used in some form in Asia Minor, Syria, Iran, India, and Malaya, and between 2500 and 2000 B.C it seems to have been combined, for the first time, with copper to bronze. The new technique which was soon to revolutionize the working methods and tools of most territories of the Ancient World, came to Egypt and Mesopotamia from peasant regions where the first invention obviously had been made.

After 3000 B.C we also find terrestrial iron that was already produced in small quantities as a by-product of gold refinement, but which was probably more valuable than is platinum today. The use of iron became more and more important after 1500 B.C. or so. The new technique to produce cheap iron does not seem to have originated in the city cultures of the Ancient East; but rather appears to have come from the territory of the peasant cultures North of the Black Sea. The Ancient Oriental city empires of the Hittites and the Mitanni only refined and intensified the working of iron, and the mining of iron ore, until finally the new methods conquered the whole world in their progress. And so both bronze and iron techniques have an early origin in townless civilizations which had occupied areas in the vicinity of particularly favourable deposits and mines of the precious metals, copper, tin, and iron. It is true that it was only the Ancient Oriental city, which came later, that made these different inventions to decisive factors in man's material culture; but the city only took over this bronze and iron technique, exactly as had been the case with the invention of the plough or the breeding of the bloodstock horse. In a sense peasant culture remained to be more creative than that of the cities for a long period to come.

Home Industries and Crafts

Preliminary remarks

The conclusions which we have drawn so far, concerned the high development and differentiation of primary production in the Neolithic Age. On the basis of the new

primary production, craftsmanship of a similarly high standard arose and appears in our finds. There developed now, from Palaeolithic beginnings, a clear and important distinction between the crafts of men and women. Etymology, finds from excavations, and ethnological parallels show us this characteristic division of labour, which was and is strongly upheld by peasant society. Certain crafts were left to women, except where they had become the distinctive symbol of the dignity and social valuation of a limited group of men. We shall call these crafts in consequence the female home industries of Neolithic village civilization. The techniques of this kind were concerned with spinning, weaving, sewing, mending, dyeing and tanning (which go with clothing); tattooing and painting to decorate the body; brewing, milking, and everything connected with the preparation of food, medicines, and poisons; basket weaving and pottery which are linked with domestic tools. Art, medicine, and magic practised by women must be included in our list as the highest achievements of the female sex in our period. They must have developed from the lower spheres of woman's activity, according to natural logic and ethnological parallels, although these activities cannot be differentiated from that of man owing to their peculiarities.

To the men fell the heavy labour, like working with wood, stone, bone, or metal, the preparing of wood, from the fallen trees right through to the many finished articles, the obtaining of bone from hunting or slaughter, of stone and metal from mining and stone breaking, and all working with these materials. Building was a masculine occupation too. It entailed the construction of ditches, canals, roads, bridges, houses, ships, graves, castles, temples, and fortifications. In the upper structure of masculine craftsmanship we find remains, again, of another sphere, one in which woman shared, but in which her activities were not as important for world history as that of man, that is man's art, science, and magic, which have been the spiritual foundations for the great symbols of all later highly developed forms of civilization.

In general, the crafts of these days were seldom anything than pure house work. Originally such tasks were traditionally the duty of each family, clan, or tribe as economic units, and were intended to meet the daily needs of these collectives, as much as possible, with individual strength. This preference for autarky in local craftsmanship is still to be found, to-day, among remote villages and native nonEuropeans, who preserve the pattern of the periods before town civilizations arose.

A sentimental trend to autarky has, in addition, survived even in very urban strata of Western, Islamic, and Russian civilizations. But there were from the beginning exceptions to this rule. It happened that raw material which was essential for certain products of craftsmanship, could not be obtained locally, although one understood well the technical methods to work it, and had the means to do so. In such cases man attempted to gain such materials, or preferably the finished articles, from near or far, with the help of trade, robbery, piracy, or war. In this way, the domestic production of certain articles developed, in a few suitable regions, to an industry which was able to produce for a market.

Usually such work, too, was carried on in the home as one among several activities of a domestic community which either sold the finished product directly or through middlemen, at a greater distance, to the last consumer. In spite of all the primitive qualities of production and market, a genuine and continuous trade with products of craftsmanship grew up in such a manner. A second break in the autarchy of the individual units of peasant economy often appeared, if a special intelligence or experience was necessary for craftsmanship or building, or even a specialized magical knowledge. On these occasions, the customer called a craftsman to his homestead and provided him with food and lodging until he had made, with raw materials belonging to the customer, a sufficient number of the special articles required from him. Here lay the beginnings of the so-called work at a customer's house, in which the craftsman still today goes from one small economic unit to the next and takes his special knowledge from place to place. He often takes over the servants of the household or uses the members of his employer's family to help him in his task, and teaches them part of his trade. It is true, however, that neither home industry nor work in a consumer's household, in townless civilizations, are suitable to supply enough income for the craftsman's needs throughout a full year. Craftsmanship had therefore usually to be treated as something secondary to agricultural occupation.

And so, a specialized and exclusive profession of resident craftsmen, with its chances for refinement, was very seldom found, even in very rich peasant regions. Such refinement of secondary production appears fully only in city civilizations with their powerfully enlarged range of consumers. In addition, if the craftsman had no part in primary production of townless civilizations, for the one or other reason, he would be completely dependent of the customers for all his material needs, and so he would be socially inferior to the peasant, as it is still the case today in remote regions. From landless craftsmen of this kind there arose very soon a free and more fluctuating landless proletariat who wandered from country to country. Each peasant culture has had such a social stratum, as we know from ethnological evidence, historical tradition, archaeological remains, and by the fact too, that Western peasant communities still have a class of this kind to-day. It was only by magical practices, which made them proteges of the gods, or superiority in fighting technique, that some of these landless wage earners were lifted to a higher social plane, and even, in some cases, were accepted in the ranks of the nobles.

Genuine craftsmanship as a profession in prehistoric townless civilizations, therefore, was generally connected with the production of rare, artistic, and luxury articles only. These were acquired by exchanging goods at a conventional payment. The principal activity in secondary production, preparation of articles which had to be used every day, remained almost completely the affair of domestic industry. Ethnological parallels show that some crafts, practised in the house community or at customer's homesteads, were transferred to the men, although they would normally have been part of the duty of the women. For the Neolithic Age the pattern of the subsequent city cultures in the

Ancient Orient strengthens the hypothesis of a development of this kind. For we find, right at the beginnings of written tradition in hieroglyphs and cuneiform characters, that, although customs varied in different regions, the men had taken over a number of women's tasks for economic reasons, without that this process had already been completed. In addition, we observe a tendency in certain peasant civilizations of the Neolithic and Bronze Ages, and in the early stages of the Ancient Oriental town civilizations, to put an individually refined product in the place of the more average and traditional one, and possibly this was connected with such a change-over also. At any rate, the well known psychological differences between men and women in peasant and town societies would fit very well to this antithesis in artistic feeling, which appears in our finds.

Female industry

Clothing, particularly woven fabric, was much more in prominence in the period which followed the Palaeolithic Age than in the hunter and foodgatherer stage. We can take it that continuous activity in this field of secondary production was required from a housewife. This tendency was particularly obvious in Northern countries. In the South, art was more to the fore, either in painting the body itself, in using cosmetics, or in tattooing; and here the peasant civilizations were not far away from the Old and Young Palaeolithic Ages. The basic materials used for clothing were wool from sheep, hair from goats and camels, furs and hides, the cocoons of the silkworm and of non-domesticated related insects, and, very rarely, asbestos. More or less the same species of fibrous plants were used, in different regions of our globe, that are still at our disposal, bast, for instance, crêpe, muslin, flax, hemp, and cotton. Fur clothing is indicated by etymology as having been used by the original Indogermans. Pieces of leather are in our museums which come from Neolithic Egypt, from Central European pile dwellings, and many other regions. They have been found together with tools for textile and leather production of all kinds, like awls and bodkins made of stone or wood, and strong bone needles. Recently some very interesting leather and woven fragments have been added, which can be assigned to the Northern Neolithic period with certainty, from pollen analysis. Tanning seems to have appeared gradually and spread slowly. At last, we have as yet no direct archaeological proof for it, before the Ancient Oriental town civilizations arose; and there is no general expression for it in Indogermanic. Tanning in the peasant civilizations may have been a specialized craft from its beginnings, that is a masculine occupation, which required a large amount of physical strength and was either practised as a home industry for many neighbouring households or in customer's households.

Woman learned to spin fibrous textiles early in our period. Spinning wheels and spindels made of wood were well known to the pile dwellers, as were shuttles. Originally the latter were made of wood too. Then woman discovered how to make them in a more practical way from clay, or more rarely from materials which women did not use a great deal, like bone, stone, amber, and rock crystal. Probably here the influence of masculine

activities can be seen. In peasant spinning today there are analogous examples of a similar traditional custom, according to which all kinds of secondary help are given to the spinning maids by their boy friends. The threads were threaded into the spindle and the shuttle, and often entwined in such an expert manner that the remains we have found still have the appearance of woven fabric. But weaving with a loom was soon known also and practised.

There are pieces of looms, combs to hold the threads, and shuttles, made of many materials, in our museums, sewing needles and knitting pins from the pile dwellings. Remains of nets for fishing and other purposes show that the women understood the technique of knotting such work. There was embroidery, too, of different kinds and in different colours, remains of which have been found in the Swiss lake dwellings. Felt appears to be early Indogermanic, if we follow etymology.

Clothing in the Neolithic Age and in all later townless civilizations was a garb, traditional and conservative, varying with race, sex, age, social position, and region, but yet very little influenced by fashion or individual taste. Buttons and bone needles were attached to hold the clothing together. In the South the normal wear of the majority of the men was a kerchief and a band around the hips, a loin cloth. But there were richly ornamented festive clothes, too, which covered the whole body. The Indogermanic men had at least two articles of apparel, a cloak, and an apron, if we follow etymology. Underclothing, shirts, and other clothes, appeared at later dates. We have not much woman's clothing from the Neolithic Age; but what little we have includes all kinds of cloaks, coats, and jackets, of the type still extant among peasant people. Woman's clothing must have been much more individual and differentiated than men's. Sacks and bags were entwined, woven or made of leather as early as during the Mesolithic and, perhaps, even the latest Young Palaeolithic Age. We have cushions from prehistoric Egypt. Shoes made of leather or the bark of the lime-tree are Indogermanic.

Painting the body, particularly the face, is shown in clay statuettes of both women and men. Cosmetics, boxes to keep them in, cosmetic pencils, and palettes were already familiar to Palaeolithic man, and thereafter to the prehistoric Egyptians, Mesopotamians, and Europeans of the Neolithic and Cuprolithic Ages. Ethnological parallels show that the techniques of make-up, painting, and decorating the body were in the hands of women, mainly, for the men too. It is true, that up to modern times woman seems to have had a special aptitude for such activities. Similarly, the products required for primitive hygiene, and many traditional medicines and poisons, were prepared by women from the materials which they used for preparing food and intoxicating beverages. The moderate and arctic climates allowed the use of soap. The subtropic and tropic regions had olive oil and many of the parfums which we know and use today. Milk, butter, thran and some alcoholic fluids were, in addition, used to keep clean and fit. Many of these products of the village stage not only survive in folklore, with primitive tribes—and with the female poisoners of our own civilization—but some prove useful in our own houses and families up to the present day.

The preparation of food was the women's duty. They already had supervised it to a great extent in the Palaeolithic Age; but the technique of preparing a meal became vastly different with the new economic forms. Cereals were roasted for longer periods, trodden down, ground for meal, or even eaten raw. There were leavened and unleavened forms of porridge. Flat cakes were baked in the hot ashes, made of wheat meal, millet, rice, or flax seeds. Bread was usually an unleavened mixture, and many examples have been found among the remains of lake dwellings. The grains were, for this purpose, pounded in wooden mortars with pestles, and subsequently ground in stone querns. Leavened bread appears to have been prepared in Egypt first; for the earliest remains come from there. Baking followed, in baking dishes of stone, but later still, many regions had baking ovens. However bread was not such a frequent item on the menus of this period, or even the Ancient Oriental, Greek, and Roman town civilizations, as porridge made from inferior grains. The reason was that the milling processes of the Ancient World were not efficient enough, at least for those varieties of grain which produce plentiful harvests, to produce flour which could be converted easily into digestible bread and cakes.

There were many different ways of preparing food, and quite a number of them has survived in our modern kitchens. We have found stone branders from the villages of the lake dwellers. Soup was known in Old Germanic times, and was quite common everywhere. Vegetables were already cooked in the Palaeolithic Age, and further methods of preparing them were invented now. Numerous expressions for cooking, baking, and roasting appear to be Indogermanic according to etymology. Honey was a popular dish, which had been used in the Palaeolithic Age, and took the place of our sugar in most regions of the Ancient World. The early Indogermans knew, and enjoyed, milk, butter, cheese, and sausage. The role which butter played in the preparation of dishes in the North was taken by olive oil and other vegetable oils in the South. The recipes were mainly inherited from grandmother to granddaughter and traditional. Many survive in folklore and modern cookery books, and are the pride of our housewives from China to America and from Siberia to South Africa and New Zealand.

Intoxicating drinks were popular also, and were usually prepared in the home by the women, a custom which prevails in rural areas up to the present time. Beer and mead were produced from every conceivable kind of material from East Asia to the Atlantic and from Siberia to the tropic lands of the Ancient World. These beverages were made of honey, or of plants which contain starch, like juniper berries and carrots, and often, of course, from cereals and malt. Scholars have often examined very carefully, and in microscopic detail, the sediments of barrels and vessels which have been found, especially in graves, and have been able to give a fairly accurate account of what they originally must have contained. Wine and vinegar were made in the Near East and were native to the Balkans, but spread very quickly over vast areas. There were many alcoholic drinks which were peculiar to limited regions throughout the Ancient World. We find them described as local specialities, from the beginnings of city culture in our written

tradition. The most important were date brandy and palmwine in Mesopotamia and Egypt, rice wine (arak) and rum made from the sugar cane in India and the Far East. Fermented cow and mare's milk and grain brandy (*vodka*) were enjoyed in Central Asia and on the shores of the Black Sea, fermented milk of the cocoa nut in the Pacific region. There was wine made from many fruits, particularly from apples, pears, and berries, in Central Europe and France. The coast lands of the Mediterranean had wines made from figs, rose hips, pomegranates and mulberries, and these recipes were imitated in regions farther North and East.

At the first glance it is surprising to notice that men and women must have smoked in the Ancient World before the discovery of America, where the tobacco had been made use of by the Red Indians for a long time before Columbus. We have excavated pipes with a head, a tube, and often with a lid, which could be opened and closed; they basically resemble our own. In the West they come from prehistoric Gaul in the Bronze Age, perhaps from the Neolithic South and Southwest of Europe, and from all the Roman provinces. The materials used for smoking must have been opium, poppy seed, hemp, or similarly intoxicating materials, and our sources indicate indeed that the smoking of opium and hashish must have begun very early.

The third sphere of secondary production in which woman was supreme was, as ethnology and archaeological finds prove, the preparation of basket work and of clay pottery. Weaving baskets, mats, cords, and ropes kept many women busy. Remains have been found here too, among other evidence, of fragments of clay vessels which had been ornamented by pressing basket work into the soft material before the pots had been fired. In the vicinity of bitumen beds, women were able to plait water tight barrels, and even to build houses and tents with the help of such techniques. We have been able to establish a relative chronology of the Mesolithic, the Neolithic, and the following prehistoric Ages by their designs in ceramics. These reveal great differences in technique and shape in the different regions of our earth according to tribe and nation, which are maintained for many generations. In my opinion, it is woman's conservatism which is responsible for this slow alteration of design which is such a blessing for dating excavated prehistoric levels. Some finds of pottery on European Young Palaeolithic sites, if these much discussed and doubted finds should be genuine, could be easily explained by Mesolithic influences from Asia owing to the wellknown time lag between the beginnings of the Mesolithic Age in Europe and the Middle East.

Clay was prepared according to methods of mixing favoured by each region. The vessels were built up in zones as a rule, and only seldom made from one lump or by pressing the material into a model form. The potter's wheel which rotated quickly and made the work much easier, is found first in the lands of the Middle East. It probably originated in Turkestan and spread very early from Central Asia to the city civilizations of the Ancient Orient and China. But it was known to some townless civilizations of Europe comparatively early too. The reason for the relatively late general acceptance of the potter's wheel is perhaps explained by the fact that such an instrument only pays,

if it continuously works for many customers. Where it was established that each family provided for their own vessels, the use of the potter's wheel was restricted. In addition, woman's conservatism may have retarded the general acceptance of this innovation. Probably the potter's wheel proved its full economic usefulness, for the first time, in the Ancient Oriental city civilizations, and only later became generally known as an essential instrument in pottery making and was found desirable by a majority. The potter's wheels were originally simple forms of tablets which could be pushed to spin around, and were made, for the most part, of wood. Such simple potter's wheels were known in the Bronze Age of Central Europe, and were probably used by some tribes as early as during the European Neolithic Age. Although the early form of the potter's wheel reached the North of Europe not much later than the Near East, the main improvements of the instrument, which we find later, were made by the craftsmen of the Ancient Oriental town cultures.

The raw pots of the prehistoric village civilizations were decorated in different ways, polished and levelled, dried in the air, and finally baked. Sometimes they were even glazed, a technique which led to the faience of Predynastic Egypt and finally to the discovery of actual glass articles. Glass is found, as far as our present knowledge goes, in the centuries after 3000 B.C. or so, first in Mesopotamia and then in Egypt. This invention may nevertheless be earlier than the Ancient Oriental town civilizations. Where pottery became more artistic and desirable in form, and established a market for its design, it probably changed over to the sphere of masculine activity in many cases. This process seems to have practically reached its conclusion in the archaic period of the city civilizations in the Ancient Orient, but is noticeable in some of the more progressive village civilizations of Bronze Age Europe also. Ceramics are our most promising source in establishing a number of migrations of tribes, cultural influences from neighbours, changes in craftsmanship and taste, for the Mesolithic, Neolithic, and all the later prehistoric epochs. From changes of design and migrations of characteristic types of vessels into other regions the prehistory scholar is able to come to certain and far reaching historic conclusions.

Home industries and crafts with preference for masculine labour

Man's industrial activities were actually different from woman's in that the men were able to expend greater strength, and use materials which were harder to work. In addition the men were less addicted to traditional design and mass production, and occupied themselves preferably with tools and buildings which society required in smaller quantities and less regularly, but which were valued higher and needed more time until they could be completed. As in all later peasant cultures, man was less often required to produce articles of craftsmanship for his family or tribe than his wife, but what he produced had to be done more thoroughly and with greater physical strength, and had to last longer. In the Neolithic Age practically all tools and weapons existed that townless civilizations still use in everyday life. The later city cultures did not add

much more, although materials, designs, and methods of production were considerably improved. The modern peasant still uses instruments which have remained with him from very early times, and to which he has adhered in a conservative way.

I will give a list here of the more important articles, mainly tools, which we have been able to discover in our excavations of sites from the Neolithic Age, and which were probably produced by men. They are: bodkins, hooks, anchors, axes, bows, drills, boomerangs, threshing flails, drills, daggers, borers, barrows, hoes, hammers, harpoons, lances, ladders (trees prepared for climbing), querns and other hand mills, wooden clubs, spoons, knives, needles, nails, ovens, arrows, awls, looms, ploughs, wheels, carts, sleighs, saws, razors especially for shaving, shields, skates, skis, hammers for breaking up clods of earth, sickles, sieves, dishes, tables and stools potter's wheels, olive and wine presses. A modern farm, which is not mechanized, is hardly any richer in equipment, the only difference probably being that the farmer buys his tools from craftsmen, and makes them only from raw materials where he has to be economic or does craftsman's jobs as a hobby. Naturally, all these instruments were not discovered in one place, exactly as, today, they are not all at hand everywhere. It is significant, that the tools of the Neolithic Age were already so extraordinarily well made, that man could frequently perform operations on the skull and similar surgical treatment, the results of which would be considered as quite satisfactory by a forum of modern practitioners. Skeletons which have come down to us, prove that at least some of the patients survived such operations for many years. In fact, knowledge of medicine, veterinary surgery, astronomy, and natural sciences, which has been of value for the experiences of many later generations, has come down from these townless civilizations together with the absurdities of magical superstition. The spiritual potential of Neolithic peasant civilization was considerable from many points of view.

As tools improved, in the Neolithic Age, man went on to take up building, following up experiments of the Mesolithic period. As opposed to the Palaeolithic Age, buildings erected in this new epoch were so stable and so numerous, that they almost surpass those of present day townless cultures. The different forms of houses are countless. They can be classified according to race and region. They were made with stone walls, woodwork, panelling, sun-dried bricks, and of burnt bricks also. In many districts the houses were built exactly according to a mason's plan, and often richly decorated. We have found store houses for the harvests, stables, single farm yards, and whole settlements, even mighty temples, and cave buildings, of Neolithic Jericho and Malta for instance, or of Nottingham in England. The lake dwellings in Southern Germany, Switzerland, and Austria are famous; they were built into a lake, on piles, mainly for reasons of security, or on moors and riverbanks where the piles would give a better foundation. Strong earthworks were erected as protection against enemy attacks. They were strengthened with towers, ditches, and palisades, and sometimes even had stone walls, as a buttress. There were simple, but useful systems of paths and roads over wide territories, and bridges were built, wells were dug, dykes and dams were built, as need

arose. The graves of the people of the Neolithic Age are often rich in deposits. Famous examples which are particularly impressive are the Dolmen and Menhirs which are spread along the Atlantic and Mediterranean coasts of a large part of the Ancient World. They are dated from different periods, as are the Megalith graves of the European North, South, and East. Numerous cemeteries have been excavated and offer most valuable evidence.

The corpses were actually mummified in some regions as early as our age. The cattle were given stables, in North and South alike, often in simple form, but under the same roof as its owner, although on a somewhat lower ground level. It is very probable that many religious buildings and some roads were placed according to astronomical positions following a fixed position of the sun or of certain stars. This practice, which becomes a characteristic of the Ancient Oriental town civilizations, seems to have originated in the townless cultures already. Shipbuilding improved in a remarkable way.

Dug-outs made from one tree, canoes, felt boats, rafts, and other floating conveyances were used according to the district, and some ships were even able to brave the high sea. In England one understood to build ships wholly from wood, fitting different parts carefully together, without using metal nails, as a recent surprising find shows. A mast seems to have appeared in the North also, as we conclude from one solitary drawing. In prehistoric Egypt and around the Cyclades masts and sails were already known early, and they were used more widely in the Bronze Age. The ship soon became an important aid to tribal migrations, war, piracy, and trade, just like the cart or the domesticated draught and riding animals.

The use of water power, by means of canals, dams, wells, cisterns, conveyances for water transport, paths for water carts, and by drainage according to planned supervision, began in the great river plains, oases, lake and sea regions of the Ancient World early in our period. This happened as soon as the economic structure enabled man to use enough labour force for such enormous projects, by providing draught animals and beasts of burden, and preparing great food stores from his agricultural surplus production. Excavations in Egypt and Mesopotamia are important here, but there are others in Malta, Iran, and Turkestan which are most instructive.

The hunter of the Young Palaeolithic Age had already gained some experience in using his tools for similar purposes, in constructing his great pits to trap wild beasts. It was only lack of labour power, and not of technical skill, which had prevented mankind from building extensively or mining deeply in the hunter and foodgatherer stage. The building activities of the townless cultures, after the end of the hunter and foodgatherer periods, were usually a communal and collective effort in which all able-bodied men of the village, tribe, or clan were obliged to participate in traditional forms, according to ethnological parallels. What was achieved here, in the peasant stage, was noteworthy enough, even compared with the much greater potentialities of the city civilizations of the Ancient World.

Jewellery, art, magical tools

To conclude this part concerning the domestic industry and craftsmanship of the village civilizations, we have to add a description of the crafts in which more spiritual needs were met. Both sexes were concerned in these activities, although ethnological parallels and modern psychological research seem to indicate, that woman had already, as a rule, the well known handicaps which were the consequence of the victory of patriarchal society and the inferiority complexes which were to follow, and that man had the heavier and more creative share in the upper spheres of peasant craftsmanship. A clear distinction of the work of both sexes would be too hypothetical in this case. Remains which are connected with spiritual activity in the Neolithic planter, animal breeder, and plough cultures, as research has uncovered them, are for the most part numerous and conclusive. Only the city cultures brought a higher standard. For they presented mankind for the first time with a leisured upper class, a growing and large minority, and in exceptional circumstances even a majority. Such a leisured class seems to be the essential condition for the creation and realization of human spiritual dreams which do not pay for the effort at once and, therefore, require much of the investments of any society.

Feminine need for adornment, and man's too, has been great at all times and still is. Even Palaeolithic man and woman had this urge to a marked degree. The wealth of our source material increases after the end of the Palaeolithic hunter and foodgatherer stage, without the general picture being in any way particularly different. Materials like the following were used for jewellery and decoration in the peasant civilizations of the Neolithic Age: There were amber, shells, feathers, cloth and fur pieces, figurines and pearls made of clay, the more usual and the precious stones of all kinds in various conditions of working, ivory, animal teeth, here and there gold, silver, natural electrum, nickel, tin, copper, natural alloys of tin and copper, or of antimony and copper, meteoric and terrestrial iron, and perhaps even platinum. Hair style, beard style, and make-up were related to the wearing of jewellery. Among the new forms I must mention pendants, combs, and hairpins, chains and rings in numerous shapes and colours, which developed from the earlier forms of the Young Palaeolithic Age. The decoration of pottery and jewellery made from clay was done by means of inlaid work or incrustations, especially with white. Various children's toys seem to have existed also. At least, we have excavated small, hollowed-out figures, of animals for the most part, which mainly have been found in graves. They had small stones inside them, to make them rattle. Work in clay which was meant to be decorated was probably, as a rule, made by women, while jewellery in harder materials is perhaps the product of masculine craftsmanship.

Art in the narrower sense, already showed all the seeds of development, which would germinate fully later in the city cultures. But it was formalized, as far as we can discover, restricted by the sacral attachment, in the same way that all peasant cultures today seem to have fixed religious symbols. We know a great number of musical instruments of prehistoric date, like trumpets, bone pipes, and once, in Mesopotamia, a harp. Metal

horns and most stringed instruments, on the other hand, like the lyre in Europe, the cithara in the Near East, appear with the Bronze Age. Yet the lyre and the cithara might have been older. The wooden flute was certainly known in the Neolithic Age, before the later bone flutes, if this instrument does not go back to the Young Palaeolithic Period. With music goes dancing. There are paintings and reliefs on walls and on vessels to tell us about this form of art that the Palaeolithic Age had already produced. Religious festivals and performances at funerals stimulated the artistic sensibility of the masses. Certain forms of poetry, like magical curses, songs for dancing and collective work, the religious and dramatized mystery play, the lament for the dead, the songs for a wedding or the initiation rites for boys and girls, the cradle and the children's songs, battle songs, and love songs too, appear in Indogermanic, Semito-Hamitic, Ural-Altaic, and early Sumerian language archaeology, and have therefore to be dated back to the Neolithicum, if not earlier.

Pictorial art showed all the forms which city cultures were to develop to such a high pitch later on. Exactly as in the case of primitive races today, taste, form, and technique changed with region, race, and social position. Plastic sense and sculpture, inspired by religion, advanced into Europe from the South, as far as we can see. The nations of the North, and of the region around the Arabian desert, originally had no pictorial representations for their gods. Idols and statues of animals migrated with the so-called "Bandkeramik" culture from the Danube into Central and Southern Germany. But they were still rare in the Nordic circle. Painting and the related arts of ornamentation and relief were, on the other hand, much more widely known and developed in Northern Europe and in Siberia than in the South. Articles made in the Neolithic North were heavy with ornament. The rock drawings of Scandinavia and neighbouring regions, and of North Africa, are famous, as are the naturalist or ornamental bone drawings of this time. There are paintings in colour too; and sketches in stone and bone which prepare the way for the classical art of relief in the later city civilizations. Some works of art of the Neolithic peasant stage and its continuations are quite close to the aesthetic ideals of modern expressionism and even surrealism. On the whole, the tendencies towards symbolism of the Old Stone Age gained now the upper hand. Naturalism was disregarded. Art was bound to the hierarchy of the day and portrayed scenic compositions of human and animal figures, the spiritual symbol of a new semi-collective age.

Instructive examples of monuments in creative art are spread over different regions and often show marked differences. There are notable examples in Egypt, Mesopotamia, Syria and Palestine, Asia Minor, Greece, the whole Near East, the whole Mediterranean Region, the Sahara, China, India and Polynesia, Scandinavia, France, England, Central Europe, Russia, and Siberia. All these remains of aesthetic interest were produced, from the economic point of view, by domestic industry or work in a customer's household. For the economic pattern of the time was not at all favourable to the principle *l'art pour l'art*, art as a form of spiritual relaxation with its main importance in itself. On the contrary, art had always a practical purpose in village civilizations, and usually was

considered as a magical and religious technique to improve the life of individuals and of society. It had, in the final instance, the same basis as in the Palaeolithic Age. There are, in our museums, amulets in the form of hooded heads, sexual symbols, and idols in the form of men and women, showing witchcraft aim. In other cases the existence of a practical purpose is not clearly understandable from the work of art itself, but becomes evident from ethnological parallels or from myths which the later city civilizations have preserved for us.

7

The Capitalist Mode of Production and the World Market

The continuous interaction of the development of production and the world market had already come out in the early stages of the rise of the capitalist social formation, and the determinant place in this interaction, moreover, was taken by capitalist industry. As a result of his thorough investigation of extensive historical data, Marx came to the conclusion that when in the 16th, and partially still in the 17th century the sudden expansion of commerce and emergence of a new world market overwhelmingly contributed to the fall of the old mode of production and the rise of capitalist production, this was accomplished conversely on the basis of the already existing capitalist mode of production.

The world market itself forms the basis for this mode of production. On the other hand, the immanent necessity of this mode of production to produce on an ever-enlarged scale tends to extend the world market continually, so that it is not commerce in this case which revolutionises industry, but industry which constantly revolutionises commerce.

Marxist political economy consequently looks upon the rise and development of the world capitalist market as a constituent of a broader process, i.e. the gradual conversion of capitalism into a universal socio-economic system. This methodological approach has enabled it to disclose the unequal character of the division of labour between countries that is inherent in this system, and the exploiter nature of commodity exchange and of capitalist international economic relations in general on the world market, based as

they are on private property in the instruments and means of production. The struggle for foreign markets has always been an urgent economic need for capitalism, since a striving to extend its sphere of influence without limit is inherently characteristic of it, in contrast to previously dominant modes of production. Because of the operation of objective economic laws, capitalist enterprise inevitably outgrows the boundaries of the community, region, or country, the economic exclusiveness and isolation of countries being broken down by commodity circulation itself.

As a result, capitalist production becomes inconceivable without foreign trade and world economic relations at a certain stage of its development, which in turn reflects growth of the social division of labour on an international scale. Marx stressed that capitalist production rests on the *value* or the transformation of the labour embodied in the product into social labour. But (his is only [possible] on the basis of foreign trade and of the world market. This is at, once the precondition and the result of capitalist production.

He expressed the same idea even more succinctly in the formula 'capitalist production does not exist at all without foreign commerce'.

All that in no way signifies, of course, that any capitalist enterprise indispensably works for an external market and sends what it produces beyond the local or national economy. At the same time it constantly experiences the effect of the already existing system of international business relations and is drawn into their orbit, if not directly, then indirectly. The capitalist, even the one who is wholly oriented on the home market, is forced to compare his own costs of production with world prices as well as with the market prices in his country. An influence of the same order has to be allowed for when an all-round appraisal is being made of the position of labour power in capitalist countries, including the spheres where its value depends wholly or predominantly on domestic factors of production.

Capitalism's international economic interconnections arose later, historically, than its domestic ones, the latter being original and basic. Their level and trend depended to a decisive degree on how far each nation developed its own productive forces, division of labour, and internal dealings. That, however, does not eliminate the undoubted fact that extension of both the home and foreign market under capitalism represents different aspects of a single process; the boundaries between them are very mobile and arbitrary. As the world market develops the problem of the limits of the home market of each country taken separately proves to lie very tightly bound up with that of extension of its external economic relations. Hence one of the essential principles of Marxian research into the development of capitalism in scope and depth is that, when analysing international relations of production (including the capitalist class's colonial policy), it is not important where the boundary between the home and foreign market is drawn; what is important is that capitalism cannot exist and develop without constantly expanding the sphere of its domination, without colonising new countries and drawing old non-capitalist countries into the whirlpool of world economy.

The rise of capitalism's international exploiter relations is inextricably linked with the forming of a system of world economy appropriate to them.

The watershed on this point between Marxian and capitalist science is clear and distinct. Although the creation of this system was completed on the whole under imperialism, it by no means follows that its present-day history can be separated from the preceding history of the development of the capitalist mode of production and that imperialist exploitation of some countries by others begins only with capitalism's transition from free enterprise to the domination of monopolies. Many of the features of the international division of labour typical of the monopoly stage had already arisen and begun to take on a stable, long-term character during the transition of capitalist society to the machine stage of production. They were not brought about by chance, transitional factors but were caused by deep-seated changes in both the home and external conditions of capitalist reproduction.

The analysis of international economic inter-relationships made in their day by the founders of Marxist political economy indicated that it was large-scale machine industry that prepared the ground for the subsequent formation of a world economy. Machine production, as Marx and Engels pithily put it, produced world history for the first time, insofar as it made all civilised nations and every individual member of them dependent for the satisfaction of their wants on the whole world, thus destroying the former natural exclusiveness of separate nations.

It is through the international division of labour on the world market that the tendency to bring countries together economically, constitutionally inherent in capitalism, is manifested. The kernel of this pattern, however, discovered by Marx, is that this tendency, while progressive in itself, led at the same time, because of capitalist methods and the forms of economic association of the different countries, to consolidation of the division of the world into industrially developed countries, and economically backward agrarian and primary commodity producing countries exploited by them. The antagonism between these two main groups of countries, which are opposite poles of the world capitalist economy, has become one of its main contradictions.

This contradiction has become specially acute in the stage of imperialism, and a very important factor in subsequent origin and development of the crisis of the colonial structure of the capitalist economy, a crisis that is irreversible for the monopoly capitalist class. The break-up of the colonial system after World War II, and the marked deepening of the internal contradictions of world capitalism, which are undermining the historically formed foundations of its system of unequal international economic relations, have put on today's agenda, with all urgency, the issue of eliminating the exploiter system of division of labour that completely dominated the world economy in the past.

It would be wrong to treat this task (which corresponds to the vital interests of all nations without exception) as the need to liquidate the progress already made in the internationalisation of social production. The anti-imperialist struggle for genuine national independence, just like the transition of countries to a higher socio-economic

system, does not eliminate the objective need to develop the social (including the international) division of labour, and leads to a change in and perfecting of the forms in which this essentially progressive process is manifested. It is self-evident, Marx wrote to Ludvig Kugelman on 11 July 1868, that this necessity of the distribution of social labour in definite proportions cannot possibly be done away with by a *particular jirir* of social production but can only change the mode of *its appearance*.

The underlying principles of a Marxist-Leninist analysis of the ways and causes of capital's internationalisation of social production have not lost their theoretical value over the years. The topicality of many of them, moreover, is growing, especially in today's situation of the coexistence and opposition of two world social systems and of the natural extension of economic and technical co-operation on the basis of mutual advantage and equal international division of labour.

In this situation capitalist ideologists are trying in every way to prove the 'erroneousness', in particular, or at least 'obsolescence' of the Marxist-Leninist political economy's analysis. Above all they endeavour to refute its conclusions about the irreversibility of tendencies toward a steady deepening of capitalism's internal contradictions. But the whole long history of capitalist economics, and of its mistakes, its constant wavering between one conception and another, its fallacies and failures, shows that theories that consider the international market relations of capitalism to be legitimate and that are abstracted from a socio-economic analysis of the production sphere, at best contain superficial analogues of the real trends of development. In the main they bring out the consequences but not the causal connections of this development. At the same time it would clearly be an oversimplification of things to consider that the theoretical findings of capitalist political economy in the sphere of the world economy's market relations are in general isolated from real analysis of the reproduction processes. In a number of cases they more or less adequately reflect important features of the backlash of these relations in the production sphere. And this effect determines many of the essential features of the capitalist economy at the various stages of its development.

In the last instance (Engels wrote to Conrad Schmidt on 27 October 1890) production is the decisive factor. But as soon as trade in products becomes independent of production proper, it has a movement of its own, which, although by and large governed by that of production, nevertheless in particulars and within this general dependence again follows laws of its own inherent in the nature of this new factor; this movement has phases of its own and in its turn reacts on the movement of production.

All this calls for close attention to research both into the ways and determinant tendencies in the genesis of the world market and into its place in the process of extended capitalist reproduction. Not only is such research of educational value historically; it is also very necessary for an analysis of current changes in the structure of the world capitalist economy. A systems analysis of its inner and external interconnections in today's international situation cannot be made sufficiently full without detailed allowance for the general patterns and long-term trends of development of these interconnections.

Lenin more than once stressed the need to study broad socio-economic phenomena in historical retrospect, since the most important thing if one is to approach this question scientifically is not to forget the underlying historical connection, to examine every question from the standpoint of how the given phenomenon arose in history and what were the principal stages in its development, and, from the standpoint of its development to examine what it has become today.

There were relatively broad trade and financial relations between separate countries and peoples, of course, in the socio-economic formations that preceded capitalism, but it was capitalism that first formed a world market which played a progressive, revolutionary role in the subsequent increase of mankind's productive forces and growth of its social production. Marx, attributing paramount importance to that, wrote that 'the specific task of bourgeois society is the establishment of a world market'. It was with the formation of a world market that the level of development of capitalism began to be marked by the degree of growth both of domestic and of international market relations.

There was a qualitatively new leap in the shaping of these relations associated with the vast increase in trade and colonial expansion of the young European capitalist class. This expansion became a decisive factor in the primitive accumulation of capital. The world market thus created important material preconditions for victory of the capitalist mode of production over the feudal.

At that time the merchant capital of European powers undoubtedly prevailed in international economic relations, but its dominance on the world market began to decline as the capitalist mode of production was consolidated in the countries that were the basis of its existence. The subordinating of merchant capital to industrial, natural to the new mode of production, was also manifested then in the realm of international trade. As a result the material conditions were laid for an ever increasing intensification of the action of the mechanism of the economic exploitation of all the countries of the world by a handful of 'advanced' countries. And the wider the gap in levels of labour productivity becomes between this handful of advanced countries and the economically backward ones, the greater is the exploitation suffered by the latter on the world market. This pattern was expressed with maximum clarity in the following formula of Marx:

The favoured country recovers more labour in exchange for less labour, although this difference, this excess is pocketed, as in any exchange between labour and capital, by a certain class.

In that way, in particular, the main antagonistic contradiction of the capitalist mode of production, i.e. that between its constantly increasing social character and the private capitalist forms of appropriating the results of social labour, thus found expression within the world market.

The putting of capitalism onto the rails of large-scale machine industry during the industrial revolution, which was completed in the second half of the nineteenth century, meant the beginning of a new, higher stage in the development of world economic relations. Large-scale industry in fact shaped the world market of the epoch of mature

capitalism. And its dependence on external economic links and the international division of labour in turn also grew immensely. As Karl Marx said in his day: Large-scale industry, detached from the national soil, depends entirely on the world market, on international exchange, on an international division of labour.

In this situation the antagonistic contradictions of the world market stemming from the basic economic contradiction of capitalist society, have become an ever more serious obstacle to a further rise of social production. One of the most convincing bits of evidence of that were the world economic crises that were converted into a constitutionally inalienable feature of the whole economic system of capitalism. Years of comparatively rapid industrial boom began to alternate with regular consistency with periods of stagnation and direct destruction of already created productive forces. This cyclicity, determined by the objective laws of capitalist production, was intensified in turn by industry's mounting dependence on the foreign market. It was their comprehensive study both of the internal, home factors of the cyclic development of capitalist business and of the international ones that enabled the authors of classical Marxist political economy to draw the conclusion, confirmed by the whole subsequent course of social development, that economic crises would remain, while capitalism existed, one of the incurable ills determining its historically transient character.

For more than a century and a half now, since 1825, capitalism has periodically suffered from cyclic crises. All the attempts of capitalist economists to work up prescriptions to cure this illness without eliminating its basis, namely its exploiter relations of production, have unfailingly proved unsound and bankrupt. The capitalists as a class have demonstrated more and more clearly with the course of time their incapacity to control the productive forces of society rationally not only within the context of the separate national economies but also and especially on the scale of the world economy.

The world cyclic crisis of 1974–75, the greatest since the war, which developed under an unprecedented sharpening of several of modern capitalism's most important world problems, viz. the monetary, energy, raw material, and food crises, was convincing evidence of that indisputable fact in our day. It swept almost all the main centres of capitalism simultaneously and in turn interrupted their economic development. During the crisis the monopoly capitalists as usual exerted immense efforts to shift its most serious consequences onto the working masses of their own countries and the peoples of the agrarian and primary producer periphery.

As capital internationalises society's productive powers the connection between the reproductive processes of the individual countries (including those that are the epicentres of world crises under the impact of the economy's cyclic movement) is becoming closer and closer. Therefore, when we are determining the objective patterns of capitalism's development, the question of the interaction of the national cycles within the context of a single world cycle (study of which was begun by Karl Marx) is posed in all its acuteness. We shall dwell in detail on the features of the cyclic development of today's

world capitalist economy later, here it is necessary to stress the following from the standpoint of the matter we are examining.

Despite the fact that the cyclic synchronousness of the movement of capitalist production and of the world market already has a history of 150 years, a line of refusing to recognise the general patterns of this movement still clearly dominates capitalist economics. When, however, the course of the world cycle is studied, its development is mainly considered as a kind of arithmetical sum total of the results of cyclic fluctuations within the national economies taken separately. As a result it comes about that the specific features of national development are, as it were, the sole decisive factor in the shaping of each world cycle. In that case the role of the system of international socio-economic connections and of the internationalisation of social production, which in essence convert these countries into links in a single economic chain, is patently underestimated.

Since the beginning of the 70s, it is true, because of the steady exacerbation of modern capitalism's international economic problems, an increasingly negative attitude toward such a 'narrowly nationalist' approach to analysis of the main world crisis processes (including its cyclic development) has been traceable in capitalist economic literature. Sometimes, while justly noting the close connection between these processes, and in particular between national overproduction crises, within the context of the capitalist economy, most capitalist economists as usual, however, prefer to identify their prime causes with the processes taking place at any definite moment in the sphere of the international market and political relations. They thus willy-nilly leave aside objective study of the system of international exploiter relations of production that underlies the deep internal contradictions of the world capitalist economy, which are inexorably undermining its main foundations.

The political dependence of most of the countries of the world on the main centres of capitalism has played an extremely important role in the long history of the building up of this system. The structure of its unequal, exploiter relations was also moulded by that dependence in the imperialist stage, the crisis of which developed with particular force in the second half of this century as a consequence of most colonial countries' winning of national independence.

FOOD MARKETING

Food marketing brings together the producer and the consumer. It is the chain of activities that brings food from "farm gate to plate." The marketing of even a single food product can be a complicated process involving many producers and companies. For example, fifty-six companies are involved in making one can of chicken noodle soup. These businesses include not only chicken and vegetable processors but also the companies that transport the ingredients and those who print labels and manufacture cans. The food marketing system is the largest direct and indirect nongovernment employer in the United States.

THE THREE HISTORICAL PHASES OF FOOD MARKETING

There are three historical phases of food marketing: the fragmentation phase (before 1870–1880), the unification phase (1880–1950), and the segmentation phase (1950 and later). In the fragmentation phase, the United States was divided into numerous geographic fragments because transporting food was expensive, leaving most production, distribution, and selling locally based. In the unification phase, distribution was made possible by railroads, coordination of sales forces was made possible by the telegraph and telephone, and product consistency was made possible by advances in manufacturing. This new distribution system was led by meat processors such as Armour and Swift in midwestern cities and by companies such as Heinz, Quaker Oats, Campbell Soup, and Coca-Cola, which sold their brands nationally. Advertising in print media and direct marketing through demonstrations at stores and public venues were among the prime marketing tools. The initial Crisco campaign, in 1911, was an example. In the segmentation phase (1950 and later) radio, television and internet advertising made it possible for a wider range of competing products to focus on different benefits and images and thus appeal to different demographic and psychographic markets. Distribution via the new national road system strengthened national brands.

Technical Processing in Agricultural Production

Above we have already had occasion to note that writers on agriculture, in classifying systems of farming according to the principal market product, assign the industrial or technical system of farming to a special category.

The essence of this system is that the agricultural product, before going into consumption (personal or productive), undergoes technical processing. The establishments which effect this processing either constitute part of the very farms on which the raw material is produced or belong to special industrialists who buy up the raw material from the peasant farmers.

From the standpoint of political economy the difference between these two types is unimportant. The growth of agricultural technical trades is extremely important as regards the development of capitalism. Firstly, this growth represents one of the forms of the development of commercial farming, and is, moreover, the form that shows most vividly the conversion of agriculture into a branch of industry of capitalist society.

Secondly, the development of the technical processing of agricultural produce is usually connected intimately with technical progress in agriculture: on the one hand, the very production of the raw material for processing often necessitates agricultural improvement (the planting of root-crops, for example); on the other hand, the waste products of the processing are frequently utilised in agriculture, thus increasing its effectiveness and restoring, at least in some measure, the equilibrium, the interdependence, between agriculture and industry, the disturbance of which constitutes one of the most profound contradictions of capitalism. We must accordingly now describe the development of technical agricultural trades in post-Reform Russia.

Distilling

Here we regard distilling only from the point of view of agriculture. Accordingly, there is no need for us to dwell on the rapid concentration of distilling in large plants (partly due to excise requirements), on the rapid progress of factory technique, with the consequent cheapening of production, and the increase in excise duties which has outstripped this cheapening of production and because of its excessive amount has retarded the growth of consumption and production.

Thus, over 9/10 of the distilleries (accounting for over 4/5 of the total output) are directly connected with agriculture. Being large capitalist enterprises, these establishments lend the same character to all the landlord farms on which they are set up (the distilleries belong almost without exception to landlords, mainly to members of the nobility).

The type of commercial farming under review is particularly developed in the central black-earth gubernias, in which are concentrated over 1/10 of the total number of distilleries in the Russian Empire (239 in 1896-97, of which 225 were agricultural and mixed), producing over a quarter of the total output of spirits (7,785,000 vedros in 1896-97, of which 6,828,000 at agricultural and at mixed distilleries). Thus in the area where labour-service predominates, the commercial character of agriculture most frequently (as compared with other areas) manifests itself in the distilling of vodka from grain and potatoes. Distilling from potatoes has undergone a particularly rapid development since the Reform, as may be seen from the following data relating to the whole of the Russian Empire:

Thus, with a general twofold increase in the quantity of crops distilled, the quantity of potatoes used increased about 15-fold. This fact strikingly corroborates the proposition established above that the enormous increase in the potato area and crop signifies the growth of precisely commercial and capitalist farming, along with improvement of agricultural technique, with the replacement of the three-field system by multi-field crop rotation, etc.

The area of the biggest development of distilling is also distinguished for the biggest (in the Russian gubernias, *i.e.*, not counting the Baltic and the western gubernias) net per-capita harvest of potatoes. Thus in the northern black-earth gubernias the figures for 1864-1866, 1870-1879 and 1883-1887 were 0.44, 0.62 and 0.60 chetverts respectively, whereas for the whole of European Russia (50 gubernias) the corresponding figures were 0.27, 0.43 and 0.44 chetverts. As far back as the beginning of the 80s the *Historico-Statistical Survey* noted that “the region marked by the greatest expansion of potato cultivation covers all the gubernias of the central and northern parts of the black-earth belt, the Volga and Transvolga gubernias and the central non-black-earth gubernias” (*loc. cit.*, p. 44). The expansion of potato cultivation by landlords and well-to-do peasants means an increase in the demand for hired labour; the cultivation of a dessiatine of potatoes absorbs much more labour than the cultivation of a dessiatine of cereals and the use of machinery in, for example, the central black-earth area is still very slight.

Thus, while the number of workers engaged in the distilling industry proper has decreased, the elimination of labour-service by the capitalist system of farming, with the cultivation of root-crops, has increased the demand for rural day labourers.

Beet-Sugar Production

The processing of sugar-beet is even more highly concentrated in big capitalist enterprises than distilling is, and is likewise an adjunct of the landlords' (mainly noblemen's) estates. The principal area of this industry is the south-western gubernias, and then the southern black-earth and central black-earth gubernias. The area under sugar-beet amounted in the 60s to about 100,000 dess., in the 70s to about 160,000 dess.; in 1886-1895 to 239,000 dess., in 1896-1898 to 369,000 dess., in 1900 to 478,778 dess., in 1901 to 528,076 dess. (*Torgovo-Promyshlennaya Gazeta*, 1901, No. 123), in 1905-06 to 483,272 dess. (*Vestnik Finansov*, 1906, No. 12).

Hence, in the period following the Reform the area cultivated has increased more than 5-fold. Incomparably more rapid has been the growth of the amount of sugar-beet harvested and processed: on an average the weight of sugar-beet processed in the Empire in the years 1860-1864 was 4.1 million berkovets; in 1870-1874—9.3 million; in 1875-1879—12.8 million; in 1890-1894—29.3 million; and in 1895-96 and 1897-98—35 million. The amount of processed sugar-beet has grown since the 60s more than 8-fold. Hence, there has been an enormous increase in the beet yield, *i.e.*, in labour productivity, on the big estates organised on capitalist lines.

The introduction of a root-plant like beet into the rotation is indissolubly linked with the transition to a more advanced system of cultivation, with improved tillage and cattle feed, etc. "The tillage of the soil for beetroot," we read in the *Historico-Statistical Survey*, "which, generally speaking, is rather complicated and difficult, has been brought to a high degree of perfection on many beet farms, especially in the south-western and Vistula gubernias. In different localities, various more or less improved implements and ploughs are used for tilling; in some cases even steam ploughing has been introduced".

This progress of large-scale capitalist farming gives rise to quite a considerable increase in the demand for agricultural wage-workers—regular and particularly day labourers—the employment of female and child labour being particularly extensive. Among the peasants of the neighbouring gubernias a special type of migration has arisen, known as migration "to sugar". It is estimated that the complete cultivation of a morg (= 2/3 dess.) of beet land requires 40 working days. The *Combined Material on the Position of the Rural Population* (published by Committee of Ministers) estimates that the cultivation of one dessiatine of beet land, when done by machine, requires 12, and when by hand 25, working days of males, not counting women and juveniles. Thus, the cultivation of the total beet area in Russia probably engages not less than 300,000 agricultural day labourers, men and women. But the increase in the number of dessiatines under beet is not enough to give a complete idea of the demand for hired labour, since

some jobs are paid for at so much per berkovets. Here, for example, is what we read in *Reports and Investigations of Handicraft Industry in Russia*.

“The female population both of the town, and of the uyezd” “think highly of work on the beet fields; in the autumn the cleaning of beets is paid at 10 kopeks per berkovets, and two women clean from six to ten berkovets a day, but some contract to work during the growing season as well, weeding and hoeing; in that case, for the full job, including digging and cleaning, they get 25 kopeks per berkovets of cleaned beets.” The conditions of the workers on the beet plantations are extremely bad. For instance, the *Vrachebnaya Khronika Kharkovskoi Gubernii* cites “a number of exceedingly deplorable facts about the conditions of those working on the red-beet plantations. Thus, the Zemstvo physician, Dr. Podolsky, of the village of Kotelva, Akhtyrka Uyezd, writes: ‘In the autumn typhus usually breaks out among young people employed on the red-beet plantations of the well-to-do peasants.

The sheds assigned for the workers’ leisure and sleeping quarters are kept by such planters in a very filthy condition; by the time the job ends the straw used for sleeping is literally converted into dung, for it is never changed: this becomes a breeding ground of infection. Typhus has had to be diagnosed immediately in the case of four or five patients brought in from one and the same plantation.’ In the opinion of this doctor, ‘most of the syphilis cases come from the red-beet plantations.’ Mr. Feinberg rightly asserts that ‘work on the plantations, which is no less injurious to the workers themselves and to the surrounding population than work in the factories, has particularly disastrous consequences, because large numbers of women and juveniles are engaged in it, and because the workers here are without the most elementary protection from society and the State’; in view of this, the author wholly supports the opinion expressed by Dr. Romanenko at the Seventh Congress of Doctors of Kharkov Gubernia that ‘in issuing compulsory regulations, consideration must also be given to the conditions of the workers on the beet plantations. These workers lack the most essential things; they live for months under the open sky and eat from a common bowl.’” Thus, the growth of beet cultivation has enormously increased the demand for rural workers, converting the neighbouring peasantry into a rural proletariat. The increase in the number of rural workers has been but slightly checked by the inconsiderable drop in the number of workers engaged in the beet-sugar industry proper.

Potato-Starch Production

From branches of technical production conducted exclusively on landlord farms let us pass to such as are more or less within the reach of the peasantry. These include, primarily, the processing of potatoes (partly also wheat and other cereals) into starch and treacle. Starch production has developed with particular rapidity in the post-Reform period owing to the enormous growth of the textile industry, which raises a demand for starch. The area covered by this branch of production is mainly the non-black-earth, the industrial, and, partly, the northern black-earth gubernias.

The *Historico-Statistical Survey* estimates that in the middle of the 60s, there were about 60 establishments with an output valued at about 270,000 rubles, while in 1880 there were 224 establishments with an output valued at 1,317,000 rubles. In 1890, according to the *Directory of Factories and Works* there were 192 establishments employing 3,418 workers, with an output valued at 1,760,000 rubles.

“In the past 25 years,” we read in the *Historico-Statistical Survey*, “the number of establishments engaged in starch production has increased 4 1/2 times and the total output 10 3/4 times; nevertheless, this productivity is far from covering the demand for starch”, as evidenced by the increased starch imports from abroad. Analysing the data for each gubernia, the *Historico-Statistical Survey* reaches the conclusion that our production of potato starch (unlike that of wheat-starch) is of an agricultural character, being concentrated in the hands of peasants and landlords. “Showing promise of extensive development” in the future, “it is even now furnishing our rural population with considerable advantages”.

We shall see in a moment who enjoys these advantages. But first let us note that two processes must be distinguished in the development of starch production: on the one hand, the appearance of new small factories and the growth of peasant production, and on the other, the concentration of production in large steam-powered factories. For instance, in 1890 there were 77 steam-powered factories, with 52% of the total number of workers and 60% of the total output concentrated in them. Of these works only 11 were established before 1870, 17 in the 70s, 45 in the 80s, and 2 in 1890.

To acquaint ourselves with the economy of peasant starch production, let us turn to local investigations. In Moscow Gubernia, in 1880-81, 43 villages in 4 uyezds engaged in starch making. The number of establishments was estimated at 130, employing 780 workers and having an output valued at not less than 137,000 rubles. The industry spread mainly after the Reform; its technique gradually improved and larger establishments were formed requiring more fixed capital and showing a higher productivity of labour. Hand graters were replaced by improved ones, then horse power appeared, and finally the *drum* was introduced, considerably improving and cheapening production. Here are data we have compiled from a house-to-house census of “handicraftsmen,” according to size of establishment: Thus we have here small capitalist establishments in which, as production expands, the employment of hired labour increases and the productivity of labour rises. These establishments bring the peasant bourgeoisie considerable profit, and also improve agricultural technique. But the situation of the workers in these workshops is very unsatisfactory, owing to the extremely insanitary working conditions and the long working day.

The peasants who own “grating” establishments farm under very favourable conditions. The planting of potatoes (on allotment, and chiefly on rented land) yields a considerably larger income than the planting of rye or oats. To enlarge their business the workshop owners rent a considerable amount of allotment land from the poor peasants. For example, in the village of Tsybino (Bronnitsy Uyezd), 18 owners of starch

workshops (out of 105 peasant families in the village) rent allotments from peasants who have left in search of employment, and also from horseless peasants, thus adding to their own 61 allotments 133 more, which they have rented; concentrated in their hands are a total of 194 allotments, *i.e.*, 44.5% of the total number of allotments in the village. “Exactly similar things,” we read in the *Returns*, “are met with in other villages where the starch industry is more or less developed” (*loc. cit.*, 42). The owners of the starch workshops have twice as much livestock as the other peasants: they average 3.5 horses and 3.4 cows per household, as against 1.5 horses and 1.7 cows among the local peasants in general. Of the 68 workshop owners (covered by the house-to-house census) 10 own purchased land, 22 rent non-allotment land and 23 rent allotment land. In short, these are typical representatives of the peasant bourgeoisie.

Exactly analogous relations are to be found in the starch making industry in the Yuryev Uyezd, Vladimir Gubernia. Here, too, the workshop owners carry on production mainly with the aid of wage-labour (out of 128 workers in 30 workshops, 86 are hired); and here, too, the workshop owners are far above the mass of the peasantry as far as stock-breeding and agriculture are concerned; they use potato pulp as feed for their cattle. Even real capitalist farmers emerge from among the peasants. Mr. Prugavin describes the farm of a peasant who owns a starch works (valued at about 1,500 rubles) employing 12 wage-workers. This peasant grows potatoes on his own farm, which he has enlarged by renting land. The crop rotation is seven-field and includes clover. For the farm work he employs from 7 to 8 workers, hired from spring to autumn (“from end to end”). The pulp is used as cattle feed, and the owner intends to use the waste water for his fields.

Mr. V. Prugavin assures us that this works enjoys “quite exceptional conditions.” Of course, in any capitalist society the rural bourgeoisie will always constitute a very small minority of the rural population, and in this sense will, if you like, be an “exception.” But this term will not eliminate the fact that in the starch-making area, as in all the other commercial farming areas in Russia, a class of rural entrepreneurs is being formed, who are organising capitalist agriculture.

Vegetable-Oil Production

The extraction of oil from linseed, hemp, sunflower and other seeds is also frequently an agricultural industry. One can gauge the development of vegetable-oil production in the post-Reform period from the fact that in 1864 the vegetable-oil output had an estimated value of 1,619,000 rubles, in 1879 of 6,486,000 rubles, and in 1890 of 12,232,000 rubles. In this branch of production, too, a double process of development is to be observed: on the one hand, small peasant (and sometimes also landlord) oil presses producing oil for sale are established in the villages. On the other hand, large steam-driven works develop, which concentrate production and oust the small establishments. Here we are interested solely in the agricultural processing of oil-bearing plants. “The owners of the hempseed oil presses,” we read in the *Historico-Statistical*

Survey, “belong to the well-to-do members of the peasantry”; they attach particular value to vegetable-oil production because it enables them to obtain excellent feed for their cattle (oil cake).

Mr. Prugavin (*loc. cit.*), noting the “extensive development of the production of linseed oil” in the Yuryev Uyezd, Vladimir Gubernia, states that the peasants derive “no little advantage” from it, that crop and stock raising is conducted on a far higher level by peasants who own oil presses than by the bulk of the peasantry and that some of the oil millers also resort to the hire of rural workers. The Perm handicraft census for 1894-95 also showed that crop raising is conducted on a much higher level by handicraft oil millers than by the bulk of the peasants (larger areas under crops, far more animals, better harvests, etc.), and that this improvement in cultivation is accompanied by the hiring of rural workers. In the post-Reform period in Voronezh Gubernia, there has been a particular development of the commercial cultivation of sunflower seed, which is crushed for oil in local presses. The area under sunflowers in Russia in the 70s was estimated at about 80,000 dess. (*Historico-Statistical Survey*, I), and in the 80s at about 136,000 dess., of which 2/3 belonged to peasants. “Since then, however, judging by certain data, the area under this plant has considerably increased, in some places by 100 per cent and even more” (*Productive Forces*, I, 37). “In the village of Alexeyevka alone” (Biryuch Uyezd, Voronezh Gubernia), we read in the *Historico-Statistical Survey*, Vol. II, “there are more than 40 oil presses, and Alexeyevka itself, solely owing to sunflowers, has prospered and grown from a wretched little hamlet into a rich town ship, with houses and shops roofed with sheet iron” (p. 41). How this wealth of the peasant bourgeoisie was reflected in the condition of the mass of the peasantry may be seen from the fact that in 1890, in the village of Alexeyevka, out of 2,273 families registered (13,386 persons of both sexes), 1,761 had no draught animals, 1,699 had no implements, 1,480 cultivated no land, and only 33 families did not engage in industries.

Tobacco Growing

In conclusion, let us make some brief observations on the development of tobacco growing. The average crop in Russia for the years 1863-1867 was 1,923,000 poods from 32,161 dess.; for 1872-1878 it was 2,783,000 poods from 46,425 dess.; for the 80s, it was 4 million poods from 50,000 dess. The number of plantations in the same periods was estimated at 75,000, 95,000 and 650,000 respectively, which evidently indicates a very considerable increase in the number of small cultivators drawn into this type of commercial farming. Tobacco growing requires a considerable number of workers. Among the types of agricultural migration note is therefore made of migration to tobacco plantations (particularly to the outer gubernias in the South, where the cultivation of tobacco has recently expanded with exceptional rapidity). Reference has already been made in publications to the fact that the workers on the tobacco plantations lead a very hard life. In the *Survey of Tobacco Growing in Russia*, there are very detailed and interesting data on tobacco growing as a branch of commercial farming. Mr. V. S.

Shcherbachov, describing tobacco growing in Malorossia, gives wonderfully precise information on three uyezds of Poltava Gubernia (Priluki, Lokhvitsa and Romny). This information, gathered by the author and arranged by the Bureau of Statistics, Poltava Gubernia Zemstvo Board, covers 25,089 peasant farms in the three uyezds that grow tobacco; they have 6,844 dessiatines under tobacco and 146,774 dessiatines under cereals. We see an enormous concentration of both the tobacco and the cereal area in the hands of the capitalist farms. Less than one-eighth of the farms (3,000 out of 25,000) hold more than half the area under cereals (74,000 dess. out of 147,000), with an average of nearly 25 dess. per farm. Almost half the area under tobacco (3,200 dess. out of 6,800) belongs to these farms, the average per farm being over 1 dessiatine, whereas for all the other groups the area under tobacco does not exceed one-to two-tenths of a dessiatine per household.

The owner of 7 dessiatines under tobacco must therefore have at least 14 workers; in other words, he must undoubtedly base his farm on wage-labour. Some grades of tobacco require not two but three seasonal workers per dessiatine, and day labourers in addition. In a word, we see quite clearly that the greater the degree to which agriculture becomes commercial, the more highly developed is its capitalist organisation.

The preponderance of small and tiny farms among the tobacco growers (11,997 farms out of 25,089 have *up to one-tenth* of a dessiatine planted) does not in the least refute the fact of the capitalist organisation of this branch of commercial agriculture; for this mass of tiny farms accounts for an insignificant share of the output (11,997, *i.e.*, nearly half the farms, have in all 522 dess. out of 6,844, or less than one-tenth). Nor do “average” figures, to which people so often confine themselves, provide a picture of the real situation (the average per farm is a little over 114 dessiatine under tobacco).

In some uyezds the development of capitalist agriculture and the concentration of production are still more marked. In the Lokhvitsa Uyezd, for example, 229 farms out of 5,957 each have 20 dessiatines and more under cereals. Their owners have 22,799 dess. under cereals out of a total of 44,751, *i.e.*, more than half. Each farmer has about 100 dess. under crops. Of the land under tobacco they have 1,126 dess. out of 2,003 dess. And if the farms are grouped according to area under tobacco, we have in this uyezd 132 farmers out of 5,957 with two and more dessiatines under tobacco. These 132 farmers have 1,441 dess. under tobacco out of 2,003, *i.e.*, 72% and more than *ten* dessiatines under tobacco per farm. At the other extreme of the same Lokhvitsa Uyezd we have 4,360 farms (out of 5,957) having up to one-tenth of a dessiatine each under tobacco, and altogether 133 dessiatines out of 2,003, *i.e.*, 6%. It goes without saying that the capitalist organisation of *production* is accompanied here by a very considerable development of *merchant's* capital and by all sorts of exploitation outside the sphere of production. The small tobacco growers have no drying sheds, are unable to give their tobacco time to ferment and to sell it (in 3 to 6 weeks) as a finished product. They sell the unfinished product at *half the price* to buyers-up, who very often plant tobacco themselves on rented land. The buyers-up “squeeze the small planters in every way” (p.

31 of cited publication). Commercial agriculture is commercial capitalist production: this relation can be clearly traced (if only one is able to select the proper methods) in this branch of agriculture too.

Agricultural Surpluses

Patrick Mulvany, Chairman of the UK Food Group, accused some governments, especially the Bush administration, of using GM food aid as a way to dispose of unwanted agricultural surpluses. The UN blamed food companies and accused them of violating human rights, calling on governments to regulate these profit-driven firms. It is true that the acceptance of biotechnology and genetically modified foods will also benefit rich research companies and could possibly benefit them more than consumers in underdeveloped nations.

Labelling

While some groups advocate the complete prohibition of GMOs, others call for mandatory labelling of genetically modified food or other products. Other controversies include the definition of patent and property pertaining to products of genetic engineering.

Underdeveloped Nations

Some groups believe that underdeveloped nations will not reap the benefits of biotechnology because they do not have easy access to these developments, cannot afford modern agricultural equipment, and certain aspects of the system revolving around intellectual property rights are unfair to undeveloped countries. For example, The CGIAR (Consultative Group of International Agricultural Research) is an aid and research organization that has been working to achieve sustainable food security and decrease poverty in undeveloped countries since its formation in 1971.

In an evaluation of CGIAR, the World Bank praised its efforts but suggested a shift to genetics research and productivity enhancement. This plan has several obstacles such as patents, commercial licenses, and the difficulty that third world countries have in accessing the international collection of genetic resources and other intellectual property rights that would educate them about modern technology. The International Treaty on Plant Genetic Resources for Food and Agriculture has attempted to remedy this problem, but results have been inconsistent. As a result, “orphan crops”, such as tef, millets, cowpeas, and indigenous plants, are important in the countries where they are grown, but receive little investment.

Private Investments

The development and implementation of policies designed to encourage private investments in research and marketing biotechnology that will meet the needs of poverty-stricken nations, increased research on other problems faced by poor nations, and joint efforts by the public and private sectors to ensure the efficient use of technology

developed by industrialized nations have been suggested. In addition, industrialized nations have not tested GM technology on tropical plants, focusing on those that grow in temperate climates, even though undeveloped nations and the people that need the extra food live primarily in tropical climates. Many European scientists are disturbed by the fact that political factors and ideology prevent unbiased assessment of the GM technology in some EU countries, with a negative effect on the whole community.

TRANSGENIC ORGANISMS

Another important controversy is the possibility of unforeseen local and global effects as a result of transgenic organisms proliferating. Some critics have raised the concern that conventionally-bred crop plants can be cross-pollinated (bred) from the pollen of modified plants. Pollen can be dispersed over large areas by wind, animals and insects. In 2007, the U.S. Department of Agriculture fined Scotts Miracle-Gro \$500,000 when modified genetic material from creeping bentgrass, a new golf-course grass Scotts had been testing, was found within close relatives of the same genus (*Agrostis*) as well as in native grasses up to 21 km (13 miles) away from the test sites, released when freshly cut grass was blown by the wind.

GM proponents point out that outcrossing, as this process is known, is not new. The same thing happens with any new open-pollinated crop variety—newly introduced traits can potentially cross out into neighbouring crop plants of the same species and, in some cases, to closely related wild relatives. Defenders of GM technology point out that each GM crop is assessed on a case-by-case basis to determine if there is any risk associated with the outcrossing of the GM trait into wild plant populations.

The fact that a GM plant may outcross with a related wild relative is not, in itself, a risk unless such an occurrence has negative consequences. If, for example, a herbicide resistance trait was to cross into a wild relative of a crop plant it can be predicted that this would not have any consequences except in areas where herbicides are sprayed, such as a farm. In such a setting the farmer can manage this risk by rotating herbicides. The European Union funds research programmes such as Co-Extra, that investigate options and technologies on the coexistence of GM and conventional farming. This also includes research on biological containment strategies and other measures, to prevent outcrossing and enable the implementation of coexistence.

If patented genes are outcrossed, even accidentally, to other commercial fields and a person deliberately selects the outcrossed plants for subsequent planting then the patent holder has the right to control the use of those crops. This was supported in Canadian law in the case of *Monsanto Canada Inc. v. Schmeiser*.

“TERMINATOR” AND “TRAITOR”

An often cited controversy is a “Technology Protection” technology dubbed ‘Terminator’. This yet-to-be-commercialized technology would allow the production

of first generation crops that would not generate seeds in the second generation because the plants yield sterile seeds. The patent for this so-called “terminator” gene technology is owned by *Delta and Pine Land Company* and the United States Department of Agriculture. Delta and Pine Land was bought by Monsanto Company in August 2006. Similarly, the hypothetical Trait-specific Genetic Use Restriction Technology, also known as ‘Traitor’ or ‘T-gut’, requires application of a chemical to genetically modified crops to reactivate engineered traits. This technology is intended both to limit the spread of genetically engineered plants, and to require farmers to pay yearly to reactivate the genetically engineered traits of their crops. Traitor is under development by companies including Monsanto and AstraZeneca.

In addition to the commercial protection of proprietary technology in self-pollinating crops such as soybean (a generally contentious issue), another purpose of the terminator gene is to prevent the escape of genetically modified traits from cross-pollinating crops into wild-type species by sterilizing any resultant hybrids. The terminator gene technology created a backlash amongst those who felt the technology would prevent reuse of seed by farmers growing such terminator varieties in the developing world and was ostensibly a means to exercise patent claims.

Use of the terminator technology would also prevent “volunteers”, or crops that grow from unharvested seed, a major concern that arose during the Starlink debacle. There are technologies evolving which contain the transgene by biological means and still can provide fertile seeds using fertility restorer functions. Such methods are being developed by several EU research programmes, among them Transcontainer and Co-Extra.

8

Economic and Trade Development

DEVELOPMENT OF ECONOMICS

FOUNDATIONS

The great revolution, from an economic standpoint, which followed the Palaeolithic Age with the beginnings of the Mesolithic and Neolithic periods, is clearly revealed by detailed economic analysis which has rarely been traced farther back than this new period by our economists. For it was then for the first time that the pattern of the so-called domestic economy of Professor Spiethoff began, and spread practically over the whole globe. Many archaeologists and ethnologists, on the other hand, have attempted to explain the various differences between the economic pattern of the Palaeolithic Age and that of the succeeding periods, and named as new and revolutionary moments in economic progress either such advances in the technique of labour as hoe agriculture, cattle economy, plough culture, the division of labour into agricultural, industrial, and trading professions, or new types of settlement which gradually developed to permanent villages. Furthermore progress could be traced in the improvements in technique which appeared in the hoe, plough, potter's wheel, cart, sledge, ship, and many other implements.

And so stress has been laid here, from the economic point of view of the historian, on labour alone as an economic factor. The refinement in technical methods of labour was used to explain the indisputable and complete revolution in the economic pattern

of the Neolithic, as opposed to the Palaeolithic Ages. In this point of view, there appear subconsciously ideas of progress and invention which are characteristic for the period between the French revolution and the beginnings of the First World War. In this antiquated pattern of the history of mankind, the technique and its inventions do not appear as contributory, but as leading factors in the development of the economy of man, as one can often read in the popular literature of the last 150 years also. In fact this structural change presents a difficulty to the historian, which is not to be easily circumvented. For a fissure appears between the development of the two periods according to our sources, which completely refutes any assumption of a gradual process of growth. In the Palaeolithic Age, we find (with some variations which are not to be considered in this general survey) a gradual and very intelligent refinement of technical working methods in man's fight for existence without the economic pattern of mankind being structurally altered. On that account Young Palaeolithic man held fast to the habits of the foodgathering age in a very conservative development of twenty millennia or so. Spiritually he sought to reconcile these limited possibilities with his own wishes, as best as he could. A striking stability, varied more on account of changes in the climate than by man's desire, lasted for a very long time. On the other hand, the almost 12,000 years which have elapsed between the innocent beginnings of the revolution and today have been characterized by continuous changes in economic patterns. Similarly with the appearance of the new pattern of behaviour at the commencement of this Mesolithic period there arose a sudden change in social relationships and in man's man made environment also. Looked at from the historical point of view, man has, from then until now, untiringly built new structures on old foundations, so that the Mesolithic and Neolithic Ages appear rather as the powerful nucleus of the greatest economic revolution of all time and are much more closely related to the subsequent periods of world history than to the slowly advancing Young Palaeolithic Age that preceded this change.

Little further use or advance was made in the methods of labour and technique of the Palaeolithic crafts after the dawn of the new Age. Even in the field of art the earlier forms of representation and expression disappear, as a rule, albeit from the point of view of the all too widespread theory of gradual technical refinement we should expect everywhere a continuous advance on the former lines. Instead new economic methods and techniques arose to replace the old, even in the case of materials which had been in use for many millennia like stone, so that experience had to be gained from quite primitive, and even not very clear beginnings, as if a new creation out of the unknown had begun.

All these facts make it likely, in my opinion, that another economic factor than that of human labour was revolutionized at the dawn of the new Age, and that the new technique was only subsequently created to meet the greater demands of this widened sphere of labour, the new technique giving nevertheless increasingly rich rewards. The economic factor of Nature was not a revolutionizing element at this time. The changes

in the climate of our globe during the Mesolithic and Neolithic Ages resulted in better conditions for man's existence in the temperate zones, it is true. The last Eurasian Ice Age was completely disappearing between 10000 and 3000 B.C., and this meant great changes in settlement and clothing methods. All this must be considered by the historian in his survey of this period. But the preceding changes during the nevertheless very static Palaeolithic Ages had been much more violent than now, several Ice Ages alternating with warm Interim Periods. The factor Nature can, under these circumstances, only be considered as an additional force influencing the development of our period to a certain extent.

Instead the main impetus towards human progress during the Mesolithic and Neolithic Ages was given by a different central force, that was to become important for the first time in man's history. It was the factor capital in its still relatively simple form of working property, that is, the creation of the first reliable way of transferring human work into something which gave rent for a longer time and even for the duration of generations. For the individual man and his society enormous additional labour forces came into existence, in this way, which greatly extended the possibilities of man's own labour. The effect as such of this creation of working property for which overwhelming evidence appears in our archaeological source material has long been known in prehistory research. But usually the scholars had not the necessary economic training to enable them to analyze their source material from an economist's point of view. The most important working property of the Mesolithic and Neolithic Ages differs in its form, as is only to be expected, to a great degree from that of the later town civilizations, but is none the less to be regarded as a genuine working capital.

It is alive. Human labour was either invested in plants which produced at least one harvest, or like trees would provide for generations. Or man's labour was invested in the domestication of cattle. This second form of working property was even more important for the progress of the Neolithic Age than the first, as has been seen by sociologists long ago. For, even more than the plantations, it not only saved man a considerable part of the labour which he had to expend in his search for food during the Hunter and Foodgatherer Ages, but enabled him, in addition, as his needs demanded, to make use of beings for many purposes which were physically much stronger than he himself.

Technique, an already well refined human gift, as we have seen from the Palaeolithic beginnings of man's advance, now had to take account of a new situation where human labour and additional labour force could be combined in exploiting Nature. An improved technique developed from this new starting point. For the second time we find a completely new beginning leading to a refined utilization of every possibility offered by the circumstances. Man began to cultivate certain plants or tend animals that varied with his tribe or the land at his disposal. His tools and methods of work underwent a continuous refining process.

Many different “oikos” civilizations influenced or superseded each other or combined, so that consequently each region developed the best economic style for its potential. This refinement over the whole world happened within a period relatively short in its history, that lasted only six thousand years or so. Very soon after the complete development of the new economic forms there followed, as early as 3600 B.C. or so, the first spiritual refinement of the new concepts of capital, in the money investing of the Ancient Orient, so that a continuous development in human culture from the beginnings of the Mesolithic and Neolithic Ages leads without a break to the situation of the present day.

TRADING DEVELOPMENT

As was the case with the economic pattern in many other spheres of the Neolithic village civilizations, the exchange of goods seems, after the end of the hunter and foodgatherer stage, to have reached an intensity and structural eminence which was very different from the primitive conditions of the Palaeolithic Age. It is true, the exchange of goods of the exclusively townless civilizations was still not developed as well as in the later city cultures, exactly as we shall have to observe for capital investment, division of labour, and social organization. For the city cultures possessed far larger and more powerful capital reserves than the peasant civilizations which enabled them to refine and develop their social pattern. But notwithstanding such considerations, the general standard of the exchange of goods in the Neolithic Age and the succeeding village civilizations was so high, that it becomes advisable, different from our treatment of the Palaeolithic Age, to outline systematically the author's concepts of the different heuristic forms in which it manifested itself. On this we must base our individual survey, if we want to present the whole picture of the period, many-sided as it is.

Under trade we understand, following Professor Ehrenberg's definition, that type of human production which forms part of the general exchange of goods, and tries to overcome any local scarcity of goods and individual needs for goods, which cannot be satisfied by production of an autark family unit. While the exchange of goods, in the Palaeolithic Age, had been mainly left to chance meetings, we find, in the Neolithic Age, a particular and planned form of production. Within the general phenomenon of trade we can distinguish, in accordance with the length of the way on which the goods were moved, internal trade or trade between neighbours, which has to do with one particular settlement only, and external or foreign trade, which may affect a large area. Trade between neighbours and external trade were basically different in the prehistoric village civilizations, much more than in city cultures, not only as far as the goods, which were moved, were concerned, but also when one considers the forms which the exchange of goods took. Modern primitive tribes show the same conditions. But we find a certain difference between internal and external trade even today, in our own society, although the internal trade of a modern state applies to much larger territories and populations than that of prehistoric times. The difference between sea trade, river

trade, and overland trade is important also, since each may have a different pattern owing to the regional conditions with which they are concerned.

If we further take into account the measure of valuation which is generally used, we can distinguish between exclusive barter, if there is no preference at all for a specialized measure of value and means of exchange; barter mitigated by preferences for certain means of valuation and exchange; and finally money trade, where there are used legally recognized and customary means of exchange and measures of valuation, for the territory and type of trade in question. When we discussed earlier the forms of capital investment of our period, we observed already that here all these three forms of trade were in existence, as soon as the planter, animal breeder, and plough cultures of the Neolithic Age developed. We even have definite proof of the existence of money trade. For there were now many different forms of money which had, in law, fixed conventional values, as can be made clear from our archaeological, etymological, and ethnological sources.

A further differentiation has to be stated, when we consider the economic position of the individuals in external trade, and this particular differentiation becomes of special importance, if we try to judge how refined the pattern of a given period actually was. In this connection we must distinguish between occasional trade, professional trade in association with other activities, and exclusively professional trade. The differences which developed in traditional customs connected with the exchange of goods are even more important for our survey of this period; and contribute to the understanding of trade in city civilizations too. Silent trade is age old and primitive. The contracting parties do not see each other personally. One party lays goods down and hides himself; the other group lays wares of similar value by their side, before retreating. The first party returns and adds or subtracts goods, as appears fair to him. The other contracting party may return also and continue the game, until finally both sides are satisfied. Long use has often established a rigid convention of gifts that must be exchanged, to shorten the proceedings. Silent trade must have developed everywhere where danger to life and limb for the foreigner made direct trade undesirable. It is, in fact, the oldest form of trade known. It combined business with acting and the playing of games, and therefore appealed strongly to the subconscious instincts of primitive peoples.

Pirate trade is another age old conventional trading game. One group of people breaks in by night and takes away the wares they wish to acquire, usually eatables. They destroy certain possessions of the owners, as tradition prescribes. But they leave a fair conventional payment in the place of the goods, they have taken away or destroyed. An even more humane and considerate form of trade develops from such games: group trading. Before breaking in, permission to do so is asked for by the group in a ceremonial manner, and they pay a certain indemnity for this permission as well as leaving certain wares in the house in exchange for those taken away. The highest primitive form of external trade between different tribes was finally market trade, which took place between members of contracting societies at a fixed time and on a fixed place. Here a definite price in the modern sense, and not a conventional price, was eventually fixed for the

different wares, as soon as their changing value could be established more securely by supply and demand. Usually a gratuity was paid by all trading parties to the human or divine owners of the ground where the market took place. This custom should finally give an important secondary impetus to the beginnings of city economy.

Supplementary to market trading developed the trade of the pedlar. A single individual or a small family went, in a wide circuit, from house to house and village to village. The pedlar or hawker of this period exchanged his wares for others which he could sell to somebody else further away. So that gradually he acquired a good selection of marketable goods; and if he was successful, he was able to acquire a plot of agricultural land in the end, and to become a small landowner of his own tribe. The pedlar's trade, the oldest form of specialized shop-keeping, was in existence as a profession, before the city cultures developed. Residential traders and merchants, as the Ancient Oriental *oikos* organizations of kings and temples, and finally the Greek coinage, made possible, could not develop in the peasant stage.

Inside the village communities themselves, the exchange of goods had a very different pattern from that of external trade. The oldest form of exchange was certainly the well known game of giving and taking presents. One partner gives the other some article which he desires as a present. The second partner replies in a similar way, but does not estimate the value of the gifts exactly, merely giving something which has approximately the same value. This exchange of presents was often part of a ritual; custom regulated its forms and fixed the days considered suitable for donations. Times of feasting and religious festivals were special occasions of this kind. In the South Seas whole groups of families formed a circle for giving and taking presents. A "chain of trade", a valuable and desirable piece of jewellery, was passed on from one household to the next. Only who might have this chain for the moment, had the permission to receive presents and to give others in exchange. Such customs of giving and taking presents soon spread among friendly neighbouring tribes also. They became a symbol of the host's good feeling towards his foreign guests, and developed to become an important form of external and internal trade. The same custom survives in our civilization today. Families who are friendly with another, exchange presents at birthdays, confirmations, weddings, the New Year, and religious festivals, more or less in the same forms as in the Neolithic Age. Even our presents are usually objects, as in prehistoric times, which the receiver would not buy for himself otherwise, or they are conventional gifts, like flowers in certain colours, certain cakes or buns, and similar objects.

A second important means to exchange goods in a peasant society was the division of communal property. The income from the grounds which were owned collectively by the tribe or clan, for example from forests, mountains, mines, quarries, and salt springs, or what the common mortal was allowed to eat from sacrifices to the tribal gods, and eventually plunder from war or piracy, were divided up among the members of the tribe in a conventional manner. For such divisions, tokens were occasionally used which entitled the bearer to his share and could fulfil the function of money when

the need arose. Even in Archaic and Classical Greece, and under the Roman Republic, the division of communal property in this way, was still important. The communal meals and arrangements for booty of the Hellenes, Italians, Israelites, Carthaginians, Hittites, Iranians, the original use which Athens made of public domain, the silver mines of Laurion, the state ownership of salt mines by Rome, all these facts show clearly how an early Indogermanic and Semitic usage and legal situation survived in the changed conditions of the city civilizations of the Iron Age. Even today, in many of our villages, which have widespread communal possessions, some of these age old customs may be found. Straightforward trade between neighbours on the basis of price valuation, which was governed by supply and demand, and was protected and regulated by the law of the land, can be assumed to have existed in internal trade, as we found it in primitive external trade, market trade, and pedlar's trade. On the other hand, internal trade in the forms of regular buying and selling in shops, which is closest to our own, was not found in townless civilizations. Here the influence of the later city cultures and their more intensive production was necessary to make internal trade, in the modern sense, possible.

The forms of trade, are well supported by ethnological, etymological, and archaeological evidence, and were certainly extant in the Neolithic Age. It is true that, for internal trade, there is very little help to be gained from archaeology, since it is difficult to estimate whether the articles we have found had circulated in trade from one neighbour to the next, or remained in the possession of the original producer. Pottery fragments have been discovered which appear to have been intentionally broken into equal pieces. They give reason to suppose that they may have been used as tokens for dividing communal property, or even as a local form of money. But the best information about early trade customs comes from etymology. We know, that there were several expressions for buying, selling, and haggling which were used by the early Indogermanic and Semito-Hamitic peoples before their division. In the Neolithic and Cuprolithic Ages, therefore, there was trade between neighbours and with pedlars in large territories of the Ancient World. The purchasing of the bride is not only an Indogermanic and Semito-Hamitic custom, but was, like the exchange of gifts at the time of the engagement, quite generally practised in peasant civilizations, as etymology, folklore, and ethnological sources prove. We know, too, of generally Indogermanic and Semito-Hamitic roots for "payment", "reward", and "price". Therefore, between the economically almost autark clans of the Indogermanic and Semito-Hamitic peoples, there must have been a continuous exchange of presents and goods, even with the help of buying and selling in some cases. The prices of such wares, it is true, were conventionally agreed upon, and did not alter with the supply and demand of the moment, if we follow ethnological parallels. Similarly the wages in kind remained fixed for long periods. A general exchange of goods as presents is proved by the laws of hospitality of the Indogermans, Semito-Hamites, and most other nations of the Ancient World. Division of communal property in times of peace, and of plunder in times of war, are well established in the earliest

preserved literature of most nations of the Old and New Worlds. We know rather more about trading over distances. For articles in our museums which have wandered widely from the places where they were made, or where the necessary raw materials for their production were to be found, show clearly how far they have travelled. Amber from the Baltic and the Black Sea is found, in the Neolithic Age, in graves of Eastern Germany, Poland, Galicia, in Scandinavia, France, Switzerland, and Britain. This trade seems to have developed, after tribes of the Baltic and North Sea regions had emigrated to the South and East. For there are characteristic wares from the amber import countries found near the amber coasts, which were obviously bartered in exchange. From Galicia, for example, a particular kind of stone axes returned to the Baltic, and there are other traces of mutual interchange between these two regions. On the other hand, the amber which has been found in the earliest levels of Assur, the later Assyrian capital, probably comes from the shores of the Black Sea, and was acquired by a combination of caravans with naval convoys, which was characteristic for the trade of the Ancient Oriental town civilizations from a very early date.

The counter-part of the Northern amber is the kawri shell which is only found in the regions of the Indian and Southern Pacific Oceans. The kawri advances, with other Southern shells, into the Europe of the "Bandceramic" civilization, and is often found in Central Germany from the Neolithic Age onwards. This luxury product reached the Baltic Sea in the Bronze Age. Egypt, the Sudan, the Middle East, and India exchanged their precious and semi-precious stones and their valuable jewellery materials very early. Probably perfumes and spices were included in this exchange. Cedar, elder, and cypress woods were exported to Egypt, where wood was scarce, at the latest between 4000 and 3000 B.C. Ivory has been found in the Central Europe of the Bandceramic, an analogy to the occurrence of the kawri shell on the same sites. Ivory was often imported to Southern Europe from Africa, Syria, and may occasionally even have come from India.

Stone knives and stone axes have been found in actual trader's depots. At Longueville near Metz, Erbesbuedesheim, Gonzenheim, and Rockenberg in Hesse, at Dorsheim and Ruebenach in the Rhine Province, Neckarsulm in Wuerttemberg, at Buesleben and Moenchspiffel in Thuringia, we find such depots in Central Europe. The Gonzenheim remains consist of a leather cover, which contained five stone axes ready for sale. Trader's depots from prehistoric times can be identified by the sites where they are found. They turn up regularly near old road-ways, but far away from early settlements, and in obvious hiding-places. They differ from the remains of prehistoric workshops, many of which have been excavated also, in that only tradable pieces are found together, and occasionally a few used-up tools which the trader had bought back and intended to repair or to renovate. We know depots of this kind from Italy, too, in great numbers. They are a definite proof for the existence of pedlar's trade in the Neolithic Age. This means, in addition, that there developed a special trading class, at this time, consisting probably of individual travelling traders, who were generally wandering craftsmen simultaneously,

or had some agricultural plots and animals at their disposal which were looked after by their wives and children. Trade specialists without any other economic activities were probably very rare, if existing at all, at this stage.

Flint was a desirable possession and was brought, by wandering traders, from the North, Schonen and Ruegen, to Hesse in Central Germany, and to the Southern and Eastern coasts of the Baltic by sea. Sometimes this type of flint even reached the pile dwellings of Lake Constance and Lake Neuenburg. On the other hand, the greater part of the wares made from Northern flint were kept within the Northern Circle, and does not seem to have crossed over, often, to the other spheres of civilization in Europe during the Neolithic Age. Some axes from the Bayerische Wald and the Fichtelgebirge reached Wuerttemberg. Thuringian stone axes reached Brandenburg and the territory North of the Lueneburg Heath. The hoes and the "*Schuhleistenkeile*" made of stone which the "Bandkeramik" produced, were exported to the North of Germany from the Danube, and were obviously considered as unusually efficient agricultural tools. Flint changed hands within the Western European regions also. The excellent flint of peculiar appearance from Grand Pressigny in Brittany near the Loire was exported to the whole of Northern France, Belgium, Burgundy, Savoy the West of Switzerland, Alsace, and Lake Constance region. Very fine pieces of jadeite were exported from France to Thuringia and Hungary, perhaps in the course of tribal migrations.

Another mineral which was much sought after at this time was obsidian. Obsidian from Hungary reached Lower Austria, Maehren and Bohemia, and advanced to Ratibor-Ostiz in Upper Silesia. There was obsidian from the Liparian islands, from Ischia, Sardinia, Corsica, Samnium, and Etruria, that covered the needs of the Italian tribes from Upper Italy to Sicily. There were obsidian and related stone materials in the Aegean, Abyssinia, Armenia, Arabia, and the Near East. They were exchanged against goods from Egypt and the whole Near East at a very early date. For instance, Armenia supplied the whole of Asia Minor and to a greater or lesser extent the lands bordering on this peninsula. Trading by sea, too, can be proved with some certainty for the Neolithic Age. We surveyed already economic and cultural connections around the Baltic and the North Sea, and between the coasts of the Mediterranean and the Atlantic which could not be understood, if oversea trade would have been uncommon in the Neolithic Age. Callais, a green coloured stone, of which we have not yet found the place of origin in the Neolithic Age, has only been discovered in places which are near to the Mediterranean or Atlantic coasts of France, Spain, and Portugal. The inland regions of France and the Pyrenean peninsula did not receive this material, as far as we can discover. In the case of this Callais stone there cannot have been trade overland; oversea trade is the logical conclusion.

So there grew up a considerable system of trading in the Neolithic Age. It is not, therefore, astonishing that word concepts for roads, like those for shipping, were quite familiar to the early Indogermanic and Semito-Hamitic peoples. Attempts to build a simple form of tracks for Neolithic trade with distant tribes, and shorter roadways, too,

for religious processions and the transport of materials from neighbouring regions, have been discovered in many parts of the Ancient World, although the evidence is not always conclusive. There were highways, paths through swamps, fords, and simple bridges, perhaps even roads with a wooden substructure. The external trade of this period included, almost exclusively, rare and valuable luxury articles and indispensable raw materials, as far as we can ascertain. The needs of every day could not at all be satisfied by trade, and therefore the individual households were restricted to a far reaching autarky. The economic sector in which trade appeared was very small, although it was important for the economic pattern of the period. Nevertheless, money trade and moderately professional trade are proved by archaeology, etymology, and ethnological parallels.

The prices of external trading in this period were usually, but probably not always, conventional. Ethnological parallels, etymology, and excavations show that a clearly defined and regular system of external trade over long distances was established in all prehistoric peasant civilizations. Many trading customs, still in modern use or known from folklore, can therefore be considered to be older than the city civilizations. The improved pattern of the forms of trade, which developed market and pedlar's trade in the Neolithic Age, produced clear and traditional legal terms for interchange and trade between different tribes, as etymology teaches us. The concepts for a "foreigner", even words for different types of aliens as legal expressions of constitutional terminology, existed in the early Indogermanic and Semito-Hamitic languages. There existed an international ritual of hospitality which protected the foreigner by giving him, for a time, privileged membership in a resident family. In the periods of Biblical Israel, pre-Mohammedan Arabia, Archaic Hellas, and primitive Italy, as well as in Germany, Russia, and India, tribes and nations preserved conservatively much of these customs. The laws of hospitality had similar importance for the life of the Neolithic peasant societies as is found among primitive agricultural peoples today. More than that, the Biblical ideas about *nokhri* and *ger*, the Greek provisions for *xenos* and metic, the Roman *clientela*, and other ancient customs, have influenced and modelled our own perceptions of hospitality and aliens' legislation up to the present day. Characteristic for the Neolithic laws and customs of hospitality is, from the economic point of view, an exchange of presents and services between the host and his guests. Examples for this type of external trade can be found in the Bible, in the Homeric poems and Greek mythology, in the Edda, among the Arabian Beduins, and in numerous experiences of explorers. The laws of hospitality even seem to have offered the legal foundation for establishing markets, which served quite a wide circle of tribes according to ethnological parallels, and could be connected with religious festivals of inter-tribal deities.

The economic pattern of the Neolithic civilizations was already so widely developed, that it was able to offer many potentialities on which the city civilizations were able to build, and to reach their highest points. We can see how important this Neolithic exchange of goods is for the modern world, when we consider the fact that modern research into

the pedigrees of domesticated animals and cultivated plants has been able to prove that many important species of grain plants and of animal breeds have followed the same prehistoric paths from China to Scandinavia and from Siberia to India and Abyssinia, as we described for exported articles like flint, obsidian, amber, kawri, stone axes, and other wares. Again there is no reason to despise the achievements of the townless civilizations.

A last form of early exchange of goods has still to be considered by us. It was as important for the economic pattern of the Neolithic Age as it is today, if we are honest with ourselves, and entailed a mighty traffic of possessions and services. Warfare, the brutal suppression of man by man, of class by class, of sex by sex, of age group by age group, of one tribe by another, are as inseparable from the life of prehistoric village civilization as they are implied by the organization of even very humanized town civilizations up to the present day. In the early period of human history the legalized transgressions of the Ten Commandments were made more often on the basis of religious than purely materialist reasoning. Perhaps one felt already that human nature and society was permanently poisoned and corrupted in this way, and tried to secure divine help to balance the dark sides of human nature. The Indogermans, the Semito-Hamites, and the Ural-Altaians, typical warrior races, often looked upon war as the normal state of affairs. Peace had to be bought, so that even a halt in warlike activities had an economic significance. Robbery by land and piracy on the sea brought honour and power. The heroic lays resounded with the exploits of gallant adventurers. Legends and myths in every tribe told of mighty deeds. The earlier parts of the Bible, Homer, and the poems of Archaic Hellas, early Iranian, Hindoo, Germanic, Celtic, Russian, Arabic, Finnish, Hungarian, Mongolian, Chinese, and Japanese literature abound with such stories. Later interpretation has often humanized the details. The importance of plunder and warfare in the life of these prehistoric times is revealed by etymology also. There is a cluster of expressions for waging war, for killing and robbery in many languages, especially those of patriarchal societies. Feuds were waged then, as now among boys and primitive tribes. There was, further, vengeance for blood guilt, punishment, ritual forms of duel between equals which took place in public on a predestined place. There were expeditions with the purpose of killing, robbery, plunder, and piracy, and the comparatively peaceful forms of mutual and collective insulting and sneering as inter-tribal duel and sport. Activities of this kind might continue, between two tribes, incessantly for a long time, without any appreciable ground being gained by either side.

The economic effects of such wars and fights can be shown by excavations, but can only be traced in detail for the somewhat later periods of village civilization, where they are recorded in the written documents produced by the neighbouring city civilizations. But if we examine the early records of Egyptian, Hittite, Mesopotamian, Indian, and Chinese history, we can see how mighty was the power wielded by the Indogermanic, Semito-Hamitic, and Ural-Altaic peoples, as early as between 4000 and

3000 B.C., how many countries had been conquered and settled by them, how many tribes were subdued or compelled to pay tribute. And so we can understand the importance of warfare for the exchange of goods and services in the Neolithic and Cuprolithic Ages, especially where the successful armies were able to divide the spoils among themselves, and this settlement attained legal permanency. The transfer of many plant forms has been connected with prehistoric wars of conquest, and not with peaceful trade, by Professor Netolitzky. He considers it likely that these plants were originally moved through conquest and tribal migrations, until a regular exchange of plants by trade could be built up. We have many comparable examples for positive agricultural changes owing to wars from historic times which support Professor Netolitzky's theory. Several varieties of millet, barley, wheat, and rye which were not native to Europe, may have travelled from Bucharra, in this way, to the Swiss lake dwellings. Cultivated millet from India spread to Africa, Egypt, and Italy. The einkorn, native to the Balkans, reached the coasts of the Atlantic. Emmer from Syria advanced to Babylonia, Iran, Egypt, Lake Constance, and even to Denmark. Oats travelled from Central Europe to Asia. The cultivation of the poppy which, again, seems to have begun in Central Europe by the lake dwellers, travelled right across two continents to China. There are certain breeds of dogs, horses, sheep, goats, and camels, which have been similarly spread by warlike migrations over the whole world. The impetus given frequently to the progress of mankind, and especially to the blessings of peace, by warfare is, indeed, one of the riddles of the philosophy of history.

ECONOMIC AND TRADE DEVELOPMENT

Adam Smith in his *Wealth of Nations* broke with the mercantilist tradition when he argued that free trade is the best policy for all trading countries. He based his argument of the gains from trade on absolute cost differences. Each country should specialise in the commodities which it could produce more efficiently than other countries and import commodities which it would produce less efficiently. Ricardo was later to widen the scope of the analysis to take account of comparative cost differences. Even if a country had an absolute disadvantage in the production of both commodities with respect to another country, mutually advantageous trade could still take place. International specialisation of factors of production would result in an overall increase in output which could be shared by all trading countries.

Specialisation which leads to real income gains can be represented by movements along a production frontier. This chapter also allows for the possibility that trade may enable a country to shift its production frontier outwards, so increasing the productive capacity of the economy. Trade can be the means by which countries gain increased access to enterprise, to physical and financial capital, and to new ideas and new technologies. All these things have the effect of increasing productive capacity. Trade in this instance is taking on a dynamic role. It is functioning as an engine of growth.

It is a significant point, however, that though mainstream economists tend to be unanimous in accepting real income gains from specialisation and exchange as in the pure theory of trade, they are much more divided on the supposed dynamic benefits of trade. The pendulum swings one way and then another. At one point of time the majority of economists may appear to believe in the likelihood of significant dynamic gains from trade. At other times they are much more sceptical.

‘LET GOODS BE HOMESPUN’

In the 1930s Keynes, well known previously for his staunch support of laissez-faire trade policies, switched his allegiance to protectionism, not only to promote domestic employment to counteract the slump in the UK, but also for what he saw as the advantages of self-sufficiency for economic growth. Some of his ideas would fit neatly today into the anti-globalisation movement. For example:

I become doubtful whether the economic cost of self-sufficiency is great enough to outweigh the other advantages of gradually bringing the producer and consumer within the ambit of the same national economic and financial organisation... as wealth increases, both primary and manufactured products play a smaller relative part in the national economy compared with houses, personal services and local amenities which are not the subject of international exchange... I sympathise with those who would minimise economic entanglement between nations. Ideas, knowledge, art, hospitality, travel - these are things which should of their nature be international. But let goods be homespun wherever it is reasonably and conveniently possible. (J. M. Keynes, *Collected Writings*)

The World Bank and dynamic benefits of trade

The *World Bank Report 2000* lists a number of dynamic benefits of trade which are particularly important for developing countries.

The World Bank believes that these benefits are ‘much larger’ than the static gains from trade. They accrue in the long run as economies adjust to technical innovations, new production structures and new patterns of consumption.

- Trade has a powerful ‘disciplinary’ effect on domestic firms. It *forces them to become more efficient*, to bring prices closer to marginal costs, thus reducing the distortions which would otherwise be created by monopolistic structures.
- Trade provides firms with access to *up-to-date capital equipment* and high-quality intermediate inputs. This enables firms to raise their productivity.
- Productivity levels rise when firms are exposed to the ‘*best practices*’ of overseas competitors. Demanding international clients force firms to re-engineer their products once they begin to sell their goods and services abroad.
- The dynamic benefits of trade set off a *cumulative chain of events* which tends to concentrate economic activity in a city or a region. For example, firms producing intermediate inputs are attracted to cities or regions producing

exports. These intermediate inputs will be tailored specifically to the needs of the export industries. This leads in a cumulative fashion to higher productivity in the booming city or region.

If we now bring together early classical ideas and the latest World Bank pronouncements, it seems that the sources of dynamic gains from trade are as follows:

- Trade *widens the extent of the market* and encourages businesses to innovate, make greater use of capital equipment and enjoy economies of scale.
- Trade has an important *educative effect*. It is the means whereby skills and technology are transferred from one economy to another. It suggests new goods to consumers and inculcates new tastes.
- Trade *promotes international competition*. It inspires entrepreneurs to look for new ways of reducing costs. It is also a most effective way of putting inefficient monopolies out of business.
- A combination of externalities and increasing returns, operating in a cumulative fashion, means that trade can often operate as an engine of growth in a particular *location*. Producers tend to operate more efficiently in proximity to a dense network of customers, suppliers, employees and information.

Any point inside the production possibility curve, at u for example, represents some surplus productive capacity in the economy.

The economy could move out to the production frontier, producing and consuming more of both goods x and y .

The development economist Myint pointed out that in the writings of Adam Smith can be found the suggestion that trade can provide a 'vent' or outlet for productive capacity which may be in surplus before a country is opened up to trade.

By opening a more extensive market for whatever part of the produce of their labour may exceed the home consumption, [trade] encourages [countries] to improve productive powers, and augment annual produce to the utmost, and thereby increase the real revenue and wealth of society. (*Wealth of Nations*, book I, 1776)

According to Myint, vent for surplus is another neglected element in classical economics. He draws a contrast here between Ricardo and Smith. For Ricardo, the gains from trade were associated with a reallocation of fully employed resources. In terms of Figure, this means a movement along the production possibility curve nn' . For Smith, trade provided a vent for labour and natural resources that were in surplus before trade. This meant that exports could increase, and this need not be at the expense of output for domestic consumption. In effect the economy is moving out from a point such as u within the production frontier, to the production frontier where all resources are fully employed.

'Vent for surplus' is a uniquely interesting way of approaching the question of the dynamic gains from trade, because it requires us to think very carefully about the relationship between trade and growth, from both a theoretical and an empirical (real-world) standpoint.

9

World Energy Resources and Consumption

The world has a number of energy resources which provide us with the ability to perform work (a narrowly-defined measure of the ability to effect physical change). Energy resources range from fossil fuels and nuclear fuels to renewable energy such as wind, hydro-and solar energy. When this energy is drawn upon to do work, and is converted to a less useful form, energy consumption is said to have occurred.

WORLD ENERGY CONSUMPTION

In 2004 the worldwide energy consumption of the human race was estimated as 15 TW (TW=10¹² Watts) by the United States Energy Information Administration. This is equivalent to 0.5 ZJ (ZJ =10²¹J) per year. There is at least 10 per cent uncertainty in this number because the underlying data is continuously changing and not all of the world's economies track their energy consumption with the same rigour. Consumption by fuel type Eighty-seven per cent of the world's energy is supplied by fossil fuel. The 15 TW total energy consumption of 2004 was divided as follows.

Ever since the advent of the industrial revolution, the worldwide energy consumption has been growing steadily. In 1890 the consumption of fossil fuels roughly equaled the amount of bio mass fuel burned by households and industry. In 1900 global energy consumption equaled 0.7 TW. The twentieth century saw a rapid twenty fold increase in fossil fuels. Between 1980 and 2004 the worldwide annual growth rate has been 2 per cent. The most significant growth of energy consumption is currently taking place

in China which has been growing at 5.5 per cent over the last 25 years. Its 1.3 billion people are currently consuming energy at a rate of 2 KW per person. The growth in energy consumption over the last quarter century.

	TW=10¹²	ZJ=10²¹J
	Watt	per year
Oil	5.6	0.18
Gas	3.5	0.11
Coal	3.8	0.12
Hydroelectric	0.9	0.03
Nuclear	0.9	0.03
Geothermal, Wind, Solar, Wood	0.2	0.006

Over the last twenty years Fossil fuels have continued to grow and increased their share of the energy supply, coal has been growing the fastest. Nuclear and hydro electric energy have stagnated due to environmental concerns. With the exception of France no major western country has ordered a new nuclear power plant since 1980.

In 2004 renewable energy supplied around 7 per cent of the world's energy consumption. Most (87 per cent) of the renewable energy was hydroelectric power. Solar and wind power provided 4 and 65 GW (GW=109 Watt) respectively. Consumption by country energy consumption broadly tracks with gross national product, although there is a significant difference between the consumption levels of the US with 11.4 KW per person, and Japan and Germany's with 6 KW per person. In developing countries such as India the per person energy use is closer to 0.5 KW. The per capita energy consumption against the GDP per person for all countries with more than 20 million inhabitants, this covers 95 per cent of the world population. Canada has the highest energy consumption per capita whereas the lowest energy consumption takes place in the third world.

Energy Consumption by Sector

Transportation, industry, residential and commercial (offices and shops) each consume approximately equal shares of the total 15 TW. A third of the world's energy is used to produce electricity. In 2005, global electricity consumption equaled 2 TW. The energy used to generate 2 TW of electricity is approximately 5 TW, as the efficiency of a typical existing power plant is around 38 per cent. The new generation of gas fired plants reaches a substantially higher efficiency of 55 per cent. Coal is the most popular fuel for the world's electricity plants.

World energy resources Fossil fuel resources. Despite several voices predicting the imminent decline of fossil fuels, there are still significant reserves of all the traditional energy components. Remaining reserves of conventional fossil fuels are estimated as:

Coal	290 ZJ
Oil	57 ZJ
Gas	30 ZJ

Again significant uncertainty exists for these numbers. This uncertainty is not surprising, since the estimation for the remaining fossil fuels on the planets depends on a detailed understanding of the earth crust. This understanding is still less than perfect. While modern drilling technology makes it possible to drill wells in up to 3000 meters of water to verify the exact composition of the geology, half of the ocean has depth exceeding 3000 meters, leaving about a third of the planet beyond the reach of detailed analysis.

The make up of the estimated 57 ZJ of oil on earth. The estimations for the world's oil reserves vary from low of 8 ZJ, consisting of currently proven and recoverable reserves, to a maximum of 110 ZJ consisting of available, but not necessary recoverable reserves, and including optimistic estimates for unconventional sources such as tar sands and oil shale. Both estimates provide oil for the foreseeable future at current oil consumption rate of 0.18 ZJ per year. There is a broad consensus among scientists that we are not close to running out of fossil fuels.

Coal is especially abundant and by itself can sustain the current energy consumption of the entire planet for the next 600 years. Renewable energy resources political and/or environmental considerations might move the world's energy consumption away from fossil fuels. For example, the European Commission has proposed in its Renewable Energy Roadmap, a binding target of increasing the level of renewable energy in the EU's overall mix from less than 7 per cent today to 20 per cent by 2020. In 2005 the Swedish government announced their intention to become the first country to break their dependence on oil and other 'fossil raw materials' by 2020. Renewable energy sources are even larger than the traditional fossil fuels and in theory can easily supply our energy needs. The sunlight interacts with the planet. There is 89,000 TW of solar energy that falls on the planet's surface. In other words, we need to capture less than 0.02 per cent of the available solar energy to meet our current energy needs. The available wind energy estimates range from 300 to 370 TW. Using the lower estimate, just 5 per cent of the available wind energy would supply the current worldwide energy needs. Most of this wind energy is available over open ocean. Not only does the ocean cover 71 per cent of the planet, wind also tends to blow stronger over open water because there are less obstructions. The above percentages for solar and wind energy ignore the formidable challenges of energy distribution and storage that need to be solved to overcome the intermittent and seasonal variations of these energy sources. The numbers do however illustrate that it is technically possible to move away from fossil fuels should society decide to make the necessary investment. The required investment to change energy sources is enormous. It is estimated that the cost to replace the global infrastructure for liquid transportation fuels alone is \$3-5 trillion.

ALTERNATIVE ENERGY PATHS

Whether a significant investment in renewable energy is wise or not, is currently the subject of much debate. Trials are taking place in several countries; Japan and Germany

have started to make some investments in solar energy, making them the largest consumers of photovoltaic cells in the world despite their unfavourable geographic location.

Denmark and Germany have installed 3 GW and 17 GW of wind power respectively. In 2005 wind generated 18.5 per cent of all the electricity in Denmark. Brazil has made a significant investment in ethanol fuel which supplies a significant part of the transportation fuel in that country. Starting in 1965, France made large investments in nuclear power and to this date three quarters of their electricity comes from their nuclear pressure water reactor plants. In the 21st century some of these different energy paths might become more mainstream and start replacing the ubiquitous fossil fuels.

EARTH'S ENERGY BUDGET

A schematic representation of the energy exchanges between the Earth's surface, the Earth's atmosphere, and outer space. The recycling of energy between the Earth's surface and the atmosphere is characteristic of the greenhouse effect. Note that the total energy entering each level is equal to the energy leaving that level as should be expected for a system in balance. The Earth can be considered as a physical system with an energy budget that includes all gains of incoming energy and all losses of outgoing energy. The planet is approximately in equilibrium, so the sum of the gains should be approximately equal to the sum of the losses.

Note: Although the term "energy budget" is widely used, the flow of energy in and out of the Earth is actually measured in units of power (watts), not units of energy (joules). Therefore, "power budget" would be a more accurate term.

THE POWER BUDGET

The total flux of power entering the Earth's atmosphere is estimated at 174 petawatts. This consists of:

- Solar radiation (99.978 per cent, or nearly 174 petawatts; or about 340 W m^{-2})
 - This is equal to the product of the solar constant, about 1366 watts per square metre, and the area of the Earth's disc as seen from the Sun, about 1.28×10^{14} square metres, averaged over the Earth's surface, which is four times larger. The solar flux averaged over just the sunlit half of the earth's surface is about 680 W m^{-2}
- Geothermal energy (0.013 per cent, or about 23 terawatts; or about 0.045 W m^{-2})
 - This is produced by stored heat and heat produced by radioactive decay leaking out of the Earth's interior.
- Tidal energy (0.002 per cent, or about 3 terawatts; or about 0.0059 W m^{-2})
 - This is produced by the interaction of the Earth's mass with the gravitational fields of other bodies such as the Moon and Sun.

- Waste heat from fossil fuel consumption (about 0.007 per cent, or about 13 terawatts; or about 0.025 W m^{-2}).

Note that the solar constant varies (by approximately 0.1 per cent over a solar cycle); and is not known absolutely to within better than about one watt per square metre. Hence the geothermal and tidal contributions are less than the uncertainty in the solar power.

Outgoing Power

An average albedo (reflectivity) of the Earth is about 0.3, which means that 30 per cent of the incident solar energy is reflected back into space, while 70 per cent is absorbed by the Earth and reradiated as infrared. The planet's albedo varies from month to month, but 0.3 is the average figure. It also varies very strongly spatially: ice sheets have a high albedo, oceans low.

The contributions from geothermal and tidal power sources are so small that they are omitted from the following calculations.

The 30 per cent reflected energy consists of:

- 6 per cent reflected from the atmosphere
- 20 per cent reflected from clouds
- 4 per cent reflected from the ground (including land, water and ice)

All of the 70 per cent absorbed energy is eventually reradiated:

- 64 per cent by the clouds and atmosphere
- 6 per cent by the ground

The same 70 per cent of absorbed energy can be split this way:

- 51 per cent absorbed by land and water, then emerging in the following ways:
 - 23 per cent transferred back into the atmosphere as latent heat by the evaporation of water
 - 7 per cent transferred back into the atmosphere by heated rising air
 - 6 per cent radiated directly into space
 - 15 per cent transferred into the atmosphere by radiation, then reradiated into space, and
- 19 per cent absorbed by the atmosphere, including:
 - 16 per cent reradiated back into space
 - 3 per cent transferred to clouds, from where it is radiated back into space

Anthropogenic Modification

Emission of greenhouse gases, and other factors (such as land-use changes), modify the energy budget slightly but significantly. The IPCC provides an estimate of these forcing, insofar as they are known. The largest and best known are from the well-mixed greenhouse gases (CO_2 , CH_4 , halocarbons, etc.), totalling an increase in forcing of 2.4 W m^{-2} relative to 1750. These are less than 1 per cent of the solar input, but contributes to the observed increase in atmospheric and oceanic temperature.

Electricity Generation

Electricity generation is the first process in the delivery of electricity to consumers. The other processes are electric power transmission and electricity distribution.

The importance of dependable electricity generation, transmission and distribution was revealed when it became apparent that electricity was useful for providing heat, light and power for human needs. Centralised power generation became possible when it was recognised that alternating current electric power lines can transport electricity at low costs across great distances by taking advantage of the ability to transform the voltage using power transformers. Electricity has been generated for the purpose of powering human technologies for at least 120 years from various sources of energy. The first power plants were run on wood, while today we rely mainly on petroleum, natural gas, coal, hydroelectric and nuclear power and a small amount from hydrogen, solar energy, tidal harnesses, wind generators, and geothermal sources.

Electricity Demand

The demand for electricity can be met in two different ways. The primary method thus far has been for public or private utilities to construct large scale centralised projects to generate and transmit the electricity required to fuel economies. Many of these projects have caused unpleasant environmental effects such as air or radiation pollution and the flooding of large areas of land.

Distributed generation creates power on a smaller scale at locations throughout the electricity network. Often these sites generate electricity as a by-product of other industrial processes such as using gas from landfills to drive turbines.

RIGHTS OVER FOREST RESOURCES

The tribal people are facing serious problems with regard to utilisation and rights over forest resources. Due to the increasing pressure on forests by various interest groups, there is a corresponding pressure on the tribals to reduce their dependency on forests. This is creating serious situations of conflict, as tribal life is symbiotic with land and forests and their livelihood and culture are based on their relationship with the natural wealth around them. The tribals are being harassed for using forestlands and being evicted in many places. Such reports have come from places like Khammam, Visakhapatnam, Vizianagaram, Adilabad, Srisailam and other places.

In Khammam in one particular village, the forest and police departments allegedly branded the tribals on their shoulders as an indication that they were destroying the forests. There are ambiguities in forest-revenue land demarcation. In some places like Nellore district, a lot of land on Velugonda hills is indicated as 'poramboku' in revenue records and as RF as per the forest department. These lands do not have any forest growth. The FD is taking up palm oil plantation in these RF lands (which is not a forest species). However, landless tribals are booked in criminal cases or prohibited from using these lands for agricultural cultivation. These lands should be given to tribals

with pattas for cultivation. In some areas like in Visakhapatnam district, the lack of clear forest boundaries is making tribals vulnerable to the exploitation of both the forest and revenue departments. A joint survey and demarcation of boundaries by both departments should be immediately taken up to arrest these conflicts. Such Joint surveys need to involve the villagers at various levels. The Forest-Revenue Boundary dispute is a perpetual problem in Adilabad and Warangal districts, leading to booking of cases by the Forest department and tension in these tribal villages. In Visakhapatnam Agency and Nallamala areas, there is the unique problem of Enclosure Villages. There were many tribal villages that were not enumerated in the forest surveys. Due to such sheer negligence, the villages were not given revenue status and to this day, they do not have pattas for their lands. They face constant harassment from local forest officials, as they do not possess land records. Recognising these enclosure villages and issuing pattas to tribals should immediately resolve this problem. In Buttapur (Adilabad district) and in Nellore Dt. (Yanadis), the tribals were given lands decades ago under the social forestry scheme and are cultivating there. But due to lack of pattas, they are being harassed by the police and forest departments and also do not have access to bank loans as they cannot prove their ownership. These tribals have to be given pattas as promised.

Attacks on tribals, their properties and livestock by wildlife are not compensated by the Forest Department. Several cases are pending where tribals have been either killed or disabled and yet have not received any monetary compensation as due to them under the Wildlife Act. The tribals should not be prohibited from entering the forests to collect NTFP for their domestic requirements, like firewood, medicinal herbs, food, or agricultural and housing material. The tribals should not be defined as 'encroachers', as is being projected by the forest department and also as indicated in the Circular of IG Forests, MoEF dated 3rd May 2002. In districts like Visakhapatnam, East Godavari and Vizianagaram, where there is high prevalence of podu cultivation, the tribals are facing threats of eviction from the forest department.

There was a notification issued by the government of A.P in 1987 ordering that pre-1980 settlements will not be evicted until further orders. This should be implemented. The JFM programme now renamed as the Community Forest Management programme of the A.P Forest Department has caused grievous violations with regard to tribal rights. One major violation is the displacement of tribals from their podu lands by reclaiming them back into the forests through the JFM programme. The official reports of the forest department and the World Bank (which has funded the project) reveal that 37,000 hectares of forestland has been reclaimed back from the people. The APFD further states its intention of displacing tribals in G.O. where it calls for voluntary surrender of lands by involving NGOs to motivate people under the CFM programme. The APFD has come up with a proposal for rehabilitation of tribals displaced from their podu lands through a monetary and schematic approach. This should be condemned as it bypasses the issue of eviction and rights of people over their podu lands. Their act of reclaiming lands under JFM programme and the APFD's forestry project under World

Bank assistance has to be scrutinised closely by the Commission. Samata and a few other NGOs raised this issue by writing to the World Bank on its violations of its own Operational Directives 4.2 and 4.3 in its financial support to the APFD. This has led to the Bank's insistence on an R&R policy to be proposed and implemented by the APFD in its CFM programme.

In Srisailem area, the Rajiv Gandhi Tiger Sanctuary has led to eviction of tribals from their original homes. They have not been properly settled so far. The concept of EDC (Eco Development Committees) that was introduced is working to the detriment of the tribals, as the objective of this programme was to reduce the forest dependency of the tribals. The tribals are given income-generating programmes and alternate sources of firewood by the forest department. This is an artificial mechanism that is not sustainable in the long term.

In the Srisailem Tiger Sanctuary area, the Chenchus, who are traditional hunter-gatherers, go into the forest everyday for all their needs. They are being harassed by the forest department for trespassing into the sanctuary. The groups who came from the chenchu area felt that the Chenchus should atleast be given identity cards to prevent harassment from forest department and the police department who mistake them for naxalites.

Another serious problem with regard to Sanctuaries is the settlement of people's rights. The Settlement Officer is the Conservator of Forests whereas the forest department is one of the interested parties in the land acquisition for the sanctuary and hence it is not appropriate to make the forest officer the settlement officer**. It should be the revenue department (District Collector) who should be authorised to settle the people's rights. G.O.No.112 which calls for involvement of private industries like ITC, Reliance and others for taking up commercial plantations through the VSS should be withdrawn immediately, as this is a backdoor method of allowing private industries to enter forest and tribal lands. Such tripartite agreements between the government, the industries and the tribals can never provide a level playing ground for the tribal people and make them more vulnerable to the exploitation of private industries.

The historical injustices to the tribal people in this region and that of the tribals in southern Orissa due to construction of several 'development' projects and industries like Nalco, HAL, Sileru, Machkund and others which have displaced tribals in large numbers without any rehabilitation, and has forced them to migrate in search of livelihood. They are now being treated as encroachers and criminals in forestlands. This definition should be condemned and Commission should recommend proper mechanisms for resolution of conflicts. The Commission should strongly recommend the orders passed by the Ministry of Environment and Forests in 1990, on the recommendations of the Commissioner of Scheduled Castes and Scheduled Tribes, to ensure that such conflicts are resolved. Infact, the first two orders alone show a way forward to settle disputes concerning forestlands. All the groups consulted strongly felt that the pressure on forests and the destruction of forests is taking place more due to

increase in industries and private commercial activities like Mining, paper-mills, timber smuggling and the growing non-tribal population settling down in Agency areas, than due to the podu cultivation practiced by tribals. Eco-Tourism is also causing destruction of forests. For example, the A.P Tourism Department built huge infrastructure close to Farhanbad, near Mannanur (Srisailam) for Tourism purposes in the middle of the protected area and in Araku, Borra and Anantagiri (Visakhapatnam). While tribals are prohibited from entering these forests for their survival needs, it is totally unjust to allow tourism, which is causing a lot of degradation. Hence, the constitutional provisions of the Fifth Schedule, the Samatha Judgement and the PESA Act should be strictly implemented following the right spirit with regard to industries and non-tribal settlers, which includes the Tourism Department.

The Tourism Department also conducts tours through tribal villages, especially in Chenchu settlements, and exhibits them like museum pieces. The groups felt that this degrading exercise further underscored the lack of empathy among the various government departments towards the tribals.

Because of a shift in the policy of the government, profits to G.C.C. are more important than benefits to tribals. Although there are 32 items in the procurement list of GCC Ltd, it procures a limited number of commodities and at very low prices compared to the market rates. The tribals face several constraints in collection of NTFP and marketing these items because of the monopoly rights enjoyed by GCC Ltd. They do not have the right to sell their produce to private traders even when their rates are higher. Hence, a long-standing demand has been for removal of the monopoly restriction of GCC Ltd and instead a minimum support price should be provided by GCC to protect the tribals from exploitation of traders. The monopoly of forest department over Beedi leaf collection and sale should be removed. The tribals or VSS should be given 100 per cent rights over income from beedi leaf and also the right of sale, with the forest department providing the support price.

RESOURCES OF THE NATIONAL FORESTS

The Forest Service began in earnest the development of all the resources of the national forests. Mature timber was sold wherever there was a demand for it and the permanent welfare of the forests and protection of the streams permitted, but always so as to prevent waste, guard against fire, protect young growth and ensure reproduction. Regulations were adopted which allowed small sales to be made without formality or delay, secured for the government the full value of timber sold and eliminated unnecessary routine. Care was taken to safeguard the interests of the government and provide for the maintenance of good technical standards. The conduct of local business was entrusted to local officers. Large transactions with general policies were controlled from Washington, but with careful *2 = District Boundaries and Numbers* provision for first-hand knowledge and close touch with the work in the field.

Business efficiency and the convenience of the public were carefully studied. In short, an organization was created capable of handling safely, speedily and satisfactorily the complex business of making useful a forest property of vast extent, scattered through sixteen different states of an aggregate area of over 1,500,000 sq. m. and with a population of 9,000,000.

The growth since the 1st of July 1897 of the area of the national forests, of the expenditures of the government for forestry and of the receipts from the national forests, is shown by the statement which follows. Though the act of June 4, 1897, became effective immediately upon its passage, the fiscal year 1899 was the first of actual administration, because the first for which Congress made the appropriation necessary to carry out the law.

Forest is ripe for the axe, the demand is strong and control by trained men makes it safe to cut more freely. The increase is marked both in small and in large sales, but a score of sales for less than \$5000 are made against one for more. The total cut is still far below the annual increment of the forests. As the demand grows restrictions must increase in order to husband the present supply until the next crop matures.

- The stumpage price would seem on the face of the figures to have risen from about one dollar to more than three dollars per thousand board-feet. The receipts, however, for any one year are not exclusively for the timber cut in that year, since payments are made in advance. In the year 1907 the average price obtained was something less than \$2.50 per thousand. It is therefore true that stumpage prices have risen greatly, although conditions new to the American lumbermen are in *Area of National Forests, Annual Expenditures of the Federal Government for Forestry and National Forest Administration and Receipts from National Forests, 1898-1909* Until 1906, the sole source of receipts was the sale of timber. In the fiscal year 1907, however, timber sales furnished less than half the receipts. The following statement concerning the timber sales of the fiscal years 1904-1907 will serve to bring out the change that followed the transfer of control to the forest service in the midst of the fiscal year 1905:
 - A large excess each year in the amount of timber sold over that cut and paid for;
 - Nine times as much timber sold at the end of the four-year period as at the beginning and three times as much cut;
 - A much higher price obtained per thousand board-feet at the end of the period than at the beginning. Each of these matters calls for comment. The sales are of stumpage only; the government does no logging on its own account.
- More timber is sold each year than is cut and paid for, because many of the sales extend over several years. With increasing sales the amount sold each year for future removal has exceeded the amount to be removed during that

year under sales of earlier years. Large sales covering a term of years are made because the national forests contain much overmature timber, which needs removal, but which is frequently too inaccessible to be saleable in small amounts. To prevent speculation the time allowed for cutting is never more than five years and cutting must begin at once and be continued steadily.

- The volume of sales has increased rapidly because much 1 The United States fiscal year ends June 30 and receives its designation from the calendar year in which it terminates. Thus, the fiscal year 1898 is the year July 1, 1897 - June 30, 1898.

Administration transferred to Bureau of Forestry, February 1, 1905. Full utilization of all merchantable material, care of young growth in felling and logging and the piling of brush, to be subsequently burned by the forest officers if burning is necessary, are among these conditions.

Timber to be cut must first be marked by the forest officers. Sales of more than \$100 in value are made only after public advertisement.

Only the simplest forms of silviculture have as yet been introduced. The vast area of the national forests, the comparatively sparse population of the West, the rough and broken character of the forests themselves and the newness of the problems which their management presents, make the general application of intensive methods for the present impracticable. Natural reproduction is secured. The selection system is most used, often under the rough and ready method of an approximate diameter limit, with the reservation of seed trees where needed. The tendency, however, is strongly towards a more flexible and effective application of the selection principle, as a better trained field force is developed and as market conditions improve. One conspicuous achievement was the reduction of loss by fires on the national forests. During the unusually dry season of 1905 there were only eight fires of any importance and the area burned over amounted only to about 16 of 1 per cent of the total area. In 1906 about 12 of 1 per cent was burned. This was accomplished by efficient patrol, co-operation of the public and by preventive measures, such as piling and burning the brush on cut-over areas.

Since the beginning of 1906 the largest source of income from the national forests was their use for grazing. Stock-raising is one of the most important industries of the West. Formerly cattle and sheep grazed freely on all parts of the public domain. In the early days of the national forests the wisdom of permitting any grazing at all upon them was sharply questioned.

Unrestricted grazing had led to friction between individuals, the deterioration of much of the range through overstocking and serious injury to the forests and stream flow. The forests of the West, however, are largely of open growth and contain many grassy parks, the results of old fires and many high mountain meadows.

Under proper regulations the grass and other forage plants which they produce in great quantity can be used without detriment to the forests themselves and with great benefit to the stock industry, which often can find summer pasturage nowhere else.

MINERAL AND ENERGY RESOURCES EXPLORATION

Emphasis on prognostication of low volume high-value minerals (gold, diamond, and platinum group of minerals) and ores and minerals of high demand (base metal, iron ore, coal, lignite etc.) have been prioritised keeping in view the thrust accorded by the Government of India for the X Plan. With the near exhaustion of surface proximal resources it has become imperative to have multi-disciplinary approach for locating concealed mineral deposits.

Multi-disciplinary approach to mineral exploration comprise large scale and detailed mapping aided by interpretative analysis of remotely sensed and aero geophysical data, ground geophysical survey, geochemical prospecting and subsurface exploration through pitting, trenching and followed by drilling. Collection of baseline geo-environmental data has been made mandatory for exploration of minerals, which would be of immense help in carrying out EIA and EMP studies during exploitation stage, thus ensuring a balance between development and environmental protection. The resource evaluation will be continued in P-I, P-II, E-I and E-II stages and will be monitored in conformity with the exploration input for G4, G3, G2 (Partly) stages of UNFC system respectively. This ensures continuous flow of projects from a preliminary stage to successive stages of detailed exploration while providing a built in scope for review and decision making after each stage of work through successor programmes.

Systematic updating of the database in the mineral resource sector to provide reliable and relevant information on mineral and other natural resources to the public and private sector entrepreneurs to sustain investment in mineral sector.

NON METALLIC MINERAL DEPOSITS

The state of Rajasthan is blessed with a rich variety of metallic as well as industrial (non-metallic) minerals (including rocks). Many of these are geologically unique deposits formed by varied genetic processes. Most of its Precambrian deposits have explicit metamorphic imprints on them. Sedimentation and secondary processes are dominant in younger deposits. These mineral resources play an important role in economic sustenance of the people of the state and the country.

Metamorphism is, however, dominant process of mineral formation and “Ore-Body Creator” for metallic as well as non-metallic minerals. These resources have been mined for centuries, rather millennia. Lead-zinc mining and smelting in Rajasthan is the oldest in and the Makrana marble, as also bronze and steatite possibly from Rajasthan, have been used during the pre-historic Indus Valley Civilization (Mohanjo-daro). Extensive utilization of these resources has, however, commenced in this century, essentially to meet the demands of modern-day life and owing to development of mechanization and infrastructure facility. Climatically, Rajasthan has a vast stretch of desert and a fairly large area of rocky/hilly tract, ravines etc., which are classified as “wasteland”. These

areas are not suitable for agriculture and forest activities. Minerals sector activity has therefore been playing significant role in sustenance of its populace. The state government has therefore laid priority and trust in exploration and development of its fairly vast mineral wealth for the “IXth Plan which will usher into the 21st century” and for the Xth Plan.

Geological Features of the Non-Metallic Mineral Resources

The geo-episodic history of Rajasthan is fairly well understood because of the exhaustive studies on the ore genesis, stratigraphy, geochronology, structure, tectonics metamorphism and sedimentation for the rocks of the state. These rocks represent varied lithologies—arenaceous, calcareous, argillaceous sediments, acid to ultramafic and alkaline plutonic to volcanic igneous and metamorphic rock, representing a fairly long period of geological history spanning ~ 3500 million years. The pre-Vindhyan rocks have undergone polyphase deformation and low to high-grade metamorphism. All these varied episode have generated a unique and rich variety of mineral resources that have attracted attention of researchers and entrepreneurs the world over.

Broadly, the mineral resources can be grouped into (% represents the share of national production): High-grade metamorphogenic deposits: Medium to low-grade metamorphogenic deposits: Medium to low-grade metamorphogenic deposits (marble 90%, talc, pyrophyllite 87%, asbestos 90%); Metasomatic; Skarn (wollastonite, 100%); Igneous: Pegmatitic; Sedimentary (limestone; sandstones 70%); Biogenic (rock phosphate 75%); Hydrothermal; Residual Concentration (clays 71%); Evaporation (gypsum 93%). Many of the deposits have polygenetic imprint on them. The state of Rajasthan is endowed with geologically and economically unique industrial minerals and rocks. The uniqueness is global as well as national.

- The minerals of international importance are: stromatolitic rock phosphate, serpentinite, talc and wollastonite.
- While those of national significance are: amphibole asbestos, clays (ball -, bentonite, china-, fire-, fuller’s earth), calcite, dolomite, feldspar, fluorspar, garnet, gypsum, ocher, pyrophyllite, quartz salts, sandstones, limestones (cement-grade, chemical-grade, flaggy-, SMS-grade), granites, marble (white, pink, green, gray), slate.
- Minor quantities of barite, brick earth, emerald, epidote garnet (semi-precious), graphite mica, and vermiculite are also produced.

Rock Phosphate

The deposits of stromatolitic phosphorite, belonging to Aravalli Supergroup, are of greatest significant and attained international status. These deposits are located at Jhamarkotra, Kanpur, Kabaria-Ka-Guda, Dakankotra, Maton, Badgaon, Manoharpura, Siserna and Neemchmata around Udaipur, are hosted by dolomitic limestone which at times become cherty and silicified towards the top (Srivastava, 1983). Roy et al. (1988),

have established tectonostratigraphy for this region. Super imposition of F2 and F4 folds on the limbs of a large F1 isoclinal fold and formation of domes and basins due to interference of F1 and F2 folds are important structural features of the Jhamarkotra deposits.

The Phosphorite ore extends for about 16 km with an average width of 15 meters and exposed in blocks A to K. The stromatolitic phosphorite occurs as columnar, brecciated, stratiform, lenticular, and in minor quantities as secondary colloidal deposit (Pandya, 1984). The high-grade ore Jhamarkotra phosphorite contain P₂O₅ content between 30 to 35% are directly usable, however, low-grade ores are needed to be beneficiated before use. Besides rock phosphate deposits, apatite occurrence of Nawania, Kikawas (Udaipur district), Kerpura and Salwari (Sikar district), phosphorite of Birmania and Fatehgarh (Jaisalmer district), Achrol (Jaipur district), Aduka and Avani (Alwar district), Sallopat and Ram-Ka-Munna (Banswara district) and Jaoda (Chittaurgarh district), occur in different lithological formations varying in age from Precambrian to Tertiary. The phosphate ore reserves in Rajasthan are estimated to be around 74 million tones.

Talc and Soapstone

Talc, steatite, soapstone deposits of Rajasthan are located in Udaipur, Bhilwara, Dungarpur, Banswara, Sawaimadhopur, Pali and Jaipur districts of Rajasthan. They are associated with either dolomites or ultramafic rocks. Those associated with dolomites are of better quality and larger in size. The talc-soapstone deposits of Udaipur-Dungarpur-Rajsamand districts occurring both in dolomites and ultramafic rocks, occurs as thin bands, lenses or pockets.

Major deposits are associated with dolomites belonging to Aravalli Supergroup in the four belts namely- (1) Oda-Devpura-Dangri-Natharia-Ki-Pal belt, (2) Ushan-Kagamdar-Jagat-Rabcha-Oden Patli belt, (3) Salumbar-Band-Sanjela-Lolagarh-Padla-Devla-Jagpura belt and (4) Kavita-Lakhawali-Parava-Rama Patli belt. Talc-Soapstone deposits of Undithal-Udarlotala-Khakar-Jharol-Dhandawali belt and Rishabadev-Kagdar-Bicchiwara-Sabla-Sisso-Kalayanpur belt are associated with metamorphosed ultramafic rocks of Jharol group of Aravalli Supergroup.

Soapstone deposits of Bhilwara district constitute talc deposits of Ghevria, Chainpura, Vagwara and Jahazpur region belong to dolomitic limestones of Bhilwara Supergroup. Talc-Soapstone deposits of Jaipur-Dausa districts extend for about 35 km in length. These deposits occur as thin bands, lenses and pocket in dolomite of Delhi Supergroup. Pyrophyllite deposits of Karoli – Sawaimadhopur districts occur in quartzite of Delhi Supergroup.

They are located at Douda, Kamalpura, Ranjeli, Garni, Pura and Marsh villages along a belt of about 10km. Pyrophyllite deposits also occur in Udaipur and Rajsamand districts in two belts, Debari- Gudali- Negaria-Nathuwas belt and Bari – Madar – Rama belt. The first belt that runs for 40 km and second for 10-12 km in length are located at the

contact of the Archean and Aravalli Supergroup rocks. The host rocks for pyrophyllite are quartzites and granitic gneisses.

Wollastonite

Major deposits of wollastonite are found in Sirohi, Udaipur, Dungarpur and Ajmer districts of Rajasthan. A rich wollastonite deposit has formed in calcic exoskarn of Belka and Khera Tarla Pahar (N24° 46': E 73° 10') in Sirohi and Udaipur district. The estimated reserve of 9887000 tonnes is being mined by M/s Wolkem Industries limited. The open cast mine produced 134763 tonnes of wollastonite in the year 2001-2002. This makes it the largest wollastonite resource production centre in India and second in the World i.e. next to China. About 90% production is utilized in India and rest is exported (57071 t in 2001-2002). Additionally, a large production of calcite is also obtained from this area. Minor occurrences of Wollastonite are found in Ajmer and Dungarpur districts that produce 200-300 tonnes of wollastonite per year.

Wollastonite Bearing Skarn Deposits of Gujarat

Besides Rajasthan there are deposit of wollastonite in Godha Taluka (Palampur district) and Chhota Udaipur Taluka (Baroda district) in Gujarat.

The estimated possible reserve is around 1.5 million tonnes. Both the above deposits of Rajasthan and Gujarat fall in the South Delhi Fold Belt (SDFB) and have been included in Sirohi belt. Geological Survey of India (Gupta et.al.,1981) have placed them at top of the Delhi Supergroup and the syntectonic intrusion of Erinpura granite resulted in the formation of skarn deposits. The crustal deformation and associated thermal regime caused large-scale anatexis and emplacement of a number of granite plutons (Erinpura Granite and its equivalents) along and adjacent to the SDFB at about 900 m. a. Some of these plutons are responsible for skarn mineral deposits as well as for Sn-W-Mo mineralization.

Near the southern extremity of the SDFB the regional metamorphism has been overprinted by thermal metamorphism (syntectonic to Erinpura granite). Siliceous marble developed diopside- forsterite assemblages and amphibole developed Hb + Labradorite + diopside assemblages. The late phase enriched the Sn and W is represented by the Balda Granite in Sirohi district.

Wollastonite Bearing Skarn Deposits Of South India

Two small deposits of wollastonite are reported in Dharmapuri and Tirunelveli districts in Tamilnadu.

Amphibole asbestos

The occurrence of amphibole asbestos is of great significance in national context due to very limited reserves of chrysotile asbestos in India, the latter is commercially procured in limited quantities from Andhra Pradesh with known occurrences in

Karnataka, Madhya Pradesh and Uttar Pradesh. Asbestos deposits of Rajasthan are located in Ajmer, Udaipur, Rajsamand, Pali, Dungarpur and Alwar districts. They are metamorphosed products of ultramafic rocks, which are intrusive in the rocks of Aravalli and Delhi Supergroups, along the prominent lineaments.

Clays:

1. Kaoline and ball clay: These resources are mainly confined to sediments of Precambrian, Jurassic and Eocene period, and are located in Bhilwara, Chittorgarh, Sawaimadhopur, Nagaur, Barmer, Jaisalmer, Pali, Kota, Jaipur and Sikar districts.
2. Fire Clay: The fire clay deposits of Rajasthan mainly occur in Bikaner, Alwar, Sawaimadhopur, Jhunjhunu and Barmer districts in rocks of Delhi Supergroup and Tertiary period. Due to high temperature resistant character these clays are used in manufacture of high temperature resistance bricks for steel industry, ceramic industry cement industry and thermal power plants.
3. Fuller's Earth: The fuller's earth deposits of Rajasthan occur in Barmer, Jaisalmer, Bikaner and Nagaur districts. Its beds up to 13 meters thick occur below the Kapurdi formations. It also occurs above and below the lignite seams. In Jaisalmer district the fuller's earth occurs below the limestone beds. This mineral is known in the industry as bleaching clay of non-swelling type that absorbs oily substances and is used in several industries including purification of vegetable oils and petroleum.
4. Bentonite: The bentonite deposits of Rajasthan are located in Barmer, Jaisalmer, Sawaimadhopur and Jhalawar districts. The high quality deposits occur in Barmer district only. The bentonite deposits lie above the Lathi Sandstone and Barmer Sandstone of Jurassic and Cretaceous age respectively. The bentonite deposits are also associated with Kapurdi formation of Eocene age and as an alteration product of Malani volcanics. It is bleaching clay of both swelling and non-swelling type constituted of montmorillonite. Its major uses are for grouting work in dams and other foundation works, petroleum drilling, cosmetics, insecticides, paint and pharmaceutical industries, bleaching of oils, sugar and petroleum.

Calcite

The calcite deposit of Rajasthan mainly occurs in Sirohi, Udaipur, Pali, Sikar, Jhunjhunu, Alwar, Jaipur and Ajmer districts.

The largest deposits are those of Sirohi district where it occurs in association with the wollastonite in skarns of Belka Pahar region. The calcite occurs in center part of the lenses spread over 300 meters long and 45 meters wide deposits.

The calcite also occurs as lenses in crystalline limestones and calc-silicate rocks of Delhi Supergroup in Udaipur and Sirohi districts. The calcite is mainly used in rubber, textile, plastic paint and ceramic industries.

Feldspar

Feldspars, quartz, mica beryl and associated minerals occur in complex pegmatites. Simple pegmatites contain only quartz and feldspar with or without mica and it is these pegmatites which produce the maximum quantity of feldspars. Feldspar deposits of Rajasthan occur in Ajmer, Rajsamansd, Jaipur, Tonk, Udaipur, Bhilwara, Pali, Sirohi, Alwar, Jhunjhunu and Sikar districts. The mineral has its maximum use in ceramic industry, and also used in glass industry, high temperature resistant refractory, abrasives and electrode industry.

Fluorspar

The deposits of fluorspar are found in Dungarpur and Jalore districts with minor occurrences in Sirohi, Alwar, Bhilwara, Banswara, Jaipur, Barmer and Sikar districts. The mineral is hosted in different rocks of the Precambrian BGC formations viz., mylonites, granite gneisses and quartzites and Malani pyroclastic rocks. The fluorspar mineralization is outcome of hydrothermal fluids of varied physico-chemical parameters. The mineral has its major use in steel metallurgy, manufacture of artificial cryolite, production of hydrofluoric acid, as a flux in and opalescent glasses, melting of metals, aluminum industry and as chloro-fluoro-carbon refrigerant.

Garnet

Garnets are mainly used as an abrasive and also as semiprecious gem mineral. It is product of regional and contact metamorphism and also found to occur in the form of placer deposits. Garnet deposits of Rajasthan occur in Tonk, Ajmer and Bhilwara districts in pegmatites and schist of Aravalli and Delhi Supergroups.

Gypsum

In Rajasthan gypsum deposits occur in rocks of tertiary period, and are considered to have formed by evaporation process. About 80% of Indian gypsum deposits are located in Jaisalmer, Barmer, Bikaner, Jodhpur, Nagur, Churu and Ganganagar districts of Rajasthan. Gypsum has its major use in the manufacture of cement, ammonium sulfate (fertilizer) and plaster of Paris. It is also used in insecticides, paper and paint industries.

Siliceous earth

It is mainly composed of silica and looks like clay and chalk. The composition of siliceous earth in Rajasthan varies between 69-81 % silica, 4-12% alumina and 3-10% water with low contents of CaO, MgO and FeO and found to occur in rocks of Cretaceous tertiary periods. It is mainly used as filler and filter, heat and sound resistant material and in ceramic industry. Filtration and cleaning of vegetable oils and animal fats and manufacture of medicines are other uses of this mineral. Rajasthan is the only state of India that produces this mineral.

Ochre

The Ochres are oxides of iron found in rocks of all the geological time period. The ochreous clays are formed as residual concentration due to alteration of igneous, sedimentary and metamorphic rocks. Commercially they are known as red ochre, yellow ochre, green ochre, and amber-brown- yellow ochre. Ochres are used in the colour, paper industry, paint industry, manufacture of red oxide paint, cement industry, fertilizer and foundry industry. The ochre deposits are found in Chittorgarh, Udaipur, Bhilwara, Jaisalmer, Bikaner, Jodhpur and Jhunjhunu districts of Rajasthan.

Quartz

Quartz is main raw material for ceramic, pottery, foundry and glass industry. Quartz occurs either in simple pegmatite along with feldspar or in complex pegmatites with feldspar, mica, beryl, tourmaline, apatite, lithium mica, fluorite and other minerals. It also occurs as veins in different rocks types of varying ages, resulting from late hydrothermal crystallization.

Salts

Halite is used in several chemical industries besides its use in our daily meals. Manufacture of soda caustic soda, metal smelting, cleaning of leather, manufacture of medicines, ceramic industry and acid industry. Rajasthan produces common salt (halite) from lakes of Sambhar, Didwana and Kuchaman in Nagaur district, Pachbhadra Lake of Barmer district and Nanwa Lake of Bikaner district. The potash salt minerals occur at the Nanwa Lake of Bikaner district. The potash salt minerals occur at the depth varying from 385 to 965 m in Nagaur, Churu, Ganganagar and Bikaner districts. Major potash minerals sylvite (KCl) and langbeinite [$K_2Mg_2(SiO_4)_2$] are used in fertilizer industry. They are also used in chemical industries and manufacture of explosives. The generally accepted views on the origin of salt deposits is by evaporation process, however considers their formation due to prolonged aridity in tectonically controlled depressions.

Barite

The deposits of barite are located in Alwar, Udaipur, Ajmer, Bahratpur, Sikar, Chittaurgarh, Jalore, Rajsamand, Bundi and Pali districts in rocks belonging to Bhilwar, Aravalli, Delhi, Vindhyan Supergroups as well as in Malani Suite of igneous. Barite occurs in a variety of host rocks in the form of fracture filling, breccia filling, replacement and residual concentration type. It is also used in paints, and chemical industry, manufacture of barium and as filler in rubber, textile, leather, glass and paper industries.

Emerald

It occurs sporadically in altered ultramafic rocks in close association of biotite schist, actinolite schist, talc-schist. Emerald develops wherever pegmatites intrude at the contact

of schist and altered ultramafic rocks. Well-known occurrences are at Rajgarh, Bubani in Ajmer district and Kaliguman, Gamgurha and Tikhi areas in Rajsamand district. The emerald belt strikes NNE-SSW and runs for about 200km. and was important source of valuable gem for centuries. At present, however, these localities do not produce any emerald.

Graphite

The graphite occurs in minor quantities in Precambrian rocks of Banswara, Ajmer, Udaipur and Alwar districts. Geologically, graphite occurs in rocks of Aravalli and Delhi Supergroups. Graphite has its application in foundry works and manufacture of crucibles because of its high temperature melting and resistant character.

Limestone

The limestone deposition takes place both by organic and inorganic processes. Major limestone deposits of Rajasthan have formed during the Precambrian (Aravalli, Delhi, Vindhyan, Marwar Supergroups) and Jurassic as well as in Eocene periods. It has application in iron and steel industry, construction works, cement, paper, glass, sugar, textile, chemical and leather industry. It is also utilized as a flux in copper and lead smelters. The limestone deposits occur in Udaipur, Dungarpur, Banswara, Rajsamand, Sirohi, Pali, Ajmer, Jaipur, Sikar, Jhunjhunu and Alwar districts. The limestone of Bhilwara and Aravalli Supergroups are of cement and chemical grade. The Vindhyan Supergroup has large deposits of cement grade limestones, which support cement plants in Chittorgarh, Bhilwara, Kota, Bundi, Sawaimadhopur and Karauli districts. The Marwar Supergroup has occurrences of steel, cement and flux grade limestones in addition, chemical grade limestone is also present. The Jurassic and Eocene limestones of western Rajasthan are of steel grade.

Dolomite

Dolomite is both sedimentary as well as metamorphic in origin. It is mainly utilized as a flux, cement, fertilizers, rubber and paint industries. The dolomite deposits occur in Ajmer, Bhilwara, Rajsamand, Banswara, Alwar, Jodhpur and Jaipur districts of Rajasthan. Geologically, dolomite occurs in Bhilwara, Aravalli, Delhi and Vindhyan Supergroups in Rajasthan.

Sandstone

The sandstone deposits of Rajasthan belong to Vindhyan Supergroup and Marwar Supergroup. At places the sandstone deposits of Tertiary period are also present. They are shallow water sedimentary rocks.

The vindhyan sandstone deposits are located in Chittorgarh, Bhilwara, kota, Bundi, Sawaimadhopur, Karauli and Dholpur districts while those of Marwar Supergroup occur in Jodhpur, Nagaur and Bikaner district. Acid proof sandstone deposits of Khimach

located in Jhalawar district are used in chemical industries. Cream colored sandstone of Khatu is popular for carving works.

Granites

There are huge deposits of granites in Rajasthan in the Aravalli Mountain Range as well as in western Rajasthan. Granites of varying ages (Precambrian) are of igneous and metamorphic in origin; the former is generally used to make slabs and tiles for decorative purposes in building constructions. The major deposits of granite are located in Jalore, Sirohi, Barmer, Pali, Sikar, Tonk, Jaipur, Ajmer, Jhunjhunu, Bhilwara, Rajsamand, Udaipur, Duingarpur and Banswara districts. Well-known trade names of granites are Jalore pink, Mokalsar green, Shivganj white, Chima pink, Bhilwara laharria and black etc.

Marble (White, Pink, Green and Gray)

Marble, sensu strict, is a metamorphic carbonate rock and compositionally classified as calcite, dolomitic, siliceous, ferruginous and impure marble. Calcitic marble is the best quality amongst the white marble. The term 'marble' is used here in a loose sense, which includes relatively softer rocks capable of taking good polish and suitable for decorative works. In this context marble shows different compositions and belongs to age range varying from the Precambrian to Jurassic. Marble occurrences in Rajasthan are located in Nagaur, Jaipur, Alwar, Ajmer, Rajsamand, Udaipur, Dungarpur, Banswara, Chittaugarh, Bhilwara, Sirohi, Pali, Jaisalmer, Jhunjhunu and Sikar districts, whose trade names are based on colour such as Makrana-white and Adaga (variegated), Babarmal pink, Kesariaji green, Rajsamand white, Bhainslana black, Selwara white & Adanga, Abu- Ambaji green, Banswara pink, Jaisalmer yellow, Andhipista, Mandlade brown etc. The state of Rajasthan contributes more than 90% of India's production of marble. Although Makrana marble was known for its use in temples, palaces and historical monuments for several centuries, its use for common men started only when sawing, cutting and polishing techniques developed in late 1970s.

Serpentine (the so-called green marble)

Serpentine has deep green colour, and yields slabs, which generally take good polish, therefore it is mostly exported. Rajasthan contributes up to 90% of Indian production from a belt called Rikhabdev (Rishabhadev or Kesariaji)- Masaron-ki-ovari-Dakchi belt in Udaipur and Dungarpur districts. The serpentine is a product of hydrothermal metamorphism of Precambrian ultramafic rocks.

Slates, Schists and Phyllite

Rocks of this category are used in roofing and decorative works. These are metamorphic argillaceous rocks belonging to the Precambrian period. In Rajasthan, shales and slates are mainly mined and processed into tiles in Mandon region of Alwar

district and also in Tonk district. Slates and schist of white, red, pink, brown and other colours are mined in the above mentioned regions. The Aravalli and Delhi Supergroups have vast potential of slates, phyllites and schists rocks in Udaipur, Bhilwara, Pali, Rajsamand, Banswara and Dungarpur districts.

NON CONVENTIONAL ENERGY

In India, non-conventional energy sources consist of those energy sources that are infinite, natural, and restorable. For example, tidal energy, solar energy, and wind energy are nonconventional sources of energy. Fascinatingly, the application of tidal energy and wind energy was operational in the form of energy sources long back when mineral oil, coal, and natural gas were not broadly introduced as conventional sources of energy.

In the beginning, windmills were utilized for taking out water and pounding grains. Running water and wind were applied for direction finding. Currently, some of the important and widely used non conventional sources of energy are tides, wind, solar geothermal heat, and biomass comprising animal waste, agricultural waste, and human body waste. For example, disposals from big metropolitan areas can work as a source of producing biogas. All these non conventional energy sources are unlimited or restorable and are essentially quite economical.

Some of the India non conventional energy sources are given below—

Tidal Energy

Tidal energy is a limitless and renewable source of energy. For producing electrical power from tidal energy, dams are built close to estuaries to make use of inward and outward water currents and surges. Subsequently, the water is used to power turbines to generate electricity via the usage of water. The tidal surge has to be minimum seven meters for cost-effective performance. In the state of Gujarat, the Gulf of Kutch is an appropriate place for electrical energy from the energy generated by tall and powerful tides moving into small brooks.

Wind Energy

Wind energy is a popular form of non-conventional energy. It is utilized for drawing water, which is an essential requirement in watering agricultural lands in the rural areas. In addition, it can be utilized for electricity generation. In India, states like Tamil Nadu, Gujarat, Orissa, and Maharashtra are regarded as superior areas with respect to this type of energy. Places that have regular and rapid wind flows are appropriate for this kind of power generation. Other than windmills, wind farms are there as well. The installed capacity of wind power in India till December 2010 was 13,065.37 MW, principally distributed throughout:

- Maharashtra—1837.85 MW
- Tamil Nadu—4132.72 MW
- Rajasthan—670.97 MW

- Karnataka—1184.45 MW
- Andhra Pradesh—122.45 MW
- Gujarat—1432.71 MW
- Kerala—23.00 MW
- Madhya Pradesh—187.69 MW
- West Bengal—1.10 MW
- Other states—3.20 MW

Solar Energy

The sun is the most abundant and unlimited source of energy. The sun functions as a global source of energy and has tremendous potential. As a result, solar energy is one of the most important non conventional sources of energy that are utilized in India. The solar cookers are quite economical and they have been a remarkable invention. These cookers assist in food preparation nearly without any expenditure. Additionally, many tiny and medium-scale solar power plants have been intended for the countryside areas in India. Until now, some of the effective usages of solar energy include water heating, food preparation, area heating, removal of salt from water, and drying of harvest. Furthermore, it has been forecasted that solar energy will become the future source of energy while fossil fuels, specifically oil and coal, would be completely used up. Solar photovoltaic systems, solar thermal systems, and solar energy centre are some of the means of generating solar energy. The biggest photovoltaic (PV) power plants of India are as follows:

- Gandhinagar Solar Plant—Installed January 21, 2011
- Thyagaraj Stadium Plant-Delhi—Installed April, 2010
- Azure Power, Sabarkantha Gujarat—Installed June 2011
- Tata—Mulshi, Maharashtra April 2011—Installed April 2011
- Tata—Mayiladuthurai, Tamil Nadu—Installed July 2011
- Moser Baer—Patan, Gujarat—Installed July 2011
- TATA—Osmanabad, Maharastra—Installed 1st Aug 2011
- REHPL—Sadeipali, (Bolangir) Orissa—Installed July 2011

Techniques to generate Non Conventional Source of Energy in India

1. *Thermal energy/Geo-thermal energy*: Geothermal energy is the application of natural temperature, which prevails on the exterior and below the earth, predominantly in the fissures and holes within the outer shell of the earth. Geothermal energy can be productively utilized for different objectives. The country is not affluent in thermal energy resource. Nonetheless, endeavors are on for the total exploitation of natural energy of the Manikaran thermal springs in the state of Himachal Pradesh. Energy generated from this can be applied for operating cold storage plants.
2. *Biomass*: Biomass is an important source of energy which represents approximately 33% of the overall volume of fuel used in the country. It is

broadly utilized in domestic circles for preparing food and warming. In countryside areas, farming disposables, timber, wood coal, and sunbaked droppings can be used as biomass. Many initiatives have been taken in India to make the right use of biomass in essentially a successful and more systematic way. The two principal segments of the biomass program are generation and usage of biomass. Smoke-free ambience, improved healthcare, and better quality of life and education are some of the salient benefits of biomass.

3. *Energy Plantation:* In India, barren areas are utilized for cultivation of plants and bushes that tend to spread quickly and have significant heat generating characteristics. Successively, they render wood coal, fuel wood, power, hay, and most notably opportunities for agricultural service. With the gasification plan, these energy plantations spanning approximately 8,000 hectares were generating almost 1.5 MW power every year.
4. *Bagasse Oriented Power Plants:* It was projected that Indian sugar mills have the capacity of generating over 2,000 MW additional electrical energy at the time of the grinding period. The energy generated by a sugar mill would initially fulfill its own energy needs and the remaining can be used in watering farming areas by supplying it into the local power system. Similar to bagasse, various other agricultural disposables like rice husk are also used for generating electrical energy in the country.
5. *Energy from Urban disposables:* In Delhi, a pilot plant for the intention of displaying has been established for processing hard municipal disposables for switching into energy. Urban disposables generate a significant volume of energy annually. In addition, wastes in metropolitan areas are utilized for generating electrical energy and gas.
6. *Animal, Agricultural, and Human Excretions:* With the usage of agricultural and animal excretions over and above human body wastes, a number of 'Gobar Gas' plants have been built in many rural areas to make them self-reliant in their energy requirements. The energy generated in this way is utilized for food preparation, lighting roads and houses, and satisfying the water supply requirements of the rural community. The plants have been established both at private and group levels.

Last but not least, the largest percentage of energy is used in the household kitchens of India. Cow dung and wood are regarded as global sources of energy. Unfortunately, the conventional 'chulhas' are uneconomical modes of preparing food. The better categories of effective and smoke-free chulhas assist in protecting wood fuel. These are some of the most well known and extensively utilized non-conventional sources of energy in India.

INDIAN METALLIC MINERS

India is rich in mineral resources which are spread over diverse geographic locations. In India, the mining sector employs nearly 1.5 million people directly/indirectly. India

produces 64 minerals: four fuel minerals, ten metallic minerals, and 50 nonmetallic minerals. The Indian mining industry is characterized by a large number of small operational mines; there are approximately 3,100 functioning mines in the country. Six hundred of these mines produce coal and lignite, 500 mines produce metallic minerals, and 2,000 mines produce nonmetallic minerals. Eight hundred of these mines are in the public sector and 2,300 mines in the private sector.[

The following popper user interface control may not be accessible. Tab to the next button to revert the control to an accessible version. Destroy user interface control1] India is self-sufficient in about 27 minerals. In world production, India ranked 2nd in barites, chromite, and talc, 3rd in coal and lignite, 4th in iron ore and kyanite, 6th in bauxite, and 7th in manganese in 2002-03. The estimated average daily employment in the mining sector is 5,60,000, 4,90,000 (87%) in the public sector and 70,000 (13%) in the private sector, 2 of which around 70,000 work in metallic mines.

The extraction of mineral resources from the earth's crust is seldom easy. Mineral production can be divided into two general phases: the extraction phase which is concerned with removing material from the earth, and the processing phase which separates the mineral component from the ore. Commonly, the word, "mining" is used to describe the phase of extraction, whereas processing describes the phase of preparing the product for human use. Usually the people involved in both are called miners or mineworkers. In its simplest definition, mining includes all of the job processes required to produce a mineral commodity in its saleable form.

The ore contains principal mineral plus trace amounts of other potentially toxic material. Ore is processed by crushing, screening, washing etc. Workers are thus exposed to dust of various potentially toxic substances during the processing of ore. The mine worker is exposed to large amount of dust for eight hours/day and six days/week over a lifetime of work. In the mining environment, dermal and inhalational exposures predominate and the conditions of work often intensify exposure due to increased respiration rates and repetitive exposure to toxicants. This results in the accumulation of toxicants in the body over many years. The most commonly studied occupational health problems among miners all over the world are silicosis, coal workers' pneumoconiosis, asbestosis, etc. but there are very few studies which describe the ill effects of toxic materials present in the mining environment. This article addresses the issues of exposure and poisoning among metallic miners in India.

Except in southern California grazing is now permitted on all national forests unless the watersheds furnish water for domestic use; but the time of entering and leaving, the number of head to be grazed by each applicant and the part of the range to be occupied are carefully prescribed. Planted areas and cut-over areas are closed to stock until the young growth is safe from harm and goats are allowed only in the brushland of the foothills.

The results of regulation, in addition to the protection of forest growth and streams, are the prevention of disputes, improved range, better stock, stable conditions in the

stock industry and the best use of the range in the interest of progress and development. The first right to graze stock on the forests is given to residents, small owners and those who have used the range before. Thus the crowding out of the weaker by the stronger and of the settler by the roving outsider has been stopped. In 1906 the forest service began to impose a moderate charge for the use of the national forest range.

The following statement shows the amount of stock grazed on the national forests 1904-09 and the receipts for the grazing charge: - A work of enormous magnitude which has now begun is planting on the national forests. At present, with low stumpage prices and incomplete utilization of forest products, clear cutting with subsequent planting is not practicable.

There are, however, many million acres of denuded land within the national forests which require planting. Such planting is still confined chiefly to watersheds which supply cities and towns with water. The first planting was done in 1892, in California. Since then similar work has been done on city watersheds in Colorado, Utah, Idaho and New Mexico. Other plantations are in the Black Hills national forest, where large areas of cut-over and burned-over land are entirely without seed trees and in the sandhill region of Nebraska. Up to 1908 about 2,000,000 seedlings had been planted, on over 2000 acres - a small beginning, but the work was entirely new and presented many hard problems.

The nursery operations of the forest service are concentrated at seven stations, located in southern California, Nebraska, Colorado, New Mexico Utah and Idaho, where stock is raised for local planting and for shipment elsewhere. These nurseries are small. Their annual productive capacity is between 8,000,000 and seedlings. Each nursery is practically an experimental forest-planting station, at which a large variety of species are grown and various methods are tried.

The organization of the administrative work of the national forests is by single forests. On the 1st of January 1908 the total number of forests was 165 with a total area of 162,023,190 acres. In charge of each forest is a forest supervisor. Under the supervisors are forest rangers and forest guards, whose duties include patrol, marking timber and scaling logs, enforcing the regulations and conducting some of the minor business arising from the use of the forests. Guards are temporary employees; rangers are employed by the year. The supervisors report directly to and receive instructions from the central office at Washington.

In this office there are four branches - operation, grazing, silviculture and products - each of which directs that part of the work which belongs to it, dealing directly with the supervisor.

For inspection purposes, however, the forests are separated into six districts, in each of which is located a chief inspector with a corps of assistants. The inspectors are without administrative authority, but assist by their counsel the supervisors and through inspection reports keep the Washington office informed of the condition of all lines of administrative work in progress. Administrative officers alternate frequently between

field and office duties. The number of forest officers in the several grades on the 1st of January 1908 were: 6 chief inspectors, 26 inspectors, 106 forest supervisors, 41 deputy forest supervisors, 820 forest rangers and 283 forest guards. The total number of employes of the forest service on the same date, including the clerical force, was 2034.

Besides the administration of the national forests, the forest service conducts general investigations, carries on an extensive educational work and co-operates with private owners who contemplate forest management upon their own tracts. This last work is undertaken because of the need of bringing forestry into practice, the lack of trained foresters outside of the employ of the government and the lack of information as to how to apply forestry and what returns may be obtained. Co-operation takes the form of advice upon the ground and, on occasion, of the making of working plans.

The educational work of the service is performed chiefly through publications, the purpose of which is to spread very widely a knowledge of the importance of forestry to the nation and of the principles upon which its practice rests. The investigations which the service conducts extend from studies of the natural distribution and classification of American forests, and of their varied silvicultural problems to statistics of lumber production and laboratory researches which bear upon the economical utilization of forest products. As examples of these researches may be mentioned tests of the strength of timber, studies of the preservative treatment of wood for various uses, wood-pulp investigations and studies in wood chemistry.

10

The Foreign Exchange Market

The foreign exchange market (currency, forex, or FX) trades currencies. It lets banks and other institutions easily buy and sell currencies. The purpose of the foreign exchange market is to help international trade and investment. A foreign exchange market helps businesses convert one currency to another. For example, it permits a U.S. business to import European goods and pay Euros, even though the business's income is in U.S. dollars.

In a typical foreign exchange transaction a party purchases a quantity of one currency by paying a quantity of another currency. The modern foreign exchange market started forming during the 1970s when countries gradually switched to floating exchange rates from the previous exchange rate regime, which remained fixed as per the Bretton Woods system. The foreign exchange market is unique because of:

- Its trading volumes,
- The extreme liquidity of the market,
- Its geographical dispersion,
- Its long trading hours: 24 hours a day except on weekends (from 22:00 UTC on Sunday until 22:00 UTC Friday),
- The variety of factors that affect exchange rates.
- The low margins of profit compared with other markets of fixed income (but profits can be high due to very large trading volumes),
- The use of leverage.

As such, it has been referred to as the market closest to the ideal perfect competition, notwithstanding market manipulation by central banks. According to the Bank for

International Settlements, average daily turnover in global foreign exchange markets is estimated at \$3.98 trillion. Trading in the world's main financial markets accounted for \$3.21 trillion of this. This approximately \$3.21 trillion in main foreign exchange market turnover was broken down as follows:

- \$1.005 trillion in spot transactions.
- \$362 billion in outright forwards.
- \$1.714 trillion in foreign exchange swaps.
- \$129 billion estimated gaps in reporting.

FINANCIAL INSTRUMENTS

SPOT

A spot transaction is a two-day delivery transaction (except in the case of trades between the US Dollar, Canadian Dollar, Turkish Lira and Russian Ruble, which settle the next business day), as opposed to the futures contracts, which are usually three months. This trade represents a “direct exchange” between two currencies, has the shortest time frame, involves cash rather than a contract; and interest is not included in the agreed-upon transaction. The data for this study come from the spot market. Spot transactions has the second largest turnover by volume after Swap transactions among all FX transactions in the Global FX market. NNM.

Forward

One way to deal with the foreign exchange risk is to engage in a forward transaction. In this transaction, money does not actually change hands until some agreed upon future date. A buyer and seller agree on an exchange rate for any date in the future, and the transaction occurs on that date, regardless of what the market rates are then. The duration of the trade can be a one day, a few days, months or years. Usually the date is decided by both parties.

Future

Foreign currency futures are exchange traded forward transactions with standard contract sizes and maturity dates — for example, \$1000 for next November at an agreed rate. Futures are standardized and are usually traded on an exchange created for this purpose.

The average contract length is roughly 3 months. Futures contracts are usually inclusive of any interest amounts.

Swap

The most common type of forward transaction is the currency swap. In a swap, two parties exchange currencies for a certain length of time and agree to reverse

the transaction at a later date. These are not standardized contracts and are not traded through an exchange.

Option

A foreign exchange option (commonly shortened to just FX option) is a derivative where the owner has the right but not the obligation to exchange money denominated in one currency into another currency at a pre-agreed exchange rate on a specified date. The FX options market is the deepest, largest and most liquid market for options of any kind in the world..

Exchange-Traded Fund

Exchange traded funds (or ETFs) are open ended investment companies that can be traded at any time throughout the course of the day. Typically, ETFs try to replicate a stock market index such as the S&P 500 (e.g. SPY), but recently they are now replicating investments in the currency markets with the ETF increasing in value when the US Dollar weakens versus a specific currency, such as the Euro.

Certain of these funds track the price movements of world currencies versus the US Dollar, and increase in value directly counter to the US Dollar, allowing for speculation in the US Dollar for US and US Dollar denominated investors and speculators.

Speculation

Controversy about currency speculators and their effect on currency devaluations and national economies recurs regularly. Nevertheless, economists including Milton Friedman have argued that speculators ultimately are a stabilizing influence on the market and perform the important function of providing a market for hedgers and transferring risk from those people who don't wish to bear it, to those who do.

Other economists such as Joseph Stiglitz consider this argument to be based more on politics and a free market philosophy than on economics. Large hedge funds and other well capitalized "position traders" are the main professional speculators. According to some economists, individual traders could act as "noise traders" and have a more destabilizing role than larger and better informed actors.

Currency speculation is considered a highly suspect activity in many countries. While investment in traditional financial instruments like bonds or stocks often is considered to contribute positively to economic growth by providing capital, currency speculation does not; according to this view, it is simply gambling that often interferes with economic policy.

For example, in 1992, currency speculation forced the Central Bank of Sweden to raise interest rates for a few days to 500% per annum, and later to devalue the krona. Former Malaysian Prime Minister Mahathir Mohamad is one well known proponent of this view. He blamed the devaluation of the Malaysian ringgit in 1997 on George Soros and other speculators. Gregory J. Millman reports on an opposing view, comparing

speculators to “vigilantes” who simply help “enforce” international agreements and anticipate the effects of basic economic “laws” in order to profit.

In this view, countries may develop unsustainable financial bubbles or otherwise mishandle their national economies, and foreign exchange speculators allegedly made the inevitable collapse happen sooner. A relatively quick collapse might even be preferable to continued economic mishandling.

Mahathir Mohamad and other critics of speculation are viewed as trying to deflect the blame from themselves for having caused the unsustainable economic conditions. Given that Malaysia recovered quickly after imposing currency controls directly against International Monetary Fund advice, this view is open to doubt.

Market Size and Liquidity

The foreign exchange market is the largest and most liquid financial market in the world. Traders include large banks, central banks, currency speculators, corporations, governments, and other financial institutions. The average daily volume in the global foreign exchange and related markets is continuously growing. Daily turnover was reported to be over US\$3.2 trillion in April 2007 by the Bank for International Settlements. Since then, the market has continued to grow. According to Euro money’s annual FX Poll, volumes grew a further 41% between 2007 and 2008.

Of the \$3.98 trillion daily global turnover, trading in London accounted for around \$1.36 trillion, or 34.1% of the total, making London by far the global centre for foreign exchange. In second and third places respectively, trading in New York accounted for 16.6%, and Tokyo accounted for 6.0%. In addition to “traditional” turnover, \$2.1 trillion was traded in derivatives.

Exchange-traded FX futures contracts were introduced in 1972 at the Chicago Mercantile Exchange and are actively traded relative to most other futures contracts.

Several other developed countries also permit the trading of FX derivative products (like currency futures and options on currency futures) on their exchanges. All these developed countries already have fully convertible capital accounts. Most emerging countries do not permit FX derivative products on their exchanges in view of prevalent controls on the capital accounts. However, a few select emerging countries have already successfully experimented with the currency futures exchanges, despite having some controls on the capital account. FX futures volume has grown rapidly in recent years, and accounts for about 7% of the total foreign exchange market volume, according to *The Wall Street Journal Europe* (5/5/06).

Foreign exchange trading increased by 38% between April 2005 and April 2006 and has more than doubled since 2001. This is largely due to the growing importance of foreign exchange as an asset class and an increase in fund management assets, particularly of hedge funds and pension funds. The diverse selection of execution venues have made it easier for retail traders to trade in the foreign exchange market. In 2006, retail traders constituted over 2% of the whole FX market volumes with an average daily

trade volume of over US\$50-60 billion. Because foreign exchange is an OTC market where brokers/dealers negotiate directly with one another, there is no central exchange or clearing house.

The biggest geographic trading centre is the UK, primarily London, which according to IFSL estimates has increased its share of global turnover in traditional transactions from 31.3% in April 2004 to 34.1% in April 2007. The ten most active traders account for almost 80% of trading volume, according to the 2008 Euro money FX survey. These large international banks continually provide the market with both bid (buy) and ask (sell) prices. The bid/ask spread is the difference between the price at which a bank or market maker will sell (“ask”, or “offer”) and the price at which a market taker will buy (“bid”) from a wholesale or retail customer. The customer will buy from the market-maker at the higher “ask” price, and will sell at the lower “bid” price, thus giving up the “spread” as the cost of completing the trade. This spread is minimal for actively traded pairs of currencies, usually 0–3 pips. For example, the bid/ask quote of EUR/USD might be 1.2200/1.2203 on a retail broker. Minimum trading size for most deals is usually 100,000 units of base currency, which is a standard “lot”.

These spreads might not apply to retail customers at banks, which will routinely mark up the difference to say 1.2100/1.2300 for transfers, or say 1.2000/1.2400 for bank notes or travellers’ checks. Spot prices at market makers vary, but on EUR/USD are usually no more than 3 pips wide (i.e., 0.0003). Competition is greatly increased with larger transactions, and pip spreads shrink on the major pairs to as little as 1 to 2 pips. Forex Trading.

Market Participants

Unlike a stock market, where all participants have access to the same prices, the foreign exchange market is divided into levels of access. At the top is the inter-bank market, which is made up of the largest commercial banks and securities dealers. Within the inter-bank market, spreads, which are the difference between the bid and ask prices, are razor sharp and usually unavailable, and not known to players outside the inner circle. The difference between the bid and ask prices widens (from 0-1 pip to 1-2 pips for some currencies such as the EUR). This is due to volume. If a trader can guarantee large numbers of transactions for large amounts, they can demand a smaller difference between the bid and ask price, which is referred to as a better spread. The levels of access that make up the foreign exchange market are determined by the size of the “line” (the amount of money with which they are trading). The top-tier inter-bank market accounts for 53% of all transactions. After that there are usually smaller banks, followed by large multi national corporations (which need to hedge risk and pay employees in different countries), large hedge funds, and even some of the retail FX-metal market makers. According to Galati and Melvin, “Pension funds, insurance companies, mutual funds, and other institutional investors have played an increasingly important role in financial markets in general, and in FX markets in particular, since the early 2000s.”

(2004) In addition, he notes, “Hedge funds have grown markedly over the 2001–2004 period in terms of both number and overall size” Central banks also participate in the foreign exchange market to align currencies to their economic needs.

Banks

The interbank market caters for both the majority of commercial turnover and large amounts of speculative trading every day. A large bank may trade billions of dollars daily. Some of this trading is undertaken on behalf of customers, but much is conducted by proprietary desks, trading for the bank’s own account. Until recently, foreign exchange brokers did large amounts of business, facilitating inter bank trading and matching anonymous counterparts for small fees. Today, however, much of this business has moved on to more efficient electronic systems. The broker squawk box lets traders listen in on ongoing inter bank trading and is heard in most trading rooms, but turnover is noticeably smaller than just a few years ago.

Commercial Companies

An important part of this market comes from the financial activities of companies seeking foreign exchange to pay for goods or services. Commercial companies often trade fairly small amounts compared to those of banks or speculators, and their trades often have little short term impact on market rates. Nevertheless, trade flows are an important factor in the long-term direction of a currency’s exchange rate. Some multinational companies can have an unpredictable impact when very large positions are covered due to exposures that are not widely known by other market participants.

Central Banks

National central banks play an important role in the foreign exchange markets. They try to control the money supply, inflation, and/or interest rates and often have official or unofficial target rates for their currencies. They can use their often substantial foreign exchange reserves to stabilize the market. Milton Friedman argued that the best stabilization strategy would be for central banks to buy when the exchange rate is too low, and to sell when the rate is too high—that is, to trade for a profit based on their more precise information. Nevertheless, the effectiveness of central bank “stabilizing speculation” is doubtful because central banks do not go bankrupt if they make large losses, like other traders would, and there is no convincing evidence that they do make a profit trading.

The mere expectation or rumor of central bank intervention might be enough to stabilize a currency, but aggressive intervention might be used several times each year in countries with a dirty float currency regime. Central banks do not always achieve their objectives. The combined resources of the market can easily overwhelm any central bank. Several scenarios of this nature were seen in the 1992–93 ERM collapse, and in more recent times in Southeast Asia.

Hedge Funds as Speculators

About 70% to 90% of the foreign exchange transactions are speculative. In other words, the person or institution that bought or sold the currency has no plan to actually take delivery of the currency in the end; rather, they were solely speculating on the movement of that particular currency. Hedge funds have gained a reputation for aggressive currency speculation since 1996. They control billions of dollars of equity and may borrow billions more, and thus may overwhelm intervention by central banks to support almost any currency, if the economic fundamentals are in the hedge funds' favour.

Investment Management Firms

Investment management firms (who typically manage large accounts on behalf of customers such as pension funds and endowments) use the foreign exchange market to facilitate transactions in foreign securities. For example, an investment manager bearing an international equity portfolio needs to purchase and sell several pairs of foreign currencies to pay for foreign securities purchases.

Some investment management firms also have more speculative specialist currency overlay operations, which manage clients' currency exposures with the aim of generating profits as well as limiting risk. Whilst the number of this type of specialist firms is quite small, many have a large value of assets under management (AUM), and hence can generate large trades.

Retail Foreign Exchange Brokers

There are two types of retail brokers offering the opportunity for speculative trading: retail foreign exchange brokers and market makers. Retail traders (individuals) are a small fraction of this market and may only participate indirectly through brokers or banks. Retail brokers, while largely controlled and regulated by the CFTC and NFA might be subject to foreign exchange scams. At present, the NFA and CFTC are imposing stricter requirements, particularly in relation to the amount of Net Capitalization required of its members. As a result many of the smaller, and perhaps questionable brokers are now gone. It is not widely understood that retail brokers and market makers typically trade against their clients and frequently take the other side of their trades. This can often create a potential conflict of interest and give rise to some of the unpleasant experiences some traders have had. A move toward NDD (No Dealing Desk) and STP (Straight Through Processing) has helped to resolve some of these concerns and restore trader confidence, but caution is still advised in ensuring that all is as it is presented.

Non-bank Foreign Exchange Companies

Non-bank foreign exchange companies offer currency exchange and international payments to private individuals and companies. These are also known as foreign exchange brokers but are distinct in that they do not offer speculative trading but currency

exchange with payments. i.e., there is usually a physical delivery of currency to a bank account. Send Money Home offer an in-depth comparison into the services offered by all the major non-bank foreign exchange companies.

It is estimated that in the UK, 14% of currency transfers/payments are made via Foreign Exchange Companies. These companies' selling point is usually that they will offer better exchange rates or cheaper payments than the customer's bank. These companies differ from Money Transfer/Remittance Companies in that they generally offer higher-value services.

Money Transfer/Remittance Companies

Money transfer companies/remittance companies perform high-volume low-value transfers generally by economic migrants back to their home country. In 2007, the Aite Group estimated that there were \$369 billion of remittances (an increase of 8% on the previous year). The four largest markets (India, China, Mexico and the Philippines) receive \$95 billion. The largest and best known provider is Western Union with 345,000 agents globally. Send Money Home is an international money transfer price comparison site that allows consumers access to a range of alternative products and rates available when remitting (transferring) money worldwide.

Trading Characteristics

There is no unified or centrally cleared market for the majority of FX trades, and there is very little cross-border regulation. Due to the over-the-counter (OTC) nature of currency markets, there are rather a number of interconnected marketplaces, where different currencies instruments are traded. This implies that there is not a *single* exchange rate but rather a number of different rates (prices), depending on what bank or market maker is trading, and where it is. In practice the rates are often very close, otherwise they could be exploited by arbitrageurs instantaneously. Due to London's dominance in the market, a particular currency's quoted price is usually the London market price. A joint venture of the Chicago Mercantile Exchange and Reuters, called Fxmarket space opened in 2007 and aspired but failed to the role of a central market clearing mechanism.

The main trading centre is London, but New York, Tokyo, Hong Kong and Singapore are all important centres as well. Banks throughout the world participate. Currency trading happens continuously throughout the day; as the Asian trading session ends, the European session begins, followed by the North American session and then back to the Asian session, excluding weekends.

Fluctuations in exchange rates are usually caused by actual monetary flows as well as by expectations of changes in monetary flows caused by changes in gross domestic product (GDP) growth, inflation (purchasing power parity theory), interest rates (interest rate parity, Domestic Fisher effect, International Fisher effect), budget and trade deficits or surpluses, large cross-border M&A deals and other macroeconomic conditions. Major news is released publicly, often on scheduled dates, so many people have access to the

same news at the same time. However, the large banks have an important advantage; they can see their customers' order flow.

Currencies are traded against one another. Each pair of currencies thus constitutes an individual product and is traditionally noted XXXYYY or YYY/XXX, where YYY is the ISO 4217 international three-letter code of the currency into which the price of one unit of XXX is expressed (called base currency). For instance, EURUSD or USD/EUR is the price of the euro expressed in US dollars, as in 1 euro = 1.5465 dollar. Out of convention, the first currency in the pair, the "base" currency, was the stronger currency at the creation of the pair.

The second currency, counter currency or "term" currency, was the weaker currency at the creation of the pair. Currencies are occasionally incorrectly quoted with the pairs inverted e.g. EUR/USD but this is incorrect. The "/" acts the same as the divide mathematical operator and derives the actual exchange rate. e.g. an amount of \$140,000 equates to €100,000. $\$140,000 / \text{€}100,000 = 1.4 = \text{USD/EUR}$ = a rate of 1.4 hence EURUSD or USD/EUR.

The factors affecting XXX will affect both XXXYYY and XXXZZZ. This causes positive currency correlation between XXXYYY and XXXZZZ.

On the spot market, according to the BIS study, the most heavily traded products were:

- EURUSD: 27%.
- USDJPY: 13%.
- GBPUSD (also called *cable*): 12%.

And the US currency was involved in 86.3% of transactions, followed by the euro (37.0%), the yen (17.0%), and sterling (15.0%). Volume percentages for all individual currencies should add up to 200%, as each transaction involves two currencies.

Trading in the euro has grown considerably since the currency's creation in January 1999, and how long the foreign exchange market will remain dollar-centred is open to debate. Until recently, trading the euro versus a non-European currency ZZZ would have usually involved two trades: EURUSD and USDZZZ. The exception to this is EURJPY, which is an established traded currency pair in the inter bank spot market. As the dollar's value has eroded during 2008, interest in using the euro as reference currency for prices in commodities (such as oil), as well as a larger component of foreign reserves by banks, has increased dramatically. Transactions in the currencies of commodity-producing countries, such as AUD, NZD, CAD, have also increased.

Determinants of FX Rates

The following theories explain the fluctuations in FX rates in a floating exchange rate regime (In a fixed exchange rate regime, FX rates are decided by its government):

- (a) International parity conditions viz; purchasing power parity, interest rate parity, Domestic Fisher effect, International Fisher effect. Though to some extent the above theories provide logical explanation for the fluctuations in exchange

rates, yet these theories falter as they are based on challengeable assumptions [e.g. free flow of goods, services and capital] which seldom hold true in the real world.

- (b) Balance of payments model. This model, however, focuses largely on tradable goods and services, ignoring the increasing role of global capital flows. It failed to provide any explanation for continuous appreciation of dollar during 1980s and most part of 1990s in face of soaring US current account deficit.
- (c) Asset market model views currencies as an important asset class for constructing investment portfolios. Assets prices are influenced mostly by people's willingness to hold the existing quantities of assets, which in turn depends on their expectations on the future worth of these assets. The asset market model of exchange rate determination states that "the exchange rate between two currencies represents the price that just balances the relative supplies of, and demand for, assets denominated in those currencies."

None of the models developed so far succeed to explain FX rates levels and volatility in the longer time frames. For shorter time frames (less than a few days) algorithm can be devised to predict prices. Large and small institutions and professional individual traders have made consistent profits from it. It is understood from above models that many macroeconomic factors affect the exchange rates and in the end currency prices are a result of dual forces of demand and supply. The world's currency markets can be viewed as a huge melting pot: in a large and ever-changing mix of current events, supply and demand factors are constantly shifting, and the price of one currency in relation to another shifts accordingly. No other market encompasses (and distills) as much of what is going on in the world at any given time as foreign exchange.

Supply and demand for any given currency, and thus its value, are not influenced by any single element, but rather by several. These elements generally fall into three categories: economic factors, political conditions and market psychology.

Economic Factors

These include: (a) economic policy, disseminated by government agencies and central banks, (b) economic conditions, generally revealed through economic reports, and other economic indicators.

1. Economic policy comprises government fiscal policy (budget/spending practices) and monetary policy (the means by which a government's central bank influences the supply and "cost" of money, which is reflected by the level of interest rates).
2. Economic conditions include.

Government Budget Deficits or Surpluses

The market usually reacts negatively to widening government budget deficits, and positively to narrowing budget deficits. The impact is reflected in the value of a country's currency.

Balance of Trade Levels and Trends

The trade flow between countries illustrates the demand for goods and services, which in turn indicates demand for a country's currency to conduct trade. Surpluses and deficits in trade of goods and services reflect the competitiveness of a nation's economy. For example, trade deficits may have a negative impact on a nation's currency.

Inflation Levels and Trends

Typically a currency will lose value if there is a high level of inflation in the country or if inflation levels are perceived to be rising [1]. This is because inflation erodes purchasing power, thus demand, for that particular currency. However, a currency may sometimes strengthen when inflation rises because of expectations that the central bank will raise short-term interest rates to combat rising inflation.

Economic Growth and Health

Reports such as GDP, employment levels, retail sales, capacity utilization and others, detail the levels of a country's economic growth and health. Generally, the more healthy and robust a country's economy, the better its currency will perform, and the more demand for it there will be.

Productivity of an Economy

Increasing productivity in an economy should positively influence the value of its currency. Its effects are more prominent if the increase is in the traded sector.

Political Conditions

Internal, regional, and international political conditions and events can have a profound effect on currency markets. All exchange rates are susceptible to political instability and anticipations about the new ruling party. Political upheaval and instability can have a negative impact on a nation's economy. For example, destabilization of coalition governments in Pakistan and Thailand can negatively affect the value of their currencies. Similarly, in a country experiencing financial difficulties, the rise of a political faction that is perceived to be fiscally responsible can have the opposite effect. Also, events in one country in a region may spur positive or negative interest in a neighbouring country and, in the process, affect its currency.

Market Psychology

Market psychology and trader perceptions influence the foreign exchange market in a variety of ways:

Flights to Quality

Unsettling international events can lead to a "flight to quality," with investors seeking a "safe haven."

There will be a greater demand, thus a higher price, for currencies perceived as stronger over their relatively weaker counterparts. The Swiss franc has been a traditional safe haven during times of political or economic uncertainty.

Long-term Trends

Currency markets often move in visible long-term trends. Although currencies do not have an annual growing season like physical commodities, business cycles do make themselves felt. Cycle analysis looks at longer-term price trends that may rise from economic or political trends.

“Buy the Rumor, Sell the Fact”

This market truism can apply to many currency situations. It is the tendency for the price of a currency to reflect the impact of a particular action before it occurs and, when the anticipated event comes to pass, react in exactly the opposite direction.

This may also be referred to as a market being “oversold” or “over bought”. To buy the rumor or sell the fact can also be an example of the cognitive bias known as anchoring, when investors focus too much on the relevance of outside events to currency prices.

Economic Numbers

While economic numbers can certainly reflect economic policy, some reports and numbers take on a talisman-like effect: the number itself becomes important to market psychology and may have an immediate impact on short-term market moves.

“What to watch” can change over time. In recent years, for example, money supply, employment, trade balance figures and inflation numbers have all taken turns in the spotlight.

Technical Trading Considerations

As in other markets, the accumulated price movements in a currency pair such as EUR/USD can form apparent patterns that traders may attempt to use. Many traders study price charts in order to identify such patterns.

Algorithmic Trading in Foreign Exchange

Electronic trading is growing in the FX market, and algorithmic trading is becoming much more common. According to financial consultancy Celent estimates, by 2008 up to 25% of all trades by volume will be executed using algorithm, up from about 18% in 2005.

GOODS MARKETS AND THE FOREIGN-EXCHANGE MARKET

The vast majority of the literature dealing with problems of balance-of-payments stability and with the effects of exchange-rate alteration on the autonomous balance of

payments makes the attempt to relate the conditions of the foreign-exchange market to the conditions in the export and import market. Although this approach is of less generality than the direct discussion of conditions in the foreign-exchange market, it is very advantageous whenever we try to apply economic theory to concrete problems of policy.

Even those who are actively concerned with the problems of external balance will have very little, if any, intuitive or experimental knowledge of the supply and demand elasticities in the foreign-exchange market.

Often, however, we hear economists assert that the elasticity of demand for imports or exports of such and such country is high or low. The nature of the products being traded may offer such insights. Thus it may be advantageous to relate the conditions in the exchange market, crucial in discussing balance-of-payments problems, to conditions in the export and import markets.

By doing this, we actually restrict our definition of the foreign exchange market to include only such supply and demand for foreign exchange as derives from current transactions. This often restricts the generality of analysis only very little. A greater complication arises from the fact that it is quite difficult to design conditions in the two economies considered that will leave the demand and supply curves for exports and imports unaltered when conditions in the exchange market are changing.

One such set of conditions is to assume that the volume of trade in comparison to total product or income of the two countries is very small. In this case, income effects, monetary effects, and structural price adjustment within the economy resulting from changing export and import prices may be assumed negligible. Also, to avoid other complications, the assumption is often made that exports and imports each consist of only one homogeneous commodity.

If exports and imports represent an appreciable fraction of national income and expenditure, relating commodity markets to the foreign-exchange market causes some difficulty. This difficulty stems from the fact that the export supply schedules can generally no longer be considered independently of import prices, or import demand schedules independently of export prices.

Hence, it is illegitimate to discuss the conditions in the export market in isolation from conditions in the import market, and vice-versa.

Of course, it would be possible to introduce such additional assumptions into a theoretical model. But this would make the analysis virtually unmanageable. Also, it might be unwarranted because the non monetary general equilibrium analysis of the export and import markets takes all such interdependencies into account.

Some writers, in an attempt to simplify the analysis, make the assumption of an infinite elasticity of supply of exports. But this is legitimate, if at all, only in a situation of less than full employment.

Yet the study of pure price effects and exchange-rate adjustment is usually conducted in a classical full-employment setting. To circumvent this difficulty, some international

economists substitute for the assumption of infinite supply elasticity the assumption of constancy of export prices, in terms of domestic currency, guaranteed by an appropriate monetary policy. Even under such conditions, however, the demand for imports will depend not only on the import price but also on the conditions in the export market, reflected by the monetary policy stabilizing the price of exports.

With these observations in mind, we may proceed to show the relation between the supply of foreign exchange for current-account transactions and the export market that generates such supply.

It will be left to the reader to establish such a relation for the demand function of foreign exchange. Later in this section we shall show rigorously the relations between the elasticities in the import and export markets, on the one hand, and the elasticities of supply and of demand for foreign exchange, on the other.

Pounds will be supplied in the foreign-exchange market, in order to purchase for dollars in the United States commodities demanded by the British consumer. In terms of pound prices, the demand curve for United States exports is fixed; in the somewhat cumbersome, but very useful, construction of it is depicted as D_x .

The vertical axis passing through O measures prices of United States exports in British currency and the horizontal axis the physical quantity of such exports. The supply curve of United States exports will be steeper or flatter in the British market, depending on whether the rate of exchange (i.e., price of pounds in terms of dollars) is low or high. Supply curves for three rates of exchange, $S_x(1)$, $S_x(2)$ and $S_x(3)$, corresponding to rates of exchange $r = 1$, $r = 2$, and $r = 3$, are shown in the diagram.

They determine three equilibrium levels of exports and three equilibrium pound prices. All that has to be done in order to obtain the supply schedule of foreign exchange is to relate the three rates of exchange to corresponding products of the pound prices and quantities of exports as obtained by the construction.

This may easily be done in the diagram itself if we observe that all the rotating supply curves have to pass through one point on the horizontal axis, O' , and that their intersections along any horizontal ray (not identical with the horizontal axis passing through O' and O) must show proportions of the different rates of exchange.

Thus the three supply curves intersect a new horizontal axis passing through O'' at points 1, 2, and 3 units of distance to the right of O'' . Using this axis for measuring the rate of exchange and plotting on the coordinate the areas of the rectangles, such as $OABC$ (the number of pounds supplied), we obtain the complete supply curve of foreign exchange, S_f . We notice that at the rate of exchange $r = 3$ the supply of pounds is maximum—the elasticity of D_x being unity and declines for higher rates of exchange.

When we want to study the relation between changes in the foreign-exchange market and changes in the export and import markets, it is often unnecessary to know the complete supply and demand schedules in the export and import markets and their relation to the foreign-exchange market. Sometimes it suffices to be able to relate elasticities of supply and demand in a particular foreign-trade market to the corresponding

elasticity of the foreign exchange supply or demand schedule. As we are going to see in the remaining part of this section, a simple relation does exist between the elasticities in the trade markets and the foreign-exchange market.

Using E_s and E_d to represent the elasticity of supply and of demand for foreign exchange, respectively, we may express these two magnitudes as where s and d represent the elasticities of supply and demand, respectively, and the subscripts x and m are used to indicate exports and imports.

The derivation of these functions in explicit form is quite straightforward. We may write the supply curve of foreign exchange (pounds) as $S = S(r)$ and the supply and demand functions of exports, generating this supply, as.

$S_x = S_x(P_s)$ and $D_x = D_x(P_{-2})$ where P_s and P_{-2} represent dollar and pound prices, respectively. In order to be able to establish equilibrium in the export market and thus derive the supply of pounds, it is necessary to express the dollar price in terms of the pound price and the rate of exchange, i.e., $P_s = P - 2 \cdot r$. Substituting relation we obtain $S_x = S_x(P - 2 \cdot r)$.

We know that the elasticity of supply of foreign exchange E_s is actually where S' is used to express a derivative of S , and S is equal to the product of pound price and the quantity of exports, i.e., using relation, $S = P_{-2} \cdot D_x(P_{-2})$.

Thus we may write $S'(r)$ as, From the equilibrium conditions of the export market, i.e., $S_x = D_x$ after differentiating with respect to r and rearranging, we obtain where "primes" again are used to indicate derivatives.

Substituting that $r = P_s / P_{-2}$ after a little reorganization of the terms in the fraction, using the fact that S_x and D_x in equilibrium are equal to exports. Thence, using the definitions of the elasticity of demand for exports, d_x , and of the elasticity of supply of exports, s_x , we may write the important result.

Stability in the Foreign-Exchange Market Defined by the Export and Import Elasticities

We have indicated how important a role the elasticities in the foreign-exchange market play in any concrete consideration of an exchange-rate alteration. We observed that if the rate of exchange were moved from its equilibrium level re downward, an autonomous deficit in the balance of payments would result for the dollar country.

This deterioration in the balance of payments is closely linked to the fact that the equilibrium at r_e is stable; if it were unstable, i.e., if the ED curve at r_o were upward-sloping, a depreciation of the foreign currency (appreciation of domestic currency) would improve, rather than worsen, the balance-of-payments position of the dollar country.

Thus, for small movements in the neighborhood of the equilibrium rate r_o , the conditions for stability and instability of the foreign-exchange market are perfectly identical with the conditions for the balance of autonomous payments to improve or deteriorate from the depreciation of the home currency. The earlier writers fully accepted

this identity of conditions for all cases and discussed problems of appreciation and devaluation in terms of the stability conditions.

As we shall see in the next section, this correspondence is somewhat modified if in the initial situation the balance of payments were out of balance; actually, it is this case that is more pertinent to the problems of exchange-rate alteration, because, in practice, there would be very little reason to depreciate or to appreciate if the balance of autonomous payments were in balance.

Nevertheless, it will be useful to consider at first the case of stability and instability as determined by the elasticities of the export and import markets and thus establish the relation between changes in the exchange rate and the balance of payments. The stability conditions require that the elasticity of supply in the exchange market be larger in algebraic value than the elasticity of demand. Assuming that only transactions on current account depend on the rate of exchange and using relations, this yields and, after regrouping and elimination of the fractions,

$$S_x \cdot S_m \cdot (1 + d_x + d_m) - d_x \cdot d_m \cdot (1 + S_x + S_m) < 0$$

We notice that the right side of the latter inequality must always be positive, except, of course, if either the export or the import commodity were an inferior good, subject to "Giffen's paradox." Thus the market could be unstable only if the left side of the last inequality were positive, i.e., if the sum of the demand elasticities were larger than. But even under such conditions stability could be found, provided only that the right side were large enough.

Thus the positive sign of the left-hand side is only a necessary, but not a sufficient, condition of instability or of balance-of-payments deterioration after devaluation of currency. To state the same proposition in reverse, the negative sign of the left-hand side is a sufficient, but not a necessary, condition of stability in the foreign-exchange market.

It is only in the special case where the elasticities of supply of exports and imports are infinite that the above-stated conditions are both necessary and sufficient.

In this case, the right side of the inequality becomes zero, and thus the market must be stable whenever the left-hand side is negative. It is the latter case that is usually referred to rightly or wrongly as the Marshall-Lerner conditions for devaluation or stability. Again from the symmetry of markets it follows that whatever proposition we have made so far about the dollar market we have also made concerning the pound market. Actually, the numerical value of the left-hand side of inequality regardless of which exchange market we consider.

The Balance of Autonomous Payments on Current Account and the Rate of Exchange

In the preceding section we have roughly established the direction of change in the balance of payments that will result from an alteration in the exchange rate. It may be more interesting, however, for practical policy considerations to evaluate exactly the

change in the balance of payments that may be expected from a given alteration in the rate of exchange. In the preceding section we have only dealt with the case in which the autonomous payments and receipts on current account are initially in equilibrium; we shall relax this assumption here and assume, more realistically, that the balance of payments on current account is out of equilibrium prior to devaluation or appreciation.

The change in the pound value of exports of the dollar country resulting from a small relative change in the rate of exchange will be equal to the initial value of exports times the elasticity of supply of foreign exchange times the assumed relative change in the exchange rate.

It is now apparent that if exports are equal to imports, all the conclusions relative to stability and balance of payments obtain. If, however, current exports and imports are not equal in the initial situation, these conditions may be altered. Thus, for example, even with the elasticity of supply of foreign exchange larger in algebraic value than the demand elasticity, a surplus balance of payments (i.e., $X > M$) may improve from appreciation of currency; or a deficit balance may improve, even though the elasticity of demand for foreign exchange exceeds in algebraic value the elasticity of supply.

It is clear that such situations may arise because the initial values of exports and imports may give different weights to the elasticities of supply and demand for foreign exchange in relation 6.3. The reader may use the method of calculation used in the preceding section to establish for himself a new relation, in terms of the elasticities of supply and demand for exports and imports, expressing the condition of balance-of-payments improvement as a result of devaluation.

Often a country will be concerned not so much about what will happen to its balance of payments as a result of exchange-rate changes, in terms of foreign currency, as in terms of its domestic currency.

Also, it may be concerned with the amplitude of exchange rate fluctuations in a free market resulting from given autonomous shifts in the supply and demand of domestic, rather than foreign, currency. In this case it will be preferable to express the relation between the exchange-rate changes and the balance of payments in domestic currency rather than in foreign currency, as we have done so far. A simple inspection of the graphical representation of the foreign-exchange market, will convince the reader that the change in the balance of payments expressed in one currency will not be proportional to the change in the balance expressed in the other currency.

This is immediately apparent from the fact that the deficit (or surplus) in terms of foreign exchange is measured on the abscissa of the diagram, while the deficit (or surplus) in terms of the domestic currency is measured as the area of the rectangle formed by the abscissa (foreign exchange) and the coordinate (the rate of exchange).

Actually, for a small change in the exchange rate, the balance might improve in terms of foreign exchange and deteriorate in terms of domestic currency. Such a situation is the more likely, the greater the initial deficit and the smaller the algebraic difference between the elasticities of supply and demand of foreign exchange.

Thus let us write the balance of payments in terms of domestic currency (\$) as $BP_{\$}$ $= r \cdot BP_{-2}$ and differentiating.

$dB P_{\$} = dr \cdot BP_{-2} + r \cdot dBP_{-2}$ because the pound values in 6.7 multiplied by r are equal to corresponding dollar values of exports and imports. Again it may be preferable to rewrite relation, using the elasticities of the exports and import markets rather than those of the foreign-exchange market. Using relations this is extremely simple and is left to the reader.

It will be noticed that if, initially, the balance of payments is in equilibrium, i.e., $X_{\$} = M_{\$}$, then relation 6.8 will be simplified to a relation resembling relation, except for the fact that the value of trade is expressed in domestic rather than foreign currency. For any small displacement from equilibrium, the change in the balance of payments will be exactly the same, whether expressed in terms of domestic or of foreign currency, provided, of course, that one is converted into the other at the original rate of exchange.

As a question of policy, it is often argued that devaluation of a currency as a tool for restoring external balance is undesirable on grounds of the low elasticities of demand for imports. From relations, it is immediately apparent that, if this is so, for any “normal” values of the elasticities of supply both the elasticities of supply and demand for foreign exchange are likely to be low and negative.

Even if this does not lead to instability in the autonomous foreign-exchange market, the slope of the excess demand curve for foreign exchange is likely to be quite high in absolute value. Under such conditions, equilibration of the accounts might require an appreciable alteration of the rate of exchange. That if income effects are also introduced into the adjustment mechanism, such difficulties in restoring the external balance through devaluation may become even more serious.

If the elasticities of demand for imports are so low as to render the foreign-exchange market unstable, then appreciation of currency rather than devaluation can be used in restoring the external balance.

If the elasticities of demand for imports are low, it also follows from the analysis of the present section that a free exchange rate will fluctuate rather widely as a result of relatively small autonomous shifts in the supply of and/or demand for foreign exchange. It may be argued, however—and on quite solid grounds—that exchange speculation and interest arbitrage might provide a sufficient stabilizer. Also, the question still remains to be answered whether the elasticities are in actuality so low.

The Exchange Rate, the Terms of Trade, and the Balance-of-Payments Adjustment

We shall turn in greater detail to the discussion of the terms of trade, i.e., to the ratio between export and import prices. Let us simply observe here that the *terms of trade* is an important index expressing the purchasing power of exports in paying for imports. If a country's terms of trade improve, less of its real product exported will be needed to purchase one unit of real product of the rest of the world and vice-versa. Some special

conditions are fulfilled (to be stated presently), the terms of trade will move in one direction or another whenever the rate of exchange changes either as a matter of deliberate policy or, in free adjustment, in response to changing conditions in the exchange market. Thus the balance of payments, if the rate of exchange is pegged at a given rate, will be related to the terms of trade and the terms of trade to the balance of payments. Also an autonomous transfer from one country to the other will affect the export and import prices of the trading partners. While we shall concentrate in greater detail on the transfer problem in the next section, our present task is to relate the terms of trade to the key concepts and variables of the foreign exchange and the export and import markets.

To illustrate the problem in a simple example, consider a case in which both trading countries, for one reason or another, are able to supply their exports at a fixed price in terms of their domestic currencies. It is easy enough to see that in this situation the terms of trade ($TT = P_x/P_m$) of either country will vary in inverse proportion with the changes in the exchange rate, defined as the price of foreign currency in terms of domestic currency. This will hold true whether the exchange-rate fluctuation is a result of official exchange rate adjustment or free rate fluctuation. Thus, if under our assumptions the dollar were devalued by 1 percent, the terms of trade, whether expressed in domestic or foreign prices, would deteriorate by 1 percent.

Similarly, if the elasticities of demand for imports in both countries were infinite, the terms of trade of each country would always move in proportion with the exchange rate. If, for example, the dollar were devalued by 10 per cent, export prices in terms of dollars would increase by the corresponding amount, while dollar import prices would remain unchanged. Thus the terms of trade would improve by 10 per cent. It is evident that the same result would be obtained for the terms of trade if pound prices were considered; however, in this case the export prices would remain unchanged, while those of imports would go down by 10 per cent.

In the "normal" case, where neither both supply nor both demand elasticities are infinite, the result is no longer so clear cut or self-evident. However, it may easily be derived. Suppose that the rate of exchange of the dollar country is changed by dr . This may happen either because of official exchange-rate adjustment or because an autonomous capital transfer takes place that does not affect the supply and demand functions of exports and imports. The pound value of imports $M_{\text{£}}$ may be written as the product of the pound price and quantity of imports supplied

$$S_m, \text{ i. e., } M_{\text{£}} = P_{\text{£}} \cdot S_m$$

The change in this value resulting from a small change in the exchange rate may be expressed as where represents the derivative of the supply of imports function with respect to the pound price of imports. But the change in value of imports also, as we saw in the preceding sections, must be equal to the product of the elasticity of demand for foreign exchange (pounds) times the value of imports times the per cent change of the exchange rate.

These results are perfectly sufficient to analyse the impact of exchange alteration on dollar and pound export and import prices and consequently on the terms of trade. Sometimes it may be preferable to express the change in the terms of trade directly as a function of the relative change in the rate and of the export and import elasticities. This will be easily done if we observe that, for sufficiently small changes of the value of the currency, the relative change in the terms of trade is simply the difference between the relative change in the export and import prices. Obviously, it is no longer important whether we consider the dollar or the pound prices, because the terms-of-trade index (as well as its alteration) is identical whether measured in terms of one set of prices or the other.

THE ROLE OF EFFICIENT MARKETS

Reserve management, and any related policy operations, should be conducted in markets that have sufficient depth and liquidity, and can process transactions in a sound and efficient manner. Reserve managers need to be certain that reserves can be liquidated in a prompt and efficient manner to provide the necessary foreign exchange for the implementation of policy objectives relating to, for example, market intervention, meeting balance of payments or debt servicing needs, or limiting external vulnerability. Policy actions, on the other hand, can also involve the purchase of foreign exchange that should be placed promptly into investment portfolios. In these situations, undertaking the relevant investment transactions in deep and liquid markets serves to ensure that they can be easily absorbed by these markets and without undue impact on investment prices received, or paid by, the reserve manager.

Policy actions invariably, however, involve transactions initiated in the reserve management entity's domestic foreign exchange market, and which have a consequential implication for the reserve manager. In these circumstances, the effectiveness of the policy action will be dependent upon the efficiency and soundness of the domestic market. Furthermore, and of particular relevance for reserve levels, any failings or operating weaknesses in that market can directly impact on either the amount of reserves required to support the policy action or the amount of foreign exchange added to reserve holdings. Undertaking transactions in deep and well-established markets ensures that reserve-related transactions can be easily absorbed at market determined prices without undue distortions, or adverse impacts on the level and availability of foreign exchange reserves.

THE EUROPEAN MONETARY SYSTEM

Since its conception in 1978, the European Monetary System (EMS) has attracted much attention as a framework for policy coordination. Disillusioned by the prospects for global monetary reform and by the performance of the floating exchange rate system, the EMS founding fathers wanted to restore a system of fixed-but-adjustable exchange

rates within as large a part as was possible of the European Community. Such a system would protect the large intra-European trade flows against the sharp shifts in competitiveness which were thought likely to occur in its absence. It would also limit the divergence of national inflation rates and permit lower, less variable inflation. This vision of the evolution of 'a zone of monetary stability' contained a strong element of damage limitation. Yet at the time it was rightly regarded as highly ambitious because it brought back into European management the currencies of some countries, France and Italy in particular, which had remained aloof from earlier efforts. The system has subsequently evolved well beyond its original intentions: the exchange rate mechanism has become tighter, monetary policy coordination is now closer and capital mobility higher than was the case in the early years of the EMS.

In the light of these developments, CEPR organized a joint conference with the Banca d'Italia and the Centro Interuniversitario di Studi Teorici per la Politica Economica (STEP) in Perugia in October. The conference took stock of the substantial existing research on the EMS and assessed the contributions it can make to our understanding of how the EMS works, how some of its effects may be measured, and how it might be improved according to certain well defined criteria. Macroeconomic theory and careful use of econometric analysis can illuminate some of the main implications of the particular blend of exchange rate management and external constraints represented by the EMS. These tools can also facilitate the evaluation of the more operational, short-term problems encountered in managing the system and the national monetary policies that underpin it.

The International Environment

In 'The EMS, the Dollar and the Yen', Rudiger Dornbusch (MIT and CEPR) criticized three characteristics of the present EMS. Dornbusch questioned whether inflation rates in Southern Europe should converge towards the near-zero level reached in Germany. He also argued that the EC should consider carefully whether to pursue full liberalization of capital movements without allowing more flexibility in the exchange rate for financial transactions.

Dornbusch also observed that the process of adjustment in international imbalances is far from complete and that the US dollar may still have some distance to fall: this may pose a major challenge to the cohesiveness of the EMS currencies. More generally, Dornbusch doubted whether a high degree of nominal exchange rate rigidity was either desirable or feasible for all present and potential members of the system.

His argument against further disinflation in Southern Europe was novel: it rested less on the output losses which would result from such a policy than on the sustainability of public debt during a period of declining inflation, when the scope for financing part of the public sector deficit by money creation was diminishing. The aim of stabilizing nominal exchange rates relative to the strong EMS currencies may allow a 'Mediterranean' government to lower real interest rates further and generate more growth

(together with the associated budgetary improvement) than it could achieve by assuming a less disinflationary stance. But if some countries, notably Italy and the new entrants in the EC, were to follow the Dornbusch prescription and openly accept a permanently higher inflation rate than Germany and other members aiming at price stability, rules would have to be found for accommodating the required exchange rate changes. Dornbusch suggested a crawling peg with frequent, almost continuous adjustments between the two groups of currencies so as to maintain unchanged competitiveness. Domestic monetary and fiscal policies would then have to ensure that inflationary instability does not develop.

Dornbusch argued that this option represented the best framework for widening exchange rate management in Europe, particularly if it were complemented by a system of dual exchange rates. Real exchange rates, or competitiveness, would then be stabilized by the crawling peg, while the rates at which financial transactions take place would not be constrained by intervention limits.

Dornbusch conceded that a moderate transactions tax on all foreign exchange purchases imposed worldwide may constitute a simpler method of deterring excessive capital mobility than a dual exchange rate system, but it would bring the added inconvenience of a tax on current account transactions. He remarked that the EMS countries today benefit from what is in effect a system of dual rates. Most of the trade of EMS participants is with other members and is carried out at tightly managed exchange rates which do not permit sudden shifts in competitiveness, while a large share of members' financial transactions involve floating currencies, notably the dollar.

Disinflation and External Adjustment

Four conference papers offered a detailed assessment of the EMS as a framework for cooperation among its members; of the implications for key macroeconomic variables, notably inflation and output; and of the determinants of the trade imbalances of the three largest participants.

Jacques Melitz (INSEE and CEPR), in his paper 'Monetary Discipline and Cooperation in the EMS: A Synthesis', explored the incentives countries have to participate in an arrangement like the EMS. Most observers have tended to focus on the disinflationary mechanism operating on the weaker participants in the system. The monetary authorities in the countries with weaker currencies gain credibility for their disinflationary stance by pegging their currencies to the less inflationary currencies over long intervals, with the prospect that they cannot fully accommodate their excess inflation by devaluing when the EMS parities are realigned. Private agents will therefore interpret an EMS commitment as a more effective constraint than a national monetary target or some other intermediate objective by which an individual country signals its disinflationary intentions.

These benefits of 'tying one's hands', to use the expression of Giavazzi and Pagano, seem attainable in the context of the EMS. Melitz noted that they are transitory; if a

weak-currency country has pursued a disinflationary stance through its link to a stable DM rate in the EMS for some length of time, and national inflation has been reduced to near-German rates, the effort could presumably have been made by the country acting alone, without the external commitment represented by the EMS.

Melitz anticipated the discussion later in the conference on the desirability of a European central bank to replace the present mechanism where countries with weak currencies draw on the reputation of the Bundesbank. Greater internal autonomy for other national central banks could facilitate the construction of a European institution, or might even be regarded as a partial alternative to it.

Melitz also analysed why Germany too has found the EMS attractive. The impetus to launch the system came after all at least as much from Germany as from France. Why does Germany prefer the EMS to the alternative of individual pursuit of price stability? In the latter case Germany would export stability to those countries that want to peg their currencies through their own efforts to the DM, but without undertaking the commitments to sustain these other currencies through intervention and accepting 'excessive' money creation and some imported inflation. Melitz developed an extended version of his model in which competitiveness enters the welfare function of both Germany and France. Both countries desire growth in the sector producing traded goods, but Germany finds this easier to achieve than other EMS members because of the steady real depreciation of the DM which results from keeping nominal exchange rates stable while inflation rates have not yet fully converged.

This argument suggested that EMS membership is attractive to Germany because it gains more in terms of gradually improving competitiveness than it loses in terms of upsets to German monetary policy. Melitz argued that these gains may prove more durable than those of Germany's partners pursuing disinflation through the EMS.

Given the emphasis on the convergence of inflation towards a low and stable rate within the EMS, it is natural that the evidence on the determinants of inflation should be examined carefully. Can part of the remarkable deceleration of inflation observable in all participating countries since 1979 or rather since inflation peaked in 1980 after the second oil price shock be attributed to the EMS? This question was approached in both theoretical and empirical terms in the papers by Giavazzi and Giovannini and by Collins.

In their paper, 'The Role of the Exchange Rate Regime in a Disinflation: Empirical Evidence on the European Monetary System', Francesco Giavazzi (University of Bologna and CEPR) and Alberto Giovannini (Columbia University and CEPR) first reviewed the recent literature on the channels through which a commitment to the EMS may help a process of disinflation, and then looked for evidence of a downward shift in inflationary expectations following the introduction of the new, supposedly tighter exchange rate regime of the EMS. They estimated a reduced-form system of equations for the quarterly behaviour of wages, prices and output for three high-inflation EMS countries (Denmark, France and Italy), for Germany, and for the United Kingdom over

the 1960s and 1970s, prior to the formation of the EMS. They compared the predictions from this model with actual observations of wages, prices and output over 1979-87. The results corresponded broadly to what one would expect: in the three countries trying to squeeze their inflation through EMS membership, wage and price inflation was lower than predicted by the model based on the pre-EMS experience; in Germany, however, inflation was higher than the pre-EMS model predicted. The evidence further suggested that there was no loss in output in Denmark and Italy (rather a gain), some loss in France between 1980 and 1984, and, surprisingly, a substantial loss in Germany.

On the whole the evidence was relatively weak; a downward shift in inflationary expectations did appear to have taken place, but only gradually and less dramatically than in the United Kingdom where the actual inflation rate dropped sharply below the predicted rate from 1981 onwards.

In her paper 'Inflation and the EMS', Susan Collins (Harvard University) reached a similar conclusion using different data and methods. Collins used the experience of a group of non-EMS countries as a benchmark and found weak evidence that since 1979 inflation rates have converged more rapidly within the EMS than outside.

This seemed to occur during the period before 1978 as well, however, and there is evidence only of a gradual effect within the EMS, from about 1983 onwards. There appear to have been two stages to EMS membership. In the initial stage the behaviour of the authorities was modified, because the policy instruments available to them worked differently in a regime with more stable exchange rates. After several years, however, the disinflationary commitment became more credible to the public.

The paper by Stefano Vona and Lorenzo Bini Smaghi (Banca d'Italia), 'Economic Growth and Exchange Rates in the EMS: Their Trade Effects in a Changing External Environment', examined the determinants of the exports and imports of goods in the three main EMS economies. Their results permitted them to address a number of questions of central importance to the sustainability of an EMS in which parity realignments have been relegated to a minor role in the adjustment of external imbalances.

Vona and Bini Smaghi found that changes in trade balances for Germany, France and Italy are strongly influenced by relative demand growth in the three countries and by growth outside the EMS. Price elasticities were significant and sufficient for a devaluation to contribute to a trade balance improvement. But devaluation was, even within the partial equilibrium approach adopted by the authors, far less effective as an instrument for shifting demand between the three countries than measures which affect the relative growth of demand more directly, notably fiscal policy.

The authors also concluded that the boom in the United States from 1983-4 and the appreciation of the dollar and several other non-EMS currencies between 1980 and 1985 improved trade balances significantly for the main EMS economies. These economies display a comparable sensitivity to events outside the EMS, but the beneficial effects of these events up to 1985-6 were more valuable to France and Italy, which were

in that period trying to disinflate and maintain their parities within the EMS. As these favourable factors have been reversed by the recent depreciation of the dollar and the convergence of US and European growth rates, difficulties have reappeared in all EMS countries except Germany, where slow growth has implied a rising trade surplus despite the unfavourable external factors.

If demand were to grow in parallel at home and abroad, Germany and Italy would experience no major trade imbalance. For France, however, the income elasticity of demand for imports seems to be well above the elasticity of demand for French goods by her trading partners in the EMS and elsewhere. If growth rates are similar, France will therefore experience a steady deterioration in its trade balance: this must be corrected through an improvement in competitiveness, via price or non-price changes. Vona and Bini Smaghi's paper suggested that tensions, in the form of growing trade imbalances, have been building up within the EMS and may threaten the system.

Capital Mobility and Monetary Coordination

Other papers presented to the conference explored the performance of the EMS in the short term and the degree of monetary coordination required to underpin the system. Most observers have concluded that the EMS has succeeded in reducing exchange rate volatility among its participants. The reduction is obvious if the comparison is made with either the pre-EMS performance of the participating currencies or with a sample of non-EMS currencies. Nor does stability inside the system appear to have been achieved at the expense of more instability vis-a-vis the currencies outside the EMS. Michael Artis (University of Manchester and CEPR) and Mark Taylor (University of Dundee and CEPR) went beyond these straightforward comparisons in their paper, 'Exchange Rates and the EMS: Assessing the Track Record'. They computed the conditional variance of EMS exchange rates, taking into account the higher degree of stability in the underlying macroeconomic determinants and without making assumptions as to the specific characteristics of the distribution of exchange rate changes. The more comprehensive approach used by Artis and Taylor suggested similar conclusions: since 1979 the monthly volatility of the nominal and real exchange rates of the DM against five other EMS currencies fell significantly, while similar measures of volatility for the dollar (and to a lesser extent sterling) have increased.

Artis and Taylor also investigated whether the EMS has merely transferred volatility from the foreign exchange markets to domestic financial markets. One plausible hypothesis is that stabilization of the exchange rate has required more activist monetary policies and more flexible use of interest rate differentials to sustain the EMS parities. This was not borne out by the empirical work of Artis and Taylor; the volatility of short-term nominal interest rates in all four of the markets considered (Germany, the Netherlands, France and Italy) had fallen since 1979, though the reduction was statistically significant only in the Netherlands and Italy. This is in contrast to what has been observed in countries with floating exchange rates.

Some concern centres on the capacity of the EMS to limit not merely short-run volatility but also longer-term misalignments, i.e. departures from a structure of intra-EMS exchange rates consistent with macroeconomic equilibrium. Under the EMS, a steady real depreciation of some currencies (DM, guilder and Belgian franc) has persisted along with real appreciation of the remaining currencies, though these misalignments are now growing more slowly. The authors concluded by noting that despite the progress achieved within the EMS, the system has not yet been successful in rendering the participating currencies close substitutes. The simple uncovered interest rate parity condition is closer to being fulfilled between the DM and the dollar than between the DM and other EMS currencies. The risk premium has not been reduced to zero even for the guilder. That suggests that financial markets still question the credibility of the longer-run commitment of the EMS countries to develop into an area of more permanently fixed exchange rates.

The papers presented by John Driffill and Maurice Obstfeld studied whether a fixed-but-adjustable exchange rate system such as the EMS was compatible with perfect capital mobility and a high degree of forward-looking behaviour in financial markets. This analytical extension is timely, because the removal of capital controls has already proceeded well beyond the stage reached during the sample period used for estimation in most of the empirical work reported at the conference. In addition, further steps towards full liberalization of capital movements and of trade in financial services have been proposed by the European Commission and are currently under debate in the Council of Ministers. Can the EMS reconcile stable exchange rates and perfect capital mobility?

John Driffill (University of Southampton and CEPR) in his paper, 'The Sustainability and Stability of the EMS with Perfect Capital Markets', suggested that an EMS without capital controls did not have particularly drastic implications for the macroeconomic policies of member countries. Inflation differentials and other indicators of divergence have fallen sufficiently to be accommodated by realignments, at roughly annual intervals, which would be modest enough not to disrupt foreign exchange markets.

Analogies to the experience of the final years of the Bretton Woods system, or the literature on collapsing exchange rate regimes inspired by Latin American experiments in currency stabilization and financial liberalization, are likely to be misleading because the residual tensions and the prospects of large speculative gains are much more modest in the present EMS. The EMS has developed a reputation as a relatively robust system. This has been achieved under conditions of less than perfect capital mobility and rests on the notion that EMS governments have gradually precommitted themselves not to allow exchange rates to jump. In order to preserve this reputation, realignments cannot become too infrequent and their timing should continue to be uncertain as well.

Driffill's discussion of the rules of monetary management which could underpin the EMS in the absence of capital controls took as its point of departure a two-country model by Buiter, where the two central banks continuously react to incipient reserve

flows by adjusting their domestic credit expansion to accommodate shifts in money demand; hence reserve flows are minimized. A less constraining mechanism can be envisaged for the EMS which has large reciprocal credit facilities for the participating central banks and in which reserves are sizeable. But the stability of the system still requires domestic policy instruments to respond promptly to reserve flows.

In his paper, entitled 'Competitiveness, Realignment and Speculation: The Role of Financial Markets', Maurice Obstfeld (University of Pennsylvania) took a more sceptical view of the compatibility of free capital flows and an exchange rate system in which parity changes can rationally be expected from time to time. Obstfeld explored what kinds of equilibria can arise when markets expect a loss of competitiveness of domestically produced goods to lead to realignment.

If such behaviour has been observed on the part of the authorities for some time and if realignments have typically restored competitiveness the type of constant real exchange rate maintained through a crawling peg, as suggested by Dornbusch fears of devaluation may force a collapse and an inflationary spiral. If, on the other hand, it is apparent that realignments typically do not fully offset cumulative excess inflation since the previous realignment, then a spiral of inflation and devaluation cannot occur. This appears to have been the case in the EMS since 1983: the realignments of April 1986 and January 1987 between the DM and the French franc were only about half the size required to remove fully the real appreciation of the franc since 1983.

Obstfeld also discussed a second model, in which multiple equilibria are possible if expected realignment is driven by expansionary fiscal policy in the home country. If past behaviour suggests to the private sector that the government will ultimately fully offset the real appreciation resulting from a fiscal expansion, an exchange rate collapse will occur when the expansion is announced but before there is any real appreciation.

This line of analysis captured some features of the early EMS experience: even though capital mobility was far from perfect, on occasion capital flight from France followed the announcement of a more expansionary fiscal policy in the 1981-3 period. This perceived risk of engaging in solitary fiscal expansion may have paralysed fiscal policy in some European countries. Obstfeld's analysis provided arguments either for undertaking fiscal action only as part of a coordinated, though possibly differentiated, package in the EMS countries, or for moving to a more tightly managed exchange rate system and ultimately a fixed-rate union in which destabilizing expectations of devaluation and inflation would be effectively curtailed. Obstfeld's paper raised important issues for the stability of the EMS, which has proceeded to full removal of capital controls before achieving an underlying convergence of inflation rates and the coordination of the macroeconomic policies which are seen to influence convergence. The discussion of Obstfeld's paper suggested that he placed too much emphasis on the likelihood of instability and an inflationary spiral. This seemed to depend more on his assumption of a high degree of foresight on the part of economic agents as to the consequences of government policies and the actions governments will take in the

EMS than it did on the assumption of high capital mobility. The uncertainty of the timing and size of EMS realignments and the practice of giving no assurance to 'inflationary' countries that they can restore competitiveness fully may well be enough to stabilize the system in practice. The paper entitled 'Interventions, Sterilization and Monetary Policy in EMS Countries, 1979-87', by Cristina Mastropasqua, Stefano Micossi and Roberto Rinaldi (Banca d'Italia), offered detailed insights into the intervention rules and practices in the EMS, as well as case studies of the interaction between the domestic and external components of money creation in four countries (Belgium, France, Germany and Italy).

Their empirical work suggested that to date, interventions have not been shared equally between Germany and the other countries studied, particularly since the Bundesbank has not until very recently participated to any major extent in 'intra-marginal' interventions. The authors attempted to measure asymmetries between member countries in the degree to which they have preserved monetary autonomy. One such indicator is the extent of sterilized intervention. Using quarterly data for 1979-86 they found that whereas Belgium, France and Italy have on average sterilized 30-40% of the net change in the foreign assets of their central banks, that proportion is 60-80% for Germany. This rough measure suggested that the Bundes bank has had twice the capacity of its EMS colleagues to sterilize its relatively more modest interventions in the foreign exchange markets in order to keep its preferred domestic monetary aggregate close to target. This difference has been observed despite the existence of some capital controls which should have facilitated sterilization in the three non-German countries. One might expect the asymmetry to widen, *ceteris paribus*, as one moves towards perfect capital mobility.

The authors' findings confirmed an asymmetric interpretation of the EMS, in which Germany pursues a domestic monetary target and sterilizes external flows which are likely to upset its pursuit of this target, while the other EMS countries conduct policy in terms of domestic credit expansion, allowing external flows to influence domestic monetary conditions. Greater symmetry will, the authors concluded, require a degree of consensus on medium-term policy objectives that simply does not exist today.

The extent of this consensus also featured in 'Monetary Policy Coordination Within the EMS: Is There a Rule?', by Massimo Russo (IMF) and Giuseppe Tullio (Banca d'Italia), which focused on procedures for decentralized or centralized monetary management in the EMS. Russo and Tullio examined the explicit or implicit adjustment mechanisms under the gold standard and the Bretton Woods system and discussed how the proposals for international monetary reform by McKinnon and by Williamson and Miller could be implemented within a reformed EMS. The simplest idea, but also the most radical, would be to fix an aggregate monetary objective for the EMS countries combined. Preliminary research reported in the paper suggested that an aggregate EMS money demand function could prove to be at least as stable as the money demand function in Germany. But would it be possible for the EMS countries to agree explicitly

on a common inflation objective, such as would be required to set a collective monetary target? How would the national components be assigned in the absence of a fully fledged European central bank? A second, very interesting possibility is to try to replicate the desirable properties of the first proposal by a decentralized rule in which all participants including Germany set a national objective for domestic credit expansion but agree to refrain from sterilizing the interventions which would be required to sustain fixed exchange rates in such a system. This would remove the asymmetry documented by Mastropasqua, Micossi and Rinaldi.

If there were no net interventions in dollars and other non-EMS currencies, aggregate domestic credit expansion in the EMS countries would be equal to total money creation and the participants might then jointly consider whether monetary growth for the area as a whole was acceptable.

The paper also considered two other, partly decentralized procedures inspired by recent proposals to reform the international monetary system: a nominal income target for each EMS participant and a monetary rule for each country, subject to a common inflation target, e.g. stable wholesale or traded goods prices for the EMS on average. These proposals have the virtue of addressing explicitly the issue of the nominal anchor for the system. The second proposal might be suitable as a medium-term objective for a European central bank. There might even be an advantage in declaring them formally in order to confirm the autonomy of such an institution and help to build its reputation. But these proposals faced operational problems in designing the feedback rules needed to link departures from price stability to the changes in national monetary growth rates.

External Market-Based Risks

Liquidity risk: The pledging of reserves as collateral with foreign financial institutions as support for loans to either domestic entities, or foreign subsidiaries of the reserve management entity, has rendered reserves illiquid until the loans have been repaid. Liquidity risks have also arisen from the direct lending of reserves to such institutions when shocks to the domestic economy led to the borrowers' inability to repay their liabilities, and impairment of the liquidity of the reserve assets.

Credit risk: Losses have arisen from the investment of reserves in high-yielding assets that were made without due regard to the credit risk associated with the issuer of the asset. Lending of reserves to domestic banks, and overseas subsidiaries of reserve management entities, has also exposed reserve management entities to credit risk.

Currency risk: Some elements of currency risk may be unavoidable with reserve asset portfolios. There have, however, been instances where large positions were taken in other countries' currencies in anticipation of favourable future changes in major cross rates, but where subsequent adverse exchange rate movements led to large losses.

Interest rate risks: Losses have arisen on reserve assets from increases in market yields that reduced the value of marketable investments below their acquisition cost. Losses have also arisen from operations involving derivative financial instruments,

including the taking of large positions, which have been subject to the effects of sharp and large adverse movements in market yields. In some instances, reserve managers may have had an inadequate understanding of all the characteristics and risks of the instruments used, and may also have lacked the technical skills required to manage exposures.

Operational Risks

Control system failure risks: There have been a few cases of outright fraud, money laundering, and theft of reserve assets that were made possible by weak or missing control procedures, inadequate skills, poor separation of duties, and collusion among reserve management staff members.

Financial error risk: Incorrect measurement of the net foreign currency position has exposed reserve management entities to large and unintended exchange rate risks, and led to large losses when exchange rate changes have been adverse. This has also occurred when risk has been measured only by reference to the currency composition of reserves directly under management by the reserve management unit, and has not included other foreign currency denominated assets and liabilities on and off the reserve management entity's balance sheet.

Financial misstatement risk: In measuring and reporting official foreign exchange reserves, some authorities have incorrectly included funds that have been lent to domestic banks, or the foreign branches of domestic banks. Similarly, placements with a reserve management entity's own foreign subsidiaries have also been incorrectly reported as reserve assets.

Loss of potential income: A failure to re-invest funds accumulating in clearing (nostro) accounts with foreign banks in a timely manner has given rise to the loss of significant amounts of potential revenue. This problem arises from inadequate procedures for monitoring and managing settlements and other cash flows, and for reconciling statements from counterparts with internal records.

Senior management needs to specify a strategic long-term portfolio that represents the best available trade-off between the different risks that the reserve management entity is facing. This is the entity's investment benchmark, which is made operational through the construction of actual benchmark portfolios that include the chosen currencies with desired weights, investment instruments with appropriate credit characteristics, and duration that reflects the desired level of interest rate risk. The benchmarks should not include transient factors or reflect short-term market expectations, but their appropriateness should be reviewed regularly. Changes to the benchmark would be triggered by changes in market structures or characteristics that alter the investment environment. The adoption of investment benchmarks is a sound practice that brings discipline to the investment process.

There are various approaches to measuring risk and controlling exposures as a result of deviations from the benchmark: Typically, such exposure is limited by the setting of

quantitative limits on the size of any deviation from the benchmark in terms of currency, duration, or credit that may be permitted by those responsible for the investment of reserves.

Currency risk is usually managed through quantitative limits on how much each individual currency, or all currencies in the benchmark taken together, may deviate from the benchmark structure. Interest rate risk can be managed by establishing benchmarks with a given duration and by limiting the actual portfolio's deviation both in terms of duration and yield curve mismatches. Credit (default) risk has traditionally been managed by placing limits on eligible issuers or counterparties based on their capital and ratings.

Methodologies can also be used to provide a quantified estimate of the maximum potential loss, with a given probability and time horizon, resulting from deviations from the benchmark portfolio.

Investment benchmarks are an important tool for assessing performance and enforcing the accountability of reserve managers: Where managers are permitted to deviate from the benchmark portfolio, performance assessment and accountability will occur through the comparison of performance of the actual portfolio with that which could have been generated by holding the benchmark portfolio. Where portfolio managers seek to replicate the benchmark, assessment of performance would also be based on comparison of actual performance versus the benchmark. In both cases, the benchmark establishes the reference point for the reserve management entity's accountability in terms of its choice of risk tolerance. A number of reserve management entities also subdivide their reserves portfolio into "tranches" according to liquidity and investment objectives and policy requirements. A "liquidity tranche" would reflect transaction and/or intervention needs based on the assessment of potential need for liquidity on demand. Such portfolios are typically invested in the most liquid and risk averse instruments. For reserves that are held to provide an additional cushion, but are less likely to be drawn upon, an "investment tranche" may be created where greater emphasis is placed on return as well as safety and liquidity. The relative size of each tranche may be determined as part of an assessment of reserves adequacy, and each tranche would have a separate benchmark reflecting the different objectives. In some countries, tranching is also used to immunize market and foreign exchange risks on the reserve balance sheet, by establishing characteristics for a particular asset portfolio that match those of a group of counterpart foreign liabilities.

The risk management framework should apply the same principles and measures to externally managed funds as it does to those managed internally: Many reserve management entities allocate part of their reserves to one or several external institutions for management. External managers may have skills that the reserve management entity lacks, or they may provide a level of safety to foreign operations that the entity is unable to achieve. They may, for example, have skills and established systems for undertaking investment activities in specialized instruments and markets for which the reserve manager does not wish to develop a capability.

Alternatively, they might provide access to new markets and activities or new investment strategies in which the reserve management entity is seeking to expand its operations. Finally, they may also assist smaller reserve management entities in managing or reducing the costs of maintaining a reserve management operation in a particular market or instruments.

Sound risk management of externally managed funds begins with the careful selection of reputable external managers, and a clear mandate for the managers to follow, together with agreed under standings of expected performance and tracking error. These arrangements should be set out clearly in separate written contracts with each appointed manager to ensure accountability.

It is also important that appointment of an external manager should not result in the reserve management entity accepting operations and risks that would not normally be considered, or are not fully understood. Appointment of external managers can also have implications for the reserve manager's choice of a custodian for its foreign securities. Generally, there should be a clear "firewall" separation between any external management and custodial functions performed by any one entity.

In principle, and in practice, there should be no difference between the risk-management and performance-monitoring framework that is applied to externally and internally managed portfolios. Most likely it will, however, be necessary to establish a separate unit, or assign a position within the middle office, to enable the reserve management entity to fully monitor the activities of the external manager, and custodians.

Risk exposures should be monitored continuously to determine whether exposures have been extended beyond acceptable limits: Monitoring is essential in identifying and limiting any cumulative losses associated with either deviations from the benchmark, any under performance of the portfolio vis-a-vis the benchmark, and increases in exposures associated with the benchmark itself due to market developments or structural changes. Since risk is inherent in both benchmark and actual portfolios, monitoring should occur regardless of whether "passive" or "active" reserve management approaches are adopted. VaR or other simulation methodologies may also form an integral part of the risk management and monitoring framework.

Reserve managers should be aware of and be able to account for potential financial losses and other consequences of the risk exposures they are prepared to accept: Active management based on expectation of movements in interest rates or exchange rates, or a choice by the reserve management entity to accept a higher risk tolerance in its benchmark portfolios, require that management is able to monitor and control any cumulative financial losses.

Implicit in such an environment is the need for specific and monitorable risk limits on the extent to which managers can deviate from the benchmark, and a reliable and timely accounting system for measuring and reporting exposures and losses. In addition to possible financial losses, other consequences can occur such as the risk of adverse signalling to participants with respect to monetary and exchange policies, damage to

the reputation of the reserve management entity, and possibly, a breach of a country's obligations in terms of the Fund's Articles of Agreement.

In this latter context, reserve management entities need to be aware of their country's obligations to collaborate with the Fund and other member countries to assure orderly exchange arrangements and consistency of policies on reserve assets in promoting better international surveillance of international liquidity. Accordingly, governance arrangements to avoid such instances might include the requirement for investment committee approval of changes to benchmark parameters or portfolio allocation policies.

The risk management framework should also address risks associated with derivative financial instruments and other foreign currency operations: As noted earlier, derivative instruments may be used as part of reserve management operations particularly in reducing risk exposures. Effective risk management requires that the reserve management entity is aware of and understands the risks and benefits of these instruments, and that staff has a sound knowledge of the underlying risks and the modalities of particular instruments used.

It also requires reserve management systems that are sufficiently developed to properly measure the values and exposures associated with operations. As the use of derivatives and other structured financial instruments grows, legal risk issues become particularly relevant. In this regard, risk mitigation could involve the use of standardized legal documentation and the performance of periodic reviews of documentation.

Beyond their reserve management operations, some reserve management entities also have other functions that involve foreign currency-denominated assets and liabilities both on-and off-balance sheet. While such operations do not fall within the definition of reserve management operations, *per se*, they may, nonetheless, represent an important part of a reserve management entity's broader mandate and involve policy choices in the utilization of the country's foreign exchange resources. The reserve management entity, therefore, may be exposed to a range of additional risks that need to be managed in a coordinated and consistent manner. In this regard, a comprehensive asset and liability risk management framework could be used to address the overall risk exposure of the reserve management entity's entire balance sheet and, in doing so, reduce the risk of loss or impairment of reserves.

To assess the risk and vulnerability of the reserve portfolio, the reserve management entity should regularly conduct stress tests to ascertain the potential effects of macroeconomic and financial variables or shocks: Stress testing can have several objectives and is often conducted using financial models ranging from simple scenario-based models, to more complex models involving sophisticated statistical and simulation techniques. One objective typically is to determine the exposure of the portfolio to changes in market factors, such as changes in exchange rates or interest rates, often by using VaR models.

A second objective concerns assessing the possible impact on the level of official foreign exchange reserves of external shocks, contingent obligations that might

materialize with such shocks, and sudden calls on reserves that may result from a reversal of short-term capital flows or closing out of the open foreign exchange position of the commercial banking system.

Stress testing in this context is qualitatively different from measuring changes in the market value of the portfolio, and can be particularly useful in the formulation of asset and liability management and strategic asset allocation policies to ensure reserve availability during times of external stress.

11

Economic Development of Foreign Trade Management

INTRODUCTION

Economists define economic development as a sustained increase in per capita income. While this definition is a good starting point, historians generally conceive economic development as a broader set of social and political changes. In the context of U.S. history, those changes include the settlement of a capacious frontier, the growth of large cities, a vast influx of immigrants, and the rapid spread of new technology. The complexity of these changes notwithstanding, the course of U.S. development can be traced through three major periods: the extensive growth of the colonial and revolutionary periods; the expansion of commerce during the first half of the nineteenth century; and the emergence of “managerial capitalism” after the Civil War.

THE COLONIAL AND REVOLUTIONARY PERIODS

The economy of colonial America grew rapidly because of sustained population growth and profitable cultivation of staple crops. Massive free and unfree immigration from Europe and Africa—coupled with a moderate climate, relatively high nutritional levels, and high fertility rates—resulted in extraordinarily rapid population growth. Between 1650 and 1770, the population of Britain’s mainland North American colonies increased from 55,000 to more than 2.2 million. Most colonists participated in a vibrant agricultural economy. High prices for wheat and tobacco—British North America’s

two main staples—encouraged the particularly rapid settlement of the middle and southern colonies in the eighteenth century. White colonists in the slave South—benefiting from slave labour and high staple prices—ranked as the wealthiest of the colonists. Rapid population growth produced substantial economic development. Because no sweeping technological or organizational advances dramatically stimulated productivity, per capita income grew quite modestly by modern standards, increasing anywhere from .3 to .6 percent per year. Yet colonists built an increasingly sophisticated commercial infrastructure that resulted in the growth of towns and cities. Such development was particularly important in New England and the Middle Colonies, where cities became major market centres. By 1770 such cities as Boston, New York, and Philadelphia provided an impressive array of mercantile services that induced future economic growth.

Government policy, both in Britain and the colonies, also aided colonial development. Unlike France or Spain, British authorities did not seek to regulate overseas migration, thus abetting the population explosion in the North American colonies. British mercantilist policies, designed to further the interests of the imperial centre, sometimes impeded colonial trade with other nations. Colonists, however, successfully evaded the most onerous regulations, while the British mercantile system enabled American merchants to export grains, lumber, livestock, and other goods. The Revolutionary War, in fact, initially had devastating economic consequences because American merchants lost many of the trading privileges they had enjoyed under the British empire.

Trade and Development

Trade is a key factor in economic development. A successful use of trade can boost a country's development. On the other hand, opening up markets to international trade may leave local producers swamped by more competitive foreign producers.

Overview

The current consensus is that trade, development, and poverty reduction are intimately linked. Sustained strong growth over longer periods is strongly associated with poverty reduction, while trade and growth are strongly linked. Countries that develop invariably increase their integration with the global economy, while export-led growth has been a key part of many countries' successful development strategies. Continents, countries and sectors that have not developed and remain largely poor have comparative advantage in three main areas:

- Natural resource exploitation, i.e. running down of natural capital such as rain forest timber;
- Low-education labour-intensive manufacturing, due to high population densities and little suitable land per person;
- Agriculture, due to low population densities and relatively large areas of suitable land per person.

Crucially for poverty reduction, the latter two at least are labour-intensive, helping to ensure that growth in these sectors will be poverty-reducing. However, low value-added, price instability and unsustainability in these commodity sectors means they should be used only temporarily as stepping stones in the path to economic development.

Agriculture

In many developing countries, agriculture employs a large proportion of the labour force, whilst food consumption accounts for a large share of household income. UNCTAD notes that this means that “even small changes in agricultural employment opportunities, or prices, can have major socio-economic effects in developing countries”. Thus whatever the development strategy a particular country adopts, the role of agriculture will often be crucial. In 1994, the agricultural sector employed over 70 % of the labour force in low-income countries, 30 % in middle-income countries, and only 4 % in high-income countries (UNCTAD 1999).

In poor countries with low population densities and enough suitable land area, which includes most countries in Africa and Latin America, agriculture is central to the economy. In poor regions and rural areas within middle-income developing countries, the concentration of poverty in rural areas of otherwise better-off developing countries makes the development of agriculture vital there. Finally, in Net Food Importing Developing Countries (NFIDCs), there is a positive link between growing agricultural exports and increases in local food production, which makes agricultural development if anything even more important, as food security and the financial stability of the government are also at stake.

In Vietnam in the 1990s, increases in production and export of coffee of 15% a year contributed to a nearly 50% rise in food production in the same period. As agricultural GDP grew 4.6% per year, rural poverty fell from 66% in 1993 to 45% in 1998 (Global Economic Prospects 2002:40).

Anderson et al. (1999) estimate annual welfare losses of \$19.8 billion for developing countries from agricultural tariffs – even after Uruguay Round reforms. This is three times the loss from OECD import restrictions on textiles and clothing.

A combination of better market access, and domestic reforms and foreign aid to enhance the ability of developing countries to take advantage of it, could have a significant impact on poverty reduction, and help to meet the Millennium Development Goals.

The largest beneficiaries of agricultural liberalization would be OECD countries themselves: welfare losses of \$62.9bn a year are estimated as resulting from the distortionary policies (Binswanger and Ernst 1999:5). Nor is the traditional objective of OECD agricultural subsidy (supporting small farmers) achieved by this system in a manner that could be characterised as efficient: most of the producer support incomes goes to better-off farmers, with the poorest 40% receiving just 8% of the support spent.

Market Access

Market Access to Developed Countries

The issue of market access to high-income countries is a thorny but crucial one. The issues fall into three main groups: first, those relating to deliberately imposed barriers to trade, such as tariffs, quotas, and tariff escalation.

Second, barriers to trade resulting from domestic and external producer support, primarily in the form of subsidies, but also including, for example, export credits.

Third, those relating to indirect barriers to trade resulting from developing countries' lack of institutional capacity to engage in the global economy and in multilateral institutions (eg the World Trade Organization) on equal terms.

Barriers to Trade

- High tariffs are imposed on agriculture: in high-income countries, the average tariff rate on agriculture is almost double the tariff for manufactures. And more than one third of the European Union's agricultural tariff lines, for instance, carry duties above 15%. Tariff peaks within agriculture occur most frequently on processed products and temperate commodities, rather than the major export crops of least developed countries (unprocessed fruits and vegetables and tropical commodities). However, many developing countries in temperate zones have the potential of competing as lower-cost producers in temperate commodities. Thus liberalization could open up new development-through-trade possibilities.
- Strong tariff escalation is typically imposed on agricultural and food products by high-income countries. This strongly discourages the development of high value added exports, and hinders diversification in particular as well as development in general. In high-income countries, tariffs on agricultural products escalate steeply, especially in the EU and Japan.
- Complex tariffs make it more difficult for developing country exporters to access industrialised-country markets because of the disadvantages developing countries face in accessing, and in their capacity to process, information. Not only are price signals distorted, they are often unclear, subject to change (for example seasonally) and difficult to interpret.
- Tariff quotas (TRQs), introduced by the Uruguay Round with the aim of securing a minimum level of market access, have performed poorly. Average fill rates have been low and declining, from 67% in 1995 to 63% in 1998, with about a quarter of TRQs filled to less than 20%. The low fill rate may reflect high in-quota rates. Overall, the UR tariffication process which produced them has not resulted in the increased market access developing countries hoped for.

Producer Support

- Support to agricultural producers remains sizable, at about five times the level of international development assistance-\$245 billion in 2000. Total support to agriculture, as defined by the OECD, reaches \$327 billion-1.3% of OECD countries' GDP. To some extent these can be justified by "multifunctionality" arguments, but it remains a priority to find means of support which effectively meet the primary objectives without the negative developmental and environmental consequences that have been seen in the past.
- The dumping of unwanted production surpluses onto the world market through export subsidies has depressed prices for many temperate agricultural commodities, with EU surpluses of exportable wheat a prime example. (Despite several CAP reforms, domestic support for wheat-as measured by OECD producer support estimates-declined only marginally from an average 52% of gross farm receipts in 1986-88, to around 48% in 1998-2000) The URAA has been relatively unsuccessful in disciplining export subsidies, with the proportion of subsidised exports in total exports increasing in many products of export interest for developing countries: for example for wheat, from 7% in 1995 to 25% in 1998. The cost to developing country production and exports is considerable, and only partially offset by the lower food prices available to NFIDC consumers. This form of transfer from high-income country taxpayers to low-income consumers is in any case rather inefficient, and the lower prices may harm production for local consumption even in NFIDCs. Agricultural reform as a whole, including the removal of export subsidies, would only result in quite small price rises for developing-country consumers.
- The counter-cyclical nature of producer support is also harmful to developing-country producers. High-income farmers are insulated from changes in world prices, making production less responsive to swings in demand. As a result, world commodity prices are more volatile, and the burden of adjustment falls disproportionately on developing-country producers.

Lack of Capacity

This includes non-tariff barriers such as food regulations and standards, which developing countries are often not (or not effectively) involved in setting, and which may be deliberately used to reduce competition from developing countries. In any case, the lack of capacity to meet implement regulations and ensure compliance with standards constitutes a barrier to trade, and must be met by increasing that capacity.

Market Access to Developing Countries

- Average applied tariffs in agriculture are higher in developing countries (although most of the very high rates, over 100%, are found in developed countries). With an increasing share of agricultural exports directed toward

other developing countries, high levels of tariff protection in the South may impede prospects for export-led growth. This may be particularly true for the export opportunities of low-income countries, which have increased export market share in agriculture.

- “Open regionalism” holds the potential to stimulate global trade and improve the efficiency of regional producers. But regional arrangements can also become a vehicle for protection, trade diversion, and unintended inefficiency. Agreements in particular between richer and poorer developing countries risk generating trade losses for the poorer ones when their imports are diverted toward the richer members whose firms are not internationally competitive. However, where regional arrangements lead to the reduction of non-tariff barriers, trade creation is likely, and the dynamic benefits of effective regional integration in terms of improved governance and regional stability are likely to outweigh diversion concerns. The World Bank suggests that key conditions to benefit from expanded trade and investment include lowering common external trade barriers, stimulating competition, reducing transaction costs, and reinforcing nondiscriminatory investment and services policies. It should be noted that the greater structural differences between North and South economies mean that North-South arrangements hold the greatest promise for economic convergence and trade creation, including in agricultural products, underlining the importance of links between South-South arrangements and northern economies.
- Trade liberalization. According to the World Bank, “most analyses suggest that unilateral reduction in barriers can produce the greatest and the quickest gains.” Some countries, such as Chile, China and Costa Rica, have undertaken domestic policy reforms. Caution must however be employed: as the case of Haiti shows, liberalization when institutions and the economy are not strong enough to face risks and opportunities can be harmful (Rodrik 2001). And while reforms may be beneficial in the long run, for example by reducing possibilities for customs corruption, in the short run they create both winners and losers. Low-income consumers, unskilled workers in sheltered industries, and previously-shielded producers may suffer in the transition period as the economy adapts to changed incentive structures. Temporary safety nets can help cushion the blow and ensure trade-led growth is pro-poor. Specific assistance to meet costs of adaptation – for example of switching to a different crop – may be appropriate.

Market Access is Vital, but not Enough

It is important to recognise that the issues facing LDCs and middle-income developing countries differ significantly. For the middle-income countries, the primary issue is market access. Many of the world’s poor live in these countries, and so market access

alone can have significant poverty-reducing effects in these countries. However, for the least-developed countries, the principal problem is not market access, but lack of production capacity to achieve new trading opportunities. This is recognised by paragraph 42 of the Doha Development Agenda: “We recognize that the integration of the LDCs into the multilateral trading system requires meaningful market access, support for the diversification of their production and export base, and trade-related technical assistance and capacity building.”

So whilst the further development of middle-income countries, and in particular the tackling of rural poverty in these countries, can be achieved most importantly through increased market access in agriculture, lower-income countries need additional help, not only to take advantage of new opportunities, but to be able to adapt to changing conditions due to the loss of preferences. This additional help must take three main forms: support for developing-country agricultural production; support for participation in trade; and support for good policies and good governance.

Support for Agricultural Production

- Support for agricultural modernization and development – investment in productive capacity in agriculture and food processing.
- Support for agricultural-related development institutions which are not trade-distorting, eg research; eg risk-management of agricultural product price fluctuations; eg diversification.

Support for Participation in Trade and the Global Economy

Cases such as Haiti’s post-1986 liberalization show that the opportunities thereby created will not be taken advantage of if macroeconomic policies, institutions, and the investment climate are not favourable. This includes

- Trade-related infrastructure: the cost of exporting must be low enough to ensure competitiveness in rapidly-expanding high-value agricultural markets where competition is stiff – such as fruits and vegetables.
- It also includes related issues that are part of the general investment climate but can be particularly important for exports, such as a weak financial sector. Here, export finance “is often a major constraint inhibiting exports in many low-income countries.”
- Other issues are more specific to exports: developing countries and their exporters may have difficulty with both the implementation of, and showing compliance with, international product standards and other multilateral agreements. Low-income developing countries need both technical and financial assistance in this area.
- Technical Assistance for negotiations is also needed to further developing-country interests in multilateral and bilateral arenas and ensure the success of future negotiations and agreements.

- Marketing of exports is also a challenge for low-income countries: product and country brands need to be built, and quality concerns met.

Given the importance of agriculture for poverty reduction, additional policies and institutional capacity are needed to ensure an effective supply response to market incentives provided by better market access. Rural infrastructure is particularly important in enabling agricultural exports in developing countries. Sufficient credit at competitive conditions is important for private sector investment in storage, transportation and marketing of agricultural products. Investment in skills and education in rural areas is needed to bolster agricultural productivity. Trade policy reforms must address any remaining anti-export bias. Efficient land policies and land tenure institutions are needed to ensure the functioning of land markets, property rights, and efficient farm structures.

World Trade Organization Negotiations

The most recent round of World Trade Organization negotiations (the Doha “Development” Round) was promoted as being directed at the interests of developing countries, addressing issues of developed country protectionism. The introduction of the (investment-related) Singapore issues together with a lack of sufficient concessions to developing countries’ interests has put the success of the negotiations in doubt.

Free Trade Zone

A free trade zone (FTZ) or export processing zone (EPZ) is one or more special areas of a country where some normal trade barriers such as tariffs and quotas are eliminated and bureaucratic requirements are lowered in hopes of attracting new business and foreign investments. It is a region where a group of countries has agreed to reduce or eliminate trade barriers. Free trade zones can be defined as labour intensive manufacturing centres that involve the import of raw materials or components and the export of factory products.

Most FTZs are located in developing countries. Bureaucracy is typically minimized by outsourcing it to the FTZ operator and corporations setting up in the zone may be given tax breaks as an additional incentive. Usually, these zones are set up in underdeveloped parts of the host country, the rationale being that the zones will attract employers and thus reduce poverty and unemployment and stimulate the area’s economy. These zones are often used by multinational corporations to set up factories to produce goods (such as clothing or shoes).

Free trade zones in Latin America date back to the early decades of the 20th century. The first free trade regulations in this region were enacted in Argentina and Uruguay in the 1920s. However, the rapid development of free trade zones across the region dates from the late 1960s and the early 1970s.

Free Trade Zones are also known as Special Economic Zone in some countries. Special Economic Zones (SEZs) have been established in many countries as testing grounds for the implementation of liberal market economy principles. SEZ are viewed

as instruments to enhance the acceptability and the credibility of the transformation policies, to attract domestic and foreign investment and also for the opening up of the economy. SEZs in India seek to promote the value addition component in exports, to generate employment as well as to mobilize foreign exchange.

Globally, many countries have initiated the Free Trade Agreements which have eventually led to a spurt in investments in the infrastructure of the Free Trade Zones (FTZs) and SEZs. A close examination of the evolution of SEZs in countries with similar economies like India eg; China, Iran, UAE and Jordan will help us to understand their success stories and further implement the factors in order to curb the SEZ bottlenecks faced by India today. The Shenzhen SEZ in China is a perfect example of a SEZ success story.

In India, the government has been proactive in the development of the Special Economic Zones (SEZs). They have formulated policies, reviewed them occasionally and have ensured that ample facilities are provided to the developers of the SEZs as well as to the companies setting up units in the SEZs. These zones are of two types i.e. 'Multiple Economic Zone' and 'Single Economic Zone'. A single SEZ can contain multiple 'specific' zones within its boundaries. One of the best examples is Sricity Multi product SEZ, part of Sricity, a satellite city being formed in between AP & TN in India.

In 1999, there were 43 million people working in about 3000 FTZs spanning 116 countries producing clothes, shoes, sneakers, electronics, and toys. The basic objectives of EPZs are to enhance foreign exchange earnings, develop export-oriented industries and to generate employment opportunities.

Criticism

The creation of special free trade zones is criticized for encouraging businesses to set up operations under the influence of other governments, and giving foreign corporations more economic liberty than is given indigenous employers who face large and sometimes insurmountable "regulatory" hurdles in developing nations. However, many countries are increasingly allowing local entrepreneurs to locate inside FTZs in order to access export-based incentives. Because the multinational corporation is able to choose between a wide range of underdeveloped or depressed nations in setting up overseas factories, and most of these countries do not have limited governments, bidding wars erupt between competing governments.

Often the government pays part of the initial cost of factory setup, loosens environmental protections and rules regarding negligence and the treatment of workers, and promises not to ask payment of taxes for the next few years. When the taxation-free years are over the corporation which set up the factory without fully assuming its costs is often able to set up operations elsewhere for less expense than the taxes to be paid, giving it leverage to take the host government to the bargaining table with more demands in order for it to continue operations in the country. Often if human rights, labour or

environmental abuses are challenged, subcontracted local entities may face consequences, but parent companies in the United States are rarely held accountable. The widespread use of free trade zones by companies such as Nike has received criticism from numerous writers such as Naomi Klein in her book *No Logo*.

Safe Trade

Safe trade is a slogan advocated by Greenpeace in its desire to “green” the World Trade Organisation and the Doha Development Round. It is designed to compete with “free trade” as a concept. Safe trade is generally seen as a single framework of rules worldwide to drastically inhibit the flow of alien organisms (e.g. Genetically modified organisms, imported animals) across the borders of ecoregions, to preserve their natural wild biodiversity. It seeks to prevent ecological disasters caused by imported organisms or untested genetic technologies, and to augment and increase local natural capital by encouraging soil remediation, precision agriculture, and local consumption of the native species, rather than imported organisms and heavy use of pesticides.

Proposal

An important achievement of safe trade advocacy is the Biosafety Protocol agreed in Montreal in January 2000. Although it relied on the weaker legal principle of Informed Consent and not the much stronger Precautionary Principle language sought by advocates, the protocol was considered by most to be a victory that could enhance both biosafety and biosecurity. Other safe trade reforms seek to advance sustainability by reducing reliance on energy subsidies and oil-based transport, and (indirectly) improves equity in economic affairs—that is, it promotes a safer political economy which is more respectful of life in general.

Safe trade is a major goal of systems of Bioregional democracy and is often advocated alongside it, e.g. by Greens. Both are also implicitly related to Community-Based Economics, as local trade in local goods with no reliance on alien organisms presents no ecological risk to its genomes, soil, or drainage basins. Accordingly, some advocates argue, local trade in any native species within an ecoregion’s borders should not be taxed at all, as it presents little or no ecological risk compared to imported goods, and so requires little or no regulation, labelling, inspection, or other expenses.

The assumption that imports carry moral hazards, and that tax, trade, tariff measures should compensate for harms done, is shared by advocates of fair trade whose programs address, in addition, more overt social justice concerns of human beings, such as the maintenance of the “human capital” of a region. Both initiatives are alternatives to free trade, which has no such controls, and generally permits and encourages free transit in goods (but not, in general, labour) across ecological and social borders.

A broader understanding of biosecurity that is emerging under threat of biological warfare, and the fear that such economically devastating events as the mad cow disease epidemic could recur, either deliberately (as an act of bioterrorism) or by accident due

to unrestricted imports, is causing some nations, notably New Zealand, to adopt relatively harsh restrictions against imported organisms. As one objective of asymmetric warfare is to cause attacks to appear initially as accidents, or blame slow responses on apparently-incompetent governments, there is some concern that spreading a virulent organism among animals would be an effective way to attack humans, damage economies, and discredit governments who are lax on biosecurity. Technologies for scanning for dangerous organisms at ports and markets are also becoming more reliable and less expensive. However, no bio-defense solution seems to be able to compete with a simple reduction of import volumes, and its corresponding reduction in risk of any accidents.

Support

Advocates point to the costs of emergency measures such as burning over one million cows suspected of having foot-and-mouth disease in the UK, smoke from which they calculated (based on dioxin levels) was to be expected to kill several hundred Britons from cancers in this generation. Safe trade, they argue, would have removed the need for any such measures, as vaccination of British beef cattle would have been possible (the burning was to prevent British exports of beef from being rejected by its trade partners, who would not have been able to tell vaccinated from infected beef), and the foot-and-mouth disease was not so dangerous to humans that it could have justified dooming so many fellow citizens to die of the dioxin-caused cancers. The burning, they argue, was justified only by bad trade rules that spread infection and advise dangerous cures that are worse than the ailment itself.

Another argument supporting safe trade rules is that there are links between primate extinction and deforestation in the regions where primates are abundant, i.e. the Amazon rainforest, African rainforest, and Sumatran rainforest. Fail to prevent devastating logging in these regions, advocates claim, and a Great Ape species will likely become extinct, causing a critical link to the human past to be permanently lost. Accordingly, preventing logs from these forests from reaching foreign markets has been a major focus of Greenpeace actions, especially in 2002.

Expansion of Commerce between the Revolution and the Civil War

Most export agriculture in the colonial and revolutionary period was confined to areas with access to water transportation. Transporting bulky crops over the nation's rutted and muddy roads was prohibitively expensive. In the period 1800–1820, private turnpike companies—which improved roads in order to collect tolls—alleviated the worst problems of overland transport, but grains and other bulky commodities remained too expensive to ship over roads. Hence proponents of economic development avidly improved rivers and built canals to extend the nation's system of waterways. The Erie Canal, completed in 1825, was by far the most important of these improvements. Built through the gently rolling landscape of upstate New York, the Erie Canal connected the Great Lakes to the Hudson River. It transformed such cities as Buffalo, Rochester, and

Syracuse into major transport, mercantile, and manufacturing hubs. Canal systems in Ohio and other Midwestern states extended the reach of the Erie system, providing farmers throughout the Middle West with an all water connection to New York City. The steamboat, meanwhile, facilitated trade on the Mississippi and Ohio River systems, providing yet another avenue of commerce.

Steam power soon had an even larger impact when applied to overland carriage. Areas in which water transportation was either unavailable or prohibitively expensive turned to the railroad. First built in the late 1820s, railroads rapidly spread across the country. By 1860, Americans had built approximately 30,000 miles of track, with four separate roads penetrating the Appalachian Mountain barrier. Railroads not only accelerated the growth of cities on the eastern seaboard, but also transformed Pittsburgh, Chicago, and other Midwestern cities into major marketing and manufacturing centres.

Although historians have discarded the view that the railroad single handedly accounted for American industrialization, it nevertheless contributed to the dramatic expansion of commerce in everyday life. In the North and Midwest, the expansion of the railroad network created a self reinforcing cycle of growth. As transportation improved, farmers produced more grains, dairy products, and other produce for urban centres. With more cash to spend on consumer goods, farm families demanded textiles, shoes, furniture, and other manufactured products. Manufacturers and merchants, responding to the larger agrarian market, increased their output and improved productivity. Firms producing such goods as ready made apparel, hats and caps, and boots and shoes, for example, increased productivity through specialization and greater division of labour. Incremental technological advances—usually the result of tinkering by rather ordinary mechanics—further increased manufacturing productivity.

The South lagged far behind in inventive activity and other measures of economic development. Its slave based economy, to be sure, produced enormous profits for many planters and farmers. The southern economy did especially well in the 1850s, when the price of cotton, tobacco, and other staples soared. Yet in almost every other measure of development—urbanization rates, inventive activity, manufacturing output, population growth—the North far outstripped the South. As the Civil War approached, the regional disparities grew. In 1860, the value of manufactured goods produced in New York City alone exceeded the combined production of the eleven southern states that would form the Confederacy. Underlying the South's failure was a lack of adequate demand to spur industrialization. Plantation slavery restrained population growth among free farmers and an unequal distribution of rural income undermined the earnings and consumption habits of farm families that had provided northern entrepreneurs with lucrative markets. However profitable to southern planters, slavery seriously impeded southern industrial development.

The limitations of southern development notwithstanding, the U.S. economy grew impressively between 1800 and 1860. Historians debate to what extent federal and state government action contributed to this remarkable economic expansion. State

governments frequently intervened in the economy in the early nineteenth century, especially when, as in New York and Pennsylvania, they operated their own canal systems. Although New York's Erie Canal succeeded, most other state owned canals failed. State and federal governments provided engineering expertise, land grants, and other subsidies to the railroads, but private investors supplied most of the capital. The most important government intervention was enforcing a stable set of property rights (including a patent system) that encouraged economic expansion and technological innovation. State governments made it easier for entrepreneurs to obtain corporate charters for banks, factories, and other enterprises, while the federal courts protected interstate commerce and used law to validate corporate charters and contracts. With a strong legal foundation in place, national markets and national enterprises came to dominate the economy.

MANAGERIAL CAPITALISM AND THE MODERN ECONOMY

The post Civil War expansion of the railroad network—including the construction of the first transcontinental lines—enabled entrepreneurs to establish national corporations. While a few large firms had flourished earlier during the market revolution, especially in the New England textile industry, these firms paled compared to the corporate giants of the late nineteenth and early twentieth century. By 1910, Standard Oil, U.S. Steel, Armour and Co., and the American Tobacco Company were among the firms with assets in excess of \$200 million. Another round of technological change soon added such newer businesses as Ford Motor Company and General Motors to the list of industrial giants. These enterprises tended to be capital intensive, using their large size to forge substantial economies of scale. Technological advances became even more important. Many large firms organized research divisions and departments that institutionalized technological change.

Borrowing from the experience of the railroads, these large industrial firms also developed complex administrative bureaucracies heavily stocked with middle managers and upper echelon executives. Entrepreneurs such as John D. Rockefeller and Andrew Carnegie used these new administrative forms to ascertain precise production costs, enabling them to identify and then to eliminate expensive bottlenecks. Rockefeller's Standard Oil Company, in particular, ruthlessly cut production costs, enabling the company to buy out or destroy less efficient competitors. By 1890, Standard Oil—which owned 20,000 oil wells, 4,000 miles of pipeline, and 5,000 specialty railroad cars—controlled more than 90 percent of the American oil and kerosene market. Whereas large corporations tended to dominate industries with relatively homogenous output (oil, steel, and tobacco), smaller firms thrived in sectors that produced specialty goods, ranging from complex machinery to such consumer goods as furniture. By the early twentieth century, America's combination of large and small firms encouraged the birth

of new industries, including automobiles and electrical manufacturing. Mass production made the marketing and distribution of consumer goods extraordinarily important. To insure a constant demand for their products, businesses invested heavily in advertising, first in print publications, later on radio and television. Innovative retailers devised new ways of selling both basic necessities and luxury goods. Department stores, fashionable retail districts, and, eventually, supermarkets as well as shopping centres and malls indelibly shaped the American landscape. By the early twentieth century, a number of firms expanded the scale of consumer borrowing (especially through instalment plans) that allowed a broad range of families to purchase automobiles and other consumer durables.

By 1900, the United States possessed the basic elements of a developed economy: growing per capita income, rapidly evolving technologies, increasingly efficient industrial producers, and sophisticated marketing and distribution channels. Although certain regions of the country—most notably Appalachia and other parts of the South—continued to lag, rapid economic development made the United States the world's largest economy for much of the twentieth century and laid the material foundations for what publisher Henry Luce called "The American Century."

Because economic development is so complicated, scholars continue to debate the ultimate source of U.S. success. Some stress the availability of such abundant natural resources as fertile soils and rich supplies of iron, coal, and oil. Others underscore the emergence of a national, integrated market that encompassed most of North America. While political and geographic barriers hindered development elsewhere, these scholars argue, U.S. businesses grew in tandem with an expanding home market. Finally, many scholars have emphasized the importance of distinctive American cultural and political attitudes, including a devotion to private property and free enterprise, and a corresponding suspicion of economic regulation that might hinder business success.

Despite widespread support of economic development, many Americans have nevertheless feared that the growing power of big business might subvert democracy. These fears, coupled with deep swings in the business cycle, have created important political movements, especially in the Populist, Progressive and New Deal Eras, that raised important questions about economic development. Should the federal government encourage certain industries with tariffs and other subsidies? Should state and federal governments dissolve or regulate corporations that threaten to become monopolies? How should government mediate conflicts among labour, business, and consumers? What kinds and levels of taxation best encourage economic development while also promoting other social goods? These and similar questions relating to economic development remained divisive as the twentieth century ended.

The advent of the desktop computer in the 1980s and the Internet and World Wide Web in the 1990s opened a new era of U.S. economic development. The "information age" also exemplified many of the same characteristics of earlier changes, including the rapid adaptation of new technologies; a complex mix of both large corporation

(including giants such as Microsoft and Intel) and smaller, more flexible firms; and attitudes and regulations generally favourable to free enterprise and hostile to government regulation.

Yet despite these links to the past, the new economy also raised new questions. Would the rapid transmission of information make global markets the driving force of continued growth? Would the new information technologies transform consumer culture? How would new technologies affect the managerial structure of “old economy” stalwarts such as the steel, automotive, and chemical industries? While these issues remained unresolved, the rapid emergence of “new economy” industries did once again illustrate the self reinforcing nature of economic development. The earlier success of the U.S. economy created deep pools of capital, high levels of per capita income, and a strong commitment to research and development that allowed new enterprises to flourish. Clearly the history of economic development remains unfinished, as it continues to influence society, politics, and culture.

Balance of Trade

The balance of trade (or *net exports*, sometimes symbolized as *NX*) is the difference between the monetary value of exports and imports of output in an economy over a certain period. It is the relationship between a nation’s imports and exports. A favourable balance of trade is known as a trade surplus and consists of exporting more than is imported; an unfavourable balance of trade is known as a trade deficit or, informally, a trade gap. The balance of trade is sometimes divided into a goods and a services balance.

Definition

The balance of trade forms part of the current account, which includes other transactions such as income from the international investment position as well as international aid. If the current account is in surplus, the country’s net international asset position increases correspondingly. Equally, a deficit decreases the net international asset position. The trade balance is identical to the difference between a country’s output and its domestic demand (the difference between what goods a country produces and how many goods it buys from abroad; this does not include money re-spent on foreign stocks, nor does it factor the concept of importing goods to produce for the domestic market).

Measuring the balance of trade can be problematic because of problems with recording and collecting data. As an illustration of this problem, when official data for all the world’s countries are added up, exports exceed imports by a few percent; it appears the world is running a positive balance of trade with itself. This cannot be true, because all transactions involve an equal credit or debit in the account of each nation. The discrepancy is widely believed to be explained by transactions intended to launder money or evade taxes, smuggling and other visibility problems. However, especially for developed countries, accuracy is likely.

Factors that can affect the balance of trade figures include:

- Prices of goods manufactured at home (influenced by the responsiveness of supply)
- Exchange rates regarded in 1933
- Trade agreements or barriers
- Offset agreements
- Other tax, tariff and trade measures
- Business cycle at home or abroad.

The balance of trade is likely to differ across the business cycle. In export led growth (such as oil and early industrial goods), the balance of trade will improve during an economic expansion. However, with domestic demand led growth (as in the United States and Australia) the trade balance will worsen at the same stage in the business cycle.

Since the mid 1980s, United States has had a growing deficit in tradeable goods, especially with Asian nations (China and Japan) which now hold large sums of U.S debt that has funded the consumption. The U.S. has a trade surplus with nations such as Australia and Canada. The issue of trade deficits can be complex. Trade deficits generated in tradeable goods such as manufactured goods or software may impact domestic employment to different degrees than trade deficits in raw materials.

Economies such as Canada, Japan, and Germany which have savings surpluses, typically run trade surpluses. China, a high growth economy, has tended to run trade surpluses. A higher savings rate generally corresponds to a trade surplus. Correspondingly, the United States with its lower savings rate has tended to run high trade deficits, especially with Asian nations.

Views on Economic Impact

Economists are sometimes divided on the economic impact of the trade deficit.

Conditions where Trade Deficits may be Considered Harmful

Those who ignore the effects of long run trade deficits may be confusing David Ricardo's principle of comparative advantage with Adam Smith's principle of absolute advantage, specifically ignoring the latter. The economist Paul Craig Roberts notes that the comparative advantage principles developed by David Ricardo do not hold where the factors of production are internationally mobile. Global labour arbitrage, a phenomenon described by economist Stephen S. Roach, where one country exploits the cheap labour of another, would be a case of absolute advantage that is not mutually beneficial.

Since the stagflation of the 1970s, the U.S. economy has been characterized by slower GDP growth. In 1985, the U.S. began its growing trade deficit with China. Over the long run, nations with trade surpluses tend also to have a savings surplus while the U.S. has been plagued by persistently lower savings rates than its trading partners which

tend to have trade surpluses with the U.S. Germany, France, Japan, and Canada have maintained higher savings rates than the U.S. over the long run. Some economists believe that GDP and employment can be dragged down by an over-large deficit over the long run. Wealth-producing primary sector jobs in the U.S. such as those in manufacturing and computer software have often been replaced by much lower paying wealth-consuming jobs such those in retail and government in the service sector when the economy recovered from recessions. Some economists contend that the U.S. is borrowing to fund consumption of imports while accumulating unsustainable amounts of debt.

In 2006, the primary economic concerns centred around: high national debt (\$9 trillion), high non-bank corporate debt (\$9 trillion), high mortgage debt (\$9 trillion), high financial institution debt (\$12 trillion), high unfunded Medicare liability (\$30 trillion), high unfunded Social Security liability (\$12 trillion), high external debt (amount owed to foreign lenders) and a serious deterioration in the United States net international investment position (NIIP) (-24% of GDP), high trade deficits, and a rise in illegal immigration. These issues have raised concerns among economists and unfunded liabilities were mentioned as a serious problem facing the United States in the President's 2006 State of the Union address. On June 26 2009, Jeff Immelt, the CEO of General Electric, called for the United States to increase its manufacturing base employment to 20% of the workforce, commenting that the U.S. has outsourced too much in some areas and can no longer rely on the financial sector and consumer spending to drive demand.

Conditions where Trade Deficits may not be Harmful

When economists contend trade deficits are not harmful, they generally refer to explanations of comparative advantage. Buyers in the receiving country send the money back. A firm in America sends dollars for Brazilian sugarcane, and the Brazilian receivers use the money to buy stock in an American company. This may lead to profits leaving the U.S. however as Americans may forfeit control. Although this is a form of capital account reinvestment, it may not be a liability on anyone in America. Such payments to foreigners have intergenerational effects: by shifting the consumption schedule over time, some generations may gain and others lose. However, a trade deficit may incur consumption in the future if it is financed by profitable domestic investment, in excess of that paid on the net foreign debts. Similarly, an excess on the current account shifts consumption to future generations, unless it raises the value of the currency, deterring foreign investment.

However, trade inequalities are not natural given differences in productivity and consumption preferences. Trade deficits have often been associated with international competitiveness. Trade surpluses have been associated with policies that skew a country's activity towards externalities, resulting in lower standards. An example of an economy which has had a positive balance of trade was Japan in the 1990s. Milton Friedman

argued that trade deficits are not necessarily important as high exports raise the value of the currency, reducing aforementioned exports, and vice versa for imports, thus naturally removing trade deficits not due to investment. Milton Friedman's son, David D. Friedman, shares this view and cites the comparative advantage concepts of David Ricardo.

Milton Friedman on Trade Deficits

In the 1980s, Milton Friedman, the Nobel Prize-winning economist and father of Monetarism, contended that some of the concerns of trade deficits are unfair criticisms in an attempt to push macroeconomic policies favourable to exporting industries. In the late 1970s and early 1980s, the U.S. had experienced high inflation and Friedman's policy positions tended to defend the stronger dollar at that time. He stated his belief that these trade deficits were not necessarily harmful to the economy at the time since the currency comes back to the country (country A sells to country B, country B sells to country C who buys from country A, but the trade deficit only includes A and B). However, it may be in one form or another including the possible tradeoff of foreign control of assets. In his view, the "worst case scenario" of the currency never returning to the country of origin was actually the best possible outcome: the country actually purchased its goods by exchanging them for pieces of cheaply-made paper. As Friedman put it, this would be the same result as if the exporting country burned the dollars it earned, never returning it to market circulation. This position is a more refined version of the theorem first discovered by David Hume. Hume argued that England could not permanently gain from exports, because hoarding gold (i.e., currency) would make gold more plentiful in England; therefore, the prices of English goods would rise, making them less attractive exports and making foreign goods more attractive imports. In this way, countries' trade balances would balance out.

Friedman believed that deficits would be corrected by free markets as floating currency rates rise or fall with time to encourage or discourage imports in favour of the exports, reversing again in favour of imports as the currency gains strength. In the real world, a potential difficulty is that currency markets are far from a free market, with government and central banks being major players, and this is unlikely to change within the foreseeable future. Nevertheless, recent developments have shown that the global economy is undergoing a fundamental shift. For many years the U.S. has bore world has lent and sold. However, as Friedman predicted, this paradigm appears to be changing.

As of October 2007, the U.S. dollar weakened against the euro, British pound, and many other currencies. For instance, the euro hit \$1.42 in October 2007, the strongest it has been since its birth in 1999. Against this backdrop, American exporters are finding quite favourable overseas markets for their products and U.S. consumers are responding to their general housing slowdown by slowing their spending. Furthermore, China, the Middle East, central Europe and Africa are absorbing more of the world's imports which in the end may result in a world economy that is more evenly balanced. All of

this could well add up to a major readjustment of the U.S. trade deficit, which as a percentage of GDP, began in 1991. Friedman and other economists have pointed out that a large trade deficit (importation of goods) signals that the country's currency is strong and desirable. To Friedman, a trade deficit simply meant that consumers had opportunity to purchase and enjoy more goods at lower prices; conversely, a trade surplus implied that a country was exporting goods its own citizens did not get to consume or enjoy, while paying high prices for the goods they actually received.

Perhaps most significantly, Friedman contended strongly that the structure of the balance of payments was misleading. In an interview with Charlie Rose, he stated that "on the books" the US is a net borrower of funds, using those funds to pay for goods and services. He pointed to the income receipts and payments showing that the US pays almost the same amount as it receives: thus, U.S. citizens are paying lower prices than foreigners for capital assets to exchange roughly the same amount of income. The reasons why the U.S. (and UK) appear to earn a higher rate of return on their foreign assets than they pay on their foreign liabilities are not clearly understood. An important contributing factor is that the U.S. has investment primarily in stocks abroad, while foreigners have invested heavily in debt instruments, such as U.S. government bonds. Other reports contend that U.S. net foreign income has deteriorated, and appears set to stay in deficit in the future. Friedman presented his analysis of the balance of trade in *Free to Choose*, widely considered his most significant popular work.

Warren Buffett on Trade Deficits

The successful American businessman and investor Warren Buffett was quoted in the Associated Press (January 20, 2006) as saying "The U.S trade deficit is a bigger threat to the domestic economy than either the federal budget deficit or consumer debt and could lead to political turmoil... Right now, the rest of the world owns \$3 trillion more of us than we own of them."

John Maynard Keynes on the Balance of Trade

In the last few years of his life, John Maynard Keynes was much preoccupied with the question of balance in international trade. He was the leader of the British delegation to the United Nations Monetary and Financial Conference in 1944 that established the Bretton Woods system of international currency management.

He was the principal author of a proposal—the so-called Keynes Plan—for an International Clearing Union. The two governing principles of the plan were that the problem of settling outstanding balances should be solved by 'creating' additional 'international money', and that debtor and creditor should be treated almost alike as disturbers of equilibrium. In the event, though, the plans were rejected, in part because "*American opinion was naturally reluctant to accept the principal of equality of treatment so novel in debtor-creditor relationships*". His view, supported by many economists and commentators at the time, was that creditor nations may be just as

responsible as debtor nations for disequilibrium in exchanges and that both should be under an obligation to bring trade back into a state of balance. Failure for them to do so could have serious consequences. In the words of Geoffrey Crowther, then editor of *The Economist*, “*If the economic relationships between nations are not, by one means or another, brought fairly close to balance, then there is no set of financial arrangements that can rescue the world from the impoverishing results of chaos.*”

These ideas were informed by events prior to the Great Depression when—in the opinion of Keynes and others—international lending, primarily by the United States, exceeded the capacity of sound investment and so got diverted into non-productive and speculative uses, which in turn invited default and a sudden stop to the process of lending. Influenced by Keynes, economics texts in the immediate postwar period put a significant emphasis on balance in trade. For example, the second edition of the popular introductory textbook, *An Outline of Money*, devoted the last three of its ten chapters to questions of foreign exchange management and in particular the ‘problem of balance’. However, in more recent years, since the end of the Bretton Woods system in 1971, with the increasing influence of Monetarist schools of thought in the 1980s, and particularly in the face of large sustained trade imbalances, these concerns—and particularly concerns about the destabilising affects of large trade surpluses—have largely disappeared from mainstream economics discourse and Keynes’ insights have slipped from view, they are receiving some attention again in the wake of the financial crisis of 2007–2009.

Physical Balance of Trade

Monetary balance of trade is different from physical balance of trade (which is expressed in amount of raw materials). Developed countries usually import a lot of primary raw materials from developing countries at low prices. Often, these materials are then converted into finished products, and a significant amount of value is added. Although for instance the EU (as well as many other developed countries) has a balanced monetary balance of trade, its physical trade balance (especially with developing countries) is negative, meaning that a lot less material is exported rather than imported.

United States Trade Deficit

The United States of America has held a trade deficit starting late in the 1960s. It was this very deficit that forced the United States in 1971 off the gold standard. Its trade deficit has been increasing at a large rate since 1997 and increased by 49.8 billion dollars between 2005 and 2006, setting a record high of 817.3 billion dollars, up from 767.5 billion dollars the previous year. It is worth noting on the graph that the deficit slackened during recessions and grew during periods of expansion. Also of note, many economists calculate trade deficits and/or current account deficits as a percentage of GDP. The US last had a trade surplus in 1991, a recession year. Every year there has been a major reduction in economic growth, it is followed by a reduction in the US

trade deficit. The investor Warren Buffett has proposed a tool called Import Certificates as a solution to the United States' problem.

International Trade Law

International trade law includes the appropriate rules and customs for handling trade between countries or between private companies across borders. Over the past twenty years, it has become one of the fastest growing areas of international law.

Overview

International trade law should be distinguished from the broader field of international economic law.

The latter could be said to encompass not only WTO law, but also law governing the international monetary system and currency regulation, as well as the law of international development.

The body of rules for transnational trade in the 21st century derives from medieval commercial laws called the *lex mercatoria* and *lex maritima* — respectively, “the law for merchants on land” and “the law for merchants on sea.” Modern trade law (extending beyond bilateral treaties) began shortly after the Second World War, with the negotiation of a multilateral treaty to deal with trade in goods: the General Agreement on Tariffs and Trade (GATT).

International trade law is based on theories of economic liberalism developed in Europe and later the United States from the 18th century onwards.

World Trade Organization

In 1995, the World Trade Organization, a formal international organization to regulate trade, was established. It is the most important development in the history of international trade law. The purposes and structure of the organization is governed by the *Agreement Establishing The World Trade Organization*, also known as the “Marrakesh Agreement”. It does not specify the actual rules that govern international trade in specific areas. These are found in separate treaties, annexed to the Marrakesh Agreement.

Trade in Goods

The GATT has been the backbone of international trade law throughout most of the twentieth century. It contains rules relating to “unfair” trading practices — dumping and subsidies.

Trade and Human Rights

The World Trade Organisation Trade Related Intellectual Property Rights (TRIPS) agreement required signatory nations to raise intellectual property rights (also known as intellectual monopoly privileges. This arguably has had a negative impact on access to essential medicines in some nations.

Dispute Settlement

Since there are no international governing judges (2004) the means of dispute resolution is determined by jurisdiction. Each individual country hears cases that are brought before them. Governments choose to be party to a dispute. And private citizens determine jurisdiction by the Forum Clause in their contract.

Besides forum, another factor in international disputes is the rate of exchange. With currency fluctuation ascending and descending over years, a lack of Commerce Clause can jeopardize trade between parties when one party becomes unjustly enriched through natural market fluctuations. By listing the rate of exchange expected over the contract life, parties can provide for changes in the market through reassessment of contract or division of exchange rate fluctuations.

Trade Facilitation

Trade facilitation looks at how procedures and controls governing the movement of goods across national borders can be improved to reduce associated cost burdens and maximise efficiency while safeguarding legitimate regulatory objectives. Business costs may be a direct function of collecting information and submitting declarations or an indirect consequence of border checks in the form of delays and associated time penalties, forgone business opportunities and reduced competitiveness. Understanding and use of the term “trade facilitation” varies in the literature and amongst practitioners. “Trade facilitation” is largely used by institutions which seek to improve the regulatory interface between government bodies and traders at national borders.

The WTO, in an online training package, once defined trade facilitation as: “The simplification and harmonisation of international trade procedures” where trade procedures are the “activities, practices and formalities involved in collecting, presenting, communicating and processing data required for the movement of goods in international trade”. In defining the term, many trade facilitation proponents will also make reference to trade finance and the procedures applicable for making payments (e.g. via a commercial banks). For example UN/CEFACT defines trade facilitation as “the simplification, standardization and harmonisation of procedures and associated information flows required to move goods from seller to buyer and to make payment”.

Occasionally, the term trade facilitation is extended to address a wider agenda in economic development and trade to include: the improvement of transport infrastructure, the removal of government corruption, the modernization of customs administration, the removal of other non-tariff trade barriers, as well as export marketing and promotion.

Examples of Regulatory Activity in International Trade

Fiscal: Collection of customs duties, excise duties and other indirect taxes; payment mechanisms

Safety and security: Security and anti smuggling controls; dangerous goods; vehicle checks; immigration and visa formalities

Environment and health: Phytosanitary, veterinary and hygiene controls; health and safety measures; CITES controls; ships' waste

Consumer protection: Product testing; labelling; conformity checks with marketing standards (e.g. fruit and vegetables)

Trade policy: Administration of quota restrictions; export refunds

Topics and Issues in Trade Facilitation

Trade facilitation has its intellectual roots in the fields of logistics and supply chain management. Trade facilitation looks at operational improvements at the interface between business and government and associated transaction costs. Trade facilitation has become a key feature in supply chain security and customs modernisation programmes. Within the context of economic development it has also come to prominence in the Doha Development Round. However, it is an equally prominent feature in unilateral and bilateral initiatives that seek to improve the trade environment and enhance business competitiveness. Reference to trade facilitation is sometimes also made in the context of "better regulation". Some organisations promoting trade facilitation will emphasise the cutting of red tape in international trade as their main objective. Propagated ideas and concepts to reforming trade and customs procedures generally resonate around the following themes:

- Simple rules and procedures
- Avoidance of duplication
- Memoranda of Understanding (MoUs)
- Alignment of procedures and adherence to international conventions
- Trade consultation
- Transparent and operable rules and procedures
- Accommodation of business practices
- Operational flexibility
- Public-service standards and performance measures
- Mechanisms for corrections and appeals
- Fair and consistent enforcement
- Proportionality of legislation and control to risk
- Time-release measures
- Risk management and trader authorisations
- Standardisation of documents and electronic data requirements
- Automation
- International electronic exchange of trade data
- Single Window System

Fair Trade Debate

Fair trade's increasing popularity has drawn criticism from both ends of the political spectrum. Different arguments are used by those who favour and by those who oppose

fair trade, or feel that more strict standards and higher fair trade prices are needed. These arguments can be divided into different broad categories:

- The price distortion argument, advocated by the Adam Smith Institute, and The Economist magazine calling fair trade a “misguided attempt to make up for market failures” encouraging market inefficiencies and overproduction.
- The creation of insider/outsider markets argument, defended by the Institute of Economic Affairs. This argument does not explicitly criticize the ideals behind fair trade, but rather current certification, production and pricing systems.
- The mainstreaming argument, defended by French author Christian Jacquiau, which criticizes segments of the fair trade movement for working within the current system (i.e. partnerships with mass retailers, multinational corporations etc.) rather than establishing a new fairer, fully autonomous trading system.

It has also been argued that the approach of the FairTrade system is too rooted in a Northern consumerist view of justice which Southern producers do not participate in setting. “A key issue is therefore to make explicit who possesses the power to define the terms of Fairtrade, that is who possesses the power to determine the need of an ethic in the first instance, and subsequently command a particular ethical vision as the truth.”

A more participatory and multi-stakeholder approach to auditing might improve the quality of the process. In addition, strategic use of labeling may help embarrass (or encourage) major suppliers into changing their practices. They may make transparent corporate vulnerabilities that activists can exploit. Or they may encourage ordinary people to get involved with broader projects of social change. Some academics have argued that rather than opposing free trade and fairtrade approaches to supporting smallholder agricultural development, these “are in fact complementary... smallholders can often benefit from both approaches.”

Price Distortion Argument

Criticism: Similar to other farm subsidies, fair trade attempts to set a price floor for a good that is in many cases above the market price and therefore encourages, as fair trade opponents claim, existing producers to produce more and new producers to enter the market, leading to excess supply. Through the laws of supply and demand, excess supply can lead to lower prices in the non-Fair Trade market.

In 2003, Cato Institute’s vice president for research Brink Lindsey referred to fair trade as a “well intentioned, interventionist scheme...doomed to end in failure.” Fair trade, according to Lindsey, is a misguided attempt to make up for market failures in which one flawed pricing structure is replaced with another. Lindsey’s comments echo the main criticisms of Fair Trade, claiming that it “leads fair trade producers to increase production.”

While benefiting a number of Fair Trade producers over the short run, fair trade critics worry about the impact on long run development and economic growth. An

article in *The Economist* claimed that the reason coffee prices are so low on the world markets is that there is too much production, and that by encouraging even more supply of coffee, fair trade makes the world price fall further making the majority of coffee producers worse off.

Henry Hazlitt explains the economic results of artificial price supports for agricultural goods stating, “but if an attempt is made to keep up the price of an agricultural commodity and no artificial restriction of output is imposed, unsold surpluses of the overpriced commodity continue to pile up until the market for that product finally collapses...”

Response: Several academics, including Hayes, Becchetti and Rosati, identify two counterarguments to this criticism.

1. First, in many cases the exchange between producers and intermediaries does not occur in a competitive framework. In such case the market price is a distortion because it does not reflect the productivity of producers but their lower market power.
2. Second, the price distortion argument does not take into account the principles of product differentiation. Coffee, for example, cannot be compared to other commodities such as oil: there is not one single type of coffee but instead many different brands that are differentiated from one another in terms of quality, blends, packaging, and now also “social responsibility” features. Consumer demand and taste defines what different market prices are acceptable for each of these products. In this sense, fair trade can be considered as a market-driven innovation in the food industry that creates a new range of products for which a growing segment of consumers are willing to pay more based on environmental and social responsibility claims.

And finally, beyond these elements, it is also important to also take into account all the other non-price related potential benefits of the fair trade value chain in terms of technical assistance, democratization of markets through increasing consumer power, crop diversification programs, etc.

IMPACT ON CONVENTIONAL PRODUCERS DEBATE

Criticism: In addition to the market not receiving the correct signals, critics such as *The Economist* claim fair trade can make non fair trade growers worse off. Critics claim that when the signal does not get through, more farmers move into coffee, “This then drives down the price of non-Fairtrade coffee even further, making non-Fairtrade farmers poorer.

Fairtrade does not address the basic problem, argues Tim Harford, author of “The Undercover Economist” (2005), which is that too much coffee is being produced in the first place. Instead, it could even encourage more production.” A further strain of this criticism is that Fairtrade principally buys from certain geographical regions, namely

Latin America. Whilst Latin America is broadly speaking poor, compared to other parts of the world, namely Africa, it is substantially richer on average. For example, one of the poorest Latin American countries, Bolivia, still ranks about half way on the UN Global Development Index. Therefore distortions in price which benefit Latin American farmers, are disproportionately disadvantageous for African farmers, for whom similar price changes carry much greater weight.

Response: The Fairtrade Foundation responds to the argument by claiming that there is absolutely no evidence that fair trade makes other farmers worse off – in fact, “cooperatives often bring competition into local markets for coffee for the first time, forcing other traders to match their prices so that more farmers benefit”.

Nicolas Eberhart, researcher at the French NGO Agronomes et Vétérinaires sans frontières, argued in his 2005 impact study “Etude d’impact du commerce équitable sur les organisations et familles paysannes et leurs territoires dans la filière café des Yungas de Bolivie” how rather than hurting conventional producers, fair trade can also have the opposite impact by raising local market prices. A case study of Bolivian coffee Fair Trade producers in 2005 concluded that Fair trade certification has had in the Yungas a positive impact on local coffee prices, thus economically benefiting all coffee producers (Fair trade certified or not). This increase was most notably attributed to the fact that many non-fair trade producers benefited from fair trade funded infrastructure and programs: processing, credit facilities, quality improvement, crop diversification, conversion to organic production etc. In the case of Bolivia, conventional producers also benefited from greater political influence as a result of reforms pushed through by stronger, NGO supported, fair trade producer organizations.

SCOPE OF FAIR TRADE DEBATE

Criticism: As *The Economist* magazine pointed out in its December 7th 2006 issue, “another objection to Fairtrade is that certification is predicated on political assumptions about the best way to organise labour. In particular, for some commodities (including coffee) certification is available only to cooperatives of small producers, who are deemed to be most likely to give workers a fair deal when deciding how to spend the Fairtrade premium. Coffee plantations or large family firms cannot be certified.

Response: On the other hand, fair trade supporters, such as Oxford University professor Alex Nicholls, argue that the program was created in the first place to address market failures affecting small farmers’ organizations. As opposed to plantations for example, perfect market information, perfect access to markets and credit, and the ability to switch production techniques and outputs in response to market information are fundamental assumptions which are fallacious in the context of small farmers’ organizations in the developing world. Fair trade is seen as an attempt to address these market failures by providing to the most in need a stable price for their crop, business support, access to premium Northern markets and better general trading conditions.

Retail Pricing Debate

Criticism: In an article published in its December 7, 2006 issue, *The Economist* magazine criticized fair trade for being “an inefficient way to get money to poor producers”. According to the magazine, retailers add their own enormous mark-ups to Fairtrade products and mislead consumers into thinking that all of the premium they are paying is passed on. *The Economist* estimated that only 10% of the premium paid for Fairtrade coffee in a coffee bar trickles down to the producer. Fairtrade coffee, like the organic produce sold in supermarkets, is used by retailers as a means of identifying price-insensitive consumers who will pay more. *Response:* Fair trade proponents such as the Fairtrade Foundation defend fair trade by arguing that under EU and US competition laws, it is illegal for them to intervene in price fixing discussions between retailers and importers. The Fairtrade Foundation counters *The Economist's* assertions by citing a range of studies conducted in 2005 concluding that the majority of retailers do not increase their profit margins on fairtrade products, for fear of losing market share in the growing market. There is at least one instance of a retailer actually reducing its profit margins in an effort to boost sales and improve its corporate image: in December 2006, British retailer Sainsbury's announced it would begin to offer only Fairtrade bananas at the same price that Sainsbury's had previously sold non-Fairtrade bananas. According to the Fairtrade Foundation, fair trade remains the “only guarantee on the market that producers are receiving an agreed and stable price and money to invest in their communities.” The amounts paid to the producer organizations are public, published in the Fairtrade standards and are checked by independent audit organizations. As explained on the Foundation's website, “it is ultimately down to all of us as individual shoppers to decide whether they feel products represent good value for them,” knowing how much commodity producers are paid for under conventional or fair trade.

Trade Justice and Fair Trade Debate

Criticism: Segments of the trade justice movement have also criticized fair trade in the past years for allegedly focusing too much on individual small producer groups while stopping short of advocating immediate trade policy changes that would have a larger impact on disadvantaged producers' lives. French author and RFI correspondent Jean-Pierre Boris championed this view in his 2005 book *Commerce inéquitable*.

The Mainstreaming of Fair Trade Debate

Some believe the fair trade system is not radical enough. French author Christian Jacquiau, in his book *Les coulisses du commerce équitable*, calls for stricter fair trade standards and criticizes the fair trade movement for working within the current system (i.e. partnerships with mass retailers, multinational corporations etc.) rather than establishing a new fairer, fully autonomous trading system. Jacquiau is also a staunch supporter of significantly higher fair trade prices in order to maximize the impact, as most producers only sell a portion of their crop under fair trade terms.

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